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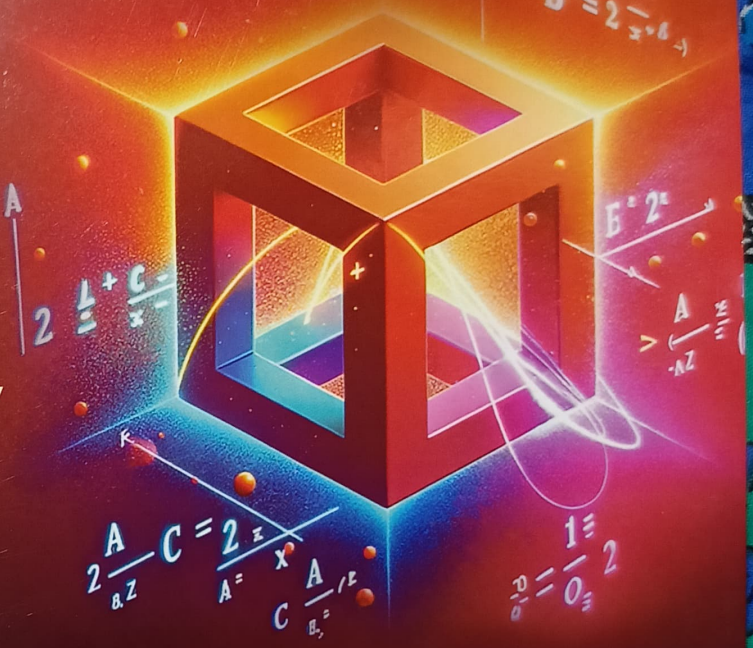
PHYSICS WALLAH

ARJUNA JEE



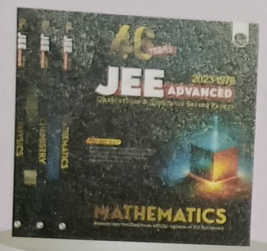
Class-XI

- Limits and Derivatives
- Statistics
- Introduction to Three Dimensional Geometry
- Probability
- Properties of Triangle
- Height & Distances

MATHEMATICS

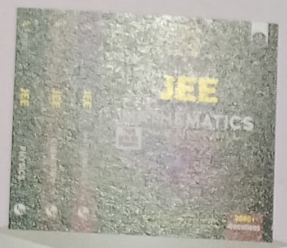
Module 4

PYQs Series



46 YEARS PYQs

- Chapterwise & topicwise solved papers of past 46 years
- Weightage analysis of Past 10 Years JEE Advanced Papers
- Question-wise difficulty level analysis in real-time with Correct (C), Wrong(W) and Unattempted (UA) questions response tagging provided by IIT-JEE



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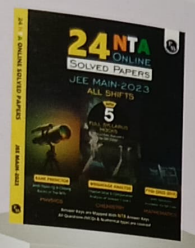

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- PYQs with solutions from 2019 to 2022 via QR code
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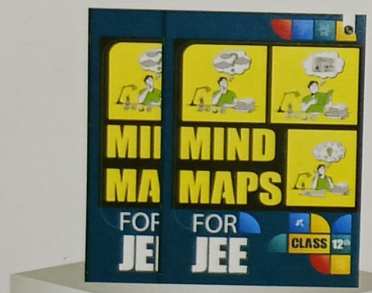

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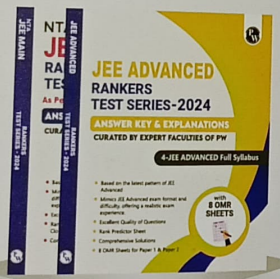

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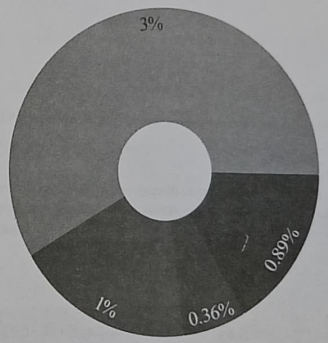
Class-XI

Chapter Wise Weightage (in %)

- Limits and Derivatives (0%)
- Statistics (3%)
- Introduction to Three Dimensional Geometry (0%)
- Probability (1%)
- Properties of Triangle (0.36%)
- Height & Distances (0.89%)



*On the basis of Past 5 Years JEE Main Papers

MATHEMATICS 4 MODULE

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PREFACE

A highly skilled professional team of Physics Wallah (PW) works arduously to ensure that the students receive the best content for the **JEE** exam.

From the beginning, the whole content team comprising Associate faculties, DTP operators, Proofreaders and others are involved in shaping the material to their best knowledge and experience to produce powerful content for the students.

Faculties have adopted a new style of presenting the content in easy-to-understand language and have provided the team with their guidance and supervision throughout the creation of this Study Material.

Physics Wallah (PW) strongly believes in conceptual and fun-based learning. PW provides highly exam-oriented content to bring quality and clarity to the students.

A plethora of **JEE Study Material** is available in the market but PW professionals are continuously working to provide the supreme Study Material for our **JEE** students.

This Study Material adopts a multi-faceted approach to master and understanding the concepts by having a rich diversity of questions asked in the examination and equip the students with the knowledge for the competitive exam.

The main objective of the study material is to provide a large number of quality problems with varying cognitive levels to facilitate the teaching-learning of concepts that are presented through the book.

It has become popular among aspirants because of its easy-to-understand language and clear illustrations with diagrams, mnemonics, flow charts and tables.

Students can benefit themselves by attempting the exercise given in this Study Material for the self-assessment and also mastering the basic techniques of problem-solving.

The questions are strictly designed in accordance with the exam relevant topics that helps to develop examination temperament within the students.

Mastering the Physics Wallah (PW) study material curated by the PW team, the students can easily qualify for the exam with a Top Rank in the **JEE**.

In each chapter, for better understanding, questions have been classified according to the latest syllabus of **JEE**

- The nature and diversity of the questions help students to ace the examination
- Quality questions to strengthen the concept of the topic at the zenith level

BOOK FEATURES

- Detailed theory with Train Your Brain and Concept Application Questions
- "Aarambh" Section for Solved Miscellaneous Examples
- School Level Problems for School Going Students
- "Prarambh" Section for Topicwise, "Prabal" Section for JEE Main Level and "Parikshit" Section for Advance Level MCQs covered in exercises
- Match the Column, Comprehension Based, Numerical Type and Multicorrect questions covered for complete practice
- JEE Past Year Questions
- Elaborated Solutions

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CHAPTER 16

Limits and Derivatives

INTRODUCTION

Limit Definition

The concept of limit is used to discuss the behaviour of a function close to a certain point.

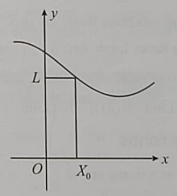
Meaning of $x \to x_0$: Let x is variable and x_0 is a constant, when x takes the values very close to x_0 we say that x approaches to x_0, but x is not equal to x_0. We denotes it by $x \to x_0$

If x approach x_0 from right hand side then we denote it by $x \to x_0^+$ and if x approach x_0 from left hand side then we denote it by $x \to x_0^-$

For e.g.: $x \to 5^- \Rightarrow x$ can take values like 4.999, 4.9999, 4.999999 ... similarly when $x \to 5^+ \Rightarrow x$ can take values like 5.01, 5.0001, 5.0000001 ...

Informal Definition of Limits

Let $f(x)$ be defined on an open interval about x_0, except possibly at x_0 itself. If $f(x)$ gets arbitrarily close to L for all x sufficiently close to x_0, we say that function approaches the limit L as x approaches x_0, and we write $\lim\limits_{x \to x_0} f(x) = L$.



This definition is "informal" because phrases like arbitrarily close and sufficiently close are imprecise, their meaning depends on the context.

The definition is clear enough and enables us to recognize and evaluate limits of specific function.

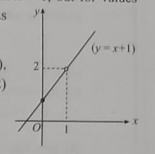
e.g., $f(x) = \dfrac{x^2 - 1}{x - 1}$

Clearly the function is not defined at $x = 1$, but for values close to $x = 1$ the function can be written as

$f(x) = x + 1$

As x approaches 1 (written as $x \to 1$), $f(x)$ approaches the value 2 (i.e., $f(x) \to 2$) we write this as

$\lim\limits_{x \to 1} f(x) = 2$



Left Hand Limit and Right Hand Limit

LHL: It is the value to which function $f(x)$ approaches as x approaches to x_0 from left hand side and we write it as $\lim\limits_{x \to x_0^-} f(x)$.

RHL: It is the value to which function $f(x)$ approaches as x approaches to x_0 from right hand side and we write it as $\lim\limits_{x \to x_0^+} f(x)$.

Existence of limit: $\lim\limits_{x \to x_0} f(x)$ is said to be exist and equal to L if LHL = RHL = L (a finite quantity).

i.e. $\lim\limits_{x \to x_0^-} f(x) = \lim\limits_{x \to x_0^+} f(x) = L$ (a finite quantity).

e.g. $f(x) = [x]$ (greatest integer function)

For any integer n.

$\lim\limits_{x \to n^-} f(x) = n - 1$, $\lim\limits_{x \to n^+} f(x) = n$

In such cases we say that $\lim\limits_{x \to n} f(x)$ does not exist.

Train Your Brain

Example 1: Consider the function $f(x) = x^3$. Let us try to find the limit of this function at $x = 1$?

Sol. We tabulate the value of f(x) at x near 1. This is given in the Table.

x	$f(x)$
0.9	0.729
0.99	0.970299
0.999	0.997002999
1.001	1.003003001
1.01	1.030301
1.1	1.331

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From this table, we deduce that value of $f(x)$ at $x = 1$ should be greater than 0.997002999 and less than 1.003003001 assuming nothing dramatic happens between

$x = 0.999$ and 1.001. It is reasonable to assume that the value of the $f(x)$ at $x = 1$ as dictated by the numbers to the left of 1 is 1 , i.e.,

$$\lim_{x \to 1^-} f(x) = 1.$$

Similarly, when x approaches 1 from the right, $f(x)$ should be taking value 1, i.e.,

$$\lim_{x \to 1^+} f(x) = 1.$$

Hence, it is likely that the left hand limit of $f(x)$ and the right hand limit of f(x) are both equal to 1. Thus,

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} f(x) = \lim_{x \to 1} f(x) = 1.$$

We note that as x approaches 1 from either right or left, the graph of the function $f(x) = x^3$ approaches the point (1,1).

We observe, again, that the value of the function at $x = 1$ also happens to be equal to 1.

Example 2: We want to find $\lim_{x \to 0} f(x)$, where

$$f(x) = \begin{cases} x-2, & x < 0 \\ 0, & x = 0 \\ x+2, & x > 0 \end{cases}$$

Sol. We make a table of x near 0 with $f(x)$. Observe that for negative values of x we need to evaluate $x - 2$ and for positive values, we need to evaluate $x + 2$.

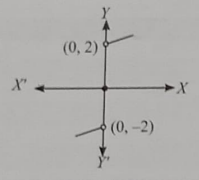
x	–0.1	–0.01	–0.001	0.001	0.01	0.1
$f(x)$	–2.1	–2.01	–2.001	2.001	2.01	2.1

From the first three entries of the Table, we deduce that the value of the function is decreasing to –2 and hence.

$$\lim_{x \to 0^-} f(x) = -2$$

From the last three entires of the table we deduce that the value of the function is increasing from 2 and hence

$$\lim_{x \to 0^+} f(x) = 2$$



Since the left and right hand limits at 0 do not coincide, we say that the limit of the function at 0 does not exist. Graph of this function is given in the figure. Here, we remark that the value of the function at $x = 0$ is well defined and is, indeed, equal to 0, but the limit of the function at $x = 0$ is not even defined.

Example 3: Evaluate the following limits

$$\lim_{x \to 2} (x + 2)$$

Sol. $x + 2$ being a polynomial in x, its limit as $x \to 2$ is given by $\lim_{x \to 2} (x + 2) = 2 + 2 = 4$

Concept Application

1. Find $\lim_{x \to 1} f(x)$, where

$$f(x) = \begin{cases} x+2 & x \neq 1 \\ 0 & x = 1 \end{cases}$$

FUNDAMENTAL THEOREMS ON LIMITS

Let b and c be real numbers, let n be a positive integer, and let f and g be functions with the following limits. $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = K$ (where L, K are finite)

1. **Sum Rule:** $\lim_{x \to c} [f(x) + g(x)] = L + K$
2. **Difference Rule:** $\lim_{x \to c} [f(x) - g(x)] = L - K$
3. **Product Rule:** $\lim_{x \to c} [f(x)\, g(x)] = LK$
4. **Quotient Rule:** $\lim_{x \to c} \frac{f(x)}{g(x)} = \frac{L}{K}$, provided $K \neq 0$
5. **Constant Multiplication Rule:** $\lim_{x \to c} [b\, f(x)] = bL$
6. **Composition Rule:** $\lim_{x \to c} f(g(x)) = f\left(\lim_{x \to c} g(x)\right) = f(K)$, provided f is continuous at $x = m$.

For example $\lim_{x \to c} \ln(f(x)) = \ln\left(\lim_{x \to a} f(x)\right) = \ln L (L > 0)$.

Indeterminate Forms

If on putting $x = a$ in $f(x)$, any one of $\frac{0}{0}, \frac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$, ∞^0, 0^0, 1^∞ form is obtained, then the limit has an indeterminate form. All the above forms are interchangeable, i.e. we can change one form to other by suitable substitutions etc.

In such cases $\lim_{x \to a} f(x)$ may exist.

Consider $f(x) = \frac{x^2 - 4}{x - 2}$.

Here $\lim_{x \to 2} x^2 - 4 = 0$ and $\lim_{x \to 2} x - 2 = 0$

$\therefore \lim_{x \to 2} f(x)$ has an indeterminate form of the type $\frac{0}{0}$.

$\lim_{x \to \infty} \frac{\ln x}{x}$ has an indeterminate form of type $\frac{\infty}{\infty}$.

$\lim_{x \to 0} (1 + x)^{1/x}$ is an indeterminate form of the type 1^∞

LIMITS OF POLYNOMIALS AND RATIONAL FUNCTIONS

A function f is said to be a polynomial function of degree n when $f(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$, where a_i's are real numbers such that $a_n \neq 0$ for some natural number n.

We know that $\lim_{x \to a} x = a$. Hence

$$\lim_{x \to a} x^2 = \lim_{x \to a} (x.x) = \lim_{x \to a} x \cdot \lim_{x \to a} x = a.a = a^2$$

An easy exercise in induction on tells us that

$$\lim_{x \to a} x^n = a^n$$

Now, let $f(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$ be a polynomial function. Thinking of each of $a_0, a_1 x, a_2 x^2, \ldots, a_n x^n$ as a function, we have

$$\lim_{x \to a} f(x) = \lim_{x \to a} [a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n]$$

$$= \lim_{x \to a} a_0 + \lim_{x \to a} a_1 x + \lim_{x \to a} a_2 x^2 + \ldots + \lim_{x \to a} a_n x^n$$

$$= a_0 + a_1 \lim_{x \to a} x + a_2 \lim_{x \to a} x^2 + \ldots + a_n \lim_{x \to a} x^n$$

$$= a_0 + a_1 a + a_2 a^2 + \ldots + a_n a^n = f(a)$$

RATIONALISATION METHOD

We can rationalise the irrational expression in numerator or denominator or in both to remove the indeterminacy.

Train Your Brain

Example 4: Which of the following limits are in indeterminant forms. Also indicate the form

(i) $\lim_{x \to 0} \frac{1}{x}$ (ii) $\lim_{x \to 1} \frac{1 - x}{1 - x^2}$

(iii) $\lim_{x \to 0} x \ln x$ (iv) $\lim_{x \to 0} \left(\frac{1}{x} - \frac{1}{x^2}\right)$

(v) $\lim_{x \to 0} (\sin x)^x$ (vi) $\lim_{x \to 0} (\ln x)^x$

(vii) $\lim_{x \to 0} (1 + \sin x)^{\frac{1}{x}}$ (viii) $\lim_{x \to 0} (1)^{\frac{1}{x}}$

Sol. (i) No (ii) Yes $\frac{0}{0}$ form

(iii) Yes $0 \times \infty$ form (iv) Yes $(\infty - \infty)$ form

(v) Yes, 0^0 form (vi) Yes ∞^0 form

(vii) Yes 1^∞ form (viii) No

Example 5: Evaluate

(i) $\lim_{x \to 3} \frac{x^2 - 2x - 3}{x^2 - 4x + 3}$

(ii) $\lim_{x \to 2} \left[\frac{1}{x - 2} - \frac{2(2x - 3)}{x^3 - 3x^2 + 2x}\right]$

Sol. (i) $\lim_{x \to 3} \frac{x^2 - 2x - 3}{x^2 - 4x + 3} = \lim_{x \to 3} \frac{(x-3)(x+1)}{(x-3)(x-1)} = 2$

(ii) $\lim_{x \to 2} \left[\frac{1}{x - 2} - \frac{2(2x - 3)}{x^3 - 3x^2 + 2x}\right]$

$= \lim_{x \to 2} \left[\frac{1}{x - 2} - \frac{2(2x - 3)}{x(x-1)(x-2)}\right]$

$= \lim_{x \to 2} \left[\frac{x(x-1) - 2(2x-3)}{x(x-1)(x-2)}\right]$

$= \lim_{x \to 2} \left[\frac{x^2 - 5x + 6}{x(x-1)(x-2)}\right]$

$= \lim_{x \to 2} \left[\frac{(x-2)(x-3)}{x(x-1)(x-2)}\right] = \lim_{x \to 2} \left[\frac{x-3}{x(x-1)}\right] = -\frac{1}{2}$

Example 6: Evaluate:

(i) $\lim_{x \to 1} \frac{4 - \sqrt{15x + 1}}{2 - \sqrt{3x + 1}}$

(ii) $\lim_{x \to 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

Sol. (i) $\lim_{x \to 1} \frac{4 - \sqrt{15x + 1}}{2 - \sqrt{3x + 1}}$

$= \lim_{x \to 1} \frac{(4 - \sqrt{15x+1})(2 + \sqrt{3x+1})(4 + \sqrt{15x+1})}{(2 - \sqrt{3x+1})(4 + \sqrt{15x+1})(2 + \sqrt{3x+1})}$

$= \lim_{x \to 1} \frac{(15 - 15x)}{(3 - 3x)} \times \frac{2 + \sqrt{3x+1}}{4 + \sqrt{15x+1}} = \frac{5}{2}$

(ii) The form of the given limit is $\frac{0}{0}$ when $x \to 0$.

Rationalising the numerator, we get

$$\lim_{x \to 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$$

$$= \lim_{x \to 0} \left[\frac{\sqrt{1+x} - \sqrt{1-x}}{x} \times \frac{\sqrt{1+x} + \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right]$$

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$$= \lim_{x\to 0}\left[\frac{(1+x)-(1-x)}{x(\sqrt{1+x}+\sqrt{1-x})}\right]$$

$$= \lim_{x\to 0}\left[\frac{2x}{x(\sqrt{1+x}+\sqrt{1-x})}\right]$$

$$= \lim_{x\to 0}\left[\frac{2}{\sqrt{1+x}+\sqrt{1-x}}\right] = \frac{2}{2} = 1$$

Example 7: Evaluate $\lim_{x\to 2}\frac{x-2}{\sqrt{x^2-4}+\sqrt{x-2}}$

Sol. $L = \lim_{x\to 2}\frac{x-2}{\sqrt{x^2-4}+\sqrt{x-2}}\cdot\frac{(\sqrt{x^2-4}-\sqrt{x-2})}{(\sqrt{x^2-4}-\sqrt{x-2})}$

$$= \lim_{x\to 2}\frac{(x-2)(\sqrt{x^2-4}-\sqrt{x-2})}{(x^2-4)-(x-2)}$$

$$= \lim_{x\to 2}\frac{(x-2)(\sqrt{x^2-4}-\sqrt{x-2})}{x^2-x-2}$$

$$= \lim_{x\to 2}\frac{(x-2)(\sqrt{x^2-4}-\sqrt{x-2})}{(x-2)(x+1)}$$

$$= \lim_{x\to 2}\frac{\sqrt{x^2-4}-\sqrt{x-2}}{x+1} = 0.$$

Concept Application

2. $\lim_{x\to 0}\frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{x} =$

(a) -1 (b) 1
(c) 2 (d) -2

3. $\lim_{x\to 3}\left\{\frac{x-3}{\sqrt{x-2}-\sqrt{4-x}}\right\} =$

(a) 1 (b) 2
(c) -1 (d) -2

4. The value of the limit of $\frac{x^3-8}{x^2-4}$ as x tends to 2 is

(a) 3 (b) $\frac{3}{2}$
(c) 1 (d) 0

5. The value of the limit of $\frac{x^3-x^2-18}{x-3}$ as x tends to 3 is

(a) 3 (b) 9
(c) 18 (d) 21

THEOREM

For any positive integer n,

$$\lim_{x\to a}\frac{x^n-a^n}{x-a} = na^{n-1}.$$

Remark: The expression in the above theorem for the limit is true even if n is any rational number and a is positive.

Proof Dividing (x^n-a^n) by $(x-a)$, we see that

$x^n-a^n = (x-a)(x^{n-1}+x^{n-2}a+x^{n-3}a^2+\ldots+xa^{n-2}+a^{n-1})$

Thus, $\lim_{x\to a}\frac{x^n-a^n}{x-a} = \lim_{x\to a}(x^{n-1}+x^{n-2}a+x^{n-3}a^2+\ldots+xa^{n-2}+a^{n-1})$

$= a^{n-1}+a\,a^{n-2}+\ldots+a^{n-2}(a)+a^{n-1}$

$= a^{n-1}+a^{n-1}+\ldots+a^{n-1}+a^{n-1}$ (n terms)

$= na^{n-1}$

Standard Limits

(a) (i) $\lim_{x\to 0}\frac{\sin x}{x} = \lim_{x\to 0}\frac{\tan x}{x} = 1$

[Where x is measured in radians]

(ii) $\lim_{x\to 0}\frac{\tan^{-1}x}{x} = \lim_{x\to 0}\frac{\sin^{-1}x}{x} = 1$

(iii) $\lim_{x\to 0}\frac{e^x-1}{x} = 1$; $\lim_{x\to 0}\frac{a^x-1}{x}$

$= \log_e a = \ln a,\ a>0$

(iv) $\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1$

(v) $\lim_{x\to a}\frac{x^n-a^n}{x-a} = na^{n-1}$

(b) If $f(x)\to 0$, when $x\to a$, then

(i) $\lim_{x\to a}\frac{\sin f(x)}{f(x)} = 1$

(ii) $\lim_{x\to a}\cos f(x) = 1$

(iii) $\lim_{x\to a}\frac{\tan f(x)}{f(x)} = 1$

(iv) $\lim_{x\to a}\frac{e^{f(x)}-1}{f(x)} = 1$

(v) $\lim_{x\to a}\frac{b^{f(x)}-1}{f(x)} = \ln b,\ (b>0)$

(vi) $\lim_{x\to a}\frac{\ln(1+f(x))}{f(x)} = 1$

(c) $\lim_{x\to a} f(x) = A>0$ and $\lim_{x\to a}\phi(x) = B$(a finite quantity), then $\lim_{x\to a}[f(x)]^{\phi(x)} = A^B$.

Remark:

(i) '0' doesn't means exact zero but represent a value approaching towards zero similarly to '1' and infinity.

(ii) $\infty+\infty=\infty$

(iii) $\infty\times\infty=\infty$

(iv) $(a/\infty)=0$ if a is finite

(v) $\frac{a}{0}$ is not defined for any $a\in R$.

(vi) $a\,b=0$, if & only if $a=0$ or $b=0$ and a & b are finite.

(vii) $\lim_{x\to 0}\frac{x}{x}$ is an indeterminate form whereas $\lim_{x\to 0}\frac{[x^2]}{x^2}$ is not an indeterminate form (where [.] represents greatest integer function)

Students may remember these forms along with the prefix 'tending to'

i.e. $\frac{\text{tending to zero}}{\text{tending to zero}}$ is an indeterminate form whereas

$\frac{\text{exactly zero}}{\text{tending to zero}}$ is not an indeterminate form, its value is zero.

Similarly (tending to one)$^{\text{tending to }\infty}$ is indeterminate form whereas (exactly one)$^{\text{tending to }\infty}$ is not an indeterminate form, its value is one.

To evaluate a limit, we must always put the value where 'x' is approaching to in the function. If we get a determinate form, then that value becomes the limit otherwise if an indeterminate form comes, we have to remove the indeterminancy, once the indeterminancy is removed the limit can be evaluated by putting the value of x, where it is approaching.

Train Your Brain

Example 8: Evaluate:

$$\lim_{x\to 0}\frac{\tan x-\sin x}{x^3}$$

Sol. $\lim_{x\to 0}\frac{\tan x-\sin x}{x^3} = \lim_{x\to 0}\frac{\tan x(1-\cos x)}{x^3}$

$$= \lim_{x\to 0}\frac{\tan x\,.\,2\sin^2\frac{x}{2}}{x^3}$$

$$= \lim_{x\to 0}\frac{1}{2}\cdot\frac{\tan x}{x}\cdot\left(\frac{\sin\frac{x}{2}}{\frac{x}{2}}\right)^2 = \frac{1}{2}.$$

Example 9: Evaluate: $\lim_{x\to 0}\frac{\sin 2x}{\sin 3x}$

Sol. $\lim_{x\to 0}\frac{\sin 2x}{\sin 3x} = \lim_{x\to 0}\left[\frac{\sin 2x}{2x}\cdot\frac{2x}{3x}\cdot\frac{3x}{\sin 3x}\right]$

$$= \left[\lim_{2x\to 0}\frac{\sin 2x}{2x}\right]\cdot\frac{2}{3}\cdot\left[\lim_{3x\to 0}\frac{3x}{\sin 3x}\right]$$

$$= 1\cdot\frac{2}{3}\times\left[\lim_{3x\to 0}\frac{\sin 3x}{3x}\right] = \frac{2}{3}\times 1 = \frac{2}{3}$$

Example 10: Show that $\lim_{x\to 0}\frac{\sin x(1-\cos x)}{x^3} = \frac{1}{2}$

Sol. $\lim_{x\to 0}\frac{\sin x(1-\cos x)}{x^3} = \lim_{x\to 0}\frac{\sin x\cdot 2\sin^2\frac{x}{2}}{x^3}$

$$= \lim_{x\to 0}4\cos\frac{x}{2}\cdot\frac{\sin^3\frac{x}{2}}{x^3} = \lim_{x\to 0}4(1)\left(\frac{1}{2}\right)^3 = \frac{1}{2}$$

Concept Application

6. If $\lim_{x\to 2}\frac{x^n-2^n}{x-2} = 80$, where n is a positive integer, then $n =$

(a) 3 (b) 5
(c) 2 (d) 1

7. $\lim_{x\to 0}\frac{1-\cos 2x}{x} =$

(a) 0 (b) 1
(c) 2 (d) 4

8. $\lim_{x\to 0}\frac{x^3\cot x}{1-\cos x} =$

(a) 0 (b) 1
(c) 2 (d) -2

DERIVATIVES

Introduction

The rate of change of one quantity with respect to some another quantity has a great importance. For example, the rate of change of displacement of a particle with respect to time is called its velocity and the rate of change of velocity w.r.t time is called its acceleration.

The rate of change of a quantity 'y' with respect to another quantity 'x' is called the derivative or differential coefficient of y with respect to x.

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Derivative at a Point

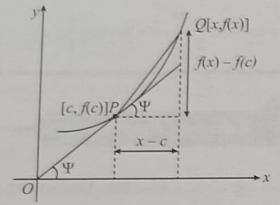
The derivative of a function at a point $x = a$ is defined by $f'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$ (provided the limit exists and is finite)

The above definition of derivative is also called derivative by first principle.

1. **Geometrical meaning of derivatives at a point:** Consider the curve $y = f(x)$. Let $f(x)$ be differentiable at $x = c$. Let $P(c, f(c))$ be a point on the curve and $Q(x, f(x))$ be a neighbouring point on the curve. Then,

   Slope of the chord $PQ = \frac{f(x)-f(c)}{x-c}$. Taking limit as $Q \to P$, *i.e.*, $x \to c$,

   we get $\lim_{Q\to P}(\text{slope of the chord } PQ) = \lim_{x\to c}\frac{f(x)-f(c)}{x-c}$ ...(*i*)



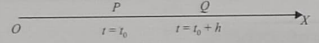
   As $Q \to P$, chord PQ becomes tangent at P.

   Therefore from (*i*), we have

   Slope of the tangent at $P = \lim_{x\to c}\frac{f(x)-f(c)}{x-c} = \left(\frac{df(x)}{dx}\right)_{x=c}$.

   Note: Thus, the derivatives of a function at a point $x = c$ is the slope of the tangent to curve, $y = f(x)$ at point $(c, f(c))$

2. **Physical interpretation at a point:** Let a particle moves in a straight line OX starting from O towards X. Clearly, the position of the particle at any instant would depend upon the time elapsed. In other words, the distance of the particle from O will be some function f of time t.



   Let at any time $t = t_0$, the particle be at P and after a further time h, it is at Q so that $OP = f(t_0)$ and $OQ = f(t_0 + h)$. Hence, the average speed of the particle during the journey from P to Q is $\frac{PQ}{h}$, *i.e.*, $\frac{f(t_0+h)-f(t_0)}{h} = f(t_0, h)$. Taking the limit of $f(t_0, h)$ as $h \to 0$, we get its instantaneous speed to be $\lim_{h\to 0}\frac{f(t_0+h)-f(t_0)}{h}$, which is simply $f'(t_0)$. Thus, if $f(t)$ gives the distance of a moving particle at time t, then the derivative of f at $t = t_0$ represents the instantaneous speed of the particle at the point P, *i.e.*, at time $t = t_0$.

Important Tips

$\frac{dy}{dx}$ is $\frac{d}{dx}(y)$ in which $\frac{d}{dx}$ is simply a symbol of operation and not 'd' divided by dx.

Finding derivatives from first principles

Consider $y = f(x)$ be a defined function of x. Let us change x by a small quantity, say h.

Then corresponding change in y can be obtained from $f(x + h) - f(x)$.

From the definition of dy/dx, given in last section, we get:

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{\text{change in } y}{\text{change in } x} = \lim_{h\to 0}\frac{f(x+h)-f(x)}{x+h-x}$$

$$= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

So finding derivative by first principles means using the following formula:

$$\frac{dy}{dx} = f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

Illustrating the Concepts

(*i*) Evaluate the derivative of $f(x) = x^n$ w.r.t x from definition of derivative. Hence find the derivative of $\sqrt{x}, 1/x, 1/\sqrt{x}, 1/x^p$ *wrt* x.

Using definition of derivative,

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$f'(x) = \lim_{h\to 0}\frac{(x+h)^n - x^n}{h} = \lim_{h\to 0}\frac{(x+h)^n - x^n}{(x+h)-x}$$

$$= \lim_{t\to x}\frac{t^n - x^n}{t-x} \quad [\text{putting } t = x+h]$$

$$= nx^{n-1}\left[\text{using } \lim_{x\to a}\frac{x^n - a^n}{x-a} = na^{n-1}\right]$$

Taking $n = \frac{1}{2}, \frac{d}{dx}\sqrt{x} = \frac{-1}{2\sqrt{x}}$

taking $n = -1, \frac{d}{dx}\left(\frac{1}{x}\right) = \frac{-1}{x^2}$

taking $n = \frac{-1}{2}, \frac{d}{dx}\left(\frac{1}{\sqrt{x}}\right) = \frac{-1}{2x\sqrt{x}}$

taking $n = -p, \frac{d}{dx}\left(\frac{1}{x^p}\right) = \frac{-p}{x^{p+1}}$

(*ii*) Find the derivative of $\sin x$ w.r.t x from first principles.

Let $f(x) = \sin x$

Using the definition of derivative,

$$f'(x) = \lim_{h\to 0}\frac{\sin(x+h)-\sin x}{h}$$

$$= \lim_{h\to 0}\frac{2\cos\left(x+\frac{h}{2}\right)\sin\frac{h}{2}}{2\frac{h}{2}} = \cos x$$

Theorems for Differentiation

Let $f(x)$, $g(x)$ and $u(x)$ be differentiable functions

1. If at all points of a certain interval. $f'(x) = 0$, then the function $f(x)$ has a constant value within this interval.
2. **Chain rule**
   (*i*) **Case I:** If y is a function of u and u is a function of x, then derivative of y with respect to x is $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$ or $y = f(u) \Rightarrow \frac{dy}{dx} = f'(u)\frac{du}{dx}$

   (*ii*) **Case II:** If y and x both are expressed in terms of t, y and x both are differentiable with respect to t then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.
3. **Sum and difference rule:** Using linear property $\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx}(f(x)) \pm \frac{d}{dx}(g(x))$
4. **Product rule:**
   (*i*) $\frac{d}{dx}(f(x)g(x)) = f(x)\frac{d}{dx}g(x) + g(x)\frac{d}{dx}f(x)$

   (*ii*) $\frac{d}{dx}(u.v.w) = u.v.\frac{dw}{dx} + v.w.\frac{du}{dx} + u.w.\frac{dv}{dx}$
5. **Scalar multiple rule:** $\frac{d}{dx}(k f(x)) = k\frac{d}{dx}f(x)$
6. **Quotient rule:**

   $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\frac{d}{dx}(f(x)) - f(x)\frac{d}{dx}(g(x))}{(g(x))^2}$, provided $g(x) \neq 0$

Some Standard Differentiation

1. **Differentiation of algebraic functions**
   (*i*) $\frac{d}{dx}x^n = nx^{n-1}, x \in R, n \in R, x > 0$

   (*ii*) $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$

   (*iii*) $\frac{d}{dx}\left(\frac{1}{x^n}\right) = -\frac{n}{x^{n+1}}$
2. **Differentiation of trigonometric functions:** The following formulae can be applied directly while differentiating trigonometric functions

   (*i*) $\frac{d}{dx}\sin x = \cos x$

   (*ii*) $\frac{d}{dx}\cos x = -\sin x$

   (*iii*) $\frac{d}{dx}\tan x = \sec^2 x$

   (*iv*) $\frac{d}{dx}\sec x = \sec x \tan x$

   (*v*) $\frac{d}{dx}\text{cosec } x = -\text{cosec } x \cot x$

   (*vi*) $\frac{d}{dx}\cot x = -\text{cosec}^2 x$

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Example 11: Differentiate $y = \sin(x^2)$

Sol. If $y = \sin(x^2)$, then the outer function is the sine function and the inner function is the squaring function, so the Chain Rule gives

$$\frac{dy}{dx} = \underbrace{\cos}_{\substack{\text{derivative}\\ \text{of outer}\\ \text{function}}}\underbrace{(x^2)}_{\substack{\text{evaluated}\\ \text{at inner}\\ \text{function}}}\underbrace{2x}_{\substack{\text{derivative}\\ \text{of outer}\\ \text{function}}} = 2x\cos x^2$$

Example 12: Differentiate $g(x) = \cos x^2 + 5\left(\frac{3}{x}+4\right)^6$

Sol. $\frac{dg}{dx} = \frac{d}{dx}\cos x^2 + 5\frac{d}{dx}(3x^{-1}+4)^6$

$= -\sin x^2 \frac{d}{dx}(x^2) + 5\left[6\left(3x^{-1}+4\right)^5 \frac{d}{dx}\left(3x^{-1}+4\right)\right]$

$= (-\sin x^2)(2x) + 30(3x^{-1}+4)^5(-3x^{-2})$

$= -2x\sin x^2 - 90x^{-2}(3x^{-1}+4)^5$

Example 13: Differentiate $g(x) = \sqrt[4]{\frac{x}{1-3x}}$.

Sol. Write $g(x) = \left(\frac{x}{1-3x}\right)^{1/4} = u^{1/4}$ where $u = \frac{x}{1-3x}$ is the inner function and $u^{1/4}$ is the outer function.

Then, $g'(x) = (u^{1/4})'u'(x) = \frac{1}{4}u^{-3/4}u'(x)$ and we have

$$g'(x) = \frac{1}{4}\left(\frac{x}{1-3x}\right)^{-3/4}\left(\frac{x}{1-3x}\right)'$$

$$= \frac{1}{4}\left(\frac{x}{1-3x}\right)^{-3/4}\left[\frac{(1-3x)(1)-x(-3)}{(1-3x)^2}\right]$$

$$= \frac{1}{4}\left(\frac{x}{1-3x}\right)^{-3/4}\left[\frac{1}{(1-3x)^2}\right] = \frac{1}{4x^{3/4}(1-3x)^{5/4}}.$$

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Example 14: If $y = x^2 \cos(\log x)$, find $\frac{dy}{dx}$

Sol. $\frac{dy}{dx} = \frac{d}{dx}\{x^2 \cos(\log x)\}$

$= x^2 \frac{d\{\cos(\log x)\}}{dx} + \cos(\log x)\frac{d(x^2)}{dx}$

$= x^2 \cdot \frac{d\cos(\log x)}{d\log x} \cdot \frac{d(\log x)}{dx} + \cos(\log x) \cdot 2x$

$= x^2(-\sin\log x)\frac{1}{x} + 2x\cos(\log x)$

$= -x\sin(\log x) + 2x\cos(\log x)$

Concept Application

9. Differentiate $(\sec x - 1)(\sec x + 1)$

10. Differentiate $\frac{3x+4}{5x^2-7x+9}$

11. Differentiate $\frac{a+b\sin x}{c+d\cos x}$

12. Differentiate $x^2 \sin x + \cos 2x$

AARAMBH (SOLVED EXAMPLES)

1. $\lim_{x\to 0}\frac{(1-\cos 2x)(3+\cos x)}{x\tan 4x}$ is equal to

Sol. $\lim_{x\to 0}\frac{(1-\cos 2x)(3+\cos x)}{x\tan 4x}$

$= \lim_{x\to 0}\frac{2\sin^2 x}{x^2}\frac{(3+\cos x)}{\frac{\sin 4x}{4x}}\frac{\cos 4x}{4} = 2$

2. $\lim_{x\to\frac{\pi}{2}}\frac{\cot x - \cos x}{(\pi-2x)^3}$ equals

Sol. $\lim_{x\to\frac{\pi}{2}}\frac{\cot x - \cos x}{-8\left(x-\frac{\pi}{2}\right)^3}$

$= \lim_{x\to\frac{\pi}{2}}\frac{\tan\left(\frac{\pi}{2}-x\right)\left(1-\cos\left(\frac{\pi}{2}-x\right)\right)}{8\left(\frac{\pi}{2}-x\right)^2}$

$= \frac{1}{8}\cdot 1\cdot\frac{1}{2} = \frac{1}{16}$

3. $\lim_{x\to 0}\frac{x\cot(4x)}{\sin^2 x\cot^2(2x)}$ is equal to:

Sol. $\lim_{x\to 0}\frac{x\cos 4x\sin^2 2x}{\sin^2 x.\cos^2 2x.\sin 4x}$

$= \lim_{x\to 0}\frac{x}{\tan 4x}\cdot\frac{1}{\sin^2 x}\cdot\frac{\tan^2 2x}{1}$

$= \lim_{x\to 0}\frac{1}{4}\frac{4x}{\tan 4x}\frac{x^2}{\sin^2 x}\cdot\frac{\tan^2 2x}{x^2}$

$= \lim_{x\to 0}\frac{1}{4}\frac{4x}{\tan 4x}\left(\frac{x}{\sin x}\right)^2\cdot\left(\frac{\tan 2x}{2x}\right)^2\frac{4}{1}$

$= \frac{1}{4}\cdot 1\cdot 1\cdot 1\cdot\frac{4}{1} = 1$

4. If $\lim_{x\to 1}\frac{x^4-1}{x-1} = \lim_{x\to k}\frac{x^3-k^3}{x^2-k^2}$, then k is:

Sol. $\lim_{x\to 1}\frac{x^4-1}{x-1} = \lim_{x\to k}\frac{x^3-k^3}{x^2-k^2}$

$\Rightarrow \lim_{x\to 1}(x+1)(x^2+1) = \frac{k^2+k^2+k^2}{2k}$

$\Rightarrow k = 8/3$

5. $\lim_{x\to 0}\frac{x+2\sin x}{\sqrt{x^2+2\sin x+1}-\sqrt{\sin^2 x-x+1}}$ is:

Sol. Rationalize

$\lim_{x\to 0}\frac{(x+2\sin x)\left(\sqrt{x^2+2\sin x+1}+\sqrt{\sin^2 x-x+1}\right)}{x^2+2\sin x+1-\sin^2 x+x-1}$

$\lim_{x\to 0}\frac{(x+2\sin x)(2)}{x^2+2\sin x-\sin^2 x+x}$

$\frac{0}{0}$ form using L' hospital

$\Rightarrow \lim_{x\to 0}\frac{(1+2\cos x)\times 2}{2x+2\cos x-2\sin x\cos x+1} = \frac{6}{3} \Rightarrow 2$

6. The value of the limit $\lim_{\theta\to 0}\frac{\tan\left(\pi\cos^2\theta\right)}{\sin\left(2\pi\sin^2\theta\right)}$

Sol. $\lim_{\theta\to 0}\frac{\tan\left(\pi\left(1-\sin^2\theta\right)\right)}{\sin\left(2\pi\sin^2\theta\right)}$

$= \lim_{\theta\to 0}\frac{-\tan\left(\pi\sin^2\theta\right)}{\sin\left(2\pi\sin^2\theta\right)}$

$= \lim_{\theta\to 0}-\left(\frac{\tan\left(\pi\sin^2\theta\right)}{\pi\sin^2\theta}\right)\left(\frac{2\pi\sin^2\theta}{\sin\left(2\pi\sin^2\theta\right)}\right)\times\frac{1}{2}$

$= \frac{-1}{2}$

7. If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, then $\frac{dy}{dx}$ at $x = 1$ is equal to

Sol. Given, $y = \sqrt{x} + \frac{1}{\sqrt{x}}$

Now, $\frac{dy}{dx} = \frac{1}{2\sqrt{x}} - \frac{1}{2x^{3/2}}$

$\therefore \left(\frac{dy}{dx}\right)_{x=1} = \frac{1}{2} - \frac{1}{2} = 0$

8. If $f(x) = \frac{x-4}{2\sqrt{x}}$, then $f'(1)$ is equal to

Sol. Given,

Now,

$f(x) = \frac{x-4}{2\sqrt{x}}$

$f'(x) = \frac{2\sqrt{x} - (x-4)\cdot 2\cdot\frac{1}{2\sqrt{x}}}{4x}$

$= \frac{2x-(x-4)}{4x^{3/2}} = \frac{2x-x+4}{4x^{3/2}}$

$= \frac{x+4}{4x^{3/2}}$

$\therefore f'(1) = \frac{1+4}{4\times(1)^{3/2}} = \frac{5}{4}$

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9. If $y=\dfrac{1+\frac{1}{x^2}}{1-\frac{1}{x^2}}$, then $\dfrac{dy}{dx}$ is equal to

Sol. Given, $y=\dfrac{1+\frac{1}{x^2}}{1-\frac{1}{x^2}} \Rightarrow y=\dfrac{x^2+1}{x^2-1}$

$\therefore \dfrac{dy}{dx}=\dfrac{(x^2-1)2x-(x^2+1)(2x)}{(x^2-1)^2}$ [by quotient rule]

$\dfrac{dy}{dx}=\dfrac{2x(x^2-1-x^2-1)}{(x^2-1)^2}$

$=\dfrac{2x(-2)}{(x^2-1)^2}=\dfrac{-4x}{(x^2-1)^2}$

10. If $y=\dfrac{\sin x+\cos x}{\sin x-\cos x}$, then $\dfrac{dy}{dx}$ at $x=0$ is equal to

Sol. Given, $y=\dfrac{\sin x+\cos x}{\sin x-\cos x}$

$\therefore \dfrac{dy}{dx}$

$=\dfrac{(\sin x-\cos x)(\cos x-\sin x)-(\sin x+\cos x)(\cos x+\sin x)}{(\sin x-\cos x)^2}$

[by quotient rule]

$=\dfrac{-(\sin x-\cos x)^2-(\sin x+\cos x)^2}{(\sin x-\cos x)^2}$

$=\dfrac{-[(\sin x-\cos x)^2+(\sin x+\cos x)^2]}{(\sin x-\cos x)^2}$

$=\dfrac{-[\sin^2 x+\cos^2 x-2\sin x\cos x+\sin^2 x+\cos^2 x+2\sin x\cos x]}{(\sin x-\cos x)^2}$

$=\dfrac{-2}{(\sin x-\cos x)^2}$

$\therefore \left(\dfrac{dy}{dx}\right)_{x=0}=-2$

11. If $y=\dfrac{\sin(x+9)}{\cos x}$, then $\dfrac{dy}{dx}$ at $x=0$ is equal to

Sol. Given,

$\therefore \dfrac{dy}{dx}=\dfrac{\cos x\cos(x+9)-\sin(x+9)(-\sin x)}{(\cos x)^2}$

[by quotient rule]

$=\dfrac{\cos x\cos(x+9)+\sin x\sin(x+9)}{\cos^2 x}$

$\therefore \left(\dfrac{dy}{dx}\right)_{x=0}=\dfrac{\cos 9}{1}$

$=\cos 9$

12. Find $\tan x^2$ derivative from first principle.

Sol. $f'(x)=\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$

$=\lim\limits_{h\to 0}\dfrac{\tan(x+h)^2-\tan x^2}{h}$

$=\lim\limits_{h\to 0}\dfrac{\sin(x+h)^2\cos x^2-\sin x^2\cos(x+h)^2}{h\cdot\cos x^2\cdot\cos(x+h)^2}$

$=\lim\limits_{h\to 0}\dfrac{\sin\left((x+h)^2-x^2\right)}{h\cdot\cos x^2\cdot\cos(x+h)^2}$

$=\lim\limits_{h\to 0}\dfrac{\sin\left(h^2+2hx\right)}{h\cdot\left(\cos x^2\right)^2}$

$=\dfrac{2x}{\cos^2\left(x^2\right)}$

$=2x\sec^2(x^2)$.

SCHOOL LEVEL PROBLEMS

SINGLE CORRECT TYPE QUESTIONS

1. The derivative of $x^2\cos x$ is

(a) $2x\sin x-x^2\sin x$ (b) $2x\cos x-x^2\sin x$

(c) $2x\sin x-x^2\cos x$ (d) $\cos x-x^2\sin x\cos x$

2. $\lim\limits_{x\to 0}\dfrac{|\sin x|}{x}$ is equal to

(a) 1 (b) -1

(c) 0 (d) does not exist

3. $\lim\limits_{x\to 0}\dfrac{(\operatorname{cosec}x-\cot x)}{x}$ is

(a) $-\dfrac{1}{2}$ (b) 1

(c) $\dfrac{1}{2}$ (d) None of these

4. $y=\dfrac{(\sin x+\cos x)}{(\sin x-\cos x)}$, $\dfrac{dy}{dx}$ at $x=0$ is

(a) -2 (b) 0

(c) $\dfrac{1}{2}$ (d) Does not exist

5. The positive integer n so that $\lim\limits_{x\to 3}\dfrac{(x^n-3^n)}{(x-3)}=108$ is

(a) 3 (b) 4 (c) -2 (d) 1

6. The value of the limit $\lim\limits_{x\to 0}(\cos x)^{\cot 2x}$ is

(a) 1 (b) e (c) $e^{1/2}$ (d) $e^{-1/2}$

7. The derivative of $\dfrac{\left[1+\left(\frac{1}{x}\right)\right]}{\left[1-\left(\frac{1}{x}\right)\right]}$ is

(a) $\dfrac{1}{(x-1)^2}$ (b) $-\dfrac{1}{(x-1)^2}$

(c) $\dfrac{2}{(x-1)^2}$ (d) $-\dfrac{2}{(x-1)^2}$

8. The value of $\lim\limits_{n\to\infty}\dfrac{\{1^2+2^2+3^2+\ldots+n^2\}}{n^4}$

(a) 0 (b) 1 (c) -1 (d) n

9. Let $f(x)=\cos x$, $x\geq 0$ and $f(x)=x+k$, when $x<0$. Find the value of k given that $\lim\limits_{x\to 0}f(x)$ exists.

(a) 0 (b) 1

(c) -1 (d) None of these

VERY SHORT ANSWER TYPE QUESTIONS

10. Evaluate the given limit: $\lim\limits_{x\to -1}\dfrac{x^{10}+x^5+1}{x-1}$.

11. What is the derivative of $\dfrac{2^x}{x}$ with respect to x?

12. Find the derivative of the expression $1+x+x^2+x^3+\ldots x^{50}$ at $x=1$.

13. What is the limit value: $\lim\limits_{h\to 0}\dfrac{e^{2h}-1}{h}$?

14. Find the derivative of expression $x^3(5+3x)$ with respect to x.

15. Evaluate: $\lim\limits_{x\to 3}\left[\dfrac{x^2-9}{x-3}\right]$

16. Evaluate: $\lim\limits_{x\to 0}\dfrac{\sin^2 4x}{x^2}$

LONG ANSWER TYPE QUESTIONS

17. Suppose $f(x)=\begin{cases}a+bx & x<1\\ 4, & x=1\\ b-ax & x>1\end{cases}$ and if $\lim\limits_{x\to 1}f(x)=f(1)$, then what are possible value of a and b?

18. Find $\lim\limits_{x\to 0}f(x)$ and $\lim\limits_{x\to 1}f(x)$, where $f(x)=\begin{cases}2x+3 & x\leq 0\\ 3(x+1) & x>0\end{cases}$

19. If the expression is given as $y=\dfrac{(1-\tan x)}{(1+\tan x)}$, then show that the derivative is $\dfrac{dy}{dx}=\dfrac{-2}{(1+\sin 2x)}$

20. Evaluate the given limit: $\lim\limits_{x\to 0}\dfrac{\sin ax+bx}{ax+\sin bx}$; $a,b,a+b\neq 0$.

21. Prove that: $\lim\limits_{x\to 0}\left(\dfrac{e^x-1}{x}\right)=1$

CASE STUDY BASED QUESTIONS

Case Study-I

A function f is said to be a rational function, if $f(x)=\dfrac{g(x)}{h(x)}$, where $g(x)$ and $h(x)$ are polynomial functions such that $h(x)\neq 0$.

Then, $\lim\limits_{x\to a}f(x)=\lim\limits_{x\to a}\dfrac{g(x)}{h(x)}=\dfrac{\lim\limits_{x\to a}g(x)}{\lim\limits_{x\to a}h(x)}=\dfrac{g(a)}{h(a)}$.

However, if $h(a)=0$, then there are two cases arise,

(i) $g(a)\neq 0$

(ii) $g(a)=0$

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In the first case, we say that the limit does not exist.
In the second case, we can find limit.
Based on above information, answer the following questions.

22. $\lim_{x\to-1}\left(\frac{x^{10}+x^5+1}{x-1}\right)$ is equal to

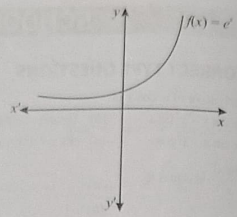
(a) $\frac{1}{2}$ (b) $\frac{-1}{2}$
(c) 2 (d) $\frac{3}{2}$

23. $\lim_{x\to0}\frac{\sqrt{1+x^3}-\sqrt{1-x^3}}{x^3}$ is equal to

(a) 1 (b) 0
(c) –1 (d) 2

Case Study-II

The great Swiss Mathematician Leonhard Euler (1707-1783) introduced the number e, whose value lies between 2 and 3. This number is useful in defining exponential function.
A function of the form of $f(x) = e^x$ is called exponential function.
The graph of the function is given below



(i) Domain of $f(x) = (-\infty, \infty)$
(ii) Range of $f(x) = (0, \infty)$
To find the limit of a function involving exponential function, we use the following theorem
Theorem $\lim_{x\to0}\frac{e^x-1}{x}=1$
Based on the above information, answer the following questions.

24. $\lim_{x\to0}\frac{e^{-x}-1}{x}$ is equal to

(a) 1 (b) –1 (c) 0 (d) 2

25. $\lim_{x\to0}\left(\frac{2e^x-3x-2}{x}\right)$ is equal to

(a) –1 (b) 0 (c) 1 (d) 2

PRARAMBH (TOPICWISE)

EXISTENCE OF LIMIT

1. If $f(x)=\begin{cases}x, & \text{when } x>1\\ x^2, & \text{when } x<1\end{cases}$, then $\lim_{x\to1} f(x) =$

(a) x^2 (b) x
(c) –1 (d) 1

2. $\lim_{x\to2}\frac{|x-2|}{x-2} =$

(a) 1
(b) –1
(c) Does not exist
(d) None of these

3. If, $f(x)=\begin{cases}\frac{2}{5-x}, & \text{when } x<3\\ 5-x, & \text{when } x>3\end{cases}$, then

(a) $\lim_{x\to3^-} f(x)=0$ (b) $\lim_{x\to3} f(x)=0$
(c) $\lim_{x\to3^-} f(x)\neq\lim_{x\to3^+} f(x)$ (d) None of these

4. Let the function f be defined by the equation $f(x)=\begin{cases}3x, & \text{if } 0\le x\le1\\ 5-3x, & \text{if } 1<x\le2\end{cases}$, then

(a) $\lim_{x\to1} f(x)=f(1)$ (b) $\lim_{x\to1} f(x)=3$
(c) $\lim_{x\to1} f(x)=2$ (d) $\lim_{x\to1} f(x)$ does not exist

5. $\lim_{x\to0}\frac{|x|}{x} =$

(a) 1 (b) –1
(c) 0 (d) Does not exist

METHODS OF EVALUATION OF LIMITS

6. $\lim_{x\to1}(3x^2+4x+5) =$

(a) 12 (b) –1
(c) Does not exist (d) None of these

7. The value of $\lim_{x\to2}\frac{3^{x/2}-3}{3^x-9}$ is

(a) 0 (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) ln 3

8. $\lim_{x\to3}\frac{x-3}{\sqrt{x-2}-\sqrt{4-x}}$ equals

(a) 1 (b) $\frac{3}{2}$
(c) $\frac{1}{4}$ (d) None of these

9. $\lim_{x\to0}\frac{x}{\sqrt{1+x}-\sqrt{1-x}}$ is equal to

(a) $\frac{1}{2}$ (b) 2
(c) 1 (d) 0

10. $\lim_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}$ equals

(a) $\frac{2a}{3\sqrt{3}}$ (b) $\frac{2}{3\sqrt{3}}$
(c) 0 (d) None of these

11. If $f(x)=\frac{2}{x-3}$, $g(x)=\frac{x-3}{x+4}$ and $h(x)=-\frac{2(2x+1)}{x^2+x-12}$ then $\lim_{x\to3}[f(x)+g(x)+h(x)]$ is

(a) –2 (b) –1
(c) $-\frac{2}{7}$ (d) 0

12. $\lim_{x\to a}\frac{(x+2)^{5/3}-(a+2)^{5/3}}{x-a} =$

(a) $\frac{5}{3}(a+2)^{2/3}$ (b) $\frac{5}{3}(a+2)^{5/3}$
(c) $\frac{5}{3}a^{2/3}$ (d) $\frac{5}{3}a^{5/3}$

TRIGONOMETRIC LIMITS

13. $\lim_{x\to0}\frac{x^3}{\sin x^2} =$

(a) 0 (b) 3
(c) 2 (d) 4

14. $\lim_{x\to0}\frac{\sin3x+\sin x}{x} =$

(a) $\frac{1}{3}$ (b) 3
(c) 4 (d) $\frac{1}{4}$

15. $\lim_{x\to1}(1-x)\tan\left(\frac{\pi x}{2}\right) =$

(a) $\frac{\pi}{2}$ (b) π
(c) $\frac{2}{\pi}$ (d) 0

16. $\lim_{x\to1}\frac{\sqrt{1-\cos2(x-1)}}{x-1}$

(a) Exists and is equal to $\sqrt{2}$
(b) Exists and is equals to $\sqrt{2}$
(c) Does not exist because $x-1\to0$
(d) Does not exist because left hand limit is not equal to right hand limit

17. $\lim_{x\to0}\frac{(1-\cos2x)\sin5x}{x^2\sin3x} =$

(a) $\frac{10}{3}$ (b) $\frac{3}{10}$ (c) $\frac{6}{5}$ (d) $\frac{5}{6}$

18. $\lim_{\theta\to\pi/2}(\sec\theta-\tan\theta) =$

(a) 0 (b) 1/2 (c) 2 (d) ∞

19. $\lim_{x\to0}\frac{\sqrt{\frac{1}{2}(1-\cos2x)}}{x} =$

(a) 1 (b) –1
(c) 0 (d) limit does not exist.

20. $\lim_{\theta\to\frac{\pi}{2}}\frac{\frac{\pi}{2}-\theta}{\cot\theta} =$

(a) 0 (b) –1 (c) 1 (d) ∞

DERIVATIVES

21. If $y=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}$ then $\frac{dy}{dx}$ is equal to

(a) y (b) $y+\frac{x^n}{n!}$
(c) $y-\frac{x^n}{n!}$ (d) $y-1-\frac{x^n}{n!}$

22. If $f(x)=\sqrt{1+\cos^2(x^2)}$, then $f'\left(\frac{\sqrt{\pi}}{2}\right)$

(a) $\frac{\sqrt{\pi}}{6}$ (b) $-\sqrt{\frac{\pi}{6}}$
(c) $\frac{1}{\sqrt{6}}$ (d) $\frac{\pi}{\sqrt{6}}$

23. If $y=a\sin x+b\cos x$, then $y^2+\left(\frac{dy}{dx}\right)^2$ is a

(a) function of x (b) function of y
(c) function of x and y (d) constant

24. If $y=\sqrt{\frac{1-x}{1+x}}$, then $(1-x^2)\frac{dy}{dx}$ is equal to

(a) y^2 (b) $1/y$ (c) $-y$ (d) $-y/x$

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25. If $y = |\cos x| + |\sin x|$, then $\frac{dy}{dx}$ at $x = \frac{2\pi}{3}$ is

(a) $\frac{1-\sqrt{3}}{2}$ (b) 0

(c) $\frac{1}{2}(\sqrt{3}-1)$ (d) None of these

26. If $f(x) = 1 + x + \frac{x^2}{2} + \ldots + \frac{x^{100}}{100}$, then $f'(1)$ is equal to

(a) $\frac{1}{100}$ (b) 100

(c) 0 (d) Does not exist

27. If $f(x) = \frac{x^n - a^n}{x-a}$ for some constant a, then $f'(a)$ is equal to

(a) 1 (b) 0

(c) $\frac{1}{2}$ (d) Does not exist

28. If $f(x) = x^{100} + x^{99} + \ldots + x + 1$, then $f'(1)$ is equal to

(a) 5050 (b) 5049 (c) 5051 (d) 50051

MISCELLANEOUS

29. Evaluate $\lim_{x \to 1/2} \frac{4x^2 - 1}{2x - 1}$.

30. Evaluate $\lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$.

31. Evaluate $\lim_{x \to a} \frac{(2+x)^{5/2} - (a+2)^{5/2}}{x-a}$.

32. Evaluate $\lim_{x \to 1} \frac{x^4 - \sqrt{x}}{\sqrt{x} - 1}$.

33. Evaluate $\lim_{x \to 0} \frac{\sqrt{1+x^3} - \sqrt{1-x^3}}{x^2}$

34. Evaluate $\lim_{x \to -3} \frac{x^3 + 27}{x^5 + 243} = \lim_{x \to -3} \frac{\frac{x^3+27}{x+3}}{\frac{x^5+243}{x+3}}$

35. Evaluate $\lim_{x \to 0} \frac{1 - \cos mx}{1 - \cos nx}$.

36. Evaluate $\lim_{x \to \pi/6} \frac{\sqrt{3}\sin x - \cos x}{x - \frac{\pi}{6}}$.

37. If $\lim_{x \to 1} \frac{x^4 - 1}{x - 1} = \lim_{x \to k} \frac{x^3 - k^3}{x^2 - k^2}$, then find the value of k.

38. $\lim_{x \to \pi} \frac{1 - \sin\frac{x}{2}}{\cos\frac{x}{2}\left(\cos\frac{x}{4} - \sin\frac{x}{4}\right)}$

39. Show that $\lim_{x \to \pi/4} \frac{|x-4|}{x-4}$ does not exist,

40. If $f(x) = \frac{\tan x}{x - \pi}$ then $\lim_{x \to \pi} f(x) =$

41. $\lim_{x \to 0}\left(\sin mx \cot\frac{x}{\sqrt{3}}\right) = 2$, then $m =$ .....

42. Differentiate this function w.r.t x $\frac{x^4 + x^3 + x^2 + 1}{x}$

43. Differentiate this function $\left(x + \frac{1}{x}\right)^3$

44. Differentiate this function $(3x + 5)(1 + \tan x)$

45. Differentiate this function $\frac{x^5 - \cos x}{\sin x}$

46. Differentiate this function $\frac{x^2 \cos\frac{\pi}{4}}{\sin x}$

47. Differentiate this function $(ax^2 + \cot x)(p + q\cos x)$

48. Differentiate this function $(\sin x + \cos x)^2$

49. Differentiate this function $(2x - 7)^2 (3x + 5)^3$

50. Differentiate this function $\sin^3 x \cos^3 x$

51. Differentiate this function $\frac{1}{ax^2 + bx + c}$

52. If $y = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$, then $\frac{dy}{dx} =$ .....

53. Find the derivative of $\sin x$ from first principle.

Answer Key

CONCEPT APPLICATION

1. [3] 2. (b) 3. (a) 4. (a) 5. (d) 6. (b) 7. (a) 8. (a) 9. $\frac{dy}{dx} = 2\tan x \sec^2 x$

10. $\frac{dy}{dx} = \frac{55 - 15x^2 - 40x}{(5x^2 - 7x + 9)^2}$ 11. $\frac{dy}{dx} = \frac{bc\cos x + ad\sin x + bd}{(c + d\cos x)^2}$ 12. $\frac{dy}{dx} = x^2\cos x + 2x\sin x - 2\sin 2x$

SCHOOL LEVEL PROBLEMS

1. (b) 2. (d) 3. (c) 4. (a) 5. (b) 6. (a) 7. (d) 8. (a) 9. (b) 22. (b)
23. (a) 24. (b) 25. (a)

PRARAMBH (TOPICWISE)

1. (d) 2. (c) 3. (c) 4. (d) 5. (d) 6. (a) 7. (c) 8. (a) 9. (c) 10. (b)
11. (c) 12. (a) 13. (a) 14. (c) 15. (c) 16. (d) 17. (a) 18. (a) 19. (d) 20. (c)
21. (c) 22. (b) 23. (d) 24. (c) 25. (c) 26. (b) 27. (d) 28. (a) 29. [2] 30. $\frac{1}{2\sqrt{x}}$
31. $= \frac{5}{2}(a+2)^{\frac{3}{2}}$ 32. [7] 33. [0] 34. $\frac{1}{15}$ 35. $\frac{m^2}{n^2}$ 36. [2] 37. $\frac{8}{3}$ 38. $\frac{1}{\sqrt{2}}$
39. LHL ≠ RHL 40. [1] 41. $m = \frac{2\sqrt{3}}{3}$ 42. $\frac{3x^4 + 2x^3 + x^2 - 1}{x^2}$ 43. $3x^2 - \frac{3}{x^2} - \frac{3}{x^4} + 3$
44. $3x\sec^2 x + 5\sec^2 x + 3\tan x + 3$ 45. $\frac{5x^4\sin x - x^5\cos x + 1}{\sin^2 x}$ 46. $\frac{x}{\sqrt{2}}\operatorname{cosec} x\left[2 - x\cot x\right]$
47. $-q\sin x\,(ax^2 + \cot x) + (p + q\cos x)(2ax - \operatorname{cosec}^2 x)$ 48. $2\cos 2x$ 49. $(2x - 7)(3x + 5)^2(30x - 43)$
50. $\frac{3}{4}\sin^2 2x\cos 2x$ 51. $\frac{-(2ax + b)}{(ax^2 + bx + c)^2}$ 52. $1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$ 53. $(\cos x)$

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CHAPTER 17

Statistics

INTRODUCTION

Variate: A quantity having a numerical value for each member of a group, especially one whose values occur according to a frequency distribution.

Frequency distributions: Frequency distributions are visual displays that organise and present frequency counts so that the information can be interpreted more easily.

Relative frequency: Relative frequency is a proportion or percentage which is calculated with the help of given frequency.

Cumulative frequency: The cumulative is calculated by adding each frequency from a frequency distribution table to sum of its predecessors.

DATA

Facts, observations and information that come from investigations. Generally three types of data are used.

(*i*) Ungrouped data, raw data or individual series.

(*ii*) Discrete frequency or ungrouped data.

Definition: Data consist of n distinct values $x_1, x_2, \ldots x_n$ occurring with frequency $f_1, f_2, \ldots, f_n$ respectively. This data in tabular form is called discrete frequency distribution.

(*iii*) Continuous frequency or grouped data:

Definition: A continuous frequency distribution is a series in which the data are classified into different class intervals without gaps along with their respective frequencies.

MEASURES OF CENTRAL TENDENCY

Averages are generally, the central part of the distribution and therefore, they are also called the measures of **Central Tendency.**

It can be divided into two groups:

(*a*) Mathematical Average: (Mean)

(*i*) Arithmetic mean or mean (*iii*) Harmonic mean

(*ii*) Geometric mean

(*b*) Positional Average:

(*i*) Median

(*ii*) Mode or positional average

ARITHMETIC MEAN

1. Individual observation or unclassified data:

If $x_1, x_2, \ldots x_n$ be n observations, then their arithmetic mean is given by

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} \quad \text{or} \quad \bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

2. Arithmetic mean of discrete frequency distribution

Let $x_1, x_2, \ldots x_n$ be n observation and let $f_1, f_2, \ldots, f_n$ be the corresponding frequencies, then their mean is

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + \ldots + f_nx_n}{f_1 + f_2 + \ldots + f_n} \quad \text{or} \quad \bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$$

Short cut method: If the values of x or (and) f are large the calculation of arithmetic mean by the previous formula used is quite tedious and time consuming. In such case we take the deviation from an arbitrary point A.

$$\bar{x} = A + \frac{\Sigma f_i d_i}{\Sigma f_i}$$

where A = Assumed mean

$d_i = x_i - A$ = deviation for each term

Step deviation method: Sometimes during the application of shortcut method of finding the mean, the deviation d_i are divisible by a common number h (say). In such case the calculation is reduced to a great extent by taking

$$u_i = \frac{x_i - A}{h},\ i = 1, 2, \ldots n \quad \therefore \quad \text{mean } \bar{x} = A + h\left(\frac{\Sigma f_i u_i}{\Sigma f_i}\right)$$

Arithmetic mean of continuous frequency distribution

3. Continuous frequency distribution

For calculating mean of continuous frequency distribution the procedure is same as for discrete frequency distribution. Here we have to obtain the mid point of various classes and find their mean.

4. Weighted arithmetic mean: If $w_1, w_2, w_3, \ldots, w_n$ are the weight assigned to the values $x_1, x_2, x_3, \ldots, x_n$ respectively then the weighted average is defined as -

$$\text{Weighted } A.M. = \frac{w_1x_1 + w_2x_2 + \ldots + w_nx_n}{w_1 + w_2 + \ldots + w_n},\ \bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

5. Combined mean: If $\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_k$ are the mean of k series of sizes $n_1, n_2, \ldots n_k$ respectively then the mean of the composite series is given by

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \ldots + n_k\bar{x}_k}{n_1 + n_2 + \ldots + n_k}$$

6. Properties of Arithmetic Mean:

(*i*) In a statistical data, the sum of the deviation of items from $A.M.$ is always zero.

i.e. $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$ or $\sum_{i=1}^{n} x_i - n\bar{x}$

or $n\bar{x} - n\bar{x} = 0 \qquad \left(\because \bar{x} = \frac{\sum x_i}{n}\right)$

(*ii*) In a statistical data, the sum of squares of the deviation of items from $A.M.$ is least i.e. $\sum_{i=1}^{n}(x_i - \bar{x})^2$ is least.

(*iii*) If each of the n given observation be doubled, then their mean is doubled.

(*iv*) If $\bar{x}$ is the mean of $x_1, x_2, \ldots, x_n$. The mean of $ax_1, ax_2, \ldots, ax_n$ is $a\bar{x}$ where a is any number different from zero.

(*v*) Arithmetic mean is independent of origin i.e. it is not effected by any change in origin.

Train Your Brain

Example 1: In a class of 40 boys and 30 girls, the average age is 16 years. If the mean age of boys is 1 year more than the mean age of girls, then find the mean age of girls.

Sol. Let mean age of 30 girls is x years so that of 40 boys is $(x + 1)$ years.

$$\therefore \quad 16 = \frac{40(x+1) + 30x}{40 + 30} = \frac{4x + 4 + 3x}{7}$$

$$\therefore \quad 7x + 4 = 112,\ 7x = 108,\ x = 15\frac{3}{7} \text{ years}$$

Example 2: If n persons donate each rupees $1, 2, 4, 8 \ldots 2^{n-1}$ respectively then the mean donation per person is

Sol. $\bar{x} = \dfrac{1 + 2 + 2^2 + 2^3 + \ldots + 2^{n-1}}{n}$

this is a $G.P.$ with common ratio as 2,

$$= \frac{(2^n - 1)}{n.(2 - 1)} = \frac{2^n - 1}{n} \text{ rupees}$$

Example 3: Find the mean of first n natural numbers whose frequencies are equal to the corresponding numbers.

Sol. Here $x_1 = 1, x_2 = 2, x_3 = 3 \ldots x_n = n$

also $f_1 = 1, f_2 = 2, f_3 = 3 \ldots f_n = n$

so $\bar{X} = \dfrac{x_1f_1 + x_2f_2 + x_3f_3 + \ldots + x_nf_n}{f_1 + f_2 + \ldots + f_n}$

$$= \frac{1.1 + 2.2 + 3.3 + \ldots + n.n}{1 + 2 + 3 \ldots + n}$$

$$= \frac{n(n+1)(2n+1)\cdot 2}{6 \cdot n(n+1)} = \frac{2n+1}{3}$$

Example 4: The mean heights of team A and team B are 6′2″ and 5′10″ respectively. One member from team A was transferred to B so that average height of team B rise by 0.5″. Then the shortest-member from team B was transferred to A so that now both teams have same average height. Find the heights of members transferred respectively. Each team consists of 6 members.

Sol. Total height of 6 members of team $A = 6'2'' \times 6 = 37'$

Total height of 6 members of team $B = 5'10'' \times 6 = 35'$

Let one member of height h' is transferred from team A to B

Now total height of 5 members of team A remaining $= 37' - h'$

Total height of 7 members of team B now $= 35' + h'$

Mean height of team $B = \dfrac{35' + h'}{7} = 5'10.5''$

$\therefore \quad h' = 7(5'\ 10.5'') - 35' = 73.5''$

$= 6'$ and $\frac{1}{8}'$,

Let one member transferred from B to A of height H'

So mean height of 6 members of team $B = \dfrac{35 + h' - H'}{6}$ feet

So mean height of 6 members of team $A = \dfrac{37 - h' + H'}{6}$ feet

$$\frac{35 + h' - H'}{6} = \frac{37 - h' + H'}{6}$$

$$2h' - 2H' = 2$$

$H' = h' - 1 = 5'$ and $\frac{1}{8}'$,

So the members transferred were of height 6 and $\frac{1}{8}$ feet, 5 and $\frac{1}{8}$ feet respectively.

Example 5: Mean of 25 observations was found to be 78.4. But later on it was found that 96 was misread as 69. The correct mean is

(*a*) 79.48 (*b*) 76.54

(*c*) 81.32 (*d*) 78.4

Sol. We know that the mean is given by

$$\bar{x} = \frac{\Sigma x}{n} \quad \text{or} \quad \Sigma x = n\bar{x}$$

Here $\bar{x} = 78.4, n = 25$

$\therefore \quad \Sigma x = 25 \times 78.4 = 1960$

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But this Σx is incorrect as 96 was misread as 69

correct $\Sigma x = 1960 - 69 + 96 = 1987$

$\therefore$ correct mean $= \frac{1987}{25} = 79.48$

Example 6: If the frequencies of first four numbers out of 1, 2, 4, 6, 8 are 2, 3, 3 , 2 respectively, then the frequency of 8 if their AM is 5, is

(*a*) 4 (*b*) 5

(*c*) 6 (*d*) None of these

Sol. Here mean $A = 5$

Let the frequency of 8 be x. Then by the formula

$$A = \frac{\Sigma xf}{\Sigma f}$$

$$5 = \frac{1.2 + 2.3 + 4.3 + 6.2 + 8x}{2 + 3 + 3 + 2 + x} = \frac{32 + 8x}{10 + x}$$

or $18 = 3x$;

$\therefore$ $x = 6$.

Example 7: In a family, there are 8 men, 7 women and 5 children whose mean ages separately are respectively 24, 20 and 6 years. The mean age of the family is

(*a*) 17.1 years (*b*) 18.1 years

(*c*) 19.1 years (*d*) None of these

Sol. Here we have three collections for which $A_1 = 24, n_1 = 8$, $A_2 = 20, n_2 = 7$ and $A_3 = 6, n_3 = 5$. Their combined mean is the required mean.

By the formula $A = \frac{n_1A_1 + n_2A_2 + n_3A_3}{n_1 + n_2 + n_3}$

$$\therefore \quad A = \frac{8 \times 24 + 7 \times 20 + 5 \times 6}{8 + 7 + 5}$$

$$= \frac{192 + 140 + 30}{20} = \frac{362}{20} = 18.1$$

Concept Application

1. If the mean of a set of observations $x_1, x_2, ...x_{10}$ is 20 then the mean of $x_1 + 4, x_2 + 8, x_3 + 12, ...x_{10} + 40$ is

(*a*) 34 (*b*) 42 (*c*) 38 (*d*) 40

2. 1, 2, 4, 6, 8, 9 transformed into another series whose mean is 12. Two middle terms of the transformed series are

(*a*) 7, 8 (*b*) 10, 14 (*c*) 9, 15 (*d*) 11, 13

3. The mean of 20 observations is 15. On checking it was found that the two observations were wrongly copied as 3 and 6. The correct values are 8 and 4, then correct mean will be given by

(*a*) 15.15 (*b*) 14.69 (*c*) 14.74 (*d*) 15.25

GEOMETRIC MEAN

1. **Individual data :** If $x_1, x_2, x_3, ..., x_n$ are n values of a variate x, none of them being zero, then the geometric mean G is defined as-

$G = (x_1 x_2 x_3 ... x_n)^{1/n}$

or $G = \text{antilog}\left(\frac{\log x_1 + \log x_2 + ... + \log x_n}{n}\right)$

or $G = \text{antilog}\left(\frac{1}{n}\sum_{i=1}^{n} \log x_i\right)$

2. **Geometric Mean of grouped data :**

Let $x_1, x_2, ...x_n$ be n observation and let $f_1, f_2, ...f_n$ be their corresponding frequency then their Geometric Mean is

$G = \left(x_1^{f_1} x_2^{f_2} .....x_n^{f_n}\right)^{1/N}$ where $N = \sum_{i=1}^{n} f_i$

$\therefore$ $G = \text{antilog}\left(\frac{\sum_{i=1}^{n} f_i \log x_i}{\sum_{i=1}^{n} f_i}\right)$

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Example 8: The geometric mean of the numbers 7, 7^2, 7^3, ..., 7^n is

Sol. $GM = \sqrt[n]{7.7^2.7^3 ... 7^n}$

$= \left[7^{1+2+3+...+n}\right]^{1/n} = \left[7^{n(n+1)/2}\right]^{1/n} = 7^{n+1/2}$

Concept Application

4. Let G_1, G_2 be the geometric means of two series $x_1, x_2, ...x_n$; $y_1, y_2 ... y_n$. If G is the geometric means of x_i/y_i, i = 1, 2.. n, then G is equal to

(*a*) $G_1 - G_2$ (*b*) Log G_1/log G_2

(*c*) Log (G_1/G_2) (*d*) None of these

5. The geometric mean G of the product of n series of data with geometric means, $G_1, G_2, ..., G_n$ respectively, then

(*a*) $\sqrt{G} = G_1.G_2...G_n$ (*b*) $G = G_1 \cdot G_2 \cdot ...G_n$

(*c*) $\sqrt{G} = -G_1.G_2...G_n$ (*d*) None of the above

6. The geometric mean of the numbers 3, 3^2, 3^3,....,3^n is

(*a*) $3^{2/n}$ (*b*) $3^{(n-1)/2}$

(*c*) $3^{n/2}$ (*d*) $3^{(n+1)/2}$

HARMONIC MEAN

Harmonic mean is reciprocal of mean of reciprocal.

1. **Individual observation:** The *H.M.* of $x_1, x_2, ..., x_n$ of n observation is given by

$H = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + ... + \frac{1}{x_n}}$ i.e. $H = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$

2. **H.M. of grouped data:** Let $x_1, x_2, ..., x_n$ be n observation and let $f_1, f_2, ..., f_n$ be their corresponding frequency then *H.M.* is

$$H = \frac{\sum_{i=1}^{n} f_i}{\sum_{i=1}^{n}\left(\frac{f_i}{x_i}\right)}$$

Relation between A.M. , G.M., and H.M.

$A.M. \geq G.M. \geq H.M.$

Equality sign holds only when all the observations in the series are same.

MEDIAN

Median is the middle most or the central value of the variate in a set of observations, when the observations are arranged either in ascending or in descending order of their magnitudes. It divides the arranged series in two equal parts.

1. **Median of an individual series:**

Let n be the number of observations-

(*i*) Arrange the data in ascending or descending order.

(*ii*) **(*a*) If n is odd then-**

Median (M) = Value of $\left(\frac{n+1}{2}\right)^{th}$ observation

(*b*) If n is even then

Median (M) = mean of $\left(\frac{n}{2}\right)^{th}$ and $\left(\frac{n}{2}+1\right)^{th}$ observation

i.e. $M = \frac{\left(\frac{n}{2}\right)^{th} \text{ observation} + \left(\frac{n}{2}+1\right)^{th} \text{ observation}}{2}$

2. **Median of the discrete frequency distribution:**

Arrange data in ascending order

Algorithm to find the median :

Step-I: Find the cumulative frequency (*C. F.*)

Step-II: Find $\frac{N}{2}$, where $N = \sum_{i=1}^{n} f_i$

Step-III: See the cumulative frequency (*C.F.*) just greater than $\frac{N}{2}$ and determine the corresponding value of the variable.

Step-IV: The value obtained in step III is the median.

3. **Median of grouped data or continuous series:**

Let the number of observations be n

(*i*) Prepare the cumulative frequency table

(*ii*) Find the median class *i.e.* the class in which the $\left(\frac{N}{2}\right)^{th}$ observation lies

(*iii*) The median value is given by the formulae

$$\text{Median } (M) = \ell + \left[\frac{\left(\frac{N}{2}\right) - F}{f}\right] \times h$$

N = total frequency = Σf_i

ℓ = lower limit of median class

f = frequency of the median class

F = cumulative frequency of the class preceding the median class

h = class interval (width) of the median class

4. **Properties of Median:**

(*i*) The sum of the absolute value of deviations of the items from median is minimum

(*ii*) It is a positional average and it is not influenced by the position of the items.

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Example 9: The median of the items 6, 10, 4, 3, 9, 11, 22, 18, is

(*a*) 9 (*b*) 10 (*c*) 9.5 (*d*) 11

Sol. Lets arrange the items in ascending order 3, 4, 6, 9, 10, 11, 18, 22.

In this data the number of items is $n = 8$, which is even.

$\therefore$ Median = M = average of $\left(\frac{n}{2}\right)$ th and $\left(\frac{n}{2}+1\right)$ th terms.

= Average of $\left(\frac{8}{2}\right)$ th and $\left(\frac{8}{2}+1\right)$ th term

= Average of 4th and 5th terms

$= \frac{9+10}{2} = \frac{19}{2} = 9.5$

Example 10: The marks obtained by 10 students in a particular subject are 96, 88, 99, 87, 92, 86, 93, 94, 96, 97. Find the median mark.

Sol. We first arrange the data in ascending order as follows 86, 87, 88, 92, 93, 94, 96, 96, 97, 99

So there are altogether 10 observations. Hence the number of observations is even.

Then the median is the average of $\frac{10}{2}$ th and $\left(\frac{10}{2}+1\right)^{th}$ i.e. 5th and 6th observations.

That is Average of 93 and 94 i.e. $\frac{93+94}{2} = 93.5$

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Example 11: In a class of 20 students, the marks obtained by them in a test are grouped as follows

Marks	No. of Students
0	1
1	3
4	2
5	6
7	2
9	3
11	1
15	1
16	1
	20

Find the median marks.

Sol.

x	f	CF
0	1	1
1	3	4
4	2	6
5	6	12
7	2	14
9	3	17
11	1	18
15	1	19
16	1	20

Here $N = 20$,

so $\frac{N}{2} = 10$

variable corresponding to CF just greater then $\frac{N}{2}$ is 5.

Hence median is 5.

Example 12: The marks obtained by 60 students in a certain examination are given below

Marks	No. of Students
10 – 20	04
20 – 30	05
30 – 40	11
40 – 50	06
50 – 60	05
60 – 70	08
70 – 80	09
80 – 90	07
90 – 100	05

Calculate the median mark.

Sol. We calculate the cumulative frequency from the following table:

Marks Obtained	Frequency (No. of students) (f)	(Cumulative Frequency) (F)
10 – 20	04	04
20 – 30	05	09
30 – 40	11	20
40 – 50	06	26
50 – 60	05	31
60 – 70	08	39
70 – 80	09	48
80 – 90	07	55
90 – 100	05	60
	n = 60	

Here we have $n = 60$, $\frac{n}{2} = 30$

We see that 30 lies in the class 50 – 60 with cumulative frequency 31.

Thus the median class 50 – 60

$\therefore$ l = lower limit of the median class i.e. the class in which $\frac{n}{2}$ th or $\frac{n+1}{2}$ th observation lies = 50

f = frequency of the median class = 05

F = 26 (cumulative frequency of the class preceding the median class)

h = class size = 60 – 50 = 10

$$\text{Median} = l + \frac{\left(\frac{n}{2} - F\right)}{f} \times h$$

$$= 50 + \frac{30-26}{5} \times 10 = 50 + 8 = 58$$

Concept Application

7. If the median of 21 observations is 40 and if the observations greater than the median are increased by then the median of the new data will be

(*a*) 40 (*b*) 46

(*c*) 46 + 40/21 (*d*) 46 – 40/21

8. The value of median of the data

Income (in ₹)	1000	1100	1200	1300	1400	1500
Number of persons	14	26	21	18	28	15

is

(*a*) 1300 (*b*) 1200 (*c*) 1250 (*d*) 1150

9. From the data given, the median of the average deposit balance of saving for the branch during March 1982 is

Average deposit balance (in ₹)	Number of deposit
Less than 100	26
100-200	68
200-300	145
300-400	242
400-500	188
500-600	65
600-700	16

(*a*) 356 (*b*) 300

(*c*) 56.2 (*d*) 356.2

MODE

Mode is that value in a series which occurs most frequently. In a frequency distribution, mode is that variate which has the maximum frequency.

1. **Computation of Mode:**
   Mode for individual series: In the case of individual series, the value which is repeated maximum number of times is the mode of the series.
2. **Mode for grouped data (discrete frequency distribution series):** In the case of discrete frequency distribution, mode is the value of the variate corresponding to the maximum frequency.
3. **Mode for continuous frequency distribution:**
   (*i*) First find the model class i.e. the class which has maximum frequency. The model class can be determined either by inspecting or with the help of grouping data.
   (*ii*) The mode is given by the formula

$$\text{Mode} = \ell + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h$$

where

$\ell \rightarrow$ lower limit of the model class

$h \rightarrow$ width of the model class

$f_m \rightarrow$ frequency of the model class

$f_{m-1} \rightarrow$ frequency of the class preceding model class

$f_{m+1} \rightarrow$ frequency of the class succeeding model class

4. **Properties of Mode:** It is not effected by presence of extremely large or small items.

Relation between Mean, Median and Mode

(*a*) **Symmetric Distribution** A distribution is a symmetric distribution if values of mean, mode and median coincide.
Mean = Model = Median

(*b*) **Skewed Distribution** A distribution which is not symmetric is called a skewed distribution. In a moderately asymmetric the interval between the mean and median is approximately one-third of the interval between the mean and the mode. We have the following **empirical relation** between them.
Mean - Mode = 3(Mean - Median)
$\Rightarrow$ Mode = 3Median - 2Mean (Approximately)

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Example 13: Calculate the mode from the following data:

Marks	No. of Students
0 - 10	04
10 - 20	12
20 - 30	24
30 - 40	36
40 - 50	20
50 - 60	16
60 - 70	08
70 - 80	05

Sol.

Value (Class)	Frequency
0 - 10	04
10 - 20	12
20 - 30	24
30 - 40	36
40 - 50	20
50 - 60	16
60 - 70	08
70 - 80	05
	$n = 125$

We choose the highest frequency class as the modal class. So here 30 - 40 is the modal class.

Now

l = lower limit of modal class = 30

h = width of the modal class = 10

f_1 = frequency of the class preceding the modal class = 24

f_2 = frequency of the class following the modal class = 20

f = frequency of the modal class = 36

$$\text{So mode} = l + \frac{(f - f_1)}{(2f - f_1 - f_2)} \times h$$

$$= 30 + \frac{36-24}{72-24-20} \times 10$$

$$= 30 + \frac{12}{28} \times 10$$

$$= 30 + \frac{30}{7} = 34.28$$

Example 14: If the value of the mode and mean is 60 and 66 respectively then the value of median is

(*a*) 60 (*b*) 64

(*c*) 68 (*d*) None of these

Sol. Median = $\frac{1}{3}$ [Mode + 2 Mean]

$$= \frac{1}{3}[60 + 2 \times 66] = 64$$

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Concept Application

10. The mode of the following distribution is

Class interval	Frequency
0-10	5
10-20	8
20-30	7
30-40	12
40-50	28
50-60	20
60-70	10
70-80	10

(a) 46 (b) 6.66
(c) 46.67 (d) None of these

11. Mode of data 3, 2, 5, 2, 3, 5, 6, 6, 5, 3, 5, 2, 5 is
(a) 6 (b) 4 (c) 5 (d) 3

12. For the data 2, 2, 3, 3, 3, 4, 4, 5, 6, 8, 8, 8, 8, the mean, median and mode are
(a) 64/13, 5, 3 (b) 64/13, 14, 3
(c) 5, 4, 3 (d) 64/13, 4, 8

MEASURES OF DISPERSION

Dispersion is the measure of the variations. The degree to which numerical data tend to spread about an average value is called the dispersion of the data.

The measures of dispersion are commonly used by:

(*i*) Range (*ii*) Mean Deviation
(*iii*) Standard Deviation (*iv*) Variance

(*i*) Range: The range is the difference between the greatest and the least observations of the distribution. Thus if A and B are the greatest and the smallest observations respectively in a distribution, then its range $= A - B$.

The co-efficient of range (or scatter) $= \dfrac{A-B}{A+B}$

(*ii*) Mean Deviation: Mean deviation is defined as the arithmetic mean of the absolute deviations of all the values taken about any central value.

(*a*) Mean deviation of individual observations:
If $x_1, x_2, ..., x_n$ are n values of a variable x, then the mean deviation from a number A is given by

$$M.D. = \frac{1}{n}\sum_{i=1}^{n} |x_i - A|$$

$$= \frac{1}{n}\sum |d_i|, \text{ where } d_i = x_i - A$$

(*b*) Mean deviation of a discrete frequency distribution:
If $x_1, x_2, ..., x_n$ are n observation with frequencies $f_1, f_2, ..., f_n$, then mean deviation from a number A is given by-

$$\text{Mean Deviation} = \frac{1}{N}\sum f_i |x_i - A|$$

where $N = \sum_{i=1}^{n} f_i$

(*c*) Mean deviation of a grouped or continuous frequency distribution:

For calculating mean deviation of a continuous frequency distribution the procedure is same as for discrete frequency distribution. The only difference is that here we have to obtain the mid-point of the various classes and take the deviations of these mid-points from the given central value (median or mean

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Example 15: For an $A.P.$ 2, 5, 8, ..., (21 terms) are given. Find the mean deviation of $A.P.$ from its mean.

Sol. $\Sigma x_i = 2 + 5 + 8 ..... + (21 \text{ terms})$

$$= \frac{21}{2}[4 + 20\times 3] = 21 \times 32 = 672$$

$$\therefore \bar{x} = \frac{672}{21} = 32$$

$$\Sigma |x_i - \bar{x}| = 2\times 3[1+2+3+...+10]$$

$$= 2\times 3\left[\frac{10\times 11}{2}\right] = 330$$

$$\therefore \text{M.D.} = \frac{330}{21} = \frac{110}{7}$$

Concept Application

13. If the mean deviation of the numbers $1, 1 + d, ... 1 + 100d$ from their mean is 255, then d is equal to
(a) 10.1 (b) 20.2
(c) 10.0 (d) 20.0

14. If the mean deviation about the median of the numbers $a, 2a, ..., 50a$ is 50 then $|a|$ equal
(a) 5 (b) 2
(c) 3 (d) 4

15. The mean deviation from mean of the observations $a, a + d, a + 2d, ..., a + 2nd$ is
(a) $\dfrac{n(n+1)d^2}{3}$ (b) $\dfrac{n(n+1)}{2}d^2$
(c) $a + \dfrac{n(n+1)d^2}{2}$ (d) None of these

VARIANCE AND STANDARD DEVIATION

The variance of a variate x is the arithmetic mean of the squares of all deviations of x from the arithmetic mean of the observations and is denoted by var (x) or σ^2

The positive square root of the variance of a variate x is known as standard deviation i.e. standard deviation $= +\sqrt{\text{var}(x)}$

(*i*) Variance of Individual observations :

If $x_1, x_2, ..., x_n$ are n values of a variable x, then by definition

$$\text{var}(x) = \frac{1}{n}\left[\sum_{i=1}^{n}(x_i - \bar{x})^2\right] = \sigma^2 \quad \ldots (i)$$

$$\text{or var}(x) = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \bar{x}^2 \quad \ldots (ii)$$

If the values of variable x are large, the calculation of variance from the above formulae is quite tedious and time consuming. In that case, we take deviation from an arbitrary point A (say) then

$$\text{var}(x) = \frac{1}{n}\sum_{i=1}^{n} d_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} d_i\right)^2 \quad \ldots (iii)$$

(*ii*) Variance of a discrete frequency distribution:

If $x_1, x_2, ..., x_n$ are n observations with frequencies $f_1, f_2, ..., f_n$ then $\text{var}(x) = \dfrac{1}{N}\left\{\sum_{i=1}^{n} f_i (x_i - \bar{x})^2\right\}$ ... (*i*)

$$\text{or var}(x) = \frac{1}{N}\sum_{i=1}^{n} f_i x_i^2 - \bar{x}^2 \quad \ldots (ii)$$

If the values of x or f are large, we take the deviations of the values of variable x from an arbitrary point A. (say)

$\therefore\ d_i = x_i - A;\ i = 1, 2, ...n$

$$\therefore\ \text{Var}(x) = \frac{1}{N}\left(\sum_{i=1}^{n} f_i d_i^2\right) - \left(\frac{1}{N}\sum_{i=1}^{n} f_i d_i\right)^2 \quad \ldots (iii)$$

where $N = \sum_{i=1}^{n} f_i$

Sometime $d_i = x_i - A$ are divisible by a common number h (say)

then $u_i = \dfrac{x_i - A}{h} = \dfrac{d_i}{h}$, $i = 1, 2, ..., n$

$$\text{then var}(x) = h^2\left[\frac{1}{N}\sum_{i=1}^{n} f_i u_i^2 - \left(\frac{1}{N}\sum_{i=1}^{n} f_i u_i\right)^2\right] \quad \ldots (iv)$$

(*iii*) Variance of a grouped or continuous frequency distribution: In a grouped or continuous frequency distribution any of the formulae discussed in discrete frequency distribution can be used.

PROPERTIES OF VARIANCE

1. If $x_1, x_2, x_3, ..., x_n$ be n values of a variable X. If these values are changed to $x_1 + a, x_2 + a, ..., x_n + a$, where $a \in R$, then the variance remains unchanged.
2. If $x_1, x_2, ..., x_n$ values of a variable X and let 'a' be a non-zero real number. Then, the variance of the observation $ax_1, ax_2, ..., ax_n$ is a^2 Var(X) and S.D is $|a|$ S.D(x)

COMBINED STANDARD DEVIATION

If there are two sets of observations containing n_1 and n_2 items with respective mean $\bar{x}_1$ & $\bar{x}_2$ and standard deviation σ_1 and σ_2, then the mean $\bar{x}$ and the standard deviations of $n_1 + n_2$ observations, taken together, are

$$\bar{x} = \frac{n_1\bar{x}_1 + n_1\bar{x}_2}{n_1 + n_2}$$

$$\sigma^2 = \frac{1}{n_1 + n_2}[n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)]$$

where $d_1 = \bar{x} - \bar{x}_1$, $d_2 = \bar{x} - \bar{x}_2$

or

$$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2} \times (\bar{x}_1 - \bar{x}_2)^2$$

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Example 16: Find the variance of first n natural numbers.

Sol. We have

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)^2$$

$$= \frac{1}{n}(1^2 + 2^2 + ... + n^2) - \left(\frac{1}{n}\cdot(1 + 2 + ... + n)\right)^2$$

$$= \frac{1}{n}\cdot\frac{n(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2$$

$$= \frac{n^2 - 1}{12}$$

Example 17: A student obtained the mean and standard deviation of 100 observations as 40 and 5.1 respectively. It was later found that he had wrongly copied down an observation as 50 instead of 40. The correct mean and standard deviation are
(a) 39.9, 6 (b) 36.4, 5
(c) 39.9, 5 (d) None of these

Sol. Let the variable be x. Here $n = 100$, incorrect $A = 40$, incorrect $s = 5.1$.

From formula $A = \dfrac{\Sigma x}{n}$, $\Sigma x = nA$.

$\therefore$ incorrect $\Sigma x = 100 \times 40 = 4000$

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$\therefore$ correct Σx = incorrect $\Sigma x - 50 + 40$

$= 4000 - 50 + 40 = 3990$

Again, from $\sigma^2 + A^2 = \frac{\Sigma x^2}{n}$,

we get $\Sigma x^2 = n(\sigma^2 + A^2)$

$\therefore$ incorrect $\Sigma x^2 = 100\{(5.1)^2 + (40)^2\}$

$= 100(26.01 + 1600) = 162601$

$\therefore$ correct Σx^2 = incorrect $\Sigma x^2 - 50^2 + 40^2$

$= 162601 - 2500 + 1600 = 161701$

$\therefore$ correct mean $A = \frac{\Sigma x}{n} = \frac{3900}{100} = 39.9$

Correct $(\sigma^2 + A^2) = \frac{\Sigma x^2}{n} = \frac{161701}{100} = 1617.01$

$\therefore$ correct $\sigma^2 = 1617.01 - (\text{correct } A^2)$

$= 1617.01 - (39.9)^2 = 1617.01 - 1592.01 = 25$

$\therefore \sigma = 5$

Concept Application

16. If $\sum_{i=1}^{18}(x_i - 8) = 9$ and $\sum_{i=1}^{18}(x_i - 8)^2 = 45$ then the standard deviation of $x_1, x_2, \ldots x_{18}$ is

(*a*) 9/4 (*b*) 9/8

(*c*) 1/6 (*d*) 3/2

17. Assuming the variance of four numbers a, b, c and d as 9, the variance of 5a, 5b, 5c and 5d is

(*a*) 45 (*b*) 5/9

(*c*) 9/5 (*d*) 225

18. A scientist is weighing each of 30 fishes. Their mean weight worked out is 30 gm and a standard deviation of 2 gm. Later, it was found that the measuring scale was misaligned and always under reported every fish weight by 2 gm. The correct mean and standard deviation (in gm) of fishes are respectively:

(*a*) 32, 2 (*b*) 32, 4

(*c*) 28, 2 (*d*) 28, 4

ANALYSIS OF FREQUENCY DISTRIBUTIONS

Measures of dispersion are unable to compare two or more series which are measured in different units even if they have the same mean. Thus, we require those measures which are independent of the units. The measure of variability which is independent of units is called Coefficient of Variation (C.V.). The coefficient of variation is defined as

$C.V. = \frac{\sigma}{\bar{X}} \times 100$

where σ and $\bar{X}$ are the standard deviation and mean of the data.

For comparing the variability of two series, we calculate the coefficient of variation for each series. the series having greater C.V. is said to be more variable or conversely less consistent, less uniform less stable or less homogeneous than the other and the series having lesser C.V. is said to be more consistent (or homogeneous) than the other.

MEAN AND VARIANCE OF BINOMIAL DISTRIBUTION

If the frequencies of the values 0, 1, 2,...., n of a variate are represented by the following coefficients of a binomial :

$q^n, {}^nC_1 q^{n-1}p, {}^nC_2 q^{n-2}p^2, \ldots, p^n$

where p is the probability of the success of the experiment (variate), q is the probability of its failure and $p + q = 1$ i.e. distribution is a binomial distribution then:

$P(x = r) = {}^nC_r q^{n-r} p^r$

mean $\bar{x} = \sum p_i x_i = np$.

Variance $\sigma^2 = npq = \bar{x}\, q$

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Example 18. Find coefficient of variation for first n even natural numbers.

Sol. $\because$ 2, 4, 6, ..., $2n$ are first n even natural no.

$$\sigma^2 = \frac{1}{n}\left(\sum_{k=1}^{n}(2k)^2\right) - \left(\frac{1}{n}\sum_{k=1}^{n}(2k)\right)^2$$

$$= \frac{1}{n} \cdot 4 \cdot \left(\sum_{k=1}^{n}k^2\right) - 4\left(\frac{1}{n}\sum_{k=1}^{n}k\right)^2$$

$$= 4\left(\frac{1}{n}\sum_{k=1}^{n}n^2 - \left(\frac{1}{n}\sum_{k=1}^{n}k\right)^2\right)$$

$$= 4 \times \frac{n^2 - 1}{12} = \frac{n^2 - 1}{3}$$

$$\therefore \sigma = \sqrt{\frac{n^2 - 1}{3}}$$

$$\bar{x} = \frac{2 + 4 + \ldots.. + 2n}{n} = \frac{2}{n} \times \frac{n(n+1)}{2} = (n+1)$$

Hence

$$C.V. = \frac{\sigma}{x} \times 100 = \frac{\sqrt{\frac{n^2 - 1}{3}}}{n+1} \times 100$$

Example 19. Find C.V. of a binomial distribution with n = 18 an p = 0.4 ?

Sol. Given n = 18

P = 0.4

$\because p + q = 1 \Rightarrow q = 1 - p$

$\therefore q = 1 - 0.4 = 0.6$

Thus, mean $(\bar{x}) = np = 18 \times 0.4$

$= 7.2$

Variance $(\sigma^2) = npq = 18 \times 0.4 \times 0.6$

$= 4.32$

S.D. $(\sigma) = 2.08$

Hence, $C.V. = \frac{\sigma}{\bar{x}} \times 100$

$= \frac{2.08}{7.2} \times 100$

$= 28.87$

Concept Application

19. Find C.V. of a binomial distribution in which a fair coin is tossed16 times for getting head

(*a*) 4

(*b*) 100

(*c*) 25

(*d*) 200

20. In the following table mean and variance of earning of male and female is given

	Mean	Variance
Male	162.6 ₹	127.69 ₹
Female	52.36 ₹	23.1361 ₹

Then

(*a*) Female are more homogeneous than male

(*b*) Male are more homogeneous than male

(*c*) S.D. of male is less than the S.D. of female.

(*d*) None of these

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AARAMBH (SOLVED EXAMPLES)

1. Let $x_1, x_2, \ldots, x_n$ be n observations, and let $\bar{x}$ be their arithmetic mean and σ^2 be the variance

Statement-1: Variance of $2x_1, 2x_2, \ldots, 2x_n$ is $4\sigma^2$.

Statement-2: Arithmetic mean $2x_1, 2x_2, ., 2x_n$ is 4 $\bar{x}$.

(1) Statement-1 is false, Statement-2 is true.

(2) Statement-1 is true, statement-2 is true; statement-2 is a correct explanation for statement-1.

(3) Statement-1 is true, statement-2 is true; statement-2 is true; statement-2 is **not** a correct explanation for statement-1.

(4) Statement-1 is true, statement-2 is false.

Sol. $\because \bar{x} = \dfrac{x_1 + x_2 + \ldots + x_n}{n}$

$$\therefore \bar{x}' = \frac{2x_1 + 2x_2 + \ldots + 2x_n}{n} = \frac{2(x_1 + x_2 + \ldots + x_n)}{n}$$

$\bar{x}' = 2\bar{x}$

Since each variate is multiply by 2

$\therefore$ new variance $= 2^2 \times$ old variance $= 4\sigma^2$

2. Find the variance and standard deviation of the following frequency distribution :

Variable (x_i)	2	4	6	8	10	12	14	16
Frequency (f_i)	4	4	5	15	8	5	4	5

If each variate is multiplied by 2 then new variance and standard deviation.

Sol. Calculation of variance and standard deviation

Variable x_i	Frequency f_i	$f_i x_i$	$x_i - \bar{x} = x_i - 9$	$(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^2$
2	4	8	–7	49	196
4	4	16	–5	25	100
6	5	30	–3	9	45
8	15	120	–1	1	15
10	8	80	1	1	8
12	5	60	3	9	45
14	4	56	5	25	100
16	5	80	7	49	245
	$N = \sum f_i = 50$	$\sum f_i x_i = 450$			$\sum f_i(x_i - \bar{x})^2 = 754$

Here N = 50, $\sum f_i x_i = 450$

$$\therefore \bar{x} = \frac{1}{N}(\sum f_i x_i) = \frac{450}{50} = 9$$

We have $\sum f_i (x_i - \bar{x})^2 = 754$

$$\therefore \text{var}(X) = \frac{1}{N}[\sum f_i (x_i - \bar{x})^2] = \frac{754}{50} = 15.08$$

S.D. $= \sqrt{Var(x)} = \sqrt{15.08} = 3.88$

$\because \sigma^2_{new} = a^2 \sigma_{old}, \sigma_{new} = |a| \sigma_{old}$

$\therefore \sigma^2_{new} = 60.32, \sigma_{new} = 7.76$

3. For two data sets, each of size 5, the variance are given to b 4 and 5 and the corresponding means are given to be 2 and respectively. The variance of the combined data set is

(*a*) $\dfrac{11}{2}$ (*b*) 6

(*c*) $\dfrac{13}{2}$ (*d*) $\dfrac{5}{2}$

Sol. Given $n_1 = 5, n_2 = 5$

$\sigma_1^2 = 4, \sigma_2^2 = 5$

$\bar{x}_1 = 2, \bar{x}_2 = 4$

Thus

$$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$$

$$= \frac{5\cdot4 + 5\cdot5}{5+5} + \frac{5\cdot5}{(5+5)^2}(2-4)^2$$

$$= \frac{45}{10} + \frac{1}{4} \times 4 = \frac{11}{2}$$

Therefore, option (*a*) is the correct answer.

4. If $\bar{x}_1 = 50$ $\sigma_1^2 = 25$, $\bar{x}_2 = 100$ $\sigma_2^2 = 50$ are mean variance of two systems. Then which system is mo homogeneous

Sol. $C.V._I = \dfrac{\sigma_1}{x_1} \times 100 = \dfrac{\sqrt{25}}{50} \times 100 = 5x_2 = 10$

$$C.V._{II} = \frac{\sigma_2}{x_2} \times 100 = \frac{\sqrt{50}}{100} \times 100 = 9\sqrt{2} = 7.07$$

$\because C.V._I > C.V._{II}$

$\therefore$ System-II is more homogeneous

5. Calculate the mean deviation about the median for following distribution.

C.I.	10-20	20-30	30-40	40-50	50-60	60-70	70-
Frequency	2	3	8	14	8	3	

Sol. Let us calculate the median and mean deviation for the given data.

C.I.	Fi	CF	Mid paint (x_i)	$\lvert x_i - M \rvert$	$F_i \lvert x_i - M \rvert$
10–20	2	2	15	30	60
20–30	3	5	25	20	60
30–40	8	13	35	10	80
40–50	14	27	45	0	0
50–60	8	35	55	10	80
60–70	3	38	65	20	60
70–80	2	40	75	30	60
Total	$\sum fi = 40$				$\sum f_i \lvert x_i - M \rvert = 400$

$N = \sum f_i = 40$

$N/2 = 40/2 = 20$

The cumulative frequency greater than or equal to $N/2$ is 27 for which the corresponding class interval is 40 – 50.

Median class = 40 – 50

Thus,

$l = 40$

N = 40

C = 13

f = 14

h = 10

Median = l + [(N/2 – C)/f] × h

= 40 + [(20 – 13)/14] × 10

= 40 + (7/14) × 10

= 40 + 5

= 45

Therefore, median (M) = 45

From the above table,

$\sum f_i |x_i - M| = 400$

$N = \sum f_i = 40$

Hence, the required mean deviation = 400/40 = 10

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SCHOOL LEVEL PROBLEMS

SINGLE CORRECT TYPE QUESTIONS

1. If the variance of the data is 121 then the standard deviation of the data is
   (a) 121 (b) 11
   (c) 12 (d) 21
2. The mean deviation from the mean for the following data: 4. 7, 8. 9. 10. 12. 13 and 17 is
   (a) 2 (b) 3
   (c) 4 (d) 5
3. The mean of 1, 3,4, 5, 7, 4 is m, the numbers 3, 2, 2, 4, 3, 3, p have mean $m-1$ and median q. Then, $p+q=$
   (a) 4 (b) 5
   (c) 6 (d) 7
4. The coefficient of variation is computed by
   (a) S.D/Mean × 100
   (b) S.D./Mean
   (c) Mean./S.D × 100
   (d) Mean/S.D.
5. If the difference of mode and median of a data is 24. Then the difference of median and mean is
   (a) 12 (b) 24
   (c) 8 (d) 36
6. Mean of the first n terms of the A.P. $a+(a+d)+(a+2d)$ +.....is
   (a) $a+nd/2$
   (b) $a+(n-1)d$
   (c) $a+(n-1)d/2$
   (d) $a+nd$
7. The mean of a group of 100 observations was found to be 20. Later on: it was found that three observations were incorrect, which was recorded as 21, 21 and 18. Then the mean, if the incorrect observations are omitted is
   (a) 18 (b) 20
   (c) 22 (d) 24
8. The variance of the given data 2, 4, 5, 6, 8,17 is 23.33. Then find the variance for the data 4, 8,10,12,16, 34.
   (a) 23.23 (b) 25.33
   (c) 46.66 (d) 93.32
9. If mode of a series exceeds its mean by 12, then mode exceeds the median by
   (a) 4 (b) 8
   (c) 6 (d) 12

VERY SHORT ANSWER TYPE QUESTIONS

10. In a test with a maximum marks 25, eleven students scored 3, 9, 5, 3, 12, 10, 17, 4, 7, 19, 21 marks respectively. Calculate the range.
11. Write the formula for
    (i) Mean Deviation (ii) Variance
12. Find the mean deviation about the mean for the data

x_i	5	7	9	10	12	15
f_i	8	6	2	2	2	6

13. If the mean of the following data is 20.6, then find the val of p.

x	10	15	p	25	35
f	3	10	25	7	5

14. If a variable takes discrete values $x+4, x-7/2, x-5/2, x-3, x-2, x+1/2, x-1/2, x$ $(x>0)$ then find median.
15. An analysis of monthly wages point to workers in two fi A and B, belonging to the same industry, given the follow result.

	Firm A	Firm B
Number of workers	500	600
Average monthly wage	Rs.1860	Rs.1750
Variance of distribution ofwages	81	100

The average of monthly wage and variance of distributio wages of all the workers in the firms A and B taken toge are

16. Two sets each of 20 observations, have the same stan deviation 5. The first set has a mean 17 and the second a mea Determine the standard deviation of the set obtained combining the given two sets.

LONG ANSWER TYPE QUESTIONS

17. Coefficient of variation of two distributions is 70 and 75, their standard deviations are 28 and 27 respectively. W are their arithmetic mean?
18. The weights of coffee in 70 jars is shown tn the follow table:

Weight (in grams)	Frequency
200-201	13
201-202	27
202-203	18
203-204	10
204-205	1
205-206	1

Determine variance and standard deviation of the a distribution.

19. Calculate the mean deviation about the mean for the following frequency distribution:

Class Interval	Frequency
0-4	4
4-8	6
8-12	8
12-16	5
16-20	2

20. Find the variance and the standard deviation for the following data: 57, 64, 43, 67,49, 59, 44, 47, 61, 59

CASE STUDY BASED QUESTIONS

Case Study-I

There are 60 students in a class. The following is the frequency distribution of the marks obtained by the students in a test:

Marks	Frequency
0	$x-2$
1	x
2	x^2
3	$(x+1)^2$
4	$2x$
5	$x+1$

where x is a positive integer.

21. Determine the mean for the given data
    (a) 2 (b) 2.8
    (c) 3.2 (d) 4.2
22. Standard deviation for the given data is
    (a) 1.12 (b) 2.2 (c) 3.2 (d) 1

Case Study-II

Transport department of a city wants to buy some electric buses for the city. For which they wants to analyse the distance travelled by existing public transport buses in a day.

The following data shows the distance travelled by 60 existing public transport buses in a day.

Daily distance travelled (in km)	Number of buses
200-209	4
210-219	14
220-229	26
230-239	10
240-249	6

Based on the above information, answer the following questions.

23. The cumulative frequency of the class preceding the median class is
    (a) 14 (b) 18 (c) 26 (d) 10
24. If the mode of the distance travelled is 223.78 km, then mean of the distance travelled by the bus is
    (a) 225 km (b) 220 km
    (c) 230.29 km (d) 224.29 km

PRARAMBH (TOPICWISE)

ARITHMETIC MEAN

1. If the mean of 3, 4, x, 7, 10 is 6, then the value of x is:
   (a) 4 (b) 5 (c) 6 (d) 7
2. The $A.M.$ of the series 1, 2, 4, 8, 16, ..., 2^n is:
   (a) $\frac{2^n-1}{n}$ (b) $\frac{2^{n+1}-1}{n+1}$ (c) $\frac{2^n+1}{n}$ (d) $\frac{2^n-1}{n+1}$
3. Mean of the first n terms of the $A.P.$ $a+(a+d)+(a+2d)+...$ is
   (a) $a+\frac{nd}{2}$ (b) $a+\frac{(n-1)d}{2}$
   (c) $a+(n-1)d$ (d) $a+nd$
4. If the mean of n observations $1^2, 2^2, 3^2, ..., n^2$ is $\frac{46n}{11}$, then n is equal to
   (a) 11 (b) 12
   (c) 23 (d) 22
5. Following table shows the weight of 12 students:

Weight (in kgs.)	67	70	72	73	75
No. of students	4	3	2	2	1

Find their mean weight
(a) 70.25 kg. (b) 70.50 kg
(c) 70.75 kg. (d) None of these

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6. A factory employs 100 workers of whom 60 work in the first shift and 40 work in the second shift. The average wage of all the 100 workers is Rs.38. If the average wage of 60 workers of the first shift is Rs.40, then the average wage of the remaining 40 workers of the second shift is:
(a) 35 (b) 40
(c) 45 (d) None of these

7. If $\bar{x}$ is the mean of a set of n observations $x_1, x_2, x_3, ..., x_n$ then $\sum_{i=1}^{n}(x_i - \bar{x})$ is equal to
(a) Mean deviation about mean
(b) Standard deviation
(c) Zero
(d) None of these

8. The mean of a set of observations is $\bar{x}$. If each observation is divided by α, $\alpha \neq 0$, and then is increased by 10 then the mean of the new set is-
(a) $\frac{\bar{x}}{\alpha}$ (b) $\frac{\bar{x}+10}{\alpha}$
(c) $\frac{\bar{x}+10\alpha}{\alpha}$ (d) $a\bar{x}+10$

9. If the mean of first n natural numbers is equal to $\frac{n+7}{3}$, then n is equal to -
(a) 10 (b) 11
(c) 12 (d) None of these

10. The mean of first three terms is 14 and mean of next two terms is 18. The mean of all the five terms is:
(a) 14.5 (b) 15.0
(c) 15.2 (d) 15.6

11. The weighted arithmetic mean of first n natural numbers whose weight are equal to the corresponding numbers is equal to
(a) $2n+1$ (b) $\frac{1}{2}(2n+1)$
(c) $\frac{1}{3}(2n+1)$ (d) $\frac{(2n+1)}{6}$

12. The mean of distribution, in which the values of X are 1, 2, ..., n the frequency of each being unity is
(a) $\frac{n(n+1)}{2}$ (b) $\frac{n}{2}$
(c) $\frac{n+1}{2}$ (d) $\frac{n(n-1)}{2}$

GEOMETRIC MEAN & HARMONIC MEAN

13. Geometric mean of the numbers $2, 2^2, 2^3, ..., 2^n$ is
(a) $2^{2/n}$ (b) $2^{n/2}$
(c) $2^{\frac{n-1}{2}}$ (d) $2^{\frac{n+1}{2}}$

14. The harmonic mean of 4, 8, 16 is:
(a) 6.4 (b) 6.7 (c) 6.85 (d) 7.8

MEDIAN

15. The median of 10, 14, 11, 9, 8, 12, 6 is
(a) 10 (b) 12
(c) 14 (d) 11

16. If a variable takes the discrete values
$\alpha+4, \alpha-\frac{7}{2}, \alpha-\frac{5}{2}, \alpha-3, \alpha-2, \alpha+\frac{1}{2}, \alpha-\frac{1}{2}, \alpha+5 (\alpha>0)$,
then the median is:
(a) $\alpha-\frac{5}{4}$ (b) $\alpha-\frac{1}{2}$
(c) $\alpha-2$ (d) $\alpha+\frac{5}{4}$

17. The median of the items 6, 10, 4, 3, 9, 11, 22, 18 is:
(a) 9 (b) 10
(c) 9.5 (d) 11

18. Find the median from the following distribution

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of students	10	20	30	50	40	30

is
(a) 30 (b) 45 (c) 36 (d) 40

MODE

19. Mode of the distribution

Marks	4	5	6	7	8
No. of students	3	5	10	6	1

is
(a) 6 (b) 10
(c) 8 (d) None of these

20. The mode of the following distribution

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of students	2	18	30	45	35	20	6	3

is
(a) 36 (b) 37.3 (c) 38 (d) 39

21. If median = (mode + 2 mean)M, then M is equal to:
(a) 3 (b) 1/3
(c) 2 (d) None of these

DISPERSION

22. The standard deviation of a variate x is σ. The standar deviation of the variable $\frac{ax+b}{c}$; a, b, c are constants, is
(a) $\left(\frac{a}{c}\right)\sigma$ (b) $\left|\frac{a}{c}\right|\sigma$
(c) $\left(\frac{a^2}{c^2}\right)\sigma$ (d) None of these

23. The scores of a batsman in ten innings are: 38, 70, 48, 34, 42, 55, 63, 46, 54, 44, then mean deviation about the median is:
(a) 8.4 (b) 8.5 (c) 8.6 (d) 8.8

24. Mean deviation from the mean for the observations $-1, 0, 4$ is
(a) $\sqrt{\frac{14}{3}}$ (b) $\frac{2}{3}$
(c) 2 (d) None of these

25. The Standard deviation of 7 scores 1, 2, 3, 4, 5, 6, 7 is
(a) 4 (b) 2
(c) $\sqrt{7}$ (d) None of these

26. Let σ be the standard deviation of n observations. Each of the n observations is multiplied by a constant c. Then the standard deviation of the resulting numbers is-
(a) σ (b) $|c| \cdot \sigma$
(c) $\sigma\sqrt{c}$ (d) None of these

27. The mean of 100 items is 50 and their Standard deviation is 4, the sum of all the items and also the sum of the squares of the items is
(a) 5000, 251600 (b) 4000, 251600
(c) 5000, 26100 (d) 3000, 26100

28. The mean and Standard deviation of distributions of 100 and 150 items are 50, 5 and 40, 6 respectively. The mean and Standard deviation of all the 250 items taken together are
(a) 44, 6.46 (b) 42, 7.46
(c) 44, 7.46 (d) 42, 6.46

29. The variance of first n natural numbers is 10 then n is
(a) 12 (b) 11 (c) 10 (d) 10 – 11

30. Mean deviation of the series $a, a+d, a+2d, ..., a+2nd$ from its mean is
(a) $\frac{(n+1)d}{2n+1}$ (b) $\frac{nd}{2n+1}$
(c) $\frac{n(n+1)d}{2n+1}$ (d) $\frac{(2n+1)d}{n(n+1)}$

PRABAL (JEE MAIN LEVEL)

1. The mean weight of 9 items is 15. If one more item is added to the series, the mean becomes 16. The value of 10th item is
(a) 35 (b) 30 (c) 25 (d) 20

2. In the frequency distribution of the discrete data given below, the frequency k against value 0 is missing.

Variable x :	0	1	2	3	4	5
Frequency f :	k	20	40	40	20	4

If the mean is 2.5, then the missing frequency k will be
(a) 0 (b) 1 (c) 3 (d) 4

3. If mean of n item is $\bar{x}$. If each r^{th} item is increased by $2r$. Then new mean will be
(a) $\bar{x}$ (b) $\bar{x}+\frac{n}{2}$
(c) $\bar{x}+\frac{n+2}{2}$ (d) $\bar{x}+n+1$

4. The SD of 15 items is 6 and if each item decreases by 1, then standard deviation will be
(a) 5 (b) 7 (c) $\frac{91}{15}$ (d) 6

5. The sum of squares of deviations for 10 observations taken from mean 50 is 250. The coefficient of variation is
(a) 10% (b) 40%
(c) 50% (d) None of these

6. Mean of n items is $\bar{x}$. If these n items are increased by $1^2, 2^2, 3^2, ..., n^2$ successively, then mean gets increased by
(a) $\frac{(n+1)(2n+1)}{6}$
(b) $\frac{n(n+1)(2n+1)}{6}$
(c) $\frac{n^2}{2}$
(d) Remains same

7. The mean value of the median and mean of the odd divisors of 360 is
(a) 13 (b) 7
(c) 6 (d) 10

8. A sample of 35 observations has the mean 80 and Standard deviation as 4. A second sample of 65 observation from the same population has mean 60 and Standard deviation 3. The Standard deviation of the combined sample is
(a) 5.85 (b) 5.58
(c) 10.12 (d) None of these

9. The mean and Standard deviation of the marks of 200 candidates were found to be 40 and 15 respectively. Later, it was discovered that a score of 40 was wrongly read as 50. The correct mean and Standard deviation respectively are
(a) 14.98, 39.95 (b) 39.95, 16.12
(c) 39.95, 224.5 (d) None of these

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10. The average of n numbers $x_1, x_2, x_3, ..., x_n$ is M. If x_1 is replaced by x, then the new average is

(a) $M - x_1 + x$ (b) $\frac{(n-1)M - x_1 + x}{n}$

(c) $\frac{(n+1)M - x_1 + x}{n}$ (d) $\frac{nM - x_1 + x}{n}$

11. The variance of α, β and γ is 9, then variance of 5α, 5β and 5γ is

(a) 45 (b) 9/5 (c) 5/9 (d) 225

12. The mean of n items is $\bar{x}$. If each item is successively increased by $3, 3^2, 3^3, ..., 3^n$, then new mean equals

(a) $\bar{x} + \frac{3^{n+1}}{n}$ (b) $\bar{x} + 3\frac{(3^n - 1)}{2n}$

(c) $\bar{x} + \frac{3^n}{n}$ (d) $\bar{x} + 3\frac{(3^n - 1)}{n}$

13. In a class of 100 students there are 70 boys whose average marks in a subject are 75. If the average marks of the complete class is 72, then what is the average of the girls?

(a) 73 (b) 65 (c) 68 (d) 74

14. The median of a set of 9 distinct observation is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set

(a) Is increased by 2
(b) Is decreased by 2
(c) Is two times the original median
(d) Remains the same as that of the original set

15. In an experiment with 15 observations on x, the following results were available

$\sum x^2 = 2830, \sum x = 170$

One observation that was 20, we found to be wrong and was replaced by the correct value 30. Then, the corrected variance is

(a) 78.00 (b) 188.66

(c) 177.33 (d) 8.33

16. In a series of $2n$ observations, half of them equal a and remaining half equal $-a$. If the standard deviation of the observations is 2, then $|a|$ equals

(a) $\frac{1}{n}$ (b) $\sqrt{2}$ (c) 2 (d) $\frac{\sqrt{2}}{n}$

17. If the standard deviation of the observations $-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$ is $\sqrt{10}$. The standard deviation of observations 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 will be

(a) $\sqrt{10} + 20$ (b) $\sqrt{10} + 10$

(c) $\sqrt{10}$ (d) None of these

18. Median of ${}^{2n}C_0, {}^{2n}C_1, {}^{2n}C_2, {}^{2n}C_3, ..., {}^{2n}C_n$ (where n is even) is

(a) ${}^{2n}C_{\frac{n}{2}}$ (b) ${}^{2n}C_{\frac{n+1}{2}}$

(c) ${}^{2n}C_{\frac{n-1}{2}}$ (d) None of these

INTEGER TYPE QUESTION

19. If the mean deviation of number $1, 1 + d, 1 + 2d, ..., 1 + 100d$ from their mean is 255, then $[d]$ is equal to (where [] is G.I.F.)

20. The means and variance of n observations $x_1, x_2, x_3, ..., x_n$ are 5 and 0 respectively. If $\sum_{i=1}^{n} x_i^2 = 400$, then the value of n is equal to

21. The mean of 30 given numbers, when it is given that the mean of 10 of them is 12 and the mean of the remaining 20 is 9, is equal to

22. The algebraic sum of the deviation of 20 observations measured from 30 is 2. Then, if mean of observations is M. Find $[M]$. (where [] is G.I.F.)

23. If the average of the numbers 148, 146, 144, 142, ... in AP be 125, then the total numbers in the series will be

24. A group of 10 items has arithmetic mean 6. If the arithmetic mean of 4 of these items is 7.5, then the mean of the remaining items is

25. Mean of 100 observation is 45. If it was later found that two observations 19 and 31 were incorrectly recorded as 91 and 13. If the correct mean is M, find $[M]$. (where [] is G.I.F.)

26. In a class of 50 students, 10 have failed and their average marks are 28. The total marks obtained by the entire class are 2800. The average marks of those who have passed, are

PYQ's (PAST YEAR QUESTIONS)

ARITHMETIC MEAN, GEOMETRIC MEAN, HARMONIC MEAN

1. The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If the mean age of the teachers in this school now is 39 years, then the age (in years) of the newly appointed teacher is: **[18 March, 2021(Shift-I)]**

2. If for some $x \in R$, the frequency distribution of the marks obtained by 20 students in a test is: **[10 April, 2019 (Shift-I)]**

Marks	2	3	5	7
Frequency	$(x+1)^2$	$2x-5$	x^2-3x	x

then the mean of the marks is:

(a) 2.8 (b) 3.2 (c) 3.0 (d) 2.5

MEDIAN, MODE AND RELATION BETWEEN MEAN, MEDIAN AND MODE

3. Consider the given data with frequency distribution

x_i	3	8	11	10	5	4
f_i	5	2	3	2	4	4

Match List-I with List-II:

	List-I	List-II	
A.	The mean of the above data is	I.	2.5
B.	The median of the above data is	II.	5
C.	The mean deviation about the mean of the above data is	III.	6
D.	The mean deviation about the median of the above data is	IV.	2.7
E.		V.	2.4

Choose the correct answer from the options given below: **[JEE Adv, 2023]**

(a) A → III; B → II; C → IV, D → V
(b) A → III; B → II; C → I; D → V
(c) A → II; B → III; C → IV; D → I
(d) A → III; B → III; C → V; D → V

4. Let $x_1, x_2, x_3, ..., x_{20}$ be in geometric progression with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data is constructed replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal to $\bar{x}$ is **[28 July, 2022 (Shift-I)]**

5. Consider the following frequency distribution:

Class	10-20	20-30	30-40	40-50	50-60
Frequency	α	110	54	30	β

If the sum of all frequencies is 584 and median is 45, then $|\alpha - \beta|$ is equal to. **[25 July, 2021 (Shift-I)]**

6. Consider the data on x taking the values $0, 2, 4, 8, ..., 2^n$ with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, ..., {}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$ then n is equal to _____.

[6 Sep, 2020 (Shift-II)]

7. The mean and the median of the following ten numbers in increasing order 10, 22, 26, 29, 34, x, 42, 67, 70, y are 42 and 35 respectively, then $\frac{y}{x}$ is equal to: **[9 April, 2019 (Shift-II)]**

(a) $\frac{7}{3}$ (b) $\frac{9}{4}$ (c) $\frac{7}{2}$ (d) $\frac{8}{3}$

MEASURES OF DISPERSION, VARIOUS MEASURES, RANGE

8. Let μ be the mean and σ be the standard deviation of the distribution

x_1	0	1	2	3	4	5
f_1	$k+2$	$2k$	k^2-1	k^2-1	k^2+1	$k-3$

where $\sum f_i = 62$. if $[x]$ denotes the greatest integer $\le x$, then $[\mu^2 + \sigma^2]$ is equal **[10 April, 2023 (Shift-II)]**

(a) 8 (b) 7 (c) 6 (d) 9

9. The mean of 10 numbers $7\times8, 10\times10, 13\times12, 16\times14, ...$ is _____. **[31 Aug, 2021 (Shift-I)]**

MEAN DEVIATION

10. Let the mean of 6 observation 1, 2, 4, 5, x and y be 5 and their variance be 10. Then their mean deviation about the mean is equal to **[11 April, 2023 (Shift-II)]**

(a) $\frac{10}{3}$ (b) $\frac{7}{3}$ (c) 3 (d) $\frac{8}{3}$

11. Let the mean of the data

x	1	3	5	7	9
Frequency(f)	4	24	28	α	8

be 5. If m and σ^2 are respectively the mean deviation about the mean and the variance of the data, then $\frac{3\alpha}{m+\sigma^2}$ is equal to _____. **[13 April, 2023 (Shift-I)]**

12. If the mean deviation about the mean of the numbers 1, 2, 3, ..., n, where n is odd, is $\frac{5(n+1)}{n}$, then n is equal to_____

[25 June, 2022 (Shift-II)]

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13. If the sum of the deviations of 50 observations from 30 is 50, then the mean of these observations is:
[12 Jan, 2019 (Shift-I)]
(a) 30 (b) 51 (c) 50 (d) 31

STANDARD DEVIATION, VARIANCE

14. Let the mean and variance of 12 observations be $\frac{9}{2}$ and 4 respectively. Later on, it was observed that two observations were considered as 9 and 10 instead of 7 and 14 respectively. If the correct variance is $\frac{m}{n}$, where m and n are co-prime, then $m + n$ is equal to [8 April, 2023 (Shift-II)]
(a) 316 (b) 314 (c) 317 (d) 315

15. If the mean and variance of the frequency distribution

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

are 9 and 15.08 respectively, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is
[6 April, 2023 (Shift-II)]

16. The mean and variance of the data 4, 5, 6, 6, 7, 8, x, y, where $x < y$, are 6 and $\frac{9}{4}$ respectively. Then $x^4 + y^2$ is equal to [27 June, 2022 (Shift-II)]
(a) 162 (b) 320 (c) 674 (d) 420

17. The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5 . Then, the correct variance is equal to. [28 June, 2022 (Shift-I)]

18. Consider three observations a, b and c such that $b = a + c$. If the standard deviation of $a + 2$, $b + 2$, $c + 2$ is d, then which of the following is true? [16 March, 2021 (Shift-I)]
(a) $b^2 = 3(a^2 + c^2) - 9d^2$ (b) $b^2 = a^2 + c^2 + 3d^2$
(c) $b^2 = 3(a^2 + c^2 + d^2)$ (d) $b^2 = 3(a^2 + c^2) + 9d^2$

19. Let the mean and variance of four numbers 3, 7, x and $y (x > y)$ be 5 and 10 respectively. Then the mean of four numbers $3 + 2x$, $7 + 2y$, $x + y$ and $x - y$ is [26 Aug, 2021(Shift-II)]

20. Let $X = \{x \in N : 1 \le x \le 17\}$ and $Y = \{ax + b : x \in X\}$ and $a, b \in R, a > 0$. If mean and variance of elements of Y are 17 and 216 respectively then $a + b$ is equal to:
[2 Sep, 2020 (Shift I)]
(a) 7 (b) –27 (c) 9 (d) –7

21. For the frequency distribution: [3 Sep, 2020 (Shift I)]
Variate $(x) : x_1, x_2, x_3 ... x_{15}$
Frequency $(f) : f_1, f_2, f_3 ... f_{15}$
Where $0 < x_1 < x_2 < x_3 < ... < x_{15} = 10$ and $\sum_{i=1}^{15} f_i > 0$, then standard deviation cannot be:
(a) 1 (b) 6 (c) 2 (d) 4

22. If the variance of the following frequency distribution:
[4 Sep, 2020 (Shift-II)]

Class	10 – 20	20 – 30	30 – 40
Frequency	2	x	2

is 50, then x is equal to _____.

23. The mean of five observations is 5 and their variance is 9.2. If three of the given five observations are 1, 3 and 8, then ratio of other two observations is: [10 Jan, 2019 (Shift-I)]
(a) 10 : 3 (b) 4 : 9 (c) 5 : 8 (d) 6 : 7

VARIOUS PROPERTIES OF VARIANCE

24. Two dice A and B are rolled, Let the numbers obtained on A and B be α and β respectively. If the variance of α – β is p/q, where p and q are coprime, then the sum of the positive divisors of p is equal to [12 April, 2023 (Shift-I)]
(a) 36 (b) 48 (c) 31 (d) 72

25. The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of the observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to.....
[26 July, 2022 (Shift-II)]

26. Consider a set of $3n$ numbers having variance 4. In this set, the mean of first $2n$ numbers is 6 and the mean of the remaining n number is 3. A new set is constructed by adding 1 in each of first $2n$ numbers, and subtracting 1 from each of the remaining n numbers. If the variance of the new set is k, then $9k$ is equal to_____. [17 March, 2021 (Shift-II)]

MISCELLANEOUS

27. The mean and variance of a set of 15 numbers are 12 and 14 respectively. The mean and variance of another set of numbers are 14 and σ^2 respectively. If the variance of all the 30 numbers in the two sets is 13, then σ^2 is equal to
[6 April, 2023 (Shift-I)]
(a) 9 (b) 12 (c) 11 (d) 10

28. The mean and variance of 7 observations are 8 and respectively. If one observation 14 is omitted a and b respectively mean and variance of remaining 6 observation then $a + 3b - 5$ is equal to [30 Jan, 2023 (Shift-I)]

29. Let $X = \{11, 12, 13, ..., 40, 41\}$ and $Y = \{61, 62, 63, ... 90, 91\}$ be the two sets of observations. If $\bar{x}$ and $\bar{y}$ are the respective means and σ^2 is the variance of all the observations in $X \cup Y$, then $\left|\bar{x} + \bar{y} - \sigma^2\right|$ is equal to _____. [29 Jan, 2023 (Shift-I)]

30. If the mean and variance of eight numbers 3, 7, 9, 12, 20, x and y be 10 and 25 respectively, then $x.y$ is equal to _____. [7 Jan, 2020 (Shift-I)]

Answer Key

CONCEPT APPLICATION

1. (b) 2. (d) 3. (a) 4. (d) 5. (b) 6. (d) 7. (a) 8. (c) 9. (d) 10. (c)
11. (c) 12. (d) 13. (a) 14. (d) 15. (d) 16. (d) 17. (d) 18. (a) 19. (c) 20. (b)

SCHOOL LEVEL PROBLEMS

1. (b) 2. (b) 3. (d) 4. (a) 5. (a) 6. (c) 7. (b) 8. (c) 9. (b) 21. (b)
22. (a) 23. (b) 24. (d)

PRARAMBH (TOPICWISE)

1. (c) 2. (b) 3. (b) 4. (a) 5. (a) 6. (a) 7. (c) 8. (c) 9. (b) 10. (d)
11. (c) 12. (c) 13. (d) 14. (c) 15. (a) 16. (a) 17. (c) 18. (c) 19. (a) 20. (a)
21. (b) 22. (b) 23. (c) 24. (c) 25. (b) 26. (b) 27. (a) 28. (c) 29. (b) 30. (c)

PRABAL (JEE MAIN LEVEL)

1. (c) 2. (d) 3. (d) 4. (d) 5. (a) 6. (a) 7. (d) 8. (c) 9. (d) 10. (d)
11. (d) 12. (b) 13. (b) 14. (d) 15. (a) 16. (c) 17. (c) 18. (a) 19. [10] 20. [16]
21. [10] 22. [30] 23. [24] 24. [5] 25. [44] 26. [63]

PYQ's (PAST YEAR QUESTIONS)

1. [35] 2. (a) 3. (a) 4. [142] 5. [164] 6. [6] 7. (a) 8. (a) 9. [398] 10. (d)
11. [8] 12. [21] 13. (d) 14. (c) 15. [25] 16. (b) 17. [17] 18. (a) 19. [12] 20. (d)
21. (b) 22. [4] 23. (b) 24. (b) 25. [238] 26. [68] 27. (d) 28. [37] 29. [603] 30. [54]

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CHAPTER 18

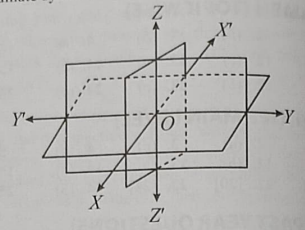
Introduction to Three Dimensional Geometry

INTRODUCTION TO THREE DIMENSIONAL GEOMETRY

You may recall that to locate the position of a point in a plane, we need two intersecting mutually perpendicular lines in the plane. These lines are called the coordinate axes and the two numbers are called the coordinates of the point with respect to the axes. In actual life, we do not have to deal with points lying in a plane only. For example, if we were to locate the position of the lowest tip of an electric bulb hanging from the ceiling of a room or the position of the central tip of the ceiling fan in a room, we will not only require the perpendicular distances of the point to be located from two perpendicular walls of the room but also the height of the point from the floor of the room. Therefore, we need not only two but three numbers representing the perpendicular distances of the point from three mutually perpendicular planes, namely the floor of the room and two adjacent walls of the room. The three numbers representing the three distances are called the coordinates of the point with reference to the three coordinate planes. So, a point in space has three coordinates.

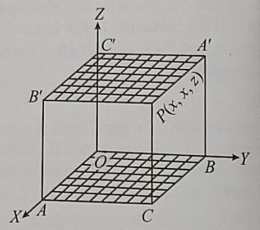
Coordinate Axes and Coordinate Planes in Three Dimensional Space

Consider three planes intersecting at a point O such that these three planes are mutually perpendicular to each other. These three planes intersect along the lines $X'OX$, $Y'OY$ and $Z'OZ$, called the x, y and z-axes, respectively. We may note that these lines are mutually perpendicular to each other. These lines constitute the rectangular coordinate system. The planes XOY, YOZ and ZOX, called, respectively the XY-plane, YZ-plane and the ZX-plane, are known as the three coordinate planes. We take the XOY plane as the plane of the paper and the line $Z'OZ$ as perpendicular to the plane XOY. If the plane of the paper is considered as horizontal, then the line $Z'OZ$ will be vertical. The distances measured from XY-plane upwards in the direction of OZ are taken as positive and those measured downwards in the direction of OZ' are taken as negative. Similarly, the distance measured to the right of ZX-plane along OY are taken as positive, to the left of ZX-plane and along OY' as negative, in front of the YZ-plane along OX as positive and to the back of it along OX' as negative. The point O is called the origin of the coordinate system.



Coordinates of a Point in Space

P be a point in space. Through P draw three planes parallel to the coordinate axes to meet the axes in A, B and C respectively. Let $OA = x$, $OB = y$ and $OC = z$. These three real numbers taken in order determined by the point P are called the coordinates of the point P, written as (x, y, z). x, y, z are positive or negative according as they are measured along positive and negative directions of the coordinate axes.

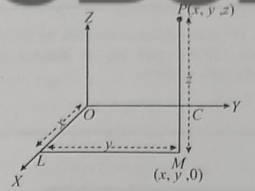


$x = OA = CB = PA'$ = perpendicular distance from P on the YOZ plane

$y = OB = A'C' = PB'$ = Perpendicular distance from P on the ZOX plane

$z = OC' = A'B = PC$ = Perpendicular distance from P on the XOY plane.

Thus, the coordinates of the point P are the perpendicular distances from P on the three mutually rectangular coordinate planes YOZ, ZOX and XOY respectively.



Alternatively, to find the coordinates of a point P in space, we first draw Perpendicular PM on the xy-plane with M as the foot of this perpendicular as shown in figure. Now, from the point M, we draw perpendicular ML on x-axis with L as the foot of this perpendicular. If $OL = x$, $LM = y$ and $PM = z$, then we say that a, b and c are x, y, and z coordinates, respectively, of the point P in space. In such a case, we say that the point P has coordinates (a, b, c).

Note: The coordinates of the origin O are $(0, 0, 0)$. The coordinates of any point on the x-axis will be as $(x, 0, 0)$ and the coordinates of any point in the YZ-plane will be as $(0, y, z)$.

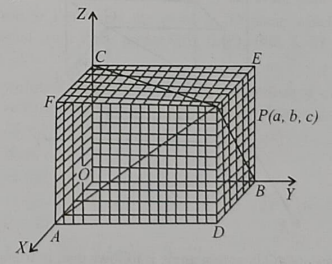
Sign of the Coordinates of a Point

The sign of the coordinates of a point determine the octant in which the point lies. The following table shows the signs of the coordinates in eight octants.

Octants / Coordinates	I	II	III	IV	V	VI	VII	VIII
x	+	−	−	+	+	−	−	+
y	+	+	−	−	+	+	−	−
z	+	+	+	+	−	−	−	−

Train Your Brain

Example 1: In figure, if the coordinates of point P are (a, b, c), then



(i) write the coordinates of points A, B, C, D, E and F.

(ii) write the coordinates of the feet of the perpendiculars from the point P to coordinate axes.

(iii) write the coordinates of the feet of the perpendicular from the point P on the coordinate planes XY, YZ and ZX.

(iv) Find the perpendicular distances of point P from XY, YZ and ZX-planes.

(v) Find the perpendicular distances of the point P from the coordinate axes,

(vi) Find the coordinates of the reflection of P in XY, YZ and ZX-planes.

Sol. (i) Since the coordinates of P are (a, b, c). Therefore, $OA = a$, $OB = b$ and $OC = c$.

Now, A lies on OX such that $OA = a$. So, the coordinates of A are $(a, 0, 0)$.

Similarly, coordinates of B and C are $(0, b, 0)$ and $(0, 0, c)$ respectively.

Since D lies in XY-plane such that $OA = a$ and $AD = OB = b$. So, the coordinates of D are $(a, b, 0)$. Point E lies in YZ-plane such that $OB = b$ and $BE = OC = c$. So, the coordinates of E are $(0, b, c)$. Similarly, coordinates of F are $(a, 0, c)$ as it lies in XZ-plane.

(ii) PA, PB and PC are perpendiculars from P on OX, OY and OZ respectively. So, A, B and C are the feet of perpendiculars from P on OX, OY and OZ respectively. Their coordinates are $A(a, 0, 0)$, $B(0, b, 0)$ and $C(0, 0, c)$ as discussed in (i).

(iii) Clearly, PD, PE and PF are the perpendiculars from P on XY, YZ and ZX-planes respectively, So, D, E and F are the feet of the perpendiculars from P on XY, YZ and ZX-planes. The coordinates of D, E and F are D $(a, b, 0)$, E $(0, b, c)$ and $F(a, 0, c)$ respectively as discussed in (i).

(iv) PD, PE and PF are the perpendicular distances of P from XY, YZ and ZX-planes respectively.

$\therefore$ $PD = OC = c$, $PE = OA = a$ and $PF = OB = b$.

Hence, the perpendicular distances of $P(a, b, c)$ from XY, YZ and ZX planes are c, a and b respectively.

(v) PA, PB and PC are the perpendicular distances of point P from OX, OY and OZ respectively.

In right-angled triangle ADP, we have

$AP^2 = AD^2 + DP^2$

$\Rightarrow PA = \sqrt{AD^2 + DP^2} = \sqrt{b^2 + c^2}$

$[\because AD = OB = b$ and $PD = OC = c]$

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In right-angled triangle PDB right angled at D, we have $PB^2 = BD^2 + PD^2$

$\Rightarrow PB = \sqrt{BD^2 + PD^2} = \sqrt{a^2 + c^2}$

$[\because BD = OA = a$ and $PD = OC = c]$

In right-angled triangle PCF right angled at F, we have

$PC^2 = PF^2 + CF^2$

$\Rightarrow PC = \sqrt{PF^2 + CF^2}$

$\Rightarrow PC = \sqrt{b^2 + a^2}$

$[\because PF = AD = OB = b$ and $CF = OA = a]$

$\Rightarrow PC = \sqrt{a^2 + b^2}$

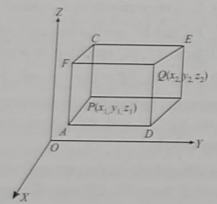
(vi) The reflection or image of $P(a, b, c)$ in xy-plane will be as much below the xy-plane as point P is above it, that is, if P' is the reflection of P in xy-plane, then $P'D = PD = c$ and $P'D$ is parallel to OZ. So, the coordinate of P' are $(a, b, -c)$.

The image of $P(a, b, c)$ in yz-plane will be as much on the back side of yz-plane as the point P is on its front side. Thus, of P'' is the image of P in yz-plane, then P'' lies on PE such that $PE = EP''$. But, $PE = OA = a$. So, the coordinates of P'' are $(-a, b, c)$.

The image of $P(a, b, c)$ in zx-plane will be as much as on the left side of xz-plane as the point P is on its right side. Thus, if P'' is the image of P in zx-plane P'' lies on PF produced such that $PF = FP''$. But, $PF = OB = b$. So, the coordinates of P'' are $(a, -b, c)$.

Example 2: Planes are drawn parallel to the coordinate planes through the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$. Find the length of the edges of the parallelopiped so formed.

Sol. Clearly, PA, PB and PC are the lengths of the edges of the parallelopiped shown in figure.



Clearly, $PBEC$, $QDAF$ are planes parallel to yz-plane such that their distances from yz-plane are x_1 and x_2 respectively. So, PA = (Distance between the planes $PBEC$ and $QDAF$) $= x_2 - x_1$.

PB is the distance between the planes $PAFC$ and $BDQE$ which are parallel to zx-plane and are at distances y_1 and y_2, respectively, from zx-plane.

$\therefore \quad PB = y_2 - y_1$

Similarly, PC is the distance between the parallel planes $PBDA$ and $CEQF$ which are at distance z_1 and z_2 respectively, from xy-plane.

$\therefore \quad PC = z_2 - z_1$

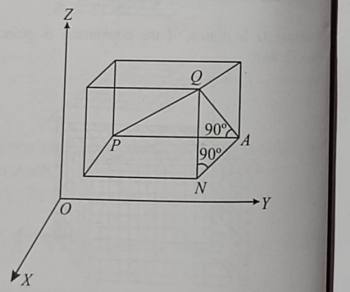
Concept Application

1. Name the octants in which the following points lie (1, 2, 3), (4, –2, 3), (4, –2, –5), (4, 2, –5), (–4, 2, –5), (–4, 2, 5), (–3, –1, 6) (–2, –4, –7).
2. Find the image of :
   (i) (–2, 3, 4) in the yz-plane.
   (ii) (–5, 4, – 3) in the xz-plane.
   (iii) (5, 2, – 7) in the xy-plane.
3. A cube of side 5 has one vertex at the point (1, 0, –1), and the three edges from this vertex are respectively, parallel to the negative x and y axes and positive z-axis. Find the coordinates of the other vertices of the cube.
4. Planes are drawn through the points (5, 0, 2) and (3, –2, 5) parallel to the coordinate planes. Find the lengths of the edges of the rectangular parallelepiped so formed.

Distance Between Two Points

We have studied about the distance between two points in two-dimensional coordinate system. Let us now extend this study to three-dimensional system.

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be two points referred to a system of rectangular axes OX, OY and OZ. Through the points P and Q draw planes parallel to the coordinate planes so as to form a rectangular parallelopiped with one diagonal PQ.



Now, since $\angle PAQ$ is a right angle it follows that, in triangle PA
$PQ^2 = PA^2 + AQ^2$ ...

Also, triangle ANQ is right triangle with $\angle ANQ$ a right angle.

Therefore $AQ^2 = AN^2 + NQ^2$ ... (ii)

From (i) and (ii), we have

$PQ^2 = PA^2 + AN^2 + NQ^2$

Now $PA = y_2 - y_1$, $AN = x_2 - x_1$ and $NQ = z_2 - z_1$

Hence $PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$

Therefore $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

This gives us the distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2).

In particular, if $x_1 = y_1 = z_1 = 0$, i.e., point P is origin O, then OQ $= \sqrt{x_2^2 + y_2^2 + z_2^2}$,

Which gives the distance between the origin O and any point Q (x_2, y_2, z_2).

Train Your Brain

Example 3: Find the distance between the points $P(1, -3, 4)$ and $Q(-4, 1, 2)$.

Sol. The distance PQ between the points $P(1, -3, 4)$ and $Q(-4, 1, 2)$ is

$PQ = \sqrt{(-4-1)^2 + (1+3)^2 + (2-4)^2}$

$= \sqrt{25+16+4} = \sqrt{45} = 3\sqrt{5}$ units

Example 4: Show that the points $P(-2, 3, 5)$, $Q(1, 2, 3)$ and $R(7, 0, -1)$ are collinear.

Sol. We know that points are said to be collinear if they lie on a line.

Now, $PQ = \sqrt{(1+2)^2 + (2-3)^2 + (3-5)^2} = \sqrt{9+1+4} = \sqrt{14}$

$QR = \sqrt{(7-1)^2 + (0-2)^2 + (-1-3)^2}$

$= \sqrt{36+4+16} = \sqrt{56} = 2\sqrt{14}$

and $PR = \sqrt{(7+2)^2 + (0-3)^2 + (-1-5)^2}$

$= \sqrt{81+9+36} = \sqrt{126} = 3\sqrt{14}$

Thus, $PQ + OR = PR$. Hence, P, Q and R are collinear.

Example 5: Determine the point in XY-plane which is equidistant from three points $A(2, 0, 3)$, $B(0, 3, 2)$ and $C(0, 0, 1)$.

Sol. we know that z-coordinate of every point on xy-plane zero. So, let $P(x, y, 0)$ be a point on xy-plane such that $PA = PB = PC$.

Now, $PA = PB$

$\Rightarrow PA^2 = PB^2$

$\Rightarrow (x-2)^2 + (y-0)^2 + (0-3)^2 = (x-0)^2 + (y-3)^2 + (0-2)^2$

$\Rightarrow 4x - 6y = 0 \Rightarrow 2x - 3y = 0$ ...(i)

$PB = PC$

$\Rightarrow PB^2 = PC^2$

$\Rightarrow (x-0)^2 + (y-3)^2 + (0-2)^2 = (x-0)^2 + (y-0)^2 + (0-1)^2$

$\Rightarrow -6y + 12 = 0 \Rightarrow y = 2$ ...(ii)

Putting $y = 2$ in (i), we obtain $x = 3$. Hence, the required point has the coordinates (3, 2, 0).

Example 6: Find the locus of the point which is equidistant from the points $A(0, 2, 3)$ and $2, -2, 1)$.

Sol. Let $P(x, y, z)$ be any point which is equidistant from $A(0, 2, 3)$ and $B(2, -2, 1)$.

Then,

$PA = PB \quad \Rightarrow \quad PA^2 = PB^2$

$\Rightarrow (x-0)^2 + (y-2)^2 + (z-3)^2 = (x-2)^2 + (y+2)^2 + (z-1)^2$

$\Rightarrow 4x - 8y - 4z + 4 = 0 \Rightarrow x - 2y - z + 1 = 0$

Hence, the required locus is $x - 2y - z + 1 = 0$.

Concept Application

5. Verify the following :
   (i) (0, 7, – 10), (1, 6, – 6) and (4, 9, –6) are the vertices of an isosceles triangle.
   (ii) (0, 7, 10), (–1, 6, 6) and (– 4, 9, 6) are the vertices of a right angled triangle.
   (iii) (–1, 2, 1), (1, –2, 5), (4, –7, 8) and (2, – 3, 4) are the vertices of a parallelogram.
6. Find the equation of the set of points which are equidistant from the points (1, 2, 3) and (3, 2, –1).
7. Are the points A (3, 6, 9), B(10, 20, 30) and C(25, –41, 5), the vertices of a right angled triangle?
8. Find the locus of P if $PA^2 + PB^2 = 2k^2$, where A and B are the points (3, 4, 5) and (–1, 3, –7), respectively.

SECTION FORMULA

Let the two given points be $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$. Let the point $R(x, y, z)$ divide PQ in the given ratio $m : n$ internally, then coordinate of point R will be

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n}\right)$$

If the point R divides PQ externally in the ratio $m : n$, then is coordinates are obtained by replacing n by $-n$ so that coordinates of point R will be

$$\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}, \frac{mz_2 - nz_1}{m-n}\right)$$

BOOKWORMCBSE

Case 1: Coordinates of the mid-point : In case R is the mid-point of PQ, then $m : n = 1 : 1$ so that $x = \frac{x_1 + x_2}{2}$, $y = \frac{y_1 + y_2}{2}$ and $z = \frac{z_1 + z_2}{2}$.

These are the coordinates of the mid point of the segment joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$.

Case 2: The coordinates of the point R which divides PQ in the ratio $k : 1$ are obtained by taking $k = \frac{m}{n}$ which are as given below:

$$\left(\frac{kx_2 + x_1}{1+k}, \frac{ky_2 + y_1}{1+k}, \frac{kz_2 + z_1}{1+k}\right)$$

Generally, this result is used in solving problems involving a general point on the line passing through two given points.

Train Your Brain

Example 7: Find the coordinates of the point which divides the line segment joining the points $(1, -2, 3)$ and $(3, 4, -5)$ in the ratio $2 : 3$ (i) internally, and (ii) externally.

Sol. (i) Let $P(x, y, z)$ be the point which divides line segment joining $A(1, -2, 3)$ and $B(3, 4, -5)$ internally in the ratio $2 : 3$. Therefore

$$x = \frac{2(3)+3(1)}{2+3} = \frac{9}{5}, \quad y = \frac{2(4)+3(-2)}{2+3} = \frac{2}{5},$$

$$z = \frac{2(-5)+3(3)}{2+3} = \frac{-1}{5}$$

Thus, the required point is $\left(\frac{9}{5}, \frac{2}{5}, \frac{-1}{5}\right)$

(ii) Let $P(x, y, z)$ be the point which divides segment joining $A(1, -2, 3)$ and $B(3, 4, -5)$ externally in the ratio $2 : 3$. Then

$$x = \frac{2(3)+(-3)(1)}{2+(-3)} = -3, \quad y = \frac{2(4)+(-3)(-2)}{2+(-3)} = -14,$$

$$z = \frac{2(-5)+(-3)(3)}{2+(-3)} = 19,$$

Therefore, the required point is $(-3, -14, 19)$.

Example 8: Using section formula, Prove that the three points $(-4, 6, 10)$, $(2, 4, 6)$ and $(14, 0, -2)$ are collinear.

Sol. Let $A(-4, 6, 10)$, $B(2, 4, 6)$ and $C(14, 0, -2)$ be the given point. Let the point P divides AB in the ratio $k : 1$.

Then coordinates of the point P are

$$\left(\frac{2k-4}{k+1}, \frac{4k+6}{k+1}, \frac{6k+10}{k+1}\right)$$

Let us examine whether for same value of k, the point P coincides with point C. on putting $\frac{2k-4}{k+1} = 14$, we get

$$k = -\frac{3}{2}$$

When $k = -\frac{3}{2}$, then $\frac{4k+6}{k+1} = \frac{4(-\frac{3}{2})+6}{-\frac{3}{2}+1} = 0$ and

$$\frac{6k+10}{k+1} = \frac{4(-\frac{3}{2})+10}{-\frac{3}{2}+1} = -2$$

Therefore, $C(14, 0, -2)$ is a point which divides AB externally in the ratio $3 : 2$ and is same as P. Hence A, B, C are collinear.

Example 9: Find the coordinates of the centroid of the triangle whose vertices are (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3).

Sol. Let ABC be the triangle. Let the coordinates of the vertices A, B, C be (x_1, y_1, z_1), (x_2, y_2, z_2) and (z_3, y_3, z_3), respectively. Let D be the mid-point of BC.

Hence coordinates of D are

$$\left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}, \frac{z_2 + z_3}{2}\right)$$

Let G be the centroid of the triangle. Therefore, it divides the median AD in the ratio $2 : 1$.

Hence, the coordinates of G are

$$\left(\frac{2\left(\frac{x_2+x_3}{2}\right)+x_1}{2+1}, \frac{2\left(\frac{y_2+y_3}{2}\right)+y_1}{2+1}, \frac{2\left(\frac{z_2+z_3}{2}\right)+z_1}{2+1}\right)$$

or $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}, \frac{z_1+z_2+z_3}{3}\right)$

Example 10: Find the ratio in which the line segment joining the points $(4, 8, 10)$ and $(6, 10, -8)$ is divided by the YZ-plane.

Sol. Let YZ-plane divides the line segment joining $A(4, 8, 10)$ and $B(6, 10, -8)$ at $P(x, y, z)$ in the ratio $k : 1$. The coordinates of P are

$$\left(\frac{4+6k}{k+1}, \frac{8+10k}{k+1}, \frac{10-8k}{k+1}\right)$$

Since P lies on the YZ-plane, its x-coordinate is zero,

i.e., $\frac{4+6k}{k+1} = 0$ or $k = -\frac{2}{3}$

Therefore, YZ-plane divides AB externally in the ratio $2 : 3$.

Concept Application

9. Find the coordinates of the point which divides the line segment joining the points $(-2, 3, 5)$ and $(1, -4, 6)$ in the ratio (i) $2 : 3$ internally, (ii) $2 : 3$ externally.

10. Given that $P(3, 2, -4)$, $Q(5, 4, -6)$ and $R(9, 8, -10)$ are collinear. Find the ratio in which Q divides PR.

11. Using section formula, show that the points $A(2, -3, 4)$, $B(-1, 2, 1)$ and $C\left(0, \frac{1}{3}, 2\right)$ are collinear.

12. Find the coordinates of the points which trisect the line segment joining the points $P(4, 2, -6)$ and $Q(10, -16, 6)$.

BOOKWORMCBSE

AARAMBH (SOLVED EXAMPLES)

1. Show that the points $A(1, 2, 3)$, $B(-1, -2, -1)$, $C(2, 3, 2)$ and $D(4, 7, 6)$ are the vertices of a parallelogram $ABCD$, but it is not rectangle.

Sol. To show $ABCD$ is a parallelogram we need to show opposite side are equal.

$$AB = \sqrt{(-1-1)^2+(-2-2)^2+(-1-3)^2} = \sqrt{4+16+16} = 6$$

$$BC = \sqrt{(2+1)^2+(3+2)^2+(2+1)^2} = \sqrt{9+25+9} = \sqrt{43}$$

$$CD = \sqrt{(4-2)^2+(7-3)^2+(6-2)^2} = \sqrt{4+16+16} = 6$$

$$DA = \sqrt{(1-4)^2+(2-7)^2+(3-6)^2} = \sqrt{9+25+9} = \sqrt{43}$$

Since $AB = CD$ and $BC = AD$, $ABCD$ is a parallelogram.

Now, it is required to prove that $ABCD$ is not a rectangle. For this, we show that diagonals AC and BD are unequal. We have

$$AC = \sqrt{(2-1)^2+(3-2)^2+(2-3)^2} = \sqrt{1+1+1} = \sqrt{3}$$

$$BD = \sqrt{(4+1)^2+(7+2)^2+(6+1)^2} = \sqrt{25+81+49} = \sqrt{155}$$

Since $AC \neq BD$, $ABCD$ is not a rectangle.

Note : We can also show that $ABCD$ is a parallelogram, using the property that diagonals AC and BD bisect each other.

2. Find the equation of the set of the points P such that its distances from the points $A(3, 4, -5)$ and $B(-2, 1, 4)$ are equal.

Sol. If $P(x, y, z)$ be any point such that $PA = PB$.

Now

$$\sqrt{(x-3)^2+(y-4)^2+(z+5)^2}$$

$$= \sqrt{(x+2)^2+(y-1)^2+(z-4)^2}$$

or $(x-3)^2+(y-4)^2+(z+5)^2 = (x+2)^2+(y-1)^2+(z-4)^2$

or $10x + 6y - 18z - 29 = 0$.

3. The centroid of a triangle ABC is at the point $(1, 1, 1)$. If the coordinates of A and B are $(3, -5, 7)$ and $(-1, 7, -6)$, respectively, find the coordinates of the point C.

Sol. Let the coordinates of C be (x, y, z) and the coordinates of the centroid G be $(1, 1, 1)$. Then

$$\frac{x+3-1}{3} = 1, \text{ i.e.,}$$

$$x = 1; \ \frac{y-5+7}{3} = 1, \text{ i.e.,}$$

$$y = 1; \ \frac{z+7-6}{3} = 1, \text{ i.e., } z = 2.$$

Hence, coordinates of C are $(1, 1, 2)$.

4. The mid-points of the sides of a triangle are $(1, 5, -1)$, $(0, 4, -2)$ and $(2, 3, 4)$. Find its vertices.

Sol. Let $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ be the vertices of the given triangle, and let $D(1, 5, -1)$, $E(0, 4, -2)$ and $F(2, 3, 4)$ be the mid-points of the sides BC, CA and AB respectively.

D is the mid-point of BC

$$\therefore \ \frac{x_2+x_3}{2} = 1, \ \frac{y_2+y_3}{2} = 5, \ \frac{z_2+z_3}{2} = -1$$

$$\Rightarrow x_2 + x_3 = 2, \ y_2 + y_3 = 10, \ z_2 + z_3 = -2 \quad ...(i)$$

E is the mid-point of CA

$$\therefore \ \frac{x_1+x_3}{2} = 0, \ \frac{y_1+y_3}{2} = 4, \ \frac{z_1+z_3}{2} = -2$$

$$\Rightarrow x_1 + x_3 = 0, \ y_1 + y_3 = 8, \ z_1 + z_3 = -4 \quad ...(ii)$$

F is the mid-point of AB

$$\therefore \ \frac{x_1+x_2}{2} = 2, \ \frac{y_1+y_2}{2} = 3, \ \frac{z_1+z_2}{2} = 4$$

$$\Rightarrow x_1 + x_2 = 4, \ y_1 + y_2 = 6, \ z_1 + z_2 = 8 \quad ...(iii)$$

Adding first three equations in (i), (ii) and (iii), we obtain

$2(x_1 + x_2 + x_3) = 2 + 0 + 4$

$\Rightarrow x_1 + x_2 + x_3 = 3$, we obtain:

$x_1 = 1, x_2 = 3, x_3 = -1$.

Adding second equation in (i), (ii) and (iii), we obtain

$2(y_1 + y_2 + y_3) = 10 + 8 + 6 \Rightarrow y_1 + y_2 + y_3 = 12$, we obtain:

$y_1 = 2, y_2 = 4, y_3 = 6$.

Solving last equations in (i), (ii) and (iii) with $z_1 + z_2 + z_3 = 1$, we obtain

$z_1 = 3, z_2 = 5, z_3 = -7$.

Thus the vertices of the triangle are $A(1, 2, 3)$, $B(3, 4, 5)$ and $C(-1, 6, -7)$.

5. Given that $P(3, 2, -4)$, $Q(5, 4, -6)$ and $R(9, 8, -10)$ are collinear. Find the ratio in which Q divide PR.

Sol. Suppose Q divides PR in the ratio $\lambda : 1$. Then coordinates of Q are $\left(\frac{9\lambda+3}{\lambda+1}, \frac{8\lambda+2}{\lambda+1}, \frac{-10\lambda-4}{\lambda+1}\right)$

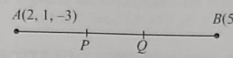
But. Coordinates of Q are $(5, 4, -6)$. Therefore,

$$\frac{9\lambda+3}{\lambda+1} = 5, \frac{8\lambda+2}{\lambda+1} = 4, \frac{-10\lambda-4}{\lambda+1} = -6$$

These three equations give $\lambda = \frac{1}{2}$. So, Q divides PR in the ratio $\frac{1}{2}$: 1 or 1 : 2.

6. Find the coordinates of the points which trisect the line segment AB, given that $A(2, 1, -3)$ and $B(5, -8, 3)$.

Sol. Let P and Q be the points which trisect AB. Then, $AP = PQ = QB$. Therefore, P divides AB in the ratio 1 : 2 and Q divides it in the ratio 2 : 1.



As P divides AB in the ratio 1 : 2, so coordinates of P are

$$\left(\frac{1\times5+2\times2}{1+2}, \frac{1\times-8+2\times1}{1+2}, \frac{1\times3+2\times-3}{1+2}\right) = (3, -2, -1)$$

Since Q divides AB in the ratio 2 : 1. So coordinates of Q are

$$\left(\frac{2\times5+1\times2}{2+1}, \frac{2\times-8+1\times1}{2+1}, \frac{2\times3+1\times-3}{1+2}\right) = (4, -5, 1)$$

7. Three vertices of a parallelogram $ABCD$ are $A(3, -1, 2)$, $B(1, 2, -4)$ and $C(-1, 1, 2)$. Find the coordinates of the fourth vertex.

Sol. Let the coordinates of fourth vertex D be (x, y, z). Since diagonals of a parallelogram bisect each other. Therefore, mid-point of AC and BD coincide.

$$\therefore \left(\frac{3-1}{2}, \frac{-1+1}{2}, \frac{2+2}{2}\right) = \left(\frac{1+x}{2}, \frac{2+y}{2}, \frac{-4+z}{2}\right)$$

$$\Rightarrow (1, 0, 2) = \left(\frac{x+1}{2}, \frac{y+2}{2}, \frac{z-4}{2}\right)$$

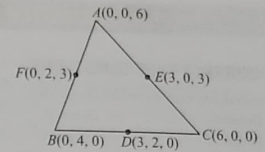
$$\Rightarrow \frac{x+1}{2} = 1, \frac{y+2}{2} = 0, \frac{z-4}{2} = 2$$

$$\Rightarrow x = 1, y = -2, \text{ and } z = 8$$

Hence, the coordinates of the fourth vertex are $(1, -2, 8)$.

8. Find the lengths of the medians of the triangle with vertices $A(0, 0, 6)$, $B(0, 4, 0)$ and $C(6, 0, 0)$.

Sol. Let D, E and F be the mid-points of sides BC, CA and AB respectively.



The coordinates of D, E and F are

$$\left(\frac{0+6}{2}, \frac{4+0}{2}, \frac{0+0}{2}\right) = (3, 2, 0), E\left(\frac{6+0}{2}, \frac{0+0}{2}, \frac{0+6}{2}\right) = (3, 0, 3)$$

and $F\left(\frac{0+0}{2}, \frac{0+4}{2}, \frac{6+0}{2}\right) = (0, 2, 3)$ respectively.

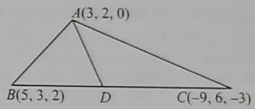
$$\therefore \ AD = \sqrt{(0-3)^2+(0-2)^2+(6-0)^2} = \sqrt{9+4+36} = 7$$

$$BE = \sqrt{(0-3)^2+(4-0)^2+(0-3)^2} = \sqrt{9+16+9} = \sqrt{34}$$

and, $CF = \sqrt{(6-0)^2+(0-2)^2+(0-3)^2} = \sqrt{36+4+9} = 7$

9. Let $A(3, 2, 0)$, $B(5, 3, 2)$, $C(-9, 6, -3)$ be three points forming a triangle. The bisector AD of $\angle BAC$ meets side BC in D. Find the coordinates of D.

Sol. The bisector AD of $\angle BAC$ divides BC in the ratio $AB : AC$.



Now, $AB = \sqrt{(3-5)^2+(2-3)^2+(0-2)^2} = 3$

and, $AC = \sqrt{(3+9)^2+(2-6)^2+(0+3)^2} = 13$

Thus, D divides BC in the ratio $AB : AC$ i.e. 3 : 13. Hence, the coordinates of D are

$$\left(\frac{3\times-9+13\times5}{3+13}, \frac{3\times6+3\times13}{3+13}, \frac{3\times-3+13\times2}{3+13}\right) = \left(\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$$

BOOKWORMCBSE

SCHOOL LEVEL PROBLEMS

SINGLE CORRECT TYPE QUESTIONS

1. In which octant does the point $(-5, 4, 3)$ lie?
   (a) Octant I (b) Octant II
   (c) Octant III (d) Octant IV

2. What is the distance between the points $(2, -1, 3)$ and $(-2, 1, 3)$?
   (a) $2\sqrt{5}$ units (b) 25 units
   (c) $4\sqrt{5}$ units (d) $\sqrt{5}$ units

3. The points $(5, 2, 4)$, $(6, -1, 2)$ and $(8, -7, k)$ are collinear, if k is equal
   (a) -2 (b) 2 (c) 3 (d) -1

4. The coordinates of a point which is equidistant from the points $(0, 0, 0)$, $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ are given by
   (a) $(a/2, b/2, c/2)$ (b) $(-a/2, -b/2, c/2)$
   (c) $(a/2, -b/2, c/2)$ (d) $(-a/2, b/2, c/2)$

5. The distance of the point $P(a, b, c)$ from the x-axis is:
   (a) $\sqrt{(a^2+c^2)}$ (b) $\sqrt{(a^2+b^2)}$
   (c) $\sqrt{(b^2+c^2)}$ (d) None of these

6. What is the ratio in which the line joining the points $(2, 4, 5)$ and $(3, 5, -4)$ is internally divided by the xy-plane?
   (a) 5 : 4 (b) 3 : 4 (c) 1 : 2 (d) 7 : 5

7. If coordinates of vertices of a triangle are $(7, 6, 4)$, $(5, 4, 6)$, $(9, 5, 8)$, find the coordinates of the centroid of the triangle.
   (a) $(7, 5, 6)$ (b) $(7, 3, 5)$
   (c) $(5, 3, 7)$ (d) $(3, 5, 7)$

8. Find the points which trisects the line joining $(4, 9, 8)$ and $(13, 27, -4)$.
   (a) $(0, 21, 10)$ (b) $(0, 21, 4)$
   (c) $(10, 21, 0)$ (d) $(4, 4, 0)$

9. If $P(2, 3, 9)$, $Q(2, 5, 5)$ and $R(8, 5, 3)$ are vertices of a triangle then find the length of the median through Q.
   (a) $\sqrt{24}$ (b) $\sqrt{38}$ (c) $\sqrt{11}$ (d) $\sqrt{53}$

VERY SHORT ANSWER TYPE QUESTIONS

10. Find the distance from the origin to $(6, -6, 9)$.

11. Show that the points $P(1, -1, 3)$, $Q(2, -4, 5)$ and $R(5, -13, 11)$ are collinear.

12. Show that if $x^2+y^2=1$, then the point $(x, y, \sqrt{1-x^2-y^2})$ is at a distance 1 unit from the origin.

13. Three consecutive vertices of a parallelogram $ABCD$ are $A(6, -2, 4)$, $B(2, 4, -8)$, $C(-2, 2, 4)$. Find the coordinates of the fourth vertex

14. Check if the triangle ABC with vertices $A(0, 4, 1)$, $B(2, 3, -1)$ and $C(4, 5, 0)$ is right angled.

15. Find the third vertex of the triangle whose centroid is the origin and two vertices are $(2, 8, 6)$ and $(0, -9, -5)$.

16. Find the centroid of the triangle whose vertices are $P(2, 6, 5)$, $Q(4, 9, 7)$ and $R(6, 10, -4)$.

LONG ANSWER TYPE QUESTIONS

17. Find the centroid of a triangle, the mid-point of whose sides are $A(1, 2, -3)$, $B(3, 0, 1)$ and $C(-1, 1, -4)$.

18. Let $A(2, 2, -3)$, $B(5, 6, 9)$ and $C(2, 7, 9)$ be the vertices of a triangle. The internal bisector of the angle A meets BC at the point D. Find the coordinates of D.

19. If the origin is the centroid of a triangle ABC having vertices $A(x, 1, 3)$, $B(-2, y, -5)$ and $C(4, 7, z)$, find the values of x, y, z.

20. Three vertices of a Parallelogram $ABCD$ are $A(1, 2, 3)$, $B(-1, -2, -1)$ and $C(2, 3, 2)$. Find the fourth vertex D.

21. The mid-point of the sides of a triangle are $(1, 5, -1)$, $(0, 4, -2)$ and $(2, 3, 4)$. Find its vertices. Also, find the centroid of the triangle.

CASE STUDY BASED QUESTIONS

Case Study-I

Read the following text and answer the following questions on the basis of the same:

Alex and Lisa are planning to meet at a park. Alex coordinates are $(2, 3, 4)$, and Lisa's coordinates are $(5, 6, 7)$. They want to know the distance between them to determine how long it will take for them to reach the park. Let's calculate the distance using the distance formula in three-dimensional geometry.

22. What is the squared distance between Alex and Lisa?
   (a) 14 (b) 27 (c) 32 (d) 45

23. What is the actual distance between Alex and Lisa?
   (a) $\sqrt{14}$ (b) $\sqrt{27}$ (c) $\sqrt{32}$ (d) $\sqrt{45}$

Case Study-II

$ABCD$ is a field in shape of parallelogram coordinate of A, B and C are $(3, -1, 2)$, $(1, 2, -4)$ and $(-1, 1, 2)$ respectively.

Based on the above information answer the following:

24. Coordinates of mid point of AC will be
   (a) $(-1, 1, 2)$ (b) $(-1, 0, -2)$
   (c) $(1, 0, 2)$ (d) $(1, 0, -2)$

25. Coordinates of D will be
   (a) $(1, -2, 8)$ (b) $(-1, -2, 8)$
   (c) $(0, -2, 8)$ (d) $(1, 2, 8)$

PRARAMBH (TOPICWISE)

1. A point P lies on the x-axis, then what is possible coordinate of point P.
   (a) $(2, 0, 0)$ (b) $(1, 0, 1)$
   (c) $(0, 3, 0)$ (d) $(0, 0, 2)$

2. A point $P(x, y, z)$ lies in the XZ-plane, then
   (a) $x = 0$ (b) $y = 0$
   (c) $z = 0$ (d) $x = 0$ and $y = 0$

3. Find the octant in which the points $(-3, 1, 2)$ and $(-3, 1, -2)$ lie.
   (a) II and V (b) II and VI
   (c) IV and VI (d) V and VIII

4. The X-axis and Y-axis taken together determine a plane known as
   (a) XY-plane (b) YZ-plane
   (c) XZ-plane (d) None of these

5. The coordinates of points in the XY-plane are of the form
   (a) $(x, 0, z)$ (b) $(x, y, 0)$
   (c) $(x, 0, 0)$ (d) $(0, y, z)$

6. Coordinate planes divide the space into _____ octants.
   (a) Eight (b) Four
   (c) Two (d) None of these

7. Find the distances of the point $P(-4, 3, 5)$ from the coordinate axes.
   (a) 10 (b) 7 (c) 8 (d) 5

8. Find the distance between the points P and Q having coordinates $(-2, 3, 1)$ and $(2, 1, 2)$.
   (a) $\sqrt{23}$ (b) 4 (c) $\sqrt{21}$ (d) $\sqrt{19}$

9. Find the coordinates of the point which divides the join of $P(2, -1, 4)$ and $Q(4, 3, 2)$ internally in the ratio 2 : 3
   (a) $\left(\frac{12}{5}, \frac{3}{5}, \frac{16}{5}\right)$ (b) $\left(\frac{14}{5}, \frac{3}{5}, \frac{16}{5}\right)$
   (c) $\left(-\frac{14}{5}, \frac{3}{5}, \frac{13}{5}\right)$ (d) $\left(\frac{12}{5}, -\frac{3}{5}, \frac{13}{5}\right)$

10. Find the ratio in which the line joining the points $(1, 2, 3)$ and $(-3, 4, -5)$ is divided by the XY-plane.
   (a) 2 : 3, externally (b) 5 : 3, internally
   (c) 3 : 5, internally (d) 1 : 2, externally

11. What is the locus of a point for which $y = 0, z = 0$?
   (a) x-axis (b) y-axis
   (c) z-axis (d) yz-plane

12. The coordinates of the foot of the perpendicular drawn from the point $P(3, 4, 5)$ on the yz-plane are
   (a) $(3, 4, 0)$ (b) $(0, 4, 5)$
   (c) $(3, 0, 5)$ (d) $(3, 0, 0)$

13. The coordinates of the foot of the perpendicular from a point $P(6, 7, 8)$ on x-axis are
   (a) $(6, 0, 0)$ (b) $(0, 7, 0)$
   (c) $(0, 0, 8)$ (d) $(0, 7, 8)$

14. The perpendicular distance of the point $P(6, 7, 8)$ from xy-plane is
   (a) 8 (b) 7 (c) 6 (d) 10

15. The length of the perpendicular drawn from the point $P(a, b, c)$ from y-axis is
   (a) $\sqrt{a^2+b^2}$ (b) $\sqrt{b^2+c^2}$
   (c) $\sqrt{a^2+c^2}$ (d) $\sqrt{a^2+b^2+c^2}$

16. Find the image of the point $(-5, 0, 3)$ in the XZ-plane.
   (a) $(-5, 0, -3)$ (b) $(-5, 0, 3)$
   (c) $(5, 0, 3)$ (d) $(-5, 0, -3)$

17. Find the equation of the set of points P, the sum of whose distances from $A(4, 0, 0)$ and $B(-4, 0, 0)$ is equal to 10.
   (a) $15x^2 - y^2 - z^2 + 192x + 609 = 0$
   (b) $15x^2 + y^2 - z^2 + 192x + 609 = 0$
   (c) $15x^2 - y^2 - z^2 + 192x - 609 = 0$
   (d) None of these

18. Planes are drawn parallel to the coordinate planes through the points $(3, 0, -1)$ and $(-2, 5, 4)$. Find the lengths of the edges of the parallelepiped so formed.
   (a) $(5, 5, 5)$ (b) $(5, 3, 0)$
   (c) $(5, 4, 3)$ (d) $(5, 4, 4)$

19. Determine the point in XY-Plane which is equidistant from three points $A(2, 0, 3)$, $B(0, 3, 2)$ and $C(0, 0, 3)$.
   (a) $(1, 2, 0)$ (b) $\left(1, \frac{2}{3}, 0\right)$
   (c) $\left(2, \frac{3}{2}, 0\right)$ (d) $(1, 3, 0)$

20. Find the coordinates of a point equidistant from the four points $O(0, 0, 0)$, $A(a, 0, 0)$, $B(0, b, 0)$ and $C(0, 0, c)$.
   (a) $\left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$ (b) $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$
   (c) (a, b, c) (d) None of these

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21. Find the ratio in which the join the $A(2, 1, 5)$ and $B(3, 4, 3)$ is divided by the plane $2x + 2y - 2z = 1$.

(a) 5 : 7, internally (b) 7 : 5, internally

(c) 5 : 7, externally (d) 3 : 2, internally

22. Determine the point on z-axis which is equidistant from the points $(1, 5, 7)$ and $(5, 1, -4)$.

(a) $\left(0, 0, \frac{1}{2}\right)$ (b) $\left(0, 0, \frac{3}{2}\right)$

(c) $\left(0, \frac{3}{2}, 0\right)$ (d) $\left(0, 0, \frac{5}{2}\right)$

23. Find the point on y-axis which is equidistant from the points $(3, 1, 2)$ and $(5, 5, 2)$.

(a) (0, 5, 0) (b) (0, 4, 0)

(c) (2, 0, 0) (d) (0, 1, 0)

24. Find the points on z-axis which are at a distance $\sqrt{21}$ from the point $(1, 2, 3)$.

(a) (0, 0, 1) (b) (0, 0, 3)

(c) (0, 0, 7) (d) (0, 0, –1)

25. Let P and Q be any two points. Find the coordinates of the point R which divides PQ externally in the ratio 2 : 1

(a) $(2x_2 - x_1, 2y_2 - y_1, 2z_2 - z_1)$

(b) $(x_2 - 2x_1, y_2 - 2y_1, z_2 - 2z_1)$

(c) $(2x_2 + x_1, 2y_2 + y_1, 2z_2 + z_1)$

(d) None of these

26. Find the ratio in which the line segment joining the points $(2, -1, 3)$ and $(-1, 2, 1)$ is divided by the plane $x + y + z = 5$.

(a) 1 : 3, externally (b) 5 : 1, internally

(c) 1 : 5, internally (d) 3 : 2, internally

27. $A(1, 2, 3)$, $B(0, 4, 1)$, $C(-1, -1, -3)$ are the vertices of a triangle ABC. Find the point in which the bisector of the angle $\angle BAC$ meets BC.

(a) $\left(-\frac{3}{10}, \frac{5}{2}, -\frac{1}{5}\right)$ (b) $\left(\frac{3}{10}, \frac{5}{2}, -\frac{2}{5}\right)$

(c) $\left(\frac{7}{10}, -\frac{5}{2}, -\frac{2}{5}\right)$ (d) $\left(\frac{7}{10}, \frac{7}{2}, -\frac{1}{5}\right)$

28. Find the centroid of a triangle, mid-points of whose sides are $(1, 2, -3)$, $(3, 0, 1)$ and $(-1, 1, -4)$.

(a) (1, 1, 3) (b) (–1, –1, –2)

(c) (1, 1, –2) (d) (–1, 0, 3)

29. The centroid of a triangle ABC is at the point $(1, 1, 1)$. If the coordinates of A and B are $(3, -5, 7)$ and $(-1, 7, -6)$ respectively, find the coordinates of the point C.

(a) (1, –1, 3) (b) (1, 1, 2)

(c) (–1, –1, –3) (d) (2, –1, 3)

30. Find the image of the point $(-4, 0, 1)$ in the XY-plane.

(a) (4, 0, 1) (b) (–4, 0, –1)

(c) (4, 0, –1) (d) (–4, 1, 1)

31. A point C with Z-coordinate 8 lies on the line segment joi the point $A(2, -3, 4)$ and $B(8, 0, 10)$ find its coordinate

(a) (6, –1, 8) (b) (–6, 1, 8)

(c) (–6, –1, 8) (d) (3, –1, 8)

32. If $P(0, 1, 2)$, $Q(4, -2, 1)$ and $O(0, 0, 0)$ are three points, t $\angle POQ$ is

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

33. If the extremities of the diagonal of a square $(1, -2, 3)$ and $(2, -3, 5)$, then the length of the side is

(a) $\sqrt{6}$ (b) $\sqrt{5}$ c) $\sqrt{3}$ (d) $\sqrt{7}$

34. The points $(5, -4, 2)$, $(4, -3, 1)$, $(7, 6, 4)$ and $(8, -7, 5)$ the vertices of

(a) A rectangle (b) A square

(c) A parallelogram (d) None of these

35. In a three dimensional space the equation $x^2 - 5x + 6 =$ represents

(a) Points (b) Planes

(c) Curves (d) Pair of straight lines

36. Let $(3, 4, -1)$ and $(-1, 2, 3)$ be the end points of a diame of a sphere. Then, the radius of the sphere is equal to

(a) 2 (b) 3 (c) 6 (d) 7

Miscellaneous

37. The coordinates of a point are $(3, -2, 5)$. Write down coordinates of seven points such that the absolute value their coordinates are the same as those of the coordinate the given point.

38. Determine the points in (*i*) xy-plane (*ii*) yz-plane (*iii*) zx-plane which are equidistant from the poi $A(1, -1, 0)$, $B(2, 1, 2)$ and $C(3, 2, -1)$.

39. Show that the three points $A(2, 3, 4)$, $B(-1, 2 - 3)$ $C(-4, 1, -10)$ are collinear and find the ratio in wh C divides AB.

Answer Key

CONCEPT APPLICATION

1. (i)-I, (ii)-IV, (iii)-VIII, (iv)-V, (v)-VI, (vi)-VI, (vii)-II, (viii)-III, (ix)-VII
2. (i)-(2, 3, 4), (ii)-(–5, –4, –3), (iii)-(5, 2, 7)
3. B: (1, 5, –1), C: (1, 0, 4), E: (–4, 0, –1), G: (–4, 0, 4), F: (–4, –5,–1), G: (–4, 0, 4), H: (–4, –5, 4)
4. $a = 2$, $b = 2$, $c = 3$
6. $x = 2z$
7. No
8. $2x^2 + 2y^2 + 2z^2 - 4x - 14y + 4z + 109 - 2k^2 = 0$
9. (i) $\left(\frac{-4}{5}, \frac{1}{5}, \frac{27}{5}\right)$, (ii) (–8, 17, 3)
10. 1 : 2
12. (6, –4, –2) and (8, –10, 2)

SCHOOL LEVEL PROBLEMS

1. (b)	2. (a)	3. (a)	4. (a)	5. (c)	6. (a)	7. (a)	8. (c)	9. (c)	22. (b)
23. (b)	24. (c)	25. (a)							

PRARAMBH (TOPICWISE)

1. (a)	2. (b)	3. (b)	4. (a)	5. (b)	6. (a)	7. (d)	8. (c)	9. (b)	10. (c)
11. (a)	12. (b)	13. (a)	14. (a)	15. (c)	16. (b)	17. (d)	18. (a)	19. (b)	20. (a)
21. (a)	22. (b)	23. (a)	24. (c, d)	25. (a)	26. (a)	27. (a)	28. (c)	29. (b)	30. (b)
31. (a)	32. (d)	33. (c)	34. (d)	35. (b)	36. (b)				

37. (3, 2, 5), (3, 2, –5), (3, –2, –5), (–3, 2, 5), (–3, –2, – 5) and (–3, –2, 5) & (–3, 2, –5)

38. (i) $\left(\frac{3}{2}, 1, 0\right)$, (ii) $\left(0, \frac{31}{16}, -\frac{3}{16}\right)$, (iii) $\left(\frac{31}{10}, 0, \frac{1}{5}\right)$ 39. 2 : 1, externally

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CHAPTER 19

Probability

INTRODUCTION

We studied about the concept of probability as a measure of uncertainty of various phenomenon. We have obtained the probability of getting an even number in throwing a die as $\frac{3}{6}$ i.e., $\frac{1}{2}$. Here the total possible outcomes are 1,2,3,4,5 and 6.

The outcomes in favour of the event of 'getting an even number' are 2, 4, 6. In general, to obtain the probability of an event, we find the ratio of the number of outcomes favourable to the event, to the total number of equally likely outcomes. This theory of probability is known as classical theory of probability.

DEFINITIONS

(*a*) **Random Experiment:** An action or operation resulting in two or more outcomes, which are unpredictable in advance.
e.g.
(*i*) Tossing of a coin
(*ii*) Throwing a die
(*iii*) Drawing a card

(*b*) **Sample space:** A set S that consists of all possible outcomes of a random experiment is called a sample space and each outcome is called a sample point. Often there will be more than one sample space that can describes outcomes of an experiment, but there is usually only one that will provide the most information. If a sample space has a finite number of points it is called finite sample space and infinite sample space if it has infinite number of points.
e.g.
(*i*) In toss of a coin, S = {H, T} where H and T are sample points representing a head and a tail respectively.
(*ii*) In throw of a die, S = {1, 2, 3, 4, 5, 6} where the numbers are the sample points representing the six faces.

(*c*) **Trial:** When an experiment is repeated under similar conditions and it does not give the same result each time but may result in any one of the several possible outcomes, the experiment is called a trial and the outcomes are called cases. The number of times the experiment is repeated is called the number of trials.

For example:

- Toss of coin is a trial when coin is tossed 5 times.
- Throw of a die is a trial when the die is thrown 4 times.

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Example 1: Write the sample space of the following experiment

(*i*) 'Three coins are tossed'.

(*ii*) 'Selection of two children from a group of 3 boys and 2 girls without replacement'.

Sol. (*i*) {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

(*ii*) {$B_1 B_2$, $B_1 B_3$, $B_1 G_1$, $B_1 G_2$, $B_2 B_3$, $B_2 G_1$, $B_2 G_2$, $B_3 G_1$, $B_3 G_2$, $G_1 G_2$}

Example 2: Consider the experiment in which a coin is tossed repeatedly until a head comes up. Describe the sample space.

Sol. In the experiment head may come up on the first toss, or the 2nd toss, or the 3rd toss and so on till head is obtained. Hence, the desired sample space is S = {H, TH, TTH, TTTH, TTTTH,...}

Concept Application

1. A bag contains one white and one red ball. A ball is drawn from the bag. If the ball drawn is white it is replaced in the bag and again a ball is drawn. Otherwise, a die is tossed. Write the sample space for this experiment.
2. A box contains 1 white and 3 identical black balls. Two balls are drawn at random in succession without replacement. Write the sample space for this experiment.

TYPE OF EVENT

(*a*) **Event:** A subset of sample space, i.e. a set of some possible outcomes of a random experiment is called an event.
e.g. Scoring a six on the throw of a die.

(*b*) **Simple Event:** Each sample point in the sample space is called an elementary event or simple event. For example occurrence of head in throw of a coin is simple event.

(*c*) **Sure Event:** The set containing all sample points is a sure event as in the throw of a die the occurrence of natural number less than 7, is a sure event.

(*d*) **Null Event:** The set which does not contain any sample point.

(*e*) **Mixed/compound Event:** A subset of sample space S containing more than one element is called a mixed event or a compound event.

(*f*) **Equally Likely Events:** Events are said to be Equally Likely when each event is as likely to occur as any other event.

ALGEBRA OF EVENTS

Compliment of an event: Let S be the sample space and E be an event then E^c or $\bar{E}$ represents complement of event E which is a subset containing all sample points in S which are not in E. It refers to the non occurrence of event E.

In every case it is set of some or all possible outcomes of the experiment. Therefore event (A) is subset of sample space (S). If outcome of an experiment is an element of A we say that event A has occurred.

- An event consisting of a single point of S is called a simple or elementary event.
- If an event has more than one simple point it is called compound event.
- ϕ is called impossible event and S (sample space) is called sure event.

MUTUALLY EXCLUSIVE EVENTS

Two events are said to be Mutually Exclusive (or disjoint or incompatible) if the occurrence of one precludes (rules out) the simultaneous occurrence of the other. If A and B are two mutually exclusive events then $P(A \cap B) = 0$.

Consider, for example, choosing numbers at random from the set {3, 4, 5, 6, 7, 8, 9, 10, 11, 12}

If, Event A is the selection of a prime number,

Event B is the selection of an odd number,

Event C is the selection of an even number,

Then A and C are mutually exclusive as none of the numbers in this set is both prime and even. But A and B are not mutually exclusive as some numbers are both prime and odd (viz. 3, 5, 7, 11).

EXHAUSTIVE EVENTS

Events A, B, C ... L are said to be Exhaustive Event if no event outside this set can result as an outcome of an experiment. For example, if A and B are two events defined on a sample space S, then A and B are exhaustive $\Rightarrow A \cup B = S \Rightarrow P(A \cup B) = 1$.

Comparative study of Equally likely (E/L), Mutually Exclusive (M/E) and Exhaustive events:

Experiment	Events	E/L	M/E	Exhaustive
1. Throwing of a die	A : throwing an odd face {1, 3, 5} B : throwing a composite {4, 6}	No	Yes	No
2. A ball is drawn from an urn containing 2W, 3R and 4G balls	E_1 : getting a W ball E_2 : getting a R ball E_3 : getting a G ball	No	Yes	Yes
3. Throwing a pair of dice	A : Throwing a doublet {11, 22, 33, 44, 55, 66} B : throwing a total of 10 or more {46, 64, 55, 56, 65, 66}	Yes	No	No
4. From a well shuffled pack of cards, a card is drawn	E_1 : getting a heart E_2 : getting a spade E_3 : getting a diamond E_4 : getting a club	Yes	Yes	Yes
5. From a well shuffled pack of a cards, a card is drawn	A = getting a heart B = getting a face card	No	No	No

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Example 3: In drawing of a card from a well shuffled ordinary deck of playing cards the events 'card drawn is spade' and 'card drawn is an ace' are

(*a*) Mutually exclusive
(*b*) Equally likely
(*c*) Forming an exhaustive system
(*d*) None of these

Sol. (*d*) A = getting spade
B = getting an ace

$$P(A) = \frac{13}{52} \Rightarrow P(A) = \frac{1}{4};$$

$$P(B) = \frac{4}{52} \Rightarrow P(B) = \frac{1}{13}$$

Example 4: In throwing a pair of dice, find whether the two events

(*i*) E_1 : 'coming up of an odd number on first dice' and E_2 : 'coming up of a total of 8'.
(*ii*) E_1 : 'coming up of 4 on first dice' and
E_2 : 'coming up of 5 on the second dice', are mutually exclusive or not

Sol. (*i*) E_1 : {(1, 1) (1, 2)...(1, 6), (3, 1)...(3, 6), (5,1)... (5, 6)}
E_2 : {: (2, 6) (3, 5) (4, 4) (5, 3) (6, 2)}
not mutually exclusive
(*ii*) E_1 : {(4, 1) (4, 2) ... (4, 6)}
E_2 : {(1, 5) (2, 5) ... (6, 5)}, not mutually exclusive

Example 5: If the letters of INTERMEDIATE are arranged, then the probability that no two *E's* occur together is

Sol. $I \to 2, N \to 1, T \to 2, E \to 3, R \to 1, M \to 1, D \to 1, A \to 1$ (3*E's* Rest 9)

First arrange rest of letters = $\frac{9!}{2!2!}$

Now 3*E's* can be placed into place in ${}^{10}C_3$ ways so favourable cases = $\frac{9!}{2!2!} \times {}^{10}C_3 = 3 \times 10!$

Total cases = $\frac{12!}{2!2!3!}$

Probability = $\frac{3\times 10!\times 2!\times 2!\times 3!}{12!} = \frac{6}{11}$

Example 6: From a group of 10 persons consisting of 5 lawyers, 3 doctors and 2 engineers, four persons are selected at random. The probability that the selection contains at least one of each category is

Sol. $n(S) = {}^{10}C_4 = 210$;
$n(E) = {}^5C_2 \times {}^3C_1 \times {}^2C_1 + {}^5C_1 \times {}^3C_2 \times {}^2C_1 + {}^5C_1 \times {}^3C_1 \times {}^2C_2 = 105.$

$\therefore P(E) = \frac{105}{210} = \frac{1}{2}.$

Concept Application

3. The number 1, 2, 3 and 4 are written separately on four slips of paper. The slips are then put in a box and mixed thoroughly. A person draws two slips from the box one after the other, without replacement. Describe the following events:
A = The number on the first slip is larger than the one on the second slip.
B = The number on the second slip is greater than 2
C = The sum of the numbers on the two slips is 6 or 7
D = The number on the second slips is twice that on the first slip.
Which pair(s) of events is (are) mutually exclusive?
4. A card is picked up from a deck of 52 playing cards.
(*i*) What is the sample space of the experiment?
(*ii*) What is the event that the chosen card is black faced card?

AXIOMATIC APPROACH TO PROBABILITY

If there are '*n*' elementary events associated with a random experiment and '*m*' of then are favourable to an event *A*, then the probability of happening or occurrence of A is denoted by $P(A)$ and is defined as the ratio $\frac{m}{n}$.

Thus, $P(A) = \frac{m}{n}$

Clearly, $0 \le m \le n$. Therefore,

$$0 \le \frac{m}{n} \le 1 \Rightarrow 0 \le P(A) \le 1$$

If $P(A) = 1$, then A is called certain event and A is called an impossible event, if $P(A) = 0$.

The number of elementary events which will ensure the non-occurrence of A i.e. which ensure the occurrence of $\bar{A}$ is $(n - m)$. Therefore,

$$P(\bar{A}) = \frac{n-m}{n}$$

$$\Rightarrow P(\bar{A}) = 1 - \frac{m}{n}$$

$$\Rightarrow P(\bar{A}) = 1 - P(A)$$

$$\Rightarrow P(A) + P(\bar{A}) = 1$$

The odds in favour of occurrence of the event A are defined by $m: (n - m)$ i.e. $P(A) : P(\bar{A})$ and the odds against the occurrence of A are defined by $n - m : m$ i.e. $P(\bar{A}) : P(A)$.

Train Your Brain

Example 7: IF the letters of the word 'ALGORITHM' are arranged at random in a row what is the probability the letters 'GOR' must remain together in same order as a unit ?

Sol. Number of letters in the word 'ALGORITHM' = 9
If 'GOR' remain together, then consider it as 1 group.
$\therefore$ Number of letters = 6 + 1 = 7
Number of word, if 'GOR' remain together = 7!
Total number of words from the letters of the word 'ALGORITHM' = 9!

$\therefore$ Required probability = $\frac{7!}{9!} = \frac{1}{72}$

Example 8: Four candidates A, B, C and D have applied for the assignment to coach a school cricket team. If A is twice as likely to be selected as B and B and C are given about the same chance of being selected, while C is twice as likely to be selected as D, then what are the probabilities that

(*i*) C will be selected ?
(*ii*) A will not be selected?

Sol. It is given that A is twice as likely to be selected as B.
$P(A) = 2P(B)$

$$\Rightarrow \frac{P(A)}{2} = P(B)$$

While C is twice as likely to be selected as D.
$P(C) = 2P(D) \Rightarrow P(B) = 2P(D)$

$$\Rightarrow \frac{P(A)}{2} = 2P(D) \Rightarrow P(D) = \frac{P(A)}{4}$$

B and C are given about the same chance of being selected.
$P(B) = P(C)$
Now, sum of probability = 1
$P(A) + P(B) + P(C) + P(D) = 1$

$$P(A) + \frac{P(A)}{2} + \frac{P(A)}{2} + \frac{P(A)}{4} = 1$$

$$\Rightarrow \frac{4P(A) + 2P(A) + 2P(A) + P(A)}{4} = 1$$

$$\Rightarrow 9P(A) = 4 \Rightarrow P(A) = \frac{4}{9}$$

(*i*) P(C will be selected) = $P(C) = P(B) = \frac{P(A)}{2} = \frac{4}{9\times 2}$

$\left[\because P(A) = \frac{4}{9}\right]$

(*ii*) P(A will not be selected) = $P(A') = 1 - P(A) = 1 - \frac{4}{9} = \frac{5}{9}$.

Example 9: If a card is drawn from a deck of 52 cards, then find the probability of getting a king or a heart or a red card.

Sol. $\because$ Number of possible event = 52
And favorable events = 4 king + 13 heart + 26 red − 13 − 2 = 28

$\therefore$ Required probability = $\frac{28}{52} = \frac{7}{13}$

Example 10: Words are formed with the letters of the word *PEACE*. Find the probability that 2*E's* come together

Sol. Total number of words which can be formed with the letters *P, E, A, C, E* = $\frac{|5}{|2} = 60$.

Number of words in which 2 *E's* come together
= $|4 = 24$ $\therefore$ required probability = $\frac{24}{60} = \frac{2}{5}$.

Example 11: A bag contains 5 red and 4 green balls. Four balls are drawn at random then find the probability that two balls are of red and two balls are of green colour.

Sol. $n(s)$ = the total number of ways of drawing 4 balls out of total 9 balls : ${}^9C_4$
A : Drawing 2 red and 2 green balls $n(A) = {}^5C_2 \times {}^4C_2$

$$\therefore P(A) = \frac{n(A)}{n(s)} = \frac{{}^5C_2 \times {}^4C_2}{{}^9C_4} = \frac{\frac{5\times 4\times 4\times 3}{2\times 2}}{\frac{9\times 8\times 7\times 6}{4\times 3\times 2}} = \frac{10}{21}$$

Concept Application

5. In shuffling a pack of 52 playing cards, four are accidently dropped; find the chance that the missing cards should be one from each suit.
6. From a deck of 52 cards, four cards are drawn simultaneously, find the chance that they will be the four honours of the same suit.
7. A committee of two persons is selected from two men and two women. What is the probability the committee will have
(*i*) No man?
(*ii*) One man?
(*iii*) Tow men?
8. In a hand at Whist, what is the probability that four kings are held by a specified player?
9. In a lottery, a person choses six different natural numbers at random from 1 to 20, and if these six numbers match with the six number already fixed by the lottery committee, he wins the prize. What is the probability of winning the prize in the game?

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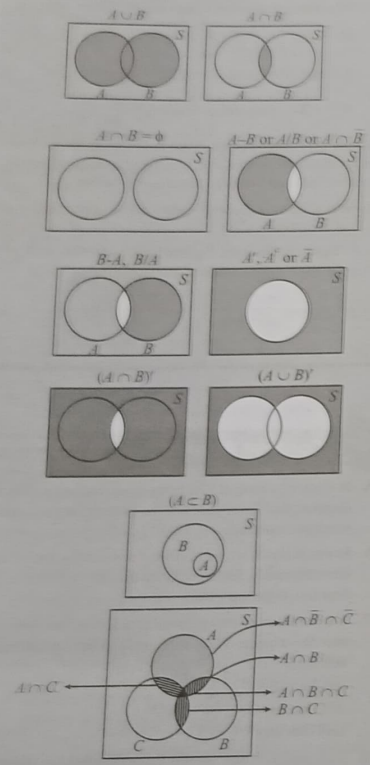
ADVANCED LEARNING

VENN DIAGRAMS

A diagram used to illustrate relationships between sets. Commonly, a rectangle represents the universal set and a circle within it represents a given set (all members of the given set are represented by points within the circle). A subset is represented by a circle within a circle and union and intersection are indicated by overlapping circles. Let S is the sample space of an experiment and A, B, C are three events corresponding to it



ADDITION THEOREM

$A \cup B = A + B = A$ or B denotes occurrence of at least A or B.

For 2 events A and B:

Note: (a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(This is known as generalized addition theorem)

$P(A+B)$
$P(A)+P(B\cap\bar{A})$
$P(A \text{ or } B)$
$P(B)+P(A\cap\bar{B})$
P (occurrence of atleast one A or B)

$\Rightarrow$

$P(A\cap\bar{B})+P(A\cap B)+P(B\cap\bar{A})$
$1-P(\bar{A}\cap\bar{B})$
$1-P(\overline{A\cup B})$

(b) $P(A \cup B) = P(A \cap \bar{B}) + P(\bar{A} \cap B) + P(A \cap B)$
$= 1 - P(\bar{A} \cap \bar{B})$

(c) Opposite of "atleast A or B" is neither A nor B
i.e. $\overline{A+B} = 1 - (A \text{ or } B) = \bar{A} \cap \bar{B}$

Note that $P(A+B) + P(\bar{A} \cap \bar{B}) = 1$

(d) If A and B are mutually exclusive then
$P(A \cup B) = P(A) + P(B)$

(e) For any two events A and B, P (exactly one of A, B occurs)
$= P(A \cap \bar{B}) + P(B \cap \bar{A}) = P(A) + P(B) - 2P(A \cap B)$
$= P(A \cup B) - P(A \cap B) = P(\bar{A} \cup \bar{B}) - P(\bar{A} \cap \bar{B})$

(f) **De Morgan's Law:** If A and B are two subsets of a universal set U, then

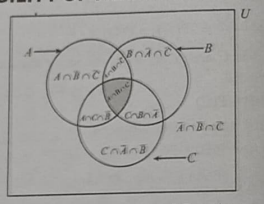
(i) $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$

(ii) $\overline{(A \cap B)} = \bar{A} \cup \bar{B}$

(iii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and

(iv) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

PROBABILITY OF THREE EVENTS



For any three events A, B and C we have

(a) $P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

(b) $P(\text{at least two of } A, B, C \text{ occur}) = P(B \cap C) + P(C \cap A) + P(A \cap B) - 2P(A \cap B \cap C)$

(c) $P(\text{exactly two of } A, B, C \text{ occur}) = P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$

(d) P (exactly one of A, B, C occurs)
$= P(A) + P(B) + P(C) - 2P(B \cap C) - 2P(C \cap A) - 2P(A \cap B) + 3P(A \cap B \cap C)$

Note: If three events A, B and C are pair wise mutually exclusive then they must be mutually exclusive i.e. $P(A \cap B) = P(B \cap C) = P(C \cap A) = 0 \Rightarrow P(A \cap B \cap C) = 0$. However the converse of this is not true.

Train Your Brain

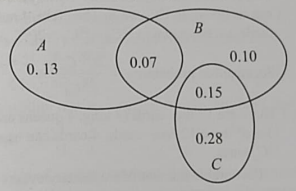
Example 12: If A and B are mutually exclusive events, $P(A) = 0.35$ and $P(B) = 0.45$, then find

(i) $P(A')$ (ii) $P(B')$
(iii) $P(A \cup B)$ (iv) $P(A \cap B)$
(v) $P(A \cap B')$ (vi) $P(A' \cap B')$

Sol. Since, it is given that, A and B are mutually exclusive events.
$\therefore \quad P(A \cap B) = 0 \quad [\because A \cap B = \phi]$
and $P(A) = 0.35$, $P(B) = 0.45$

(i) $P(A') = 1 - P(A) = 1 - 0.35 = 0.65$
(ii) $P(B') = 1 - P(B) = 1 - 0.45 = 0.55$
(iii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= 0.35 + 0.45 - 0 = 0.80$
(iv) $P(A \cap B) = 0$
(v) $P(A \cap B') = P(A) - P(A \cap B) = 0.35 - 0 = 0.35$
(vi) $P(A \cap B') = P(A) - P(A \cap B)$
$= 0.35 - 0 = 0.35$
(vii) $P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B)$
$= 1 - 0.8 = 0.2$

Example 13: The accompanying Venn diagram shows three events, A, B and C and also the probabilities of the various intersections [for instance, $P(A \cap B) = 0.07$]. Determine



(i) P(A)
(ii) $P(B \cap \bar{C})$
(iii) $P(A \cup B)$
(iv) Probability of exactly one of the three occurs.

Sol. Same the above Venn diagram,

(i) $P(A) = 0.13 + 0.07 = 0.20$

(ii) $P(B \cap \bar{C}) = P(B) - P(B \cap C)$
$= 0.07 + 0.10 + 0.15 - 0.15 = 0.07 + 0.10 = 0.17$

(iii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= 0.13 + 0.07 + 0.07 + 0.10 + 0.15 - 0.07$
$= 0.13 + 0.07 + 0.10 + 0.15$
$= 0.45$

(iv) P(exactly one of the three occurs)
$= 0.13 + 0.10 + 0.28$
$= 0.51$

Example 14: A bag contains 6 white, 5 black and 4 red balls. Find the probability of getting either a white or a black ball in a single draw.

Sol. Let A = event that we get a white ball, B = event that we get a black ball
So, the events are mutually exclusive

$P(A) = \frac{{}^6C_1}{{}^{15}C_1}$, $P(B) = \frac{{}^5C_1}{{}^{15}C_1}$

So $P(A+B) = P(A) + P(B) = \frac{6}{15} + \frac{5}{15} = \frac{11}{15}$

Concept Application

10. If A, B and C are three arbitrary events. Find the expression for the events noted below, in the context of A, B and C.

(i) Only A occurs
(ii) Both A and B, but not C occur
(iii) All the three events occur
(iv) At least one occurs
(v) At least two occur
(vi) One and no more occurs
(vii) Two and no more occur
(viii) None occurs
(ix) Not more than two occur

11. For the three events A, B and C, P (exactly one of the events A or B occurs) = P (exactly one of the events B or C occurs) = P (exactly one of the events C and A occurs) = p and P (all the three events occur simultaneously) = p^2, where $0 < p < 1/2$. Then, find the probability of occurrence of at least one of the three events A, B, and C.

BOOKWORMCBSE

AARAMBH (SOLVED EXAMPLES)

1. One card is drawn from a pack of 52 card, each of the 52 cards being equally likely to be drawn. Find the probability that the card drawn is:

(*i*) An ace (*ii*) Red

(*iii*) Either red or king (*iv*) Red and king

Sol. Out of 52 cards, one card can be drawn in $^{52}C_1$ ways.

$\therefore$ Total number of elementary events $= {}^{52}C_1 = 52$.

(*i*) There are four aces in a pack of 52 cards, out of which one ace can be drawn in $^4C_1$ ways. So, required probability $= \frac{4}{52} = \frac{1}{13}$.

(*ii*) There are 26 red cards, out of which one red card can be drawn in $^{26}C_1 = 26$.

$\therefore$ Favourable number of elementary events $= {}^{26}C_1 = 26$.

So, required probability $= \frac{26}{52} = \frac{1}{2}$.

(*iii*) There are 26 red cards including 2 red kings and there are 2 more kings. Therefore, there are 28 cards which are either red or king, out of 28 cards, one can be drawn in $^{28}C_1$ ways.

$\therefore$ Favourable number of elementary events $= {}^{28}C_1 = 28$.

So, required probability $= \frac{28}{52} = \frac{7}{13}$.

(*iv*) There are 2 cards which are red and king i.e., red kings.

$\therefore$ Favourable number of elementary events $= {}^2C_1 = 2$

So, required probability $= \frac{2}{52} = \frac{1}{26}$.

2. Four cards are drawn at random from a pack of 52 playing cards. Find the probability of getting

(*i*) All the four cards of the same suit

(*ii*) All the four cards of the same number

(*iii*) One card from each suit

(*iv*) Two red cards and two black cards

(*v*) All cards of the same colour

(*vi*) All face cards

Sol. Four cards can be drawn from a pack of 52 cards in $^{52}C_4$ ways.

So, total number cards of elementary events $= {}^{52}C_4$

(*i*) There are four suits viz. club, spade, heart and diamond, each of 13 cards. All the four cards are of the same suit means that either four cards drawn are club cards or spade cards or heart cards diamond cards. So, the total number of ways of getting all the four cards of the same suit is $^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 = 4(^{13}C_4)$

So, required probability $= \frac{4(^{13}C_4)}{^{52}C_4} = \frac{198}{20825}$

(*ii*) Four cards drawn can be of the same number in any one of the following ways:

(1, 1, 1, 1), (2, 2, 2, 2), (3, 3, 3, 3),..., (13, 13, 13, 1)

$\therefore$ Favourable number of elementary events $= 13$.

So, required probability $= \frac{13}{^{52}C_4} = \frac{13}{270725}$

(*iii*) There are four suits each of 13 cards. One card fro each suit means that there is one diamond card, o club card, one spade card and one heart card. There 13 diamond cards, out of which one can be selected $^{13}C_1$ ways, Similarly, one spade and one heart, each c be selected in $^{13}C_1$ ways.

$\therefore$ The number of ways of selecting 4 cards, one fr each suit $= {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1$

So, required probability $= \frac{(^{13}C_1)^4}{^{52}C_4} = \frac{2197}{20825}$

(*iv*) There are 26 red cards and 26 black cards. Out of red cards, 2 cards can be drawn in $^{26}C_2$ ways. Simila 2black cards can be drawn in $^{26}C_2$ ways. Therefo 2 red and 2 black cards can be drawn in $^{26}C_2 \times {}^{26}C_2$ way

So, required probability $= \frac{^{26}C_2 \times {}^{26}C_2}{^{52}C_4}$

(*v*) There are two colours viz. red and black. Out of 26 r colour cards can be drawn in $^{26}C_4$ ways. 4 black ca can be drawn in $^{26}C_4$ ways. Therefore, 4 red or 4 bla cards can be drawn in $^{26}C_4 + {}^{26}C_4 = 2(^{26}C_4)$ ways.

So, required probability $= \frac{2(^{26}C_4)}{^{52}C_4}$

(*vi*) There are 12 face cards (4 king, 4 queens and 4 jack Out of these 12 face cards, 4 cards can be selected $^{12}C_4$ ways,

$\therefore$ Favourable number of elementary events $= {}^{12}C_4$

So, required probability $= \frac{^{12}C_4}{^{52}C_4}$

3. Five marbles are drawn from a bag which contains 7 bl marbles and 4 black marbles. What is the probability tha

(*i*) All will be blue?

(*ii*) 3 will be blue and 2 black?

Sol. There are 7 + 4 = 11 marbles in the bag out of which 5 marb can be drawn in $^{11}C_5$ ways.

$\therefore$ Total number of elementary events $= {}^{11}C_5$.

(*i*) There are 7 blue marbles out of which 5 blue marb can be drawn in $^7C_5$ ways.

$\therefore$ Favourable number of elementary events $= {}^7C_5$

Hence, required probability

$= \frac{^7C_5}{^{11}C_5} = \frac{7!}{2!5!} \times \frac{5!6!}{11!} = \frac{1}{22}$

(*ii*) Three blue out of 7 blue marbles and 2 black out of 4 black marbles can be drawn in $^7C_3 \times {}^4C_2$ ways.

$\therefore$ Favourable number of elementary events $= {}^7C_3 \times {}^4C_2$

Hence, required probability

$= \frac{^7C_3 \times {}^4C_2}{^{11}C_5} = \frac{7!}{3!4!} \times \frac{4!}{2!2!} \times \frac{5!\times 6!}{11!} = \frac{5}{11}$

4. A card is drawn from an ordinary pack of 52 cards and a gambler bets that, it is a spade or an ace, What are the odds against his winning this bet?

Sol. Let A be the event of getting a spade or an ace from a pack of 52 cards. Then, Total number of elementary events $= {}^{52}C_1 = 52$

Since there are 13 spade cards including an ace of spade and three aces other than an ace of

So, $P(A) = \frac{16}{52} = \frac{4}{13}$.

Hence, odds against A are $P(\bar{A}) : P(A) = \frac{9}{13} : \frac{4}{13} = 9 : 4$

5. Six new employees, two of whom are married to each other, are to be assigned six desks that are lined up in a row. If the assignment of employees to desks is made randomly, what is the probability that the married couple will have non-adjacent desks?

Sol. Let the couple occupied adjacent desks consider those two as 1.

There are (4 + 1) *i.e.*, 5 persons to be assigned.

$\therefore$ Number of ways of assigning these five person $= 5! \times 2!$

Total number of ways of assigning 6 persons $= 6!$

$\therefore$ Probability that the couple has adjacent desk

$= \frac{5!\times 2!}{6!} = \frac{2}{6} = \frac{1}{3}$

Probability that the married couple will have non-adjacent desks $= 1 - \frac{1}{3} = \frac{2}{3}$

6. A team of medical students doing their internship have to assist during surgeries at a city hospital. The probabilities of surgeries rates as very complex, complex, routine, simple or very simple are respectively, 0.15, 0.20, 0.31, 0.26 and 0.08. Find the probabilities that a particular surgery will be rated

(*a*) Complex or very complex

(*b*) Neither very complex nor very simple

(*c*) Routine or complex

(*d*) Routine or simple

Sol. (*a*) Let E_1, E_2, E_3, E_4 and E_5 be the event that surgeries are rated as very complex, complex, routine, simple or very simple, respectively.

$\therefore$ P(complex or very Complex) $= P(E_1 \text{ or } E_2)$

$= P(\text{Complex or very Complex}) = P(E_1 \text{ or } E_2)$

$= P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$= 0.15 + 0.20 - 0$

$\because$ $P(E_1 \cap E_2) = 0$

[Because all events are independent] $= 0.35$

(*b*) P(neither very complex nor very simple),

$P(E'_1 \cap E'_5) = P(E_1 \cup E_5)' = 1 - P(E_1 \cup E_5)$

$= 1 - [P(E_1) + P(E_5)]$

$= 1 - 0.23 = 0.77$

(*c*) P(routine or complex)

$P(E_3 \cup E_2) = P(E_3) + P(E_2)$

$= 0.31 + 0.20 = 0.51$

(*d*) P(routine or simple) $= P(E_3 \cup E_4)$

$= P(E_3) + P(E_4)$

$= 0.31 + 0.26 = 0.57$

7. Determine the probability p, for the each of the following events.

(*i*) An odd number appears in a single toss of a fair die.

(*ii*) Atleast one head appears in two tosses of a fair coin.

(*iii*) A king, 9 of hearts or 3 of spades appears in drawing a single card from a well shuffled ordinary deck of 52 cards.

(*iv*) The sum of 6 appears in a single toss of a pair of fair dice.

Sol. (*i*) When a die is throw the possible outcomes are

$S = \{1, 2, 3, 4, 5, 6\}$ out of the which 1, 3, 5 are odd,

$\therefore$ Required probability $= \frac{3}{6} = \frac{1}{2}$

(*ii*) When a fair coin is tossed two times the sample space is S = {HH, HT, TH, TT}

In at least one head favourable events are

HH, HT, TH

$\therefore$ Required probability $= \frac{3}{4}$

(*iii*) Total cards = 52

Favourable = 4 king + 2 of heart + 3 spade

$= 4 + 1 + 1 = 6$

$\therefore$ Required probability $= \frac{6}{52} = \frac{3}{26}$

(*iv*) When a pair of dice is rolled total sample parts are 36. Out of which (1, 5), (5, 1), (2, 4), (4, 2) and (3, 3).

$\therefore$ Required probability $= \frac{5}{36}$

8. There are three events A, B, C, one of which must, and only one can happen; the odds are 8 to 3 against A, 5 to 2 against B : find the odds against C.

Sol. $P(A) = \frac{3}{11}$; $P(B) = \frac{2}{7}$; P(C) = ? ; P(A) + P(B) + P(C) = 1

$\Rightarrow$ $P(C) = \frac{34}{77}$.

9. Three persons A, B and C speak at a function along with 5 other persons. If the persons speak at random, find the probability that A speaks before B and B speaks before C

Sol. A ~ B ~ C; $P = \frac{8!}{3!\,8!} = \frac{1}{6}$

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❖ If outcomes is 5, then they will toss a coin. If it shows "Head", the family will do 5 minutes of rope skipping. If there is "Tail", family will do 20 minutes of Yoga

22. How many elements are there in the sample space?
(a) 6 (b) 9 (c) 7 (d) 8

23. What is the probability of doing walking?
(a) $\frac{5}{6}$ (b) $\frac{3}{7}$ (c) $\frac{4}{7}$ (d) $\frac{1}{6}$

Case Study-II

One day, a sangeet Mahotsav is to be organized in an open area of Rajasthan. In recent years, it has rained only 6 days each year. Also, it is given that when it actually rains, the weatherman correctly forecasts rain 80% of the time. When it doesn't rain, he incorrectly forecasts rain 20% of the time. If leap year is considered, answer the following questions.

24. The probability that the weatherman predicts correctly
(a) $\frac{5}{6}$ (b) $\frac{7}{8}$ (c) $\frac{4}{5}$ (d) $\frac{1}{5}$

25. The probability that it will rain on the chosen day weatherman predict rain for that day, is
(a) 0.0625 (b) 0.0725
(c) 0.0825 (d) 0.0925

PRARAMBH (TOPICWISE)

1. If a dice is thrown twice, then the probability of getting 1 in the first throw only is
(a) $\frac{1}{36}$ (b) $\frac{3}{36}$ (c) $\frac{5}{36}$ (d) $\frac{1}{6}$

2. The probability of getting a total of 5 or 6 in a single throw of 2 dice is
(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{6}$

3. In a non-leap year, the probability of having 53 Tuesday or 53 Wednesday is
(a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) None of these

4. Three numbers are chosen from 1 to 20. Find the probability that they are not consecutive
(a) $\frac{186}{190}$ (b) $\frac{187}{190}$ (c) $\frac{188}{190}$ (d) $\frac{18}{{}^{20}C_3}$

5. Two dice are thrown simultaneously. What is the probability of obtaining a multiple of 2 on one of them and a multiple of 3 on the other
(a) $\frac{5}{36}$ (b) $\frac{11}{36}$ (c) $\frac{1}{6}$ (d) $\frac{1}{3}$

6. A coin is tossed 4 times. The probability that at least one head turns up is
(a) $\frac{1}{16}$ (b) $\frac{2}{16}$ (c) $\frac{14}{16}$ (d) $\frac{15}{16}$

7. While shuffling a pack of 52 playing cards, 2 are accidentally dropped. Find the probability that the missing cards to be of different colours.
(a) $\frac{29}{52}$ (b) $\frac{1}{2}$ (c) $\frac{26}{51}$ (d) $\frac{27}{51}$

8. If a coin be tossed n times then probability that the h comes odd times is
(a) $\frac{1}{2}$ (b) $\frac{1}{2^n}$ (c) $\frac{1}{2^{n-1}}$ (d) 2^{n-1}

9. A bag contains 3 red, 7 white and 4 black balls. If three b are drawn from the bag, then the probability that all of t are of the same colour is
(a) $\frac{6}{71}$ (b) $\frac{7}{81}$ (c) $\frac{10}{91}$ (d) $\frac{7}{91}$

10. If seven persons are to be seated in a row. Then, the probab that two particular persons sit next to each other is
(a) $\frac{1}{3}$ (b) $\frac{1}{6}$ (c) $\frac{2}{7}$ (d) $\frac{1}{2}$

11. If with out repetition of the numbers, four-digit numbers formed with the numbers 0, 2, 3 and 5 then the probab of such a number divisible by 5 is
(a) $\frac{1}{5}$ (b) $\frac{4}{5}$ (c) $\frac{1}{30}$ (d) $\frac{5}{9}$

12. The letter of the word 'ASSASSIN' are written d at random in a row. The probability that no two S a together is
(a) $\frac{1}{35}$ (b) $\frac{1}{14}$ (c) $\frac{4}{13}$ (d) $\frac{2}{35}$

13. A bag contains 4 white and 3 red balls. Two draws of ball each are made without replacement. Then the probab that both the balls are red is
(a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{4}{7}$

14. If A and B are mutually exclusive events, then
(a) $P(A) \le P(\overline{B})$ (b) $P(A) \ge P(\overline{B})$
(c) $P(A) < P(\overline{B})$ (d) None of these

15. If a committee of 3 is to be chosen from a group of 38 people of which you are a member. What is the probability that you will be on the committee
(a) ${}^{38}C_3$ (b) ${}^{37}C_2$ (c) $\frac{{}^{37}C_2}{{}^{38}C_3}$ (d) $\frac{666}{8436}$

16. If $P(A \cup B) = P(A \cap B)$ for any two events A and B, then
(a) $P(A) = P(B)$ (b) $P(A) > P(B)$
(c) $P(A) < P(B)$ (d) None of these

17. If A and B are two events such that $P(A) = 0.4$, $P(A \cup B) = 0.7$ and $P(A \cap B) = 0.2$, then $P(B) =$
(a) 0.1 (b) 0.3 (c) 0.5 (d) 0.4

18. A card is drawn at random from a pack of cards. The probability of this card being a red or a queen is
(a) $\frac{1}{13}$ (b) $\frac{1}{26}$ (c) $\frac{1}{2}$ (d) $\frac{7}{13}$

19. If A and B are two events such that $P(A \cup B) + P(A \cap B) = \frac{7}{8}$ and $P(A) = 2P(B)$, then $P(A) =$
(a) $\frac{7}{12}$ (b) $\frac{7}{24}$ (c) $\frac{5}{12}$ (d) $\frac{17}{24}$

20. The probability that at least one of A and B occurs is 0.9. If A and B occur simultaneously with probability 0.3, then $P(A') + P(B') =$
(a) 0.9 (b) 0.2 (c) 0.8 (d) 1.2

21. Let A and B are two events and $P(A') = 0.3$ $P(B') = 0.4$, $P(A \cap B') = 0.5$, then $P(A \cup B')$ is
(a) 0.5 (b) 0.8 (c) 1 (d) 0.6

22. If 6 boys and 6 girls sit in a row at random, then the probability that all the girls sit together is
(a) $\frac{12}{132}$ (b) $\frac{12}{431}$ (c) $\frac{1}{132}$ (d) None of these

23. If a single letter is selected at random from the word 'PROBABILITY', then the probability that it is a vowel is
(a) $\frac{1}{3}$ (b) $\frac{4}{11}$ (c) $\frac{2}{11}$ (d) $\frac{3}{11}$

24. If the probability for A to fail in an examination is 0.2 and that for B is 0.3, then the probability that either A or B fails is
(a) > 0.5 (b) 0.5 (c) ≤ 0.5 (d) 0

25. The probability that at least one of the events A and B occurs is 0.6. If A and B occur simultaneously with probability 0.2, then $P(\overline{A}) + P(\overline{B})$ is equal to
(a) 0.4 (b) 0.8 (c) 1.2 (d) 1.6

26. If M and N are any two events, the probability that at least one of them occurs is
(a) $P(M) + P(N) - 2P(M \cap N)$
(b) $P(M) + P(N) - P(M \cap N)$
(c) $P(M) + P(N) + P(M \cap N)$
(d) $P(M) + P(N) + 2P(M \cap N)$

27. A 9 digit number using the digits 1, 2, 3, 4, 5, 6, 7, 8 and 9 is written randomly without repetition. The probability that the number will be divisible by 9 is:
(a) 1/9 (b) 1/2 (c) 1 (d) $9!/9^9$

28. Two dies are rolled simultaneously. The probability that the sum of the two numbers on the top faces will be atleast 10 is:
(a) 1/6 (b) 1/12 (c) 1/18 (d) 1/16

29. Three numbers are chosen at random without replacement from {1, 2, 3 ..., 10}. The probability that the minimum of the chosen numbers is 3 or the maximum is 7, is
(a) 1/2 (b) 1/3 (c) 1/4 (d) 11/40

30. If atleast one child in a family with 3 children is a boy then the probability that 2 of the children are boys, is
(a) 3/7 (b) 4/7 (c) 1/3 (d) 3/8

PRABAL (JEE MAIN LEVEL)

1. Three dice are thrown together. Find the probability of getting a total of at least 6.
(a) $\frac{101}{108}$ (b) $\frac{102}{108}$ (c) $\frac{103}{108}$ (d) $\frac{104}{108}$

2. Tickets numbered from 1 to 20 are mixed up together and then a ticket is draw at random. What is the probability that the ticket has a number which is a multiple of 3 or 7?
(a) $\frac{3}{5}$ (b) $\frac{2}{5}$ (c) $\frac{1}{5}$ (d) $\frac{4}{5}$

3. An urn contains 9 red, 7 white and 4 black balls, if two balls are drawn at random, then the probability that:
(a) Both the balls are red, is 18/95
(b) One ball is white, is 90/190
(c) The balls are of the same colour, is 61/190
(d) One is white and other red, is 60/190

4. Four persons are to be chosen at random from a group of 3 men, 2 women and 4 children. Then the probability of electing:
(a) 1 man, 1 woman and 2 children is $\frac{2}{7}$
(b) Exactly 2 children is $\frac{11}{21}$

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10. Tickets are numbered from 1 to 100. One ticket is picked up at random. Then find the probability that the ticket picked up has a number which is divisible by 5 or 8.

Sol. Let A (or B) be the event that the number on the ticket is divisible by 5(or 8). Then

$A = \{5, 10, 15, 20, \ldots, 95, 100\}$; $B = \{8, 16, 24, 32, \ldots, 88, 96\}$

$\Rightarrow A \cap B = \{40, 80\}$; $n(A) = \frac{100}{5} = 20$,

$n(B) = 12$, $n(A \cap B) = 2$

The reqd. prob. $= P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{20}{100} + \frac{12}{100} - \frac{2}{100} = \frac{3}{10}$

11. If $P(A) = 0.7$ and $P(A \cap B) = 0.5$ find

(*i*) $P(A - \bar{B})$

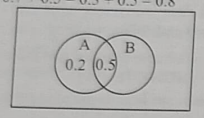
(*ii*) $P(\bar{A} \cup B)$

Sol. (*i*) $P(A - \bar{B}) = P(A \cap B) = 0.5$

(*ii*) $P(\bar{A} \cup B) = 1 - P(\overline{\bar{A} \cup B}) = 1 - \{P(A \cap \bar{B})\}$

$= 1 - \{P(A) - P(A \cap B)\} = 1 - P(A) + P(A \cap B)$

$= 1 - 0.7 + 0.5 = 0.3 + 0.5 = 0.8$



12. From a group of 2 boys and 3 girls, two children are selected at random. Describe the events.

(*i*) A = both selected children are girls.

(*ii*) B = the selected group consists of one boy and one girl.

(*iii*) C = at least one boy is selected.

Which pair (s) of events is (are) mutually exclusive?

Sol. Let B_1, B_2 be two boys and G_1, G_2, G_3 be three girls. Then the sample space associated with the random experiment is

$S = [B_1B_2, B_1G_1, B_1G_2, B_1G_3, B_2G_1, B_2G_2, B_2G_3, G_1G_2, G_1G_3, G_2G_3]$

(*i*) We have,

B = The selected children are girls $= [G_1G_2, G_1G_3, G_2G_3]$

(*ii*) We have,

B = The selected group consists of one boy and one girl

$\Rightarrow B = \{B_1G_1, B_1G_2, B_1G_3, B_2G_1, B_2G_2, B_2G_3\}$

(*iii*) We have,

C = At least one boy is selected $= [B_1B_2, B_1G_1, B_1G_2, B_1G_3, B_2G_1, B_2G_2, B_2G_3]$

$\Rightarrow A \cap B = \phi$ and $A \cap C = \phi$. So, A and B, A and C are two pairs of mutually exclusive.

SCHOOL LEVEL PROBLEMS

SINGLE CORRECT TYPE QUESTIONS

1. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. The probability that none of the balls drawn is blue

(*a*) $\frac{10}{21}$ (*b*) $\frac{11}{21}$ (*c*) $\frac{2}{7}$ (*d*) $\frac{5}{7}$

2. Two dice are thrown simultaneously. Find probability of getting a total of at least 10

(*a*) $\frac{1}{3}$ (*b*) $\frac{1}{4}$ (*c*) $\frac{1}{5}$ (*d*) $\frac{1}{6}$

3. A coin is tossed and if the head comes up, a die is thrown. But if the tail comes up, the coin is tossed again. Write the sample space of this experiment. The total no. of elements in the sample space will be

(*a*) 4 (*b*) 24 (*c*) 8 (*d*) 18

4. If four dices are rolled simultaneously, what is the total number of possible outcomes?

(*a*) 6 (*b*) 36 (*c*) 216 (*d*) 1296

5. Any subset of sample space is known as _____

(*a*) Event (*b*) Empty set

(*c*) Sample space (*d*) Probability

6. When eight coins are tossed at the same time, the probability of obtaining six or more heads is

(*a*) 57/64 (*b*) 229/56

(*c*) 7/64 (*d*) 37/256

7. Two persons *A* and *B* appear in an interview for two vacancies. If the probability of their selection are 1/4 and 1/6 respectively, then probabilities that none of them is selected is:

(*a*) $\frac{1}{24}$ (*b*) $\frac{5}{12}$ (*c*) $\frac{5}{8}$ (*d*) $\frac{19}{12}$

8. Three identical dices are rolled. What is the probability that the same number will appear on each of them?

(*a*) $\frac{1}{6}$ (*b*) $\frac{1}{36}$ (*c*) $\frac{1}{18}$ (*d*) $\frac{3}{18}$

9. The probability of getting 53 Sundays in a leap year is

(*a*) $\frac{1}{7}$ (*b*) $\frac{2}{7}$

(*c*) $\frac{3}{7}$ (*d*) None of these

VERY SHORT ANSWER TYPE QUESTIONS

10. 20 cards are numbered from 1 to 20. One card is then drawn at random. What is the probability of a prime number?

11. A letter is chosen at random from the word 'ASSASSINATION', find the probability that letter is a consonant.

12. There are four men and six women on the city council. If one council member is selected for a committee at random, how likely is it that it is a woman?

13. 4 cards are drawn from a well snuffled deck of 52 cards what is the probability of obtaining 3 diamonds and one spade?

14. A couple has 2 children. Find the probability that both are boys, if it is known that the older child is a boy.

15. Describe the sample space if a coin is tossed and a die is thrown

16. A box contains 1 white and 3 identical black balls. Two balls are drawn at random in succession without replacement. Write the sample space for this experiment.

LONG ANSWER TYPE QUESTIONS

17. *A* and *B* are events such that $P(A) = 0.42$, $P(B) = 0.48$, and $P(A \text{ and } B) = 0.16$.

Determine (i) $P(\text{not } A)$ (ii) $P(\text{not } B)$ (iii) $P(A \text{ or } B)$

18. In a town of 6000 people 1200 are over 50 yr. old and 2000 are females. It is known that 30% of the females are over 50 yr. what is the probability, that a randomly chosen individual from the town is either female or over 50 yr.?

19. Two students Anil and Ashima appeared in an examination. The probability that Anil will qualify the examination is 0.05 and that Ashima will qualify the examination is 0.10. The probability that both will qualify the examination is 0.02 find the probability that

(*a*) Both Anil and Ashima will qualify the examination

(*b*) At least one of them will not qualify the examination

(*c*) Only one of them will qualify the examination.

20. In a single throw of three dices, find the probability, of getting (*i*) A total of 5 (*ii*) A total of at most 5.

21. 20 cards are numbered from 1 to 20. One card is drawn at random what is the probability, that the number on the card drawn is

(*i*) A prime number

(*ii*) An odd number

(*iii*) A multiple of 5

(*iv*) Not divisible by 3.

CASE STUDY BASED QUESTIONS

Case Study-I

To make a healthy routine and to do some physical exercise during lockdown a family decided to roll a dice and based on the outcomes, they will decide activities to be done.

- If the outcomes is 2, 4 or 6, they will do 30 minutes walk on the roof.
- If it shows 1 or 3 on the dice, 15 minutes meditation to be done.

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(c) 2 women is $\frac{1}{5}$

(d) None of these

5. In a single throw of three dice, the probability of getting

(a) A total of 5 is $\frac{1}{38}$

(b) A total of al most 5 is $\frac{5}{108}$

(c) A total of at least 5 is $\frac{53}{55}$

(d) None of these

6. The odds in favour of an event are 3 : 5. Find the probability of occurrence of the event.

(a) $\frac{3}{8}$ (b) $\frac{5}{8}$ (c) $\frac{1}{8}$ (d) $\frac{7}{8}$

7. A typical PIN (personal identification number) is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits. If all PINs are equally likely, what is the probability that a randomly chosen PIN contains a repeated symbol ?

(a) 0.1283 (b) 0.1383 (c) 0.1883 (d) 0.1583

8. If an integer from 1 through 1000 is chosen at random, then find the probability that the integer is a multiple of 2 or a multiple of 9.

(a) 0.556 (b) 0.356 (c) 0.256 (d) 0.156

9. An experiment consists of rolling a die until a 2 appears, then

(a) Elements of the sample space correspond to the events that the 2 appears on the k^{th} roll of the die = 5^{k-1}

(b) Elements of the sample space correspond to the events that the 2 appears on the k^{th} roll of the die = 5^{k-2}

(c) Elements of the sample space correspond to the events the 2 appears not tater than the k^{th} roll of the die = 5^{k-1}

(d) None of these

10. A die is loaded in such a way that each odd number is twice as likely to occur as each even number. Find $P(G)$, where G is that a number greater than 3 occurs on a single roll of the die.

(a) $\frac{4}{9}$ (b) $\frac{2}{9}$ (c) $\frac{5}{9}$ (d) $\frac{1}{9}$

11. In a large metropolitan area , the probabilities are 0.87, 0.36, 0.30 that a family (randomly chosen for a sample survey) owns a colour television set, a black and white television set or both kinds of sets. What is the probability that a family owns either anyone or both kinds of sets ?

(a) 0.53 (b) 0.73 (c) 0.33 (d) 0.93

12. One of the four persons john, Rita, Aslam or Gurpreet will be promoted next month. Consequently, the simple space of four elementary outcomes S = {John promoted, Rita, promoted, Aslam promoted, Gurpreet promoted}. You told that the chances of john's promotion is same as that Gurpreet. Rita's Chances of promotion are twice as lik as Johns. Aslam's chances are four times that of John.

(a) P(John promoted) + P(Rita promoted) = P(Aslam promoted) + P(Gurpreet promoted).

(b) If A = {John promoted or Gurpreet promoted}, th P(A) = $\frac{1}{4}$

(c) If A = {John promoted or Gurpreet promoted}, th P(A) = $\frac{1}{5}$

(d) None of these

13. One urn contains two black balls (Labelled B_1 and B_2) a two white ball. A second urn contains one black ball and t white balls (labelled W_1 and W_2). Suppose the follow experiment is performed. One of the two urns is chose random. Next a ball is randomly chosen from the urn. Th a second ball is chosen at random from the same urn with replacing the first ball, then

(a) the probability that two black balls are chosen = $\frac{1}{6}$

(b) the probability that two balls of opposite colour chosen = $\frac{2}{5}$

(c) the probability that two black balls are chosen = $\frac{1}{5}$

(d) None of these

14. A bag contains 8 red and 5 white balls. Three balls are dra at random, then the probability that

(a) All the three balls are white is $\frac{6}{143}$

(b) All the three balls are red is $\frac{28}{143}$

(c) One ball is red and two balls are white is $\frac{50}{143}$

(d) None of these

15. If the letters of the word 'ASSASSINATION' are arran at random, then the probability that

(a) Four S's come consecutively in the word = $\frac{3}{143}$

(b) Two I's and two N's come together = $\frac{5}{143}$

(c) All A's are not coming together = $\frac{6}{143}$

(d) No. two A's are coming together = $\frac{15}{26}$

16. A sample space consists of 9 elementary outcomes E_1 ... E_9 whose probabilities are

$P(E_1) = P(E_2) = 0.08$, $P(E_3) = P(E_4) = P(E_5) = 0.1$

$P(E_6) = P(E_7) = 0.2$, $P(E_8) = P(E_9) = 0.07$

Suppose $A = \{E_1, E_5, E_8\}$, $B = \{E_2, E_5, E_8, E_9\}'$, then choose the correct from following

(a) P(A) + P(B) = P(A ∩ B)

(b) P(A ∩ B) = 0.40

(c) $P(\bar{B}) = 0.78$

(d) None of these

17. If the letters of the word BANANA are arranged randomly, then find the probability that the word thus formed does not contain the pattern BAN.

(a) $\frac{4}{5}$ (b) $\frac{1}{5}$ (c) $\frac{3}{5}$ (d) $\frac{2}{5}$

18. Nine cards are labelled 0, 1, 2, 3, 4, 5, 6, 7, 8. Two cards are drawn at random and put on a table in a successive order, and then the resulting number is read say 07 (seven), 42 (forty two) and so on. Find the probability that the number is even.

(a) $\frac{4}{9}$ (b) $\frac{5}{9}$ (c) $\frac{1}{9}$ (d) $\frac{2}{9}$

19. A rectangle is randomly selected from the grid of equally spaced squares as shown, then choose the correct from following.

(a) The probability that the rectangle is a square = $\frac{6}{15}$

(b) Three of the six vertices of a regular hexagon are chosen at random. Then the probability that the triangle with three vertices is equilateral is 'p' then $100p = \frac{1}{10}$

(c) Three of the six vertices of a regular hexagon are chosen at random. Then the probability that the triangle with three vertices is equilateral is 'p' then $100p = \frac{1}{12}$

(d) None of these

20. If $P(A) = 0.4$, $P(B) = 0.48$ and $P(A \cap B) = 0.16$, choose the correct from following :

(a) $P(A \cup B) = 0.72$ (b) $P(A' \cap B) = 0.35$

(c) $P(A' \cap B') = 0.5$ (d) $P(A \cap B') \cup (A' \cap B)) = 0.23$

21. Two dice are thrown and the sum of the numbers which come up on the dice is noted. Let us consider the following events.

A = The sum is even, B = The sum is multiple of 3, C = The sum is less than 4.

D = The sum is greater than 11 then choose the correct from following

(a) A and B are not mutually Exclusive Events

(b) C and D are not mutually Exclusive Events

(c) A and C are not mutually Exclusive Events

(d) None of these

INTEGER TYPE QUESTIONS

22. In throwing of a pair of dice, if the probability of the event: total is 'not 8' and 'not 11' is $\frac{a}{b}$ (a, b are co-primes) then find $|a - b|$

23. If $P(A - B) = P(A) - P(A \cap B) = P(A \cup B) - P(B) = P(A \cap \bar{B}) = q - P(\bar{A} \cup B)$, then q =

24. There are three clubs A, B, C in a town with 40, 50, 60 members respectively 10 people are members of all the three clubs, 70 members belong to only one club. A member is randomly selected. If the probability that he has membership of exactly two clubs is $\frac{a}{b}$ (a, b are co-primes) then find $|a - b|$

PYQ's (PAST YEAR QUESTIONS)

GENERAL INTRODUCTION AND DEFINITIONS

1. Let M be the maximum value of the product of two positive integers when their sum is 66. Let the sample space $S = \left\{x \in Z : x(66 - x) \geq \frac{5}{9}M\right\}$ and the event $A = \{x \in S : x \text{ is a multiple of } 3\}$. Then $P(A)$ is equal to **[25 Jan, 2023 (Shift-I)]**

(a) $\frac{15}{44}$ (b) $\frac{1}{3}$ (c) $\frac{1}{5}$ (d) $\frac{7}{22}$

2. Three dice are rolled. If the probability of getting different numbers on the three dice is $\frac{p}{q}$, where p and q are co-prime, then $q - p$ is equal to **[6 April, 2023 (Shift-II)]**

(a) 4 (b) 3 (c) 1 (d) 2

3. Out of 60% female and 40% male candidates appearing in an exam, 60% candidates qualify it. The number of females qualifying the exam is twice the number of males qualifying it. A candidate is randomly chosen from the qualified candidates. The probability, that the chosen candidate is a female, is: **[28 July, 2022 (Shift-I)]**

(a) $\frac{2}{3}$ (b) $\frac{11}{16}$ (c) $\frac{23}{32}$ (d) $\frac{13}{16}$

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4. A seven digit number is formed using digits 3,3,4,4,4,5,5. The probability, that number so formed is divisible by 2, is: **[26 Feb, 2021 (Shift-II)]**

(a) $\frac{3}{7}$ (b) $\frac{4}{7}$ (c) $\frac{1}{7}$ (d) $\frac{6}{7}$

5. Let A denote the event that a 6-digit integer formed by 0, 1, 2, 3, 4, 5, 6 without repetitions, by divisible by 3. Then probability of event A is equal to: **[16 March, 2021 (Shift-II)]**

(a) $\frac{4}{9}$ (b) $\frac{11}{27}$ (c) $\frac{9}{56}$ (d) $\frac{3}{7}$

6. The probability that a randomly selected 2-digit number belongs to the set $\{n \in N: (2^n - 2) \text{ is a multiple of } 3\}$ is equal to: **[27 July, 2021 (Shift-I)]**

(a) $\frac{2}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{6}$ (d) $\frac{1}{3}$

7. The probability that a randomly chosen 5-digit number is made from exactly two digits is: **[3 Sep, 2020 (Shift II)]**

(a) $\frac{150}{10^4}$ (b) $\frac{134}{10^4}$ (c) $\frac{121}{10^4}$ (d) $\frac{135}{10^4}$

8. In a class of 60 students, 40 opted for NCC, 30 opted for NSS and 20 opted for both NCC and NSS. If one of these students is selected at random, then the probability that the student selected has opted neither for NCC nor for NSS is: **[12 Jan, 2019 (Shift-II)]**

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

9. Let S = {1, 2, ... 20}. A subset B is S is said to be "nice", if the sum of the elements of B is 203. Then the probability that a randomly chosen subset of S is 'nice' is: **[11 Jan, 2019 (Shift-II)]**

(a) $\frac{7}{2^{20}}$ (b) $\frac{5}{2^{20}}$ (c) $\frac{4}{2^{20}}$ (d) $\frac{6}{2^{20}}$

10. Let N denotes the sum of the numbers obtained when two dice are rolled. If the probability that $2^N < N!$ is $\frac{m}{n}$, where m and n are coprime, then $4m - 3n$ is equal to **[10 April, 2023 (Shift-I)]**

(a) 8 (b) 16 (c) 10 (d) 12

11. When a certain biased die is rolled, a particular face occurs with probability $\frac{1}{6} - x$ and its opposite face occurs with probability $\frac{1}{6} + x$. All other faces occur with probability $\frac{1}{6}$. Note that opposite faces sum to 7 in any die. If $0 < x < \frac{1}{6}$, and the probability of obtaining total sum = 7, when such a die is rolled twice, is $\frac{13}{96}$, then the value of x is: **[27 Aug, 2021 (Shift-I)]**

(a) $\frac{1}{9}$ (b) $\frac{1}{16}$ (c) $\frac{1}{8}$ (d) $\frac{1}{12}$

ADDITION AND MULTIPLICATION THEOREM ON PROBABILITY, GEOMETRICAL PROBABILITY

12. A bag contains 6 white and 4 black balls. A die is rolled once and the number of balls equal to the number obtained on the die are drawn from the bag at random. The probability that all the balls drawn are white is: **[15 April, 2023 (Shift-I)]**

(a) $\frac{1}{4}$ (b) $\frac{9}{50}$

(c) $\frac{1}{5}$ (d) $\frac{11}{50}$

13. Let the probability of getting head for a biased coin be 1/4. It is tossed repeatedly until a head appears. Let N be the number of tosses required. If the probability that the equation $64x^2 + 5Nx + 1 = 0$ has no real root is p/q, where p and q are co-prime, then $q - p$ is equal to **[11 April, 2023 (Shift-I)]**

14. Let S be the sample space of all five digit numbers. If p is the probability that a randomly selected number from S is a multiple of 7 but not divisible by 5, then $9p$ is equal to **[27 July, 2022 (Shift-I)]**

(a) 1.0146 (b) 1.2085

(c) 1.0245 (d) 1.1521

15. Five number x_1, x_2, x_3, x_4, x_5 are randomly selected from the number 1, 2, 3,..., 18 and are arranged in the increasing order $(x_1 < x_2 < x_3 < x_4 < x_5)$. The probability that $x_2 = 7$ and $x_4 = 11$ is: **[27 June, 2022 (Shift-I)]**

(a) $\frac{1}{136}$ (b) $\frac{1}{72}$

(c) $\frac{1}{68}$ (d) $\frac{1}{34}$

16. Let a biased coin be tossed 5 times. If the probability of getting 4 heads is equal to the probability of getting 5 heads, then the probability of getting almost two heads is: **[26 June, 2022 (Shift-I)]**

(a) $\frac{275}{6^5}$ (b) $\frac{36}{5^4}$

(c) $\frac{181}{5^5}$ (d) $\frac{46}{6^4}$

17. If the numbers appeared on the two throws of fair six faced die are α and β, then the probability that $x^2 + \alpha x + \beta > 0$, for all $x \in R$, is: **[25 July, 2022 (Shift-I)]**

(a) $\frac{17}{36}$ (b) $\frac{4}{9}$

(c) $\frac{1}{2}$ (d) $\frac{19}{36}$

18. Let E_1, E_2, E_3 be three mutually exclusive events such that $P(E_1) = \frac{2+3p}{6}$, $P(E_2) = \frac{2-p}{8}$ and $P(E_3) = \frac{1-p}{2}$. If the maximum and minimum values of p are p_1 and p_2, then $(p_1 + p_2)$ is equal to: **[26 July, 2022 (Shift-I)]**

(a) $\frac{2}{3}$ (b) $\frac{5}{3}$ (c) $\frac{5}{4}$ (d) 1

19. The probability, that in a randomly selected 3-digit number at least two digits are odd, is **[28 June, 2022 (Shift-I)]**

(a) $\frac{19}{36}$ (b) $\frac{15}{36}$ (c) $\frac{13}{36}$ (d) $\frac{23}{36}$

20. Let $S = \{1, 2, 3, ..., 2022\}$. Then the probability, that a randomly chosen number n from the set S such that $HCF(n, 2022) = 1$, is: **[29 July, 2022 (Shift-I)]**

(a) $\frac{128}{1011}$ (b) $\frac{166}{1011}$

(c) $\frac{127}{337}$ (d) $\frac{112}{337}$

21. If the probability that a randomly chosen 6-digit number formed by using digits 1 and 8 only is a multiple of 21 is P, then $96p$ is equal to..... **[26 June, 2022 (Shift-II)]**

22. Let $S = \{E_1, E_2, ..., E_8\}$ be a sample space of a random experiment such that $P(E_n) = \frac{n}{36}$ for every $n = 1, 2, ..., 8$. Then the number of elements in the set $\left\{A \subseteq S : P(A) \geq \frac{4}{5}\right\}$ is **[27 June, 2022 (Shift-II)]**

23. Two dices are rolled. If both dices have six faces numbered 1, 2, 3, 5, 7 and 11, then the probability that the sum of the numbers on the top faces is less than or equal to 8 is: **[17 March, 2021 (Shift-I)]**

(a) $\frac{1}{2}$ (b) $\frac{17}{36}$

(c) $\frac{4}{9}$ (d) $\frac{5}{12}$

24. Words with or without meaning are to be formed using all the letters of the word EXAMINATION. The probability that the letter M appears at the fourth position in any such word is: **[20 July, 2021 (Shift-I)]**

(a) $\frac{1}{9}$ (b) $\frac{1}{11}$

(c) $\frac{2}{11}$ (d) $\frac{1}{66}$

25. The probability that two randomly selected subsets of the set {1,2,3,4,5} have exactly two elements in their intersection is: **[24 Feb, 2021 (Shift-II)]**

(a) $\frac{65}{2^7}$ (b) $\frac{35}{2^7}$

(c) $\frac{135}{2^9}$ (d) $\frac{65}{2^8}$

26. The probability of selecting integers $a \in [-5, 30]$ such that $x^2 + 2(a+4)x - 5a + 64 > 0$, for all $x \in R$ is: **[20 July, 2021 (Shift-I)]**

(a) $\frac{2}{9}$ (b) $\frac{1}{6}$

(c) $\frac{7}{36}$ (d) $\frac{1}{4}$

27. A fair coin is tossed a fixed number of times. If the probability of getting 7 heads is equal to probability of getting 9 heads, then the probability of getting 2 heads is: **[26 Feb, 2021 (Shift-I)]**

(a) $\frac{15}{2^{14}}$ (b) $\frac{15}{2^{12}}$

(c) $\frac{15}{2^8}$ (d) $\frac{15}{2^{13}}$

28. The coefficients a, b and c of the quadratic equation, $ax^2 + bx + c = 0$ are obtained by throwing a dice three times. The probability that this equation has equal roots is: **[25 Feb, 2021 (Shift-I)]**

(a) $\frac{1}{54}$ (b) $\frac{1}{72}$

(c) $\frac{1}{36}$ (d) $\frac{5}{216}$

29. Let A be a set of all 4-digit natural numbers whose exactly one digit is 7 . Then the probability that a randomly chosen element of A leaves remainder 2 when divided by 5 is: **[25 Feb, 2021 (Shift-II)]**

(a) $\frac{2}{9}$ (b) $\frac{97}{297}$

(c) $\frac{122}{297}$ (d) $\frac{1}{5}$

30. A student appeared in an examination consisting of 8 true – false type questions. The student guesses the answers with equal probability. The smallest value of n, so that the probability of guessing at least 'n' correct answers is less than $\frac{1}{2}$, is **[27 July, 2021 (Shift-II)]**

(a) 5 (b) 6

(c) 3 (d) 4

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PW CHALLENGERS

1. In four throws with a pair of dice, what is the chance of throwing doublets twice at least?

2. If $6n$ tickets numbered 0, 1, 2, ... $6n-1$ are placed in a bag, and three are drawn out, then the chance that the sum of the numbers on them is equal to $6n$ is:

   (a) $\frac{3n}{(6n-1)(6n-3)}$ (b) $\frac{3n}{(6n-2)(6n-1)}$

   (c) $\frac{3n}{(6n-3)(6n-2)}$ (d) $\frac{3n}{(6n-1)(6n-2)}$

3. A purse contains 4 coins which are either sovereigns or shillings; 2 coins are drawn and found to be shillings: if these are replaced then what is the chance that another drawing will give a sovereign?

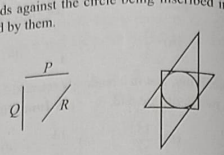
4. Two independent witnesses, A and B, whose probabilities of speaking the truth are p and p' respectively, agree in making a certain statement: what is the probability that the statement is true?

5. From bag containing n balls, all either white or black, all numbers of each being equally likely, a ball is drawn which turns out to be white: this is replaced, and another ball is drawn, which also turns out to be white. If this ball is replaced, the chance of the next draw giving a black ball is:

   (a) $\frac{1}{2}(n-1)(2n+1)^{-1}$ (b) $\frac{1}{3}(n-1)(2n+1)^{-1}$

   (c) $\frac{1}{4}(n-1)(2n+1)^{-1}$ (d) $\frac{1}{2}(n-2)(2n+1)^{-1}$

6. Three tangents are drawn at random to a given circle, find the odds against the circle being inscribed in the triangle formed by them.



7. A box is divided into m equal compartments into which balls are thrown at random; find the probability that there will be p compartments each containing a balls, q compartments each containing b balls, r compartments each containing balls, and so on, where $pa + qb + rc + \ldots = n$.

8. A number consists of 7 digits whose sum is 59; find the chance of its being divisible by 11.

9. If n integers taken at random are multiplied together, find the chance of the last digit of the product being

   (a) 1, 3, 7, or 9 is $\frac{2^n}{5^n}$ (b) 2, 4, 6 or 8 is $\frac{4^n-2^n}{5^n}$

   (c) 5 is $\frac{5^n-4^n}{10^n}$ (d) 0 is $\frac{10^n-8^n-5^n+4^n}{10^n}$

10. If on a straight line of length $a+b+c$ two lengths a are measured at random, find the chance of their having a common part which shall not exceed d.

Answer Key

CONCEPT APPLICATION

1. S = {(W, W), (W, R), (R, 1), (R, 2) (R, 3), (R, 4), (R, 5), (R, 6)}
2. S = {WB, BW, BB}
3. $A = [(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)]$, $B = [(1, 3), (2, 3), (1, 4), (2, 4), (3, 4), (4, 3)]$, $C = [(2, 4), (3, 4), (4, 2), (4, 3)]$, $D = [(1, 2), (2, 4)]$ A and D form a pair of mutually exclusive events.
4. (i) The sample space is the set of 52 cards; (ii) Required event is the set of jack, king and queen of spades and clubs.
5. $\frac{2197}{20825}$, 6. $\frac{4}{270725}$, 7. (i) $\frac{1}{6}$, (ii) $\frac{2}{3}$, (iii) $\frac{1}{6}$, 8. $\frac{11}{4165}$, 9. $= \frac{1}{38760}$,
10. (i) $A \cap \bar{B} \cap \bar{C}$, (ii) $A \cap B \cap \bar{C}$, (iii) $A \cap B \cap C$, (iv) $A \cup B \cup C$, (v) $(A \cup B) \cup (B \cap C) \cup (A \cap C) \cup (A \cap B \cap C)$,
(vi) $(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)$, (vii) $(A \cap B \cap \bar{C}) \cup (\bar{A} \cap B \cap C) \cup (A \cap \bar{B} \cap C)$
(viii) $\bar{A} \cap \bar{B} \cap \bar{C} = \overline{A \cup B \cup C}$, (ix) $(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C) \cup (A \cap B \cap \bar{C}) \cup (\bar{A} \cap B \cap C) \cup (A \cap \bar{B} \cap C)$.
11. $\frac{3p+2p^2}{2}$

SCHOOL LEVEL PROBLEMS

1. (a) 2. (d) 3. (c) 4. (d) 5. (a) 6. (d) 7. (c) 8. (b) 9. (b) 22. (c)
23. (b) 24. (c) 25. (a)

PRARAMBH (TOPICWISE)

1. (c) 2. (b) 3. (a) 4. (b) 5. (b) 6. (d) 7. (c) 8. (a) 9. (c) 10. (c)
11. (d) 12. (b) 13. (a) 14. (a) 15. (c) 16. (a) 17. (c) 18. (d) 19. (a) 20. (c)
21. (b) 22. (c) 23. (b) 24. (c) 25. (c) 26. (b) 27. (c) 28. (a) 29. (d) 30. (a)

PRABAL (JEE MAIN LEVEL)

1. (c) 2. (b) 3. (a) 4. (a) 5. (b) 6. (a) 7. (d) 8. (a) 9. (a) 10. (a)
11. (d) 12. (b) 13. (a) 14. (b) 15. (d) 16. (d) 17. (a) 18. (b) 19. (b) 20. (a)
21. (a) 22. [7] 23. [1] 24. [16]

PYQ's (PAST YEAR QUESTIONS)

1. (b) 2. (a) 3. (a) 4. (a) 5. (a) 6. (b) 7. (d) 8. (a) 9. (b) 10. (a)
11. (c) 12. (c) 13. [27] 14. (b) 15. (c) 16. (d) 17. (a) 18. (b) 19. (a) 20. (d)
21. [33] 22. [19] 23. (b) 24. (b) 25. (c) 26. (a) 27. (d) 28. (d) 29. (b) 30. (a)

PW CHALLENGERS

1. $\frac{19}{44}$ 2. (d) 3. $\frac{1}{4}$ 4. $\frac{pp'}{pp'+(1-p)(1-p')}$ 5. (a) 6. [3 : 1]
7. $\frac{\lfloor m \rfloor n}{m^n(\lfloor a)^p (\lfloor b)^q (\lfloor c)^r \ldots \lfloor p \rfloor q \lfloor r \ldots \lfloor m-p-q-r-\ldots}$ 8. $\frac{4}{21}$ 9. (a,b,c,d)
10. $\frac{(c+d)^2}{(c+a)(c+b)}$, where d is less than either a or b.

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CHAPTER 20

Properties of Triangle

INTRODUCTION

In any Δ, the three sides and the three angles are generally called the elements of the triangle.

A triangle which does not contain a right angle is called an oblique triangle.

In any ΔABC, the measures of the angles $\angle BAC$, $\angle CBA$ and $\angle ACB$ are denoted by the letters A, B and C respectively, and the sides BC, AC and AB opposite to the angles A, B and C are respectively denoted by a, b and c. These six elements of a triangle are not independent and are connected by the relations.

(*i*) $A + B + C = \pi$

(*ii*) $a + b > c, b + c > a, c + a > b$

(*iii*) $|a - b| < c, |a - c| < b, |b - c| < a$

SINE FORMULA

In any triangle the sides are proportional to the sines of the opposite angles i.e.

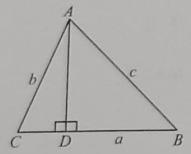
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

(1) Let the triangle ABC be acute-angled.

From A draw AD perpendicular to the opposite side; then

$AD = AB \sin(\angle ABD) = c \sin B$ and $AD = AC \sin(\angle ACD) = b \sin C$

$\therefore\ b \sin C = c \sin B$ i.e. $\frac{b}{\sin B} = \frac{c}{\sin C}$



(2) Let the triangle ABC have an obtuse angle at B

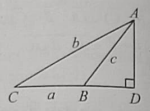
Draw AD perpendicular to CB produced; then

$AD = AC \sin \angle ACD = b \sin C$ and

$AD = AB \sin \angle ABD$

$= c \sin(180° - B) = c \sin B$;

$\therefore\ b \sin C = c \sin B$ i.e. $\frac{b}{\sin B} = \frac{c}{\sin C}$



In a similar manner it may be proved that either of the ratios is equal to $\frac{a}{\sin A}$

Thus $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Train Your Brain

Example 1: If the angles of a ΔABC are $\frac{\pi}{7}, \frac{2\pi}{7}$ and $\frac{4\pi}{7}$ and R is the radius of the circumcircle then $a^2 + b^2 + c^2$ has the value equal to

Sol. $a^2 + b^2 + c^2 = 4R^2(\sin^2 A + \sin^2 B + \sin^2 C) = 2R^2\left[1 - \cos\frac{2\pi}{7} + 1 - \cos\frac{4\pi}{7} + 1 - \cos\frac{8\pi}{7}\right]$

$= 2R^2[3 - (\cos\theta + \cos 2\theta + \cos 4\theta)]$ where $\theta = 2\pi/7$

now let $S = \cos\theta + \cos 2\theta + \cos 3\theta\ (\cos 4\theta = \cos 3\theta)$

$2 \sin\frac{\theta}{2} S = \sin\frac{3\theta}{2} - \sin\frac{\theta}{2} + \sin\frac{5\theta}{2} - \sin\frac{3\theta}{2}$

$\sin\frac{7\theta}{2} - \sin\frac{5\theta}{2} = \sin\frac{7\theta}{2} - \sin\frac{\theta}{2}$

$= \sin\pi - \sin\frac{\theta}{2}$

$= -\sin\frac{\theta}{2}$, $S = -\frac{1}{2}$

$\Rightarrow a^2 + b^2 + c^2 = 2R^2(3 + 1/2) = 7R^2$

Example 2: In a triangle ABC, $a \cos A + b \cos B + c \cos C = s$. Prove that the triangle is equilateral.

Sol. The given result can be written as $2a \cos A + 2b \cos B + 2c \cos C = a + b + c$

Using sine rule we get $2 \sin A \cos A + 2 \sin B \cos B + 2 \sin C \cos C = \sin A + \sin B + \sin C$

$\Rightarrow \sin 2A + \sin 2B + \sin 2C = \sin A + \sin B + \sin C$

$\Rightarrow 4 \sin A \sin B \sin C = 4 \cos\frac{A}{2} \cos\frac{B}{2} \cos\frac{C}{2}$

$\Rightarrow 8 \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2} = 1$

$\Rightarrow 4\left[\cos\frac{A-B}{2} - \cos\frac{A+B}{2}\right]\sin\frac{C}{2} = 1$

$\Rightarrow 4 \sin^2\frac{C}{2} - 4 \cos\frac{A-B}{2} \sin\frac{C}{2} + 1 = 0$. This is a quadratic equation in $\sin\frac{C}{2}$ which must have real roots.

Hence $16 \cos^2\frac{A-B}{2} \geq 16$

$\Rightarrow \cos^2\frac{A-B}{2} \geq 1$. But $\cos^2\frac{A-B}{2} \leq 1$

$\Rightarrow \cos^2\frac{A-B}{2} = 1 \Rightarrow A = B$

Similarly it can be prove that $B = C \Rightarrow A = B = C$

Example 3: In any ΔABC, prove that $\frac{a+b}{c} = \frac{\cos\left(\frac{A-B}{2}\right)}{\sin\frac{C}{2}}$.

Sol. Since $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$ (let)

$\Rightarrow a = k \sin A$, $b = k \sin B$ and $c = k \sin C$

$\therefore$ L.H.S. $= \frac{a+b}{c}$

$= \frac{k(\sin A + \sin B)}{k \sin C}$

$= \frac{\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{\sin\frac{C}{2}\cos\frac{C}{2}} = \frac{\cos\frac{C}{2}\cos\left(\frac{A-B}{2}\right)}{\sin\frac{C}{2}\cos\frac{C}{2}}$

$= \frac{\cos\left(\frac{A-B}{2}\right)}{\sin\frac{C}{2}}$ = R.H.S.

Hence L.H.S. = R.H.S. Proved

Concept Application

1. If A, B, C are in AP and $b : c = \sqrt{3} : \sqrt{2}$, then what is the value of $\sin C$?

(*a*) 1 (*b*) $\frac{1}{\sqrt{3}}$

(*c*) $\sqrt{3}$ (*d*) $\frac{1}{\sqrt{2}}$

2. In a ΔABC, $a + b = 3(1 + \sqrt{3})$ cm and $a - b = 3(1 - \sqrt{3})$ cm. If $\angle A$ is 30°, then what is the measure $\angle B$ of ?

(*a*) 120° (*b*) 90°

(*c*) 75° (*d*) 60°

3. How many triangle can be constructed with the data: $a = 5, b = 7, \sin A = \frac{3}{4}$

(*a*) 0 (*b*) 1

(*c*) 2 (*d*) None of these

COSINE FORMULA

To find an expression for one side (c) of a triangle in terms of other to sides and the included angle (C).

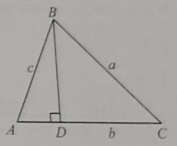
(1) Let C be an acute angle.

Draw BD perpendicular to AC ;

$AB^2 = BC^2 + CA^2 - 2AC \cdot CD$;

$\therefore\ c^2 = a^2 + b^2 - 2ba \cos C$

$= a^2 + b^2 - 2ab \cos C$.



(2) Let C be an obtuse angle.

Draw BD perpendicular to AC produced ;

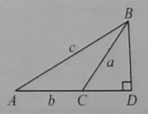
$AB^2 = BC^2 + CA^2 + 2AC \cdot CD$;

$\therefore\ c^2 = a^2 + b^2 + 2ba \cos BCD$

$= a^2 + b^2 + 2ab \cos(180° - C)$

$= a^2 + b^2 - 2ab \cos C$

Hence in each case, $c^2 = a^2 + b^2 - 2ab \cos C$



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Similarly it may be shown that

$a^2 = b^2 + c^2 - 2bc \cos A$ and $b^2 = c^2 + a^2 - 2ac \cos B$

From the above result we obtain

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc};$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca};$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

These results enable us to find the cosines of the angles when the numerical values of the sides are given.

Train Your Brain

Example 4: In a ΔABC, prove that $a(b\cos C - c\cos B) = b^2 - c^2$

Sol. $\because$ We have to prove $a(b\cos C - c\cos B) = b^2 - c^2$

$\because$ from cosine rule we know that

$$\cos C = \frac{a^2+b^2-c^2}{2ab} \text{ \& } \cos B = \frac{a^2+c^2-b^2}{2ac}$$

$\therefore$ L.H.S.

$$= a\left\{b\left(\frac{a^2+b^2-c^2}{2ab}\right) - c\left(\frac{a^2+c^2-b^2}{2ac}\right)\right\}$$

$$= \frac{a^2+b^2-c^2}{2} - \frac{(a^2+c^2-b^2)}{2}$$

$= (b^2 - c^2) =$ R.H.S.

Example 5: If in ΔABC, $\angle A = 60°$ then find the value of $\left(1+\frac{a}{c}+\frac{b}{c}\right)\left(1+\frac{c}{b}-\frac{a}{b}\right)$.

Sol. $\because \angle A = 60°$

$$\because \left(1+\frac{a}{c}+\frac{b}{c}\right)\left(1+\frac{c}{b}-\frac{a}{b}\right) = \left(\frac{c+a+b}{c}\right)\left(\frac{b+c-a}{b}\right)$$

$$= \frac{(b+c)^2 - a^2}{bc} = \frac{(b^2+c^2-a^2)+2bc}{bc}$$

$$= \frac{b^2+c^2-a^2}{bc} + 2 = 2\left(\frac{b^2+c^2-a^2}{2bc}\right) + 2$$

$= 2\cos A + 2 = 3$

$\left(\because \angle A = 60° \Rightarrow \cos A = \frac{1}{2}\right)$

$$\therefore \left(1+\frac{a}{c}+\frac{b}{c}\right)\left(1+\frac{c}{b}-\frac{a}{b}\right) = 3$$

Example 6: If the sides a, b, c of a ΔABC satisfy the relation $a^4 + b^4 + c^4 = 2c^2(a^2 + b^2)$, find the possible values of the angle C.

Sol. Solving as a quadratic equation in c we get, $c^2 = a^2 + b^2 \pm \sqrt{2}ab$ or $a^2 + b^2 - c^2 = \pm\sqrt{2}ab \Rightarrow \frac{a^2+b^2-c^2}{2ab} = \pm\frac{1}{\sqrt{2}}$

$$\Rightarrow C = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

Concept Application

4. In a triangle ABC, $a = 4$, $b = 3$, $\angle BAC = 60°$, then the equation for which c is the root, is

(a) $c^2 + 3c + 7 = 0$ (b) $c^2 + 3c - 7 = 0$

(c) $c^2 - 3c + 7 = 0$ (d) $c^2 - 3c - 7 = 0$

5. In a ΔABC, $(c + a + b)(a + b - c) = ab$. The measure of the angle C is:

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$

(c) $\frac{2\pi}{3}$ (d) None of these

6. The angle between the two sides of the triangle having length 6 cm and 8 cm is 60°, find the third side of the triangle.

(a) $\sqrt{26}$ cm (b) 12 cm

(c) $\sqrt{52}$ cm (d) 13 cm

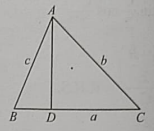
PROJECTION FORMULA

To express one side of a triangle in terms of the adjacent angles and the other two sides.

(1) Let ABC be an acute-angled triangle

Draw AD perpendicular to BC;

then $BC = BD + CD = AB\cos\angle ABD + AC\cos\angle ACD$



i.e. $a = c\cos B + b\cos C$

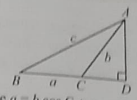
(2) Let the triangle ABC have an obtuse angle C.

Draw AD perpendicular to BC produced; then

$BC = BD - CD = AB\cos\angle ABD - AC\cos\angle ACD$;

$\therefore a = \cos B - b\cos(180° - C)$

$= c\cos B + b\cos C$



Thus in each case $a = b\cos C + c\cos B$.

Similarly it may be shown that

$b = c\cos A + a\cos C$, and $c = a\cos B + b\cos A$

Train Your Brain

Example 7: In a ΔABC prove that $(b + c)\cos A + (c + a)\cos B + (a + b)\cos C = a + b + c$

Sol. $\because$ L.H.S. $= (b + c)\cos A + (c + a)\cos B + (a + b)\cos C$

$= b\cos A + c\cos A + c\cos B + a\cos B + a\cos C + b\cos C$

$= (b\cos A + a\cos B) + (c\cos A + a\cos C)$

$+ (c\cos B + b\cos C) = a + b + c =$ R.H.S.

Example 8: In any triangle ABC, find.

$$(a+b)^2\sin^2\frac{C}{2} + (a-b)^2\cos^2\frac{C}{2}$$

Sol. $a^2\left(\sin^2\frac{C}{2} + \cos^2\frac{C}{2}\right) + b^2\left(\sin^2\frac{C}{2} + \cos^2\frac{C}{2}\right)$

$$-2ab\left(\cos^2\frac{C}{2} - \sin^2\frac{C}{2}\right)$$

$= a^2 + b^2 - 2ab\cos C = c^2$

Concept Application

7. Prove that $a(b\cos C - c\cos B) = b^2 - c^2$

8. If in a triangle a $a\cos^2\frac{C}{2} + c\cos^2\frac{A}{2}, = \frac{3b}{2}$ then find the relation between the sides of the triangle.

NAPIER'S ANALOGY - TANGENT RULE

(i) $\tan\frac{B-C}{2} = \frac{b-c}{b+c}\cot\frac{A}{2}$

(ii) $\tan\frac{C-A}{2} = \frac{c-a}{c+a}\cot\frac{B}{2}$

(iii) $\tan\frac{A-B}{2} = \frac{a-b}{a+b}\cot\frac{C}{2}$

Train Your Brain

Example 9: Find the unknown elements of the ΔABC in which $a = \sqrt{3} + 1$, $b = \sqrt{3} - 1$, $C = 60°$.

Sol. $\because a = \sqrt{3} + 1, b = \sqrt{3} - 1, C = 60°$

$\because A + B + C = 180°$

$\therefore A + B = 120°$ ... (i)

$\because$ From law of tangent, we know that $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\frac{C}{2}$

$$= \frac{(\sqrt{3}+1)-(\sqrt{3}-1)}{(\sqrt{3}+1)+(\sqrt{3}-1)}\cot 30°$$

$$= \frac{2}{2\sqrt{3}}\cot 30°$$

$$\Rightarrow \tan\left(\frac{A-B}{2}\right) = 1$$

$$\therefore \frac{A-B}{2} = \frac{\pi}{4} = 45° \Rightarrow A - B = 90° \quad \text{... (ii)}$$

From equation (i) and (ii), we get

$A = 105°$ and $B = 15°$

Now,

$\because$ From **sine-rule,**

we know that $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

$$\therefore c = \frac{a\sin C}{\sin A} = \frac{(\sqrt{3}+1)\sin 60°}{\sin 105°} = \frac{(\sqrt{3}+1)\frac{\sqrt{3}}{2}}{\frac{\sqrt{3}+1}{2\sqrt{2}}}$$

$$\because \sin 105° = \frac{\sqrt{3}+1}{2\sqrt{2}} \Rightarrow c = \sqrt{6}$$

$\therefore c = \sqrt{6}$, $A = 105°$, $B = 15°$

Example 10: Prove that $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+a}\cot\frac{A}{2}$

Sol. In any triangle ABC we have

$$\Rightarrow \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\Rightarrow \frac{b}{c} = \frac{\sin B}{\sin C}$$

$$\Rightarrow \left(\frac{b-c}{b+c}\right) = \frac{\sin B - \sin C}{\sin B + \sin C}$$

[Applying Dividendo and Componendo]

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$$\Rightarrow \left(\frac{b-c}{b+c}\right)=\frac{2\cos\left(\frac{B+C}{2}\right)\sin\left(\frac{B-C}{2}\right)}{2\sin\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)}$$

[Since, $A+B+C=\pi \Rightarrow \frac{B+C}{2}=\frac{\pi}{2}-\frac{A}{2}$]

$$\left(\frac{b-c}{b+c}\right)=\tan\frac{A}{2}\tan\left(\frac{B-C}{2}\right)$$

$$\left(\frac{b-c}{b+c}\right)=\frac{\tan\frac{B-c}{2}}{\cot\frac{A}{2}}$$

Therefore, $\tan\left(\frac{B-C}{2}\right)=\left(\frac{b-c}{b+c}\right)\cot\frac{A}{2}$.

Concept Application

9. In a triangle ABC, $\angle A = 60°$ and $b:c=\sqrt{3}+1:2$, then find the value of $(\angle B-\angle C)$.

10. If one side of a triangle is double of the other, and the angles on opposite sides are differ by 60°, then the triangle is

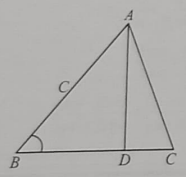
(*a*) Equilateral
(*b*) Obtuse angled
(*c*) Right angled
(*d*) Acute angled

AREA OF A TRIANGLE

To find the area of a triangle. Let Δ denote the area of the triangle ABC. Draw AD perpendicular to BC.

$$\therefore \Delta=\frac{1}{2}(\text{base}\times\text{altitude})$$

$$=\frac{1}{2}BC\,.\,AD=\frac{1}{2}\,BC\,.\,AB\sin B=\frac{1}{2}\,ca\sin B$$



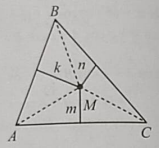
Train Your Brain

Example 11: Find the area of a triangle with angles α, β and γ knowing that the distances from an arbitrary point M taken inside the triangle to its sides are equal to m, n and k. (fig.)

Sol. The area S of the triangle ABC can be found by the formula $S=\frac{1}{2}AC.BC.\sin\gamma$, by sin rule

$$\frac{AC}{\sin\beta}=\frac{BC}{\sin\alpha}=\frac{AB}{\sin\gamma},$$

whence we find that $AC=\frac{a\sin\beta}{\sin\alpha}$ and $AB=\frac{a\sin\gamma}{\sin\alpha}$.



We have:

$$S=\frac{1}{2}AC\,.\,BC\,.\,\sin\gamma=\frac{1}{2}\,\frac{a\sin\beta}{\sin\alpha}\,.\,a\sin\gamma$$

$$=\frac{a^2\sin\beta\sin\gamma}{2\sin\alpha}.$$ On the other hand,

$$S=S_{AMB}+S_{BMC}+S_{AMC}$$

$$=\frac{1}{2}AB\cdot k+\frac{1}{2}BC\cdot n+\frac{1}{2}\cdot AC\cdot m$$

$$=\frac{1}{2}\cdot\frac{a\sin\gamma}{\sin\alpha}\cdot k+\frac{1}{2}an+\frac{1}{2}\cdot\frac{a\sin\beta}{\sin\alpha}\cdot m$$

$$=\frac{a(k\sin\gamma+n\sin\alpha+m\sin\beta)}{2\sin\alpha}$$

Hence, $\frac{a^2\sin\beta\sin\gamma}{2\sin\alpha}=\frac{a(k\sin\gamma+n\sin\alpha+m\sin\beta)}{2\sin\alpha}$,

we get: $a=\frac{k\sin\gamma+n\sin\alpha+m\sin\beta}{\sin\beta\sin\gamma}$.

Substituting this value of '*a*' into the first of the above formulas for the area of the triangle *ABC*, we obtain:

$$S\Rightarrow\frac{a^2\sin\beta\sin\gamma}{2\sin\alpha}=\frac{(k\sin\gamma+n\sin\alpha+m\sin\beta)^2}{2\sin\alpha\sin\beta\sin\gamma}.$$

Example 12: The two adjacent sides of a cyclic quadrilateral are 2 and 5 and the angle between them is 60°. If the area of the quadrilateral is $4\sqrt{3}$, find the remaining two sides.

Sol. Let *ABCD* be the cyclic quadrilateral in which $AB=2$ and $BC=5$, $\angle ADC=180°-60°$

Area of cyclic quadrilateral *ABCD* = Area of ΔABC + Area of ΔACD

$$\Rightarrow 4\sqrt{3}=\frac{1}{2}\,AB\,BC\sin60°+\frac{1}{2}CD\,DA\sin120°$$

$$=\frac{1}{2}2\times5\times(\sqrt{3}/2)+\frac{1}{2}xy(\sqrt{3}/2)$$

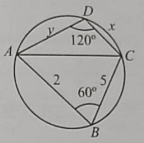
where $CD=x$, $AD=y$

$\therefore xy=6$ ...(i)

From ΔABC, we get

$AC^2=AB^2+BC^2-2AB\cdot BC\cdot\cos60°$

$=4+25-20(1/2)=19$ ...(ii)



Also from ΔACD,

$AC^2=CD^2+DA^2-2CD\cdot DA\cos120°$

$=x^2+y^2+xy=x^2+y^2+6$ [using Eq. (i)] ...(iii)

Now from Eqs. (ii) and (iii),

we have $x^2+y^2+6=19$ or $x^2+y^2=13$ ...(iv)

Solving Eqs. (ii) and (iv), we get

$x^4-13x^2+36=0\Rightarrow x^2=4,9\Rightarrow x=2,3\Rightarrow y=3,2$

Hence, the other two sides of the cyclic quadrilateral are 3 and 2.

Concept Application

11. A cyclic quadrilateral *ABCD* of area $3\sqrt{3}/4$ is inscribed in a unit circle. If one of its sides $AB=1$ and $\angle A$ is acute and the diagonal $BD=\sqrt{3}$, find the lengths of the other sides.

12. Let *P* be a point inside the triangle *ABC* such that $\angle APB=\angle BPC=\angle CPA$. Prove that $PA+PB+PC=\sqrt{\frac{a^2+b^2+c^2}{2}+2\sqrt{3}\Delta}$, where *a*, *b*, *c*, Δ are the sides and the area of triangle *ABC*.

TRIGONOMETRIC FUNCTIONS OF HALF ANGLES

(*i*) $\sin\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{bc}}$;

$\sin\frac{B}{2}=\sqrt{\frac{(s-c)(s-a)}{ca}}$;

$\sin\frac{C}{2}=\sqrt{\frac{(s-a)(s-b)}{ab}}$

(*ii*) $\cos\frac{A}{2}=\sqrt{\frac{s(s-a)}{bc}}$;

$\cos\frac{B}{2}=\sqrt{\frac{s(s-b)}{ca}}$;

$\cos\frac{C}{2}=\sqrt{\frac{s(s-c)}{ab}}$

(*iii*) $\tan\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}=\frac{\Delta}{s(s-a)}$

where $s=\frac{a+b+c}{2}$ & Δ = area of triangle

(*iv*) Area of triangle $=\sqrt{s(s-a)(s-b)(s-c)}$

Train Your Brain

Example 13: If *a*, *b*, *c* are in A.P., then the numerical value of $\tan\frac{A}{2}\tan\frac{C}{2}$ is

Sol. Given $2b=a+c$

$\Rightarrow 3b=2s=a+b+c$

$$\tan\frac{A}{2}\tan\frac{C}{2}=\frac{\Delta}{s(s-a)}\cdot\frac{\Delta}{s(s-c)}\cdot\frac{s-b}{s-b}$$

$$=\frac{2s-2b}{2s}=\frac{b}{3b}=\frac{1}{3}$$

Example 14: With usual notions, prove that in a triangle ABC, $\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}=\frac{s^2}{\Delta}$.

Sol. Using $\cot\frac{A}{2}=\frac{s(s-a)}{\Delta}$ etc.

$$LHS=\frac{s(s-a)+s(s-b)+s(s-c)}{\Delta}$$

$$=\frac{3s^2-s(a+b+c)}{\Delta}=\frac{3s^2-2s^2}{\Delta}=\frac{s^2}{\Delta}$$

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Concept Application

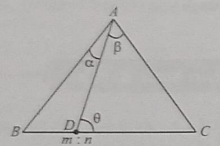
13. In any ΔABC, if $\cot\frac{A}{2}, \cot\frac{B}{2}, \cot\frac{C}{2}$ are in A. P., then a, b, c are in
(a) A.P. (b) G.P.
(c) H.P. (d) None of these

14. In ΔABC, if $\sin^2\frac{A}{2}, \sin^2\frac{B}{2}$ and $\sin^2\frac{C}{2}$ are in HP., then a, b and c will be in
(a) A.P. (b) G.P.
(c) H.P. (d) None of these

15. In a triangle ΔABC, 1 - tan (A/2) tan (B/2) is equal to
(a) $\frac{2a}{b+c-a}$ (b) $\frac{2a}{c+a-b}$
(c) $\frac{2c}{a+b-c}$ (d) $\frac{2c}{a+b+c}$

m-n RULE

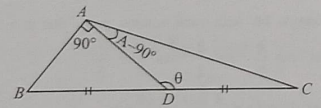
In any triangle, $(m+n)\cot\theta = m\cot\alpha - n\cot\beta = n\cot B - m\cot C$



Train Your Brain

Example 15: If the median AD of a triangle ABC is perpendicular to AB, prove that $\tan A + 2\tan B = 0$.

Sol. From the figure, we see that $\theta = 90° + B$ (as θ is external angle of ΔABD)



Now if we apply m-n rule in ΔABC, we get

$(1+1)\cot(90°+B) = 1.\cot 90° - 1.\cot(A-90°)$

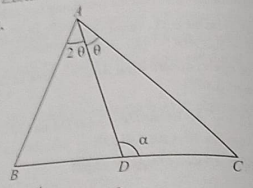
$\Rightarrow -2\tan B = \cot(90°-A)$

$\Rightarrow -2\tan B = \tan A$

$\Rightarrow \tan A + 2\tan B = 0$ **Hence proved.**

Example 16: In ΔABC, the median AD divides $\angle BAC$ such that $\angle BAD : \angle CAD = 2:1$. Then $\cos(A/3)$ is equal to

Sol.



Let $\frac{A}{3} = \angle CAD = \theta$

Now, by m-n theorem,

$(1+1)\cot\alpha = 1\cot 2\theta - 1\cot\theta$

$\Rightarrow 2\cot(B+2\theta) = \cot 2\theta - \cot\theta$

$\Rightarrow \cot(B+2\theta) + \cot\theta = \cot 2\theta - \cot(B+2\theta)$

$\Rightarrow \frac{\sin(B+3\theta)}{\sin(B+2\theta)\sin\theta} = \frac{\sin B}{\sin(B+2\theta)\sin 2\theta}$

$\Rightarrow \frac{\sin(B+A)}{\sin\theta} = \frac{\sin B}{\sin 2\theta}$

$\Rightarrow \sin C = \frac{\sin B}{2\cos\theta} \Rightarrow \cos\frac{A}{3} = \frac{\sin B}{2\sin C}$

Concept Application

16. Prove that the median through A divides angle A into two parts whose cotangents are, $2\cot A + \cot C$ and $2\cot A + \cot B$ makes on angle with the side BC whose cotangent is $\frac{1}{2}(\cot B - \cot C)$.

17. Find the angles of a triangle in which the altitude and median drawn from the same vertex divide the angle at that vertex into 3 equal parts.

RADIUS OF CIRCUMCIRCLE

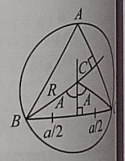
Circumcircle is the circle passing through vertices of triangle ABC. Its radius is known as circumradius (R) & centre is circumcentre (O).

Circumcentre is the point of intersection of perpendicular bisectors of sides.

In ΔBCD

$\frac{\frac{a}{2}}{R} = \sin A \Rightarrow a = 2R\sin A$

or $R = \frac{a}{2\sin A}$



Area of triangle and circumradius (R)

we know that

$\Delta = \frac{1}{2}a.b.\sin c \qquad \left(\because \sin c = \frac{c}{2R}\right)$

$\Rightarrow \Delta = \frac{1}{2}a.b.\frac{c}{2R}$

or $R = \frac{abc}{4\Delta}$

Train Your Brain

Example 17: In a ΔABC, if $a = 18$ cm., $b = 24$ cm. and $c = 30$ cm., then find its circum-radius

Sol. Clearly, the triangle is right angled.
($\because 18^2 + 24^2 = 30^2$)

Thus, the area of the triangle

$= \frac{1}{2} \times 24 \times 18 = 12 \times 18$

Therefore, the circum-radius = R

$= \frac{abc}{4\Delta} = \frac{18\times24\times30}{4\times12\times18} = 15$ cm

Example 18: In an equilateral triangle of side $2\sqrt{3}$ cm., then find the circum-radius

Sol. As we know that,

$\frac{a}{\sin A} = 2R$

$\Rightarrow 2R = \frac{2\sqrt{3}}{\sin(60°)}$

$R = \frac{\sqrt{3}}{\frac{\sqrt{3}}{2}} = 2$

Hence, the circum-radius is 2cm.

Example 19: If the length of the sides of a triangle are 3, 4 and 5 units, then find its circum-radius R.

Sol. Let a = 3, b = 4 and c = 5. Clearly, it is a right angled triangle

Thus, $\Delta = \frac{1}{2} \times 4 \times 3 = 6$ sq. u

Hence, the circum-radius R

$= \frac{abc}{4\Delta}$

$= \frac{3\times4\times5}{6} = 10$ unit

Concept Application

18. Find the radius of the circumcircle of a triangle, whose sides are given as $a = 8$, $b = 10$ and $c = 6$ units respectively.

19. In a triangle ABC $\angle A = 60°$, $\angle B = 40°$ and $\angle C = 80°$ If P is the centre of the circumcircle of triangle ABC with radius unity, then the radius of the circumcircle of triangle BPC is

20. If in a triangle ABC, $AB = 5$ units, $\angle B = \cos^{-1}\left(\frac{3}{5}\right)$ and radius of circumcircle of ΔABC is 5 units, then the area (in sq. units) of ΔABC is

RADIUS OF THE INCIRCLE

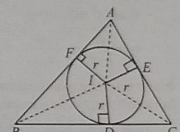
To find the radius of the circle inscribed in a triangle. Let ℓ be the circle inscribed in the triangle ABC, and D, E, F the points of contact; then ID, IE, IF are perpendicular to the sides.

Now Δ = sum of the areas of the triangles BIC, CIA, AIB

$= \frac{1}{2}ar + \frac{1}{2}br + \frac{1}{2}cr$

$= \frac{1}{2}(a+b+c)r = sr$

$\Rightarrow r = \frac{\Delta}{s}$



(a) $r = \frac{\Delta}{s}$ where $s = \frac{a+b+c}{2}$

(b) $r = (s-a)\tan\frac{A}{2}$

$= (s-b)\tan\frac{B}{2} = (s-c)\tan\frac{C}{2}$

(c) $r = \frac{a\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{A}{2}}$ & so on

(d) $r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

RADIUS OF THE EX-CIRCLES

A circle which touches one side of a triangle and the produced of other two sides is said to be an escribed circle of the triangle. Thus the triangle ABC has three escribed circles, one touching BC, and produced AB and AC, a second touching CA, and produced BC and AB; a third touching AB, and produced CA, CB.

To find the radius of an escribed circle of a triangle. Let I_1 be the centre of the circle touching the side BC and the two sides AB and AC produced. Let D_1, E_1, F_1 be the points of contact; then the lines joining I_1 to these points are perpendicular to the sides.

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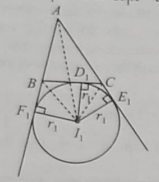
Let r_1 be the radius; then

Δ = area ABC

= area ABI_1C – area BI_1C

= area BI_1A + area CI_1A – area BI_1C



$$= \frac{1}{2}cr_1 + \frac{1}{2}br_1 - \frac{1}{2}ar_1$$

$$= \frac{1}{2}(c+b-a)\,r_1 = (s-a)\,r_1$$

$$\therefore \quad r_1 = \frac{\Delta}{s-a}$$

Similarly, if r_2, r_3 be the radii of the escribed circles opposite to the angles

B and C respectively

$$r_2 = \frac{\Delta}{s-b},\ r_3 = \frac{\Delta}{s-c}.$$

Many important relations connecting a triangle and its circles may be established by elementary geometry.

With the notation of previous articles, since tangents to a circle from the same point are equal.

we have, $AF = AE$, $BD = BF$, $CD = CE$;

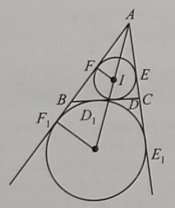
$\therefore$ $AF + (BD + CD)$ = half the sum of the sides;

$\therefore$ $AF + a = s$

$\therefore$ $AF = s - a = AE$

Similarly, $BD = BF = s - b$, $CD = CE = s - c$.

Also, $r_1 = AF \tan\frac{A}{2} = (s-a)\tan\frac{A}{2}$



Similarly, $r_2 = (s-b)\tan\frac{B}{2}$,

$r_3 = (s-c)\tan\frac{C}{2}$.

Again, $AF_1 = AE_1$, $BF_1 = BD_1$, $CE_1 = CD_1$

$\therefore$ $2AF_1 = AF_1 + AE_1 = (AB + BD_1) + (AC + CD_1)$

= sum of the sides

$\therefore$ $AF_1 = s = AE_1$

$\therefore$ $BD_1 = BF_1 = s - c$, $CD_1 = CE_1 = s - b$

Also $r_1 = AF_1 \tan\frac{A}{2} = s\tan\frac{A}{2}$

Similarly, $r_2 = s\tan\frac{B}{2}$, $r_3 = s\tan\frac{C}{2}$.

Note: $r_1 = \dfrac{a\cos\frac{B}{2}\cos\frac{C}{2}}{\cos\frac{A}{2}}$

$= 4R\sin\frac{A}{2}\cdot\cos\frac{B}{2}\cdot\cos\frac{C}{2}$ & so on

Train Your Brain

Example 20: With usual notation in a triangle ABC, prove that $r^2 + s^2 + 4Rr = ab + bc + ca$.

Sol. $r^2 + s^2 + 4Rr$

$$= \frac{\Delta^2}{s^2} + s^2 + \frac{abc}{\Delta}\cdot\frac{\Delta}{s} \qquad \left(r = \frac{\Delta}{s}, R = \frac{abc}{4\Delta}\right)$$

$$= \frac{s(s-a)(s-b)(s-c)}{s.s} + s^2 + \frac{abc}{s}$$

$$= \frac{(s-a)(s-b)(s-c) + s^3 + abc}{s}$$

$$= \frac{2s^3 - (a+b+c)s^2 + (ab+bc+ca)s - abc + abc}{s}$$

$= ab + bc + ca$ (proved)

Example 21: In a ΔABC, prove that $\sin A + \sin B + \sin C = \frac{s}{R}$

Sol. In a ΔABC, we know that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\therefore \quad \sin A = \frac{a}{2R},$$

$$\sin B = \frac{b}{2R} \text{ and } \sin C = \frac{c}{2R}.$$

$$\therefore \quad \sin A + \sin B + \sin C = \frac{a+b+c}{2R} = \frac{2s}{2R}$$

$\because$ $a + b + c = 2s$

$$\Rightarrow \quad \sin A + \sin B + \sin C = \frac{s}{R}.$$

Example 22: In a ΔABC if $a = 13$ cm, $b = 14$ cm and $c = 15$ cm, then find its circumradius.

Sol. $\because$ $R = \dfrac{abc}{4\Delta}$ ... (i)

$\because$ $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$\because$ $s = \dfrac{a+b+c}{2} = 21$ cm

$\therefore$ $\Delta = \sqrt{21\times8\times7\times6} = \sqrt{7^2\times4^2\times3^2}$

$\Rightarrow$ $\Delta = 84$ cm^2

$\therefore$ $R = \dfrac{13\times14\times15}{4\times84} = \dfrac{65}{8}$ cm $\therefore$ $R = \dfrac{65}{8}$ cm.

Example 23: In a ΔABC, prove that $s = 4R\cos\frac{A}{2}\cdot\cos\frac{B}{2}\cdot\cos\frac{C}{2}$.

Sol. In a ΔABC,

$\because$ $\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$, $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$

and $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$ and $R = \frac{abc}{4\Delta}$

$\because$ R.H.S. $= 4R\cos\frac{A}{2}\cdot\cos\frac{B}{2}\cdot\cos\frac{C}{2}$.

$$= \frac{abc}{\Delta}\cdot s\sqrt{\frac{s(s-a)(s-b)(s-c)}{(abc)^2}} = s = \text{L.H.S.}$$

($\because$ $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$)

Hence R.H.S = L.H.S. proved.

Example 24: The radii r_1, r_2, r_3 of escribed circles of a triangle ABC are in harmonic progression. If its area is 24 sq. cm and its perimeter is 24 cm, find the lengths of its sides.

Sol. $\frac{s-a}{\Delta}, \frac{s-b}{\Delta}, \frac{s-c}{\Delta}$ are in A.P.

$\Rightarrow$ a, b, c are in A.P.

$\Rightarrow$ $2b = a + c$

$\Rightarrow$ $2s = 24 \Rightarrow s = 12$

$\Rightarrow$ $\sqrt{12(12-a)4(12-16+a)} = 24$

$\Rightarrow$ $12\times4(12-a)(a-4) = 24\times24$

$\Rightarrow$ $-a^2 + 16a - 48 = 12$

$\Rightarrow$ $a^2 - 16a + 60 = 0$

$\Rightarrow$ $(a-10)(a-6) = 0$

$\Rightarrow$ $a = 10$ or $a = 6$

Hence, the lengths of side are 6, 8, 10 cm.

Concept Application

21. Find the radius of the incircle of the triangle whose sides measure 10, 24 and 26 m?

22. In an isosceles triangle, if one angle is 120° and radius of its incircle is $\sqrt{3}$, then the area of the triangle in square units is

23. If A, A_1, A_2, A_3 are the areas of incircle and the ex-circles of a triangle, then $\frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}} =$

(a) $\frac{2}{\sqrt{A}}$ (b) $\frac{1}{\sqrt{A}}$ (c) $\frac{1}{2\sqrt{A}}$ (d) $\frac{3}{\sqrt{A}}$

24. If the diameter of any excircle of a triangle is equal to its perimeter, then the triangle is

(a) Equilateral (b) Isosceles

(c) Right angled (d) Scalene

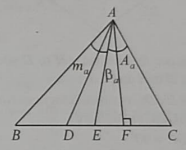
LENGTH OF ANGLE BISECTORS, MEDIANS & ALTITUDE

(i) Length of an angle bisector from the angle $A = \beta_a$

$$= \frac{2bc\cos\frac{A}{2}}{b+c}.$$

(ii) Length of median from the angle $A = m_a$

$$= \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$



(iii) Length of altitude from the angle $A = A_a = \frac{2\Delta}{a}$.

Note: $m_a^2 + m_b^2 + m_c^2 = 3/4\,(a^2 + b^2 + c^2)$

To find the radius of the circle circumscribing a triangle. Let S be the centre of the circle circumscribing the triangle ABC, and R its radius. Bisect $\angle BSC$ by SD, which will also bisect BC at right angles.

$\angle BSC = 2A$

and $\frac{a}{2}$ = BD = BS sin BSD = R sin A

$$\therefore \quad R = \frac{a}{2\sin A}$$

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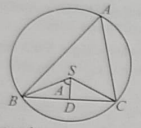
Thus $\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=2R$ or $a = 2R\sin A$, $b = 2R\sin B$, $c = 2R\sin C$



The circum-radius may be expressed in a form not involving the angles, as

$$R=\frac{a}{2\sin A}=\frac{abc}{2bc\sin A}=\frac{abc}{4\Delta}$$

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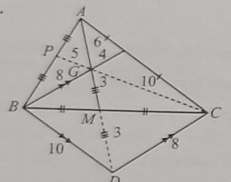
Example 25: Show that $2R^2 \sin A \sin B \sin C = \Delta$.

Sol. L.H.S. $=\frac{1}{2}.2R\sin A.2R\sin B.\sin C=\frac{1}{2}ab\sin C=\Delta$

Example 26: The medians of a triangle ABC are 9 cm, 12 cm and 15 cm respectively. Then the area of the triangle is

Sol.



Produce the median AM to D such that $GM = MD$. Join D to B and C.

Now $GBDC$ is a parallelogram. Note that the sides of the ΔGDC are 6, 8, 10

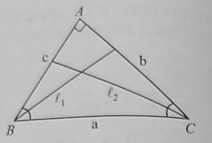
$\Rightarrow \angle GDC = 90°$

$$\left.\begin{aligned}\text{Area of a } \Delta ADC &= \frac{12.8}{2}=48\\ \text{Area of a } \Delta MDC &= \frac{3.8}{2}=12\end{aligned}\right]$$

$\Rightarrow$ Area of $\Delta AMC = 36$

$\Rightarrow$ Area of $\Delta ABC = 72$ cm^2

Example 27: The ratios of the lengths of the sides BC & AC of a triangle ABC to the radius of a circumscribed circle are equal to 2 & 3/2 respectively. Show that the ratio of the lengths of the bisectors of the interior angles B & C is, $\frac{7(\sqrt{7}-1)}{9\sqrt{2}}$



Sol. $\frac{a}{R}=2;\frac{b}{R}=\frac{3}{2}\therefore\frac{2R\sin A}{R}=2\sin B=\frac{3}{4}$

$\Rightarrow \sin A = 1;\ C^2 = 4R^2-\frac{9R^2}{4}\Rightarrow A = 90°;\ C=\frac{\sqrt{7}}{2}R$

Now, $I_1=\frac{2ac}{a+c}\cos\frac{B}{2}=\frac{2ac}{a+c}\sqrt{\frac{1+\cos B}{2}}$ and

$I_2=\frac{2ab}{a+b}\cos\frac{c}{2}=\frac{2ab}{a+b}\sqrt{\frac{1+\cos C}{2}}$

$$\therefore\frac{I_1}{I_2}=\frac{a+b}{a+c}\cdot\frac{c}{b}\sqrt{\frac{1+\cos B}{1+\cos C}}=\frac{c(a+b)}{b(a+c)}\sqrt{\frac{1+\frac{c}{a}}{1+\frac{b}{a}}}$$

$$=\frac{c}{b}\sqrt{\frac{a+b}{a+c}}$$

Substituting $a = 2R$; $b=\frac{3}{2}R$ & $c=\frac{\sqrt{7}}{2}R$, we get the desired result.

Concept Application

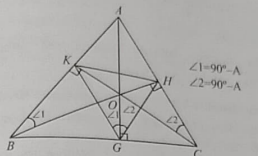
25. AD is a median of the ΔABC. If AE and AF are medians of the triangles ABD and ADC respectively, and $AD = m_1$, $AE = m_2$,, then prove that $m_1^2+m_2^2=\frac{a^2}{B}$

26. In ΔABC, in the usual notation, the area is $\frac{1}{2}bc$ sq. units AD is the median to BC. Prove that $\angle ABC=\frac{1}{2}\angle ADC$

27. In a ΔABC, the bisector of the angle A meets the side BC in D and the circumscribed circle in E. Show that $DE=\frac{a^2\sec\frac{A}{2}}{2(b+c)}$

ORTHOCENTRE AND PEDAL TRIANGLE

Let G, H, K be the feet of the perpendiculars from the angular points on the opposite sides of the triangle ABC, then GHK is called the Pedal triangle of ABC. The three perpendiculars AG, BH, CK meet in a point O which is called the orthocentre of the triangle ABC.



To find the sides and angles of the pedal triangle.

In the figure, the points K, O, G, B are concyclic:

$\therefore \angle OGK = OBK = 90° - A$

Also the points H, O, G, C are concyclic:

$\therefore \angle OGH = \angle OCH = 90° - A$

$\therefore \angle KGH = 180° - 2A$

Thus the angles of the pedal triangle are $180° - 2A$, $180° - 2B$, $180° - 2C$

Again, the triangles AKH, ABC are similar:

$\therefore \frac{HK}{BC}=\frac{AK}{AC}=\cos A$

$\therefore$ HK = a cos A

Thus the sides of the pedal triangle are $a\cos A$, $b\cos B$, $c\cos C$.

In terms of R, the equivalent forms become R sin 2A, R sin 2B, R sin 2C.

If the angle ACB of the given triangle is obtuse, the expression $180° - 2C$, and c cos C are both negative, and the values we have obtained required some modification. In this case the angles are 2A, 2B, $2C - 180°$, and the sides a cos A, b cos B, $-c$ cos C.

Remarks:

(*i*) The distances of the orthocentre from the angular points of the ΔABC are 2 R cos A, 2R cos B and 2R cos C.

(*ii*) The distances of orthocentre from sides are 2R cos B cos C, 2R cos C cos A & 2R cos B cos C.

(*iii*) To find the area and circum-radius of the pedal triangle.

Area $=\frac{1}{2}$ (product of two sides) × (sine of included angle)

$=\frac{1}{2}$ R sin 2B . R sin 2C . sin $(180° - 2A)$

$=\frac{1}{2}R^2\sin 2A\sin 2B\sin 2C$. The circum-radius

$=\frac{HK}{2\sin HGK}=\frac{2\sin 2A}{2\sin(180°-2A)}=\frac{R}{2}$.

EXCENTRAL TRIANGLE

Let ABC be a triangle and I_1, I_2, I_3 its ex-centres; then triangle $I_1I_2I_3$ is called the Ex-central triangle of ABC. Let I be the in-centre; then from the construction for finding the positions of the in-centre and ex-centres, it follows that:

(*i*) The points I, I_1 lie on the line bisecting the angle BAC; the points I, I_2 lie on the line bisecting the angle ABC; the points I, I_3 lie on the line bisecting the angle ACB.

(*ii*) The points I_2, I_3 lie on the line bisecting the angle BAC externally; the points I_3, I_1 lie on the line bisecting the angle ABC externally; the points I_1, I_2 lie on the line bisecting the angle ACB externally.

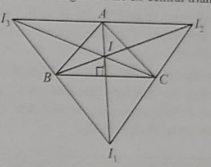
(*iii*) The line AI_1 is perpendicular to I_2I_3; the line BI_1 is perpendicular to I_3I_1; the line CI_1 is perpendicular to I_1I_2. Thus the triangle ABC is the Pedal triangle of its ex-central triangle $I_1I_2I_3$.

(*iv*) The angles IBI_1 and ICI_1 are right angles; hence the points B, I, C, I_1 are concyclic . Similarly, the points C, I, A, I_2, and the points A, I, B, I_3 are concyclic.

(*v*) The lines AI_1, BI_2, CI_3 meet at the in-centre I, which is therefore the orthocentre of the ex-central triangle $I_1I_2I_3$.

(*vi*) Each of the four points I, I_1, I_2, I_3 is the orthocentre of the triangle formed by joining the other three points.

To find the sides and angles of the ex-central triangle.



With the figure of the-last article.

$\angle BI_1C = \angle BI_1I + \angle CI_1I$

$=\angle BCI + \angle CBI = \frac{C}{2}+\frac{B}{2}$

$=90°-\frac{A}{2}$. Thus the angles are

$90°-\frac{A}{2}, 90°-\frac{B}{2}, 90°-\frac{C}{2}$.

$\therefore$ the triangles $I_1I_2I_3$ = supplement of $\angle I_1BC$ are similar

$\therefore \frac{I_2I_3}{BC}=\frac{I_1I_3}{I_1C}=\sec\left(90°-\frac{A}{2}\right)=\operatorname{cosec}\frac{A}{2}$

$\therefore I_2I_3 = a\operatorname{cosec}\frac{A}{2}=4R\cos\frac{A}{2}$

Thus the sides are $4R\cos\frac{A}{2}, 4R\cos\frac{B}{2}, 4R\cos\frac{C}{2}$

To find the area and circum-radius of the ex-central triangle.

AREA:

$=\frac{1}{2}$ (product of two sides) × (sine of included angle)

$=\frac{1}{2}\times 4R\cos\frac{B}{2}\times 4R\cos\frac{C}{2}\times\sin\left(90°-\frac{A}{2}\right)$

$=8R^2\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

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THE CIRCUM-RADIUS

$$= \frac{I_2I_3}{2\sin I_2I_1I_3} = \frac{4R\cos\frac{A}{2}}{2\sin\left(90^\circ - \frac{A}{2}\right)} = 2R$$

To find the distance between the in-centre and ex-centres.

The angles $\angle IBI_1$, $\angle ICI_1$ are right angles $\therefore$ II_1 is the diameter of the circum-circle of the triangle BCI_1

$$\therefore II_1 = \frac{BC}{\sin BI_1C} = \frac{a}{\cos\frac{A}{2}} = 4R\sin\frac{A}{2}.$$

Thus the distances are

$$4R\sin\frac{A}{2},\ 4R\sin\frac{B}{2},\ 4R\sin\frac{C}{2}$$

We have proved that OG, OH, OK bisect the angles HGK, KHG, GHK respectively, so that O is the in-centre of the triangle GHK. Thus the orthocentre of a triangle is the in-centre of the pedal triangle.

Again, the line CGB which is at right angles to OG bisect $\angle HGK$ externally. Similarly the lines AHC and BKA bisect $\angle KHG$ and $\angle GKH$ externally, so that ABC is the ex-central triangle of its pedal triangle GHK.

N. DISTANCE OF SPECIAL POINTS FROM VERTICES AND SIDES OF A TRIANGLE

(*i*) Circumcentre (O): $OA = R$ & $O_a = R\cos A$

(*ii*) Incentre (I): $IA = r\operatorname{cosec}\frac{A}{2}$ & $I_a = r$

(*iii*) Excentre (I_1): $I_1A = r_1\operatorname{cosec}\frac{A}{2}$ & $I_{1a} = r_1$

(*iv*) Orthocentre (H): HA

$= 2R\cos A$ & $H_a = 2R\cos B\cos C$

(*v*) Centroid (G): GA

$$= \frac{1}{3}\sqrt{2b^2+2c^2-a^2} \text{ \& } G_a = \frac{2\Delta}{3a}$$

DISTANCE BETWEEN SPECIAL POINTS

(*a*) The distance between circumcentre and orthocentre is $= R\cdot\sqrt{1-8\cos A\cos B\cos C}$

(*b*) The distance between circumcentre and incentre is

$$= \sqrt{R^2-2Rr} = R\sqrt{1-8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}$$

(*c*) Distance between circumcentre and centroid

$$OG = \sqrt{R^2-\frac{1}{9}(a^2+b^2+c^2)}$$

(*d*) The distance between incentre and orthocentre $\sqrt{2r^2-4R^2\cos A\cos B\cos C}$

To find the distance between the in-centre and circum-centre.

Let S be the circum-centre and I the in-centre. Produce AI to meet the circum-circle in H; join CH and CI.

Draw IE perpendicular to AC. Produce HS to meet the circumference in L, and join CL. Then

$$\angle HIC = \angle IAC + \angle ICA = \frac{A}{2}+\frac{C}{2};$$

$$\angle HCI = \angle ICB + \angle BCH = \frac{C}{2} + \angle BAH = \frac{C}{2}+\frac{A}{2};$$

$\therefore$ $\angle HCI = \angle HIC$;

$\therefore$ $HI = HC = 2R\sin\frac{A}{2}$.

Also $AI = IE\operatorname{cosec}\frac{A}{2} = r\operatorname{cosec}\frac{A}{2}$

$\therefore$ $AI\,.\,IH = 2Rr$

Produce SI to meet the circumference in M and N. Similarly MIN, AIH are chords of the circle.

$AI.IH = MI.\,IN = (R+SI)(R-SI)$;

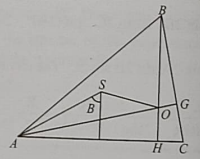
$\therefore$ $2Rr = R^2 - SI^2$; $SI^2 = R^2 - 2Rr$

To find the distance of the orthocentre from the circum-centre

With the usual notation, we have

$SO^2 = SA^2 + AO^2 - 2SA\,.\,AO\cos SAO$.

Now $AS = R$; $AO = AH\operatorname{cosec} C = c\cos A\operatorname{cosec} C$



$= 2R\sin C\cos A\operatorname{cosec} C = 2R\cos A$;

$\angle SAO = \angle SAC - \angle OAC = (90^\circ - B) - (90^\circ - C) = C - B$

$\therefore$ $SO^2 = R^2 + 4R^2\cos^2 A - 4R^2\cos A\cos(C-B)$

$= R^2 - 4R^2\cos A\{\cos(B+C) + \cos(C-B)\}$

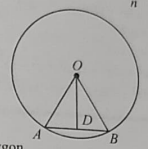
$= R^2 - 8R^2\cos A\cos B\cos C$.

INSCRIBED AND CIRCUMSCRIBED POLYGONS

To find the perimeter and area of a regular polygon of n sides inscribed a circle.

Let r be the radius of the circle, and AB a side of the polygon. Join OA, OB, and draw OD bisecting $\angle AOB$; then AB is bisected at right angles in D.

And $\angle AOB = \frac{1}{n}$ (four right angles) $= \frac{2\pi}{n}$



Perimeter of polygon

$= nAB = 2nAD = 2nAO\sin\Delta AOD = 2nr\sin\frac{\pi}{n}$

Area of polygon = n (area of triangle AOB)

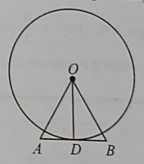
$= \frac{1}{2}nr^2\sin\frac{2\pi}{n}$.

To find the perimeter and area of a regular polygon of n sides circumscribed about a given circle.

Let r be the radius of the circle, and AB a side of the polygon. Let AB touch the circle at D. Join OA, OB, OD; then OD bisects AB at right angles, and also bisects $\angle AOB$.

Perimeter $= nAB = 2nAD = 2nOD\tan\angle AOD = 2nr\tan\frac{\pi}{n}$.

Area of polygon $= n$ (area of triangle AOB)



$= nOD\,.\,AD = nr^2\tan\frac{\pi}{n}$

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Example 28: If f, g, h denote sides, the pedal triangle of a ΔABC, then show that

$$f\cdot\frac{(b^2-c^2)}{a^2} + g\cdot\frac{(c^2-a^2)}{b^2} + h\cdot\frac{(a^2-b^2)}{c^2} = 0$$

Sol. Sides are a cos A, b cos B, c cos C. Hence LHS

$$\left(\frac{b^2-c^2}{a}\right)\cos A + \left(\frac{c^2-a^2}{b}\right)\cos B + \left(\frac{a^2-b^2}{c}\right)\cos C$$

Put the values of cos A etc. get the result.

Example 29: Vertex A of a variable triangle ABC, inscribed in a circle of radius R, is a fixed point. If the angles subtended by the side BC at orthocentre (H), circumcentre (O) and incentre (I) are equal than identify the locus of orthocentre of triangle ABC.

Sol. The angles subtended by the side BC at points H, O and I are $B + C$, $2A$ and $\pi - \left(\frac{B+C}{2}\right)$ respectively.

$\Rightarrow$ $B + C = 2A = 180 - \left(\frac{B+C}{2}\right)$

$\Rightarrow$ $A = \frac{\pi}{3}$ and $B + C = \frac{2\pi}{3}$.

Also in triangle ABC, $HA = 2R\cos A = R$

$\Rightarrow$ HA is constant

$\Rightarrow$ locus of orthocentre is a circle having centre at the vertex A.

Example 30: If I is the incentre and I_1, I_2, I_3 are the centre of escribed circles of the ΔABC. Prove that

(*i*) $II_1.II_2.II_3 = 16R^2r$

(*ii*) $II_1^2 + I_2I_3^2 = II_2^2 + I_3I_1^2 = I_1I_2^2 + II_3^2$

Sol. (*i*) $\because$ We know that

$II_1 = a\sec\frac{A}{2}$, $II_2 = b\sec\frac{B}{2}$ and $II_3 = c\sec\frac{C}{2}$

$\because$ $I_1I_2 = c,\operatorname{cosec}\frac{C}{2}$,

$I_2I_3 = a\operatorname{cosec}\frac{A}{2}$ and $I_3I_1 = b\operatorname{cosec}\frac{B}{2}$

$\because$ $II_1.II_2.II_3 = abc\sec\frac{A}{2}\sec\frac{B}{2}\sec\frac{C}{2}$ ... (*i*)

$\because$ $a = 2R\sin A$, $b = 2R\sin B$ and $c = 2R\sin C$

$\therefore$ equation (i) becomes

$\because$ $II_1.II_2.II_3 = (2R\sin A)(2R\sin B)$

$(2R\sin C)\sec\frac{A}{2}\sec\frac{B}{2}\sec\frac{C}{2}$

$$= 8R^3\cdot\frac{\left(2\sin\frac{A}{2}\cos\frac{A}{2}\right)\left(2\sin\frac{B}{2}\cos\frac{B}{2}\right)\left(2\sin\frac{C}{2}\cos\frac{C}{2}\right)}{\cos\frac{A}{2}\cdot\cos\frac{B}{2}\cdot\cos\frac{C}{2}}$$

$= 64R^3\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$\because$ $r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$\therefore$ $II_1.II_2.II_3 = 16R^2r$

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(ii) $\Pi_1^2 + I_2I_3^2 = I\,I_2^2 + I_3I_1^2 = \Pi_3^2 + I_1I_2^2$

$\because\ \Pi_1^2 + I_2I_3^2$

$= a^2\sec^2\frac{A}{2} + a^2\cdot\operatorname{cosec}^2\frac{A}{2} = \frac{a^2}{\sin^2\frac{A}{2}\cos^2\frac{A}{2}}$

$\because\ a = 2R\sin A = 4R\sin\frac{A}{2}\cos\frac{A}{2}$

$\therefore\ \Pi_1^2 + I_2I_3^2 = \frac{16R^2\sin^2\frac{A}{2}\cdot\cos^2\frac{A}{2}}{\sin^2\frac{A}{2}\cdot\cos^2\frac{A}{2}} = 16R^2$

Hence $\Pi_1^2 + I_2I_3^2 = \Pi_2^2 + I_3I_1^2 = \Pi_3^2 + I_1I_2^2$

Example 31: If r and R are radii of the incircle and circumcircle of ΔABC, prove that

$8rR\left(\cos^2\frac{A}{2}+\cos^2\frac{B}{2}+\cos^2\frac{C}{2}\right)$

$= 2bc + 2ca + 2ab - a^2 - b^2 - c^2$

Sol. LHS $= \frac{8\Delta}{s}\frac{abc}{4\Delta}\Sigma\cos^2\frac{A}{2} = \frac{abc}{s}\left[\Sigma 2\cos^2\frac{A}{2}\right]$

$= \frac{abc}{s}\ [(1+\cos A)+(1+\cos B)+(1+\cos C)]$

$= \frac{abc}{s}\left[\frac{(b+c)^2-a^2}{2bc}+\frac{(c+a)^2-b^2}{2ca}+\frac{(a+b)^2-c^2}{2ab}\right]$

$= \frac{abc}{s}\frac{(2s)}{2abc}\ [a(b+c-a)+b(c+a-b)+c(a+b-c)]$

$= 2bc + 2ca + 2ab - a^2 - b^2 - c^2$

Example 32: If a, b, c denote the sides of a ΔABC, show that the value of the expression,

$a^3(p-q)(p-r) + b^2(q-r) + b^2(q-r)(q-p) + c^2(r-p)(r-q)$ cannot be negative where $p, q, r \in R$.

Sol. Let $p > q > r$ and $\left.\begin{matrix} p-q=y>0 \\ q-r=z>0 \end{matrix}\right\} \Rightarrow p-r=y+z>0$

Consider $E = a^2y(y+z) - b^2zy + c^2z(y+z)$

$= a^2y^2 + c^2z^2 + yz(a^2+c^2-b^2)$

Now $b < a + c$

$\Rightarrow\ b_{max} = a + c$

$E_{least} = a^2y^2 + c^2z^2 + yz[a^2+c^2-(a+c)^2]$

$= a^2y^2 + c^2z^2 - 2acyz = (ay - cz)^2 \geq 0$

Concept Application

28. If in ΔABC, the distances of the vertices from the orthocenter are x, y, and z, then prove that $\frac{a}{x}+\frac{b}{y}+\frac{c}{z} = \frac{abc}{xyz}$

29. In ΔABC, let L, M, N be the feet of the altitudes. Then prove that $\sin(\angle MLN) + \sin(\angle LMN) + \sin(\angle LMN)$ $= 4\sin A\sin B\sin C$

30. Let ABC be an acute triangle whose orthocenter is at H. If altitude from A is produced to meet the circumcircle of triangle ABC at D, then find HD

31. In a triangle ABC if $2\cos\frac{B}{2}\cos\frac{C}{2} = \frac{1}{2} + \left(\frac{b+c}{a}\right)\sin\frac{A}{2}$ then find the measure of angle A.

AARAMBH (SOLVED EXAMPLES)

1. If in a triangle ABC, $\sin A = \sin^2 B$ and $2\cos^2 A = 3\cos^2 B$, then the ΔABC is -

(a) right angled (b) obtuse angled

(c) isosceles (d) equilateral

Sol. $\sin A = \sin^2 B$

$2\cos^2 A = 3\cos^2 B$ ... (i)

from (i) and (ii) ... (ii)

$2(1-\sin^2 A) = 3(1-\sin A)$

$\Rightarrow\ 2\sin^2 A - 3\sin A + 1 = 0$

$\Rightarrow\ (2\sin A - 1)(\sin A - 1) = 0$

$\Rightarrow \sin A = \frac{1}{2}$ or $\sin A = 1$

But $\sin A \neq 1$

$\Rightarrow\ \angle A = 30°$

from (i) $\sin^2 B = \frac{1}{2}$

$\angle B = 45°$

and $\angle C = 180° - 75° = 105°$

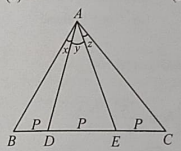
ΔABC is obtuse angled.

2. Points D, E are taken on the side BC of a triangle ABC such that $BD = DE = EC$. If $\angle BAD = x$, $\angle DAE = y$, $\angle EAC = z$, then the value of $\frac{\sin(x+y)\sin(y+z)}{\sin x\sin z} =$

(a) 1 (b) 2

(c) 4 (d) none

Sol.



$\because \frac{AD}{BD} = \frac{\sin B}{\sin x}$ and $\frac{AD}{DC} = \frac{\sin C}{\sin(y+z)}$ ... (i)

and $\frac{AE}{BE} = \frac{\sin B}{\sin(x+y)}$ and $\frac{AE}{EC} = \frac{\sin C}{\sin z}$ ... (ii)

From (i) $\frac{DC}{BD} = \frac{\sin B}{\sin C}\cdot\frac{\sin(y+z)}{\sin x}$ ... (iii)

From (ii) $\frac{BE}{EC} = \frac{\sin C}{\sin B}\cdot\frac{\sin(x+y)}{\sin z}$ ... (iv)

$\because \frac{DC}{BD} = 2$ and $\frac{BE}{EC} = 2$

Then from (iii) and (iv), we get

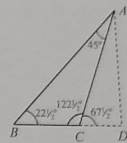
$\frac{\sin(x+y)\sin(y+z)}{\sin x\sin z} = 4$

3. If the base angles of a triangle are 22.5° and 112.5°, the height of the triangle is equal to -

(a) half the base (b) the base

(c) twice the base (d) four times the base

Sol.



From sine rule, we have

$\frac{BC}{\sin 45°} = \frac{AC}{\sin 22.5°}$ ...(i)

Also $\frac{AD}{AC} = \sin 67.5°$

$AD = AC\cos 22.5°$

$\Rightarrow AD = \frac{BC}{\sin 45°}\sin 22.5°\cos 22.5°$ from (i)

$\Rightarrow AD = \frac{BC}{2}\cdot\frac{\sin 45°}{\sin 45°}$

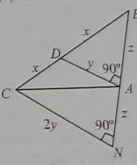
$\Rightarrow AD = \frac{1}{2}BC = \frac{1}{2}$ of base

4. In a ΔABC if median AD is perpendicular to AB, then $\tan A + 2\tan B$ is equal to-

(a) 1 (b) 3

(c) 0 (d) $\frac{1}{2}$

Sol.



Given that $BD = DC$; $\angle DAB = 90°$

Draw a perpendicular CN at extend BA

In ΔBCN; $DA = \frac{1}{2}CN = y$ and $AB = AN = z$

$\therefore\ \tan A = \tan(\pi - \angle CAN) = -\tan(\angle CAN)$

$= -\frac{CN}{AN} = -\frac{2y}{z}$

$\Rightarrow \tan A = -\frac{2AD}{AB}$

$\Rightarrow\ \tan A = -2\tan B$

$\Rightarrow\ \tan A + 2\tan B = 0$

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5. In a triangle, $a^2+b^2+c^2=ca+ab\sqrt{3}$, then the triangle is -
(a) equilateral
(b) right-angled and isosceles
(c) right-angled
(d) None of these

Sol. Write $a^2=\frac{a^2}{4}+\frac{3a^2}{4}$

then, $\left(\frac{a^2}{4}+c^2-ca\right)+\left(\frac{3a^2}{4}-ab\sqrt{3}+b^2\right)=0$

$\Rightarrow \left(\frac{a}{2}-c\right)^2+\left(\frac{\sqrt{3}a}{2}-b\right)^2=0$

$\Rightarrow c=\frac{a}{2}$ and $b=\frac{\sqrt{3}a}{2}$

then $b^2+c^2=\frac{3a^2}{4}+\frac{a^2}{4}=a^2$

triangle is right angle triangle, $A=90°$

$\sin C=\frac{c}{a}=\frac{1}{2}\ \because C=30°,\ \therefore B=60°$.

6. If in a ΔABC, $a^2\cos^2 A=b^2+c^2$,then-

(a) $A<\frac{\pi}{2}$ (b) $\frac{\pi}{4}<A<\frac{\pi}{2}$

(c) $A>\frac{\pi}{2}$ (d) $A=\frac{\pi}{2}$

Sol. Given $a^2\cos^2 A=b^2+c^2$

we know that $\cos A=\frac{b^2+c^2-a^2}{2bc}$

$\Rightarrow \cos A=\frac{a^2\cos^2 A-a^2}{2bc}$

or $\cos A=-\frac{a^2\sin^2 A}{2bc}=-ve$

$A>\frac{\pi}{2}$

7. In a ΔABC, if $a^4+b^4+c^4=2c^2(a^2+b^2)$, then $\angle C$ is equal to-
(a) 60° (b) 135° (c) 90° (d) 75°

Sol. $a^4+b^4+c^4-2a^2c^2-2b^2c^2=0$

$a^4+b^4+c^4-2a^2c^2-2b^2c^2+2a^2b^2=2a^2b^2$

$\Rightarrow (a^2+b^2-c^2)^2=2a^2b^2$

$\Rightarrow \left(\frac{a^2+b^2-c^2}{2ab}\right)^2=\frac{1}{2}$

$\Rightarrow \cos^2 C=\frac{1}{2}$

$\Rightarrow \cos C=\pm\frac{1}{\sqrt{2}} \Rightarrow C=45°, 135°$

8. In ΔABC, $a=5$, $b=4$ and $\tan\frac{C}{2}=\sqrt{\frac{7}{9}}$. The third side is-
(a) 6 (b) 3 (c) 2 (d) None

Sol. $\because \cos C=\frac{1-\tan^2\frac{C}{2}}{1+\tan^2\frac{C}{2}}$

$\because \tan\frac{C}{2}=\sqrt{\frac{7}{9}}$

$\Rightarrow \cos C=\frac{1-\frac{7}{9}}{1+\frac{7}{9}}$

$\Rightarrow \cos C=\frac{1}{8}$

$\because \cos C=\frac{a^2+b^2-c^2}{2ab}$

$\because a=5, b=4$

$\Rightarrow \frac{1}{8}=\frac{25+16-c^2}{40}$

$\Rightarrow c^2=36 \Rightarrow c=6$

9. In a ΔABC, $A=\frac{2\pi}{3}, b-c=3\sqrt{3}$ cm and ar (ΔABC) $=\frac{9\sqrt{3}}{2}$cm^2. Then a is -
(a) $6\sqrt{3}$ cm (b) 9 cm
(c) 18 cm (d) None of these

Sol. Given $A=\frac{2\pi}{3}, b-c=3\sqrt{3}$ & $\text{ar}(\Delta ABC)=\frac{9\sqrt{3}}{2}$

$\because \Delta=\frac{1}{2}bc\sin A \Rightarrow \frac{1}{2}bc\sin\frac{2\pi}{3}=\frac{9\sqrt{3}}{2}$

$\Rightarrow bc=18$ and $b-c=3\sqrt{3}$

$\Rightarrow b^2+c^2=27+2bc=27+36$

$\Rightarrow b^2+c^2=63$

$\because \cos A=\frac{b^2+c^2-a^2}{2bc}$

$\Rightarrow \cos\frac{2\pi}{3}=\frac{63-a^2}{2\times 18} \Rightarrow -\frac{1}{2}=\frac{63-a^2}{36}$

$\Rightarrow a^2=63+18=81 \Rightarrow a=9$

10. In a ΔABC, $\cos B.\cos C+\sin B.\sin C\sin^2 A=1$. Then the triangle is-
(a) right-angle isosceles
(b) isosceles whose equal angles are greater than $\frac{\pi}{4}$
(c) equilateral
(d) none of these

Sol. $\sin^2 A=\frac{1-\cos B\cos C}{\sin B\sin C}\le 1$ ...(i)

$\Rightarrow 1\le \cos B\cos C+\sin B\sin C$

$\Rightarrow 1\le \cos(B-C)$

$\Rightarrow 1=\cos(B-C)$ $(\because \cos\theta\le 1)$

$\Rightarrow B-C=0$

$\Rightarrow B=C$

then from (i)

$\sin^2 A=\frac{1-\cos^2 B}{\sin^2 B}=1$

$\Rightarrow \sin A=1$

$\Rightarrow A=90°$

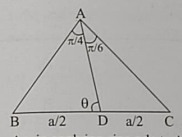
$B=C=45°$

triangle is right angled isosceles.

11. In a ΔABC, the median to the side BC is of length $\frac{1}{\sqrt{11-6\sqrt{3}}}$ and it divides angle A into the angles of measure $\frac{\pi}{6}$ and $\frac{\pi}{4}$. Then the length of side BC is.

Sol. By m–n theorem, $2\cot\theta=\cot\frac{\pi}{6}-\cot\frac{\pi}{4}$

$\Rightarrow 2\cot\theta=\sqrt{3}-1 \Rightarrow \cot\theta=\frac{\sqrt{3}-1}{2}$ is acute



Again applying sine rule to ΔABD, we have

$\frac{AD}{\sin B}=\frac{AB}{\sin\theta}=\frac{BD}{\sin\pi/4}$

$\Rightarrow \frac{AD}{\sin\left(\pi-\frac{\pi}{4}-\theta\right)}=\frac{c}{\sin\theta}=\frac{a/2}{1/\sqrt{2}}$

$\Rightarrow \frac{AD}{\sin\left(\frac{\pi}{4}+\theta\right)}=\frac{c}{\sin\theta}=\frac{a}{\sqrt{2}}$

$\Rightarrow AD=\frac{a}{\sqrt{2}}\sin\left(\frac{\pi}{4}+\theta\right)$

$\Rightarrow AD=\frac{a}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}\cos\theta+\frac{1}{\sqrt{2}}\sin\theta\right]$

$\Rightarrow AD=\frac{a}{2}(\sin\theta+\cos\theta)$

$\Rightarrow AD=\frac{a}{2}\left[\frac{2}{4-\sqrt{3}}+\sqrt{\frac{4-2\sqrt{3}}{8-2\sqrt{3}}}\right]$

$\Rightarrow AD=\frac{a}{2}\left[\frac{2}{\sqrt{8-\sqrt{3}}}+\frac{\sqrt{4-2\sqrt{3}}}{\sqrt{8-2\sqrt{3}}}\right]$

$\Rightarrow AD=\frac{a}{2}\left[\frac{2+\sqrt{4-2\sqrt{3}}}{\sqrt{8-2\sqrt{3}}}\right]$

$\Rightarrow AD=\frac{a}{2}\left[\frac{\sqrt{3}+1}{\sqrt{8-2\sqrt{3}}}\right]$

$\Rightarrow BC=a=\frac{2AD\left(\sqrt{8-2\sqrt{3}}\right)}{\left(\sqrt{3}+1\right)}$

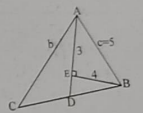
$=2\times\frac{1}{\sqrt{11-6\sqrt{3}}}\times\frac{\sqrt{8-2\sqrt{3}}}{\left(\sqrt{3}+1\right)}$

$=\frac{2\sqrt{8-2\sqrt{3}}}{\sqrt{\left(11-6\sqrt{3}\right)\left(4+2\sqrt{3}\right)}}$

$=2\sqrt{\frac{8-2\sqrt{3}}{8-2\sqrt{3}}}=2$ Thus $BC=2$

12. In ΔABC, $AC>AB$, the internal angle bisector of angle A meets BC at D and E is the foot of the perpendicular from B onto AD. Suppose AB = 5 and BE = 4, find the value of the expression $\left(\frac{AC+AB}{AC-AB}\right)(ED)$.

Sol. $\cos\frac{A}{2}=\frac{3}{5}(AE=3)$



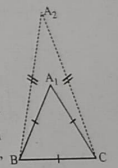
$x=\frac{2bc}{b+c}\cos\frac{A}{2} \Rightarrow x=\frac{2b\cdot 5}{b+5}\cdot\frac{3}{5}$

$x=\frac{6b}{b+5}=AD;\ ED=AD-AE$

$ED=\frac{6b}{b+5}-3=\frac{6b-3b-15}{b+5}=\frac{3(b-5)}{(b+5)}$

Expression $=\frac{AC+AB}{AC-AB}\cdot ED=\left(\frac{b+5}{b-5}\right)\cdot 3\left(\frac{b-5}{b+5}\right)=3$

13. The length of two sides of an equilateral triangle are doubled, creating an isosceles triangle, as shown in the diagram. The two longer sides are doubled again creating a third isosceles triangle (all three triangles have the same base BC). This process is continued indefinitely. If the measure of the vertex angle of each triangle is represented by A1, A2, A3, ..., and $\sum_{n=1}^{\infty}(1-\cos A_n)$



$=\frac{p}{q}(p, q\in N)$, then find the minimum value of $(p+q)$.

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Sol.

$$\sin\frac{A_1}{2}=\frac{1}{2}\left(A_1=60^\circ\right)$$

$$\sin\frac{A_2}{2}=\frac{a}{2\times 2a}=\frac{1}{4}$$

$$\sin\frac{A_3}{2}=\frac{a}{2\times 4a}=\frac{1}{8}$$

$$\vdots$$

$$\sin\frac{A_n}{2}=\frac{a}{2\times 2^{n-1}a}=\frac{1}{2^n}$$

$$\sum_{n=1}^{\infty}\left(1-\cos A_n\right)=2\sum_{n=1}^{\infty}\left(\sin^2\frac{A_n}{2}\right)$$

$$=\text{Lim}_{n\to\infty}2\left(\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{8^2}+\ldots\ldots+\frac{1}{2^{2n}}\right)=\frac{2\times\frac{1}{2^2}}{1-\frac{1}{4}}=\frac{2}{3}$$

14. In a triangle ABC, if α, β, γ are the altitudes and r is the inradius then minimum value of $\frac{\alpha+r}{\alpha-r}+\frac{\beta+r}{\beta-r}+\frac{\gamma+r}{\gamma-r}$ is

Sol. $AD=\alpha, BE=\beta, CF=\gamma.$

$$\Delta=\frac{1}{2}a\alpha\Rightarrow\alpha=\frac{2\Delta}{a}$$

Similarly $\beta=\frac{2\Delta}{b}$ and $\gamma=\frac{2\Delta}{c}$

$$\frac{\alpha+r}{\alpha-r}+\frac{\beta+r}{\beta-r}+\frac{\gamma+r}{\gamma-r}$$

$$\text{Now}=\frac{\frac{2\Delta}{a}+\frac{\Delta}{s}}{\frac{2\Delta}{a}-\frac{\Delta}{s}}+\frac{\frac{2\Delta}{b}+\frac{\Delta}{s}}{\frac{2\Delta}{b}-\frac{\Delta}{s}}+\frac{\frac{2\Delta}{c}+\frac{\Delta}{s}}{\frac{2\Delta}{c}-\frac{\Delta}{s}}$$

$$=\frac{2s+a}{2s-a}+\frac{2s+b}{2s-b}+\frac{2s+c}{2s-c}$$

$$=\left(\frac{2s+a}{2s-a}+1\right)+\left(\frac{2s+b}{2s-b}+1\right)+\left(\frac{2s+c}{2s-c}+1\right)-3$$

$$=\frac{4s}{2s-a}+\frac{4s}{2s-b}+\frac{4s}{2s-c}-3$$

Now $AM\geq HM$

$$\frac{1}{3}\left(\frac{4s}{2s-a}+\frac{4s}{2s-b}+\frac{4s}{2s-c}\right)\geq\frac{3}{\left(\frac{2s-a}{4s}+\frac{2s-b}{4s}+\frac{2s-c}{4s}\right)}$$

$$\Rightarrow\frac{4s}{2s-a}+\frac{4s}{2s-b}+\frac{4s}{2s-c}\geq 9$$

$$=\left\{\frac{\alpha+r}{\alpha-r}+\frac{\beta+r}{\beta-r}+\frac{\gamma+r}{\gamma-r}\right\}\geq 6$$

$$\Rightarrow\frac{\alpha+r}{\alpha-r}+\frac{\beta+r}{\beta-r}+\frac{\gamma+r}{\gamma-r}\geq 6$$

SCHOOL LEVEL PROBLEMS

SINGLE CORRECT TYPE QUESTIONS

1. In ΔABC, if $A=30^\circ$, $b=8$, $a=6$, $B=\sin^{-1}x$, then $x=$
(a) 1/2 (b) 1/3 (c) 2/3 (d) 1

2. In triangle ABC, if $r_1=3$, $r_2=10$, $r_3=15$, then $c=$
(a) 5 (b) 12 (c) 13 (d) 13/2

3. If in ΔABC, $\frac{1}{a+c}+\frac{1}{b+c}=\frac{3}{a+b+c}$ then $\angle C=$
(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{6}$

4. In ΔABC, $\frac{\cos C+\cos A}{c+a}+\frac{\cos B}{b}=$
(a) $\frac{1}{a}$ (b) $\frac{1}{b}$ (c) $\frac{1}{c}$ (d) $\frac{c+a}{b}$

5. In a ΔABC, $\sum(b+c)\tan\frac{A}{2}\tan\left(\frac{B-C}{2}\right)=$
(a) a (b) b (c) c (d) 0

6. In a ΔABC, If $A=60^\circ$, then the value of $\left(1+\frac{a}{c}+\frac{b}{c}\right)\left(1+\frac{c}{b}-\frac{a}{b}\right)=$
(a) 3 (b) 2 (c) 1 (d) 0

7. If the radius of the incircle of a triangle with its sides $5k$, $6k$ and $5k$ is 6, then k is equal to
(a) 3 (b) 4 (c) 5 (d) 6

8. If the angles of a triangle are 30° and 45° and the included side is $\left(\sqrt{3}+1\right)$ cm, then what is the area of the triangle?
(a) $\frac{\sqrt{3}+1}{2}\text{cm}^2$ (b) $2\left(\sqrt{3}+1\right)\text{cm}^2$
(c) $\frac{\sqrt{3}+1}{3}\text{cm}^2$ (d) $\frac{\sqrt{3}-1}{2}\text{cm}^2$

9. Consider the following statements
I. There exists no ΔABC for which sin A + sin B = sin C.
II. If the angles of a triangle are in the ratio 1 : 2 : 3, then its side will be in the ratio $1:\sqrt{3}:2$.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

VERY SHORT ANSWER TYPE QUESTIONS

10. In a triangle ABC, $\sin A:\sin B:\sin C=1:2:3$. Find the perimeter of the triangle if $b=4$ cm.

11. Find the value of the $r\cot\frac{B}{2}\cot\frac{C}{2}$.

12. In a triangle ABC, $b=16$ cm, $c=25$ cm, and $B=33^\circ15'$. Find the angle C.

13. In any triangle ABC, prove that $4\Delta\cot A=b^2+c^2-a^2$.

14. If the exradii of a triangle are in HP, the corresponding sides.

15. In any triangle ABC, prove that $\frac{a^2-b^2}{2}\cdot\frac{\sin A\sin B}{\sin(A-B)}=\Delta$.

16. Find the ratio of the circumradius and inradius of an equilateral triangle.

LONG ANSWER TYPE QUESTIONS

17. In a triangle ABC, if $\cot\frac{A}{2}\cot\frac{B}{2}=c$, $\cot\frac{B}{2}\cot\frac{C}{2}=a$ and $\cot\frac{C}{2}\cot\frac{A}{2}=b$, then find the value of $\frac{1}{s-a}+\frac{1}{s-b}+\frac{1}{s-c}$.

18. In any triangle ABC, show that
$\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}=\frac{a+b+c}{a+b-c}\cot\frac{C}{2}$

19. In any triangle ABC, prove that
$a\cos A+b\cos B+c\cos C=2a\sin B\sin C=\frac{8\Delta^2}{abc}$.

20. Find the value of $\frac{b-c}{r_1}+\frac{c-a}{r_2}+\frac{a-b}{r_3}$.

21. The angle of elevation of the top point P of the vertical tower PQ of height h from a point A is 45° and from a point B, the angle of elevation is 60°, where B is a point at a distance d from the point A measured along the line AB which makes an angle 30° with AQ. Prove that $h=d\left(\sqrt{3}-1\right)$.

CASE STUDY BASED QUESTIONS

Case Study-I

Consider a ΔABC satisfying

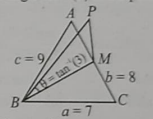
$$2a\sin^2\left(\frac{C}{2}\right)+2c\sin^2\left(\frac{A}{2}\right)=2a+2c-3b$$

22. The sides of the triangle are in
(a) GP
(b) AP
(c) HP
(d) Neither in GP nor AP nor in HP

23. If a, b and c are the sides of a ΔABC, then $a^{\frac{1}{p}}+b^{\frac{1}{p}}-c^{\frac{1}{p}}$, where $p>1$, is
(a) Always negative
(b) Always positive
(c) Always zero
(d) Positive, if $1<p<2$ and negative, if $p<2$

Case Study-II

A lamp post is situated at the middle point M of the side AC of a triangular plot ABC with $BC=7$ m, $CA=8$ m and $AB=9$ m. This lamp post subtends an angle $\tan^{-1}(3)$ at the point B.



24. What will be the cos C?
(a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{4}{7}$

25. Determine the height of the lamp post (PM)?
(a) 7 m (b) 14 m (c) 21 km (d) 28 m

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PRARAMBH (TOPICWISE)

1. If the angles of a triangle ABC be in A.P., then
(a) $c^2 = a^2 + b^2 - ab$ (b) $b^2 = a^2 + c^2 - ac$
(c) $a^2 = b^2 + c^2 - ac$ (d) $b^2 = a^2 + c^2$

2. In ΔABC, $a \sin(B - C) + b \sin(C - A) + c \sin(A - B) =$
(a) 0 (b) $a + b + c$
(c) $a^2 + b^2 + c^2$ (d) $2(a^2 + b^2 + c^2)$

3. In ΔABC, $(b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C =$
(a) 0 (b) $a^2 + b^2 + c^2$
(c) $2(a^2 + b^2 + c^2)$ (d) $\frac{1}{2abc}$

4. If the sides of a triangle are p, q and $\sqrt{p^2 + pq + q^2}$, then the biggest angle is
(a) $\pi/2$ (b) $2\pi/3$
(c) $5\pi/4$ (d) $7\pi/4$

5. In a triangle ABC, if B = 3C, then the values of $\sqrt{\left(\frac{b+c}{4c}\right)}$ and $\left(\frac{b-c}{2c}\right)$ are
(a) $\sin C, \sin\frac{A}{2}$ (b) $\cos C, \sin\frac{A}{2}$
(c) $\sin C, \cos\frac{A}{2}$ (d) $\cos C, \cos\frac{A}{2}$

6. In a ΔABC, if $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$, then cos C =
(a) $\frac{7}{5}$ (b) $\frac{5}{7}$ (c) $\frac{17}{36}$ (d) $\frac{16}{17}$

7. In a triangle ABC, $b = \sqrt{3}$, $c = 1$ and $\angle A = 30°$, then the largest angle of the triangle is
(a) 135° (b) 90°
(c) 60° (d) 120°

8. In a ΔABC, $2ac \sin\left(\frac{A - B + C}{2}\right)$ is equal to
(a) $a^2 + b^2 - c^2$ (b) $c^2 + a^2 - b^2$
(c) $b^2 - c^2 - a^2$ (d) $c^2 - a^2 - b^2$

9. If in triangle ABC, $\cos A = \frac{\sin B}{2\sin C}$, then the triangle is
(a) Equilateral (b) Isosceles
(c) Right angled (d) Scalene

10. If a, b and c are the sides of a triangle such that $a^4 + b^4 + c^4 = 2c^2(a^2 + b^2)$ then the angles opposite to the side c is
(a) 45° or 135° (b) 30° or 100°
(c) 50° or 100° (d) 60° or 120°

11. In a triangle ABC, $(a + b + c)(b + c - a) = \lambda bc$ if -
(a) $\lambda < 0$ (b) $\lambda > 0$
(c) $0 < \lambda < 4$ (d) $\lambda > 4$

12. Let ABC be a triangle such that $\angle A = 45°$, $\angle B = 75°$ then $a + c\sqrt{2}$ equal to -
(a) 0 (b) b (c) $2b$ (d) $-b$

13. Angles A, B and C of a triangle ABC are in A.P. If $\frac{b}{c} = \frac{\sqrt{3}}{\sqrt{2}}$ then angle A is equal to -
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{5\pi}{12}$ (d) $\frac{\pi}{2}$

14. If in a triangle ABC, $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A - B)}{\sin(A + B)}$ then the triangle is -
(a) Right angled or isosceles
(b) Right angled and isosceles
(c) Equilateral
(d) None of these

15. In any triangle ABC,
$\frac{a^2\sin(B - C)}{\sin B + \sin C} + \frac{b^2\sin(C - A)}{\sin C + \sin A} + \frac{c^2\sin(A - B)}{\sin A + \sin B}$ is equal to
(a) $a + b + c$ (b) $a + b - c$
(c) $a - b + c$ (d) 0

16. The expression $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$
is equal to
(a) $\cos^2 A$ (b) $\sin^2 A$
(c) $\cos A \cos B \cos C$ (d) None of these

Projection Formula, Napier's Analogy & Area of Triangle Half-Angled Formulae m-n theorem

17. If in a triangle ABC, $b = \sqrt{3}$, c = 1 and B – C = 90° the $\angle A$ is
(a) 30° (b) 45° (c) 75° (d) 15°

18. If a^2, b^2, c^2 are in A. P. then which of the following are also in A.P.
(a) $\sin A, \sin B, \sin C$
(b) $\tan A, \tan B, \tan C$
(c) $\cot A, \cot B, \cot C$
(d) $\operatorname{cosec} A, \operatorname{cosec} B, \operatorname{cosec} C$

19. If in a triangle ABC, $\cos A \cos B + \sin A \sin B \sin C = 1$, then the sides are proportional to
(a) $1:1:\sqrt{2}$ (b) $1:\sqrt{2}:1$ (c) $\sqrt{2}:1:1$ (d) $1:1:1$

20. If in the ΔABC, AB = 2BC, then $\tan\frac{B}{2} : \cot\left(\frac{C - A}{2}\right)$
(a) 3 : 1 (b) 2 : 1 (c) 1 : 2 (d) 1 : 3

21. If in a triangle ABC, $a = 5$, $b = 4$, $A = \frac{\pi}{2} + B$, then C =
(a) $\tan^{-1}\left(\frac{1}{9}\right)$ (b) $\tan^{-1}\frac{1}{40}$
(c) Cannot be evaluated (d) $2\tan^{-1}\left(\frac{1}{9}\right)$

22. If A is the area and $2s$ the sum of 3 sides of triangle, then
(a) $A \le \frac{s^2}{3\sqrt{3}}$ (b) $A \le \frac{s^2}{2}$
(c) $A > \frac{s^2}{\sqrt{3}}$ (d) $A \ge \frac{s^2}{3\sqrt{3}}$

23. If in a triangle ABC right angled at B, $s - a = 3$, $s - c = 2$, then the values of a and c are respectively
(a) 2, 3 (b) 3, 4 (c) 4, 3 (d) 6, 8

24. In a ΔABC a, c, A are given and b_1, b_2 are two values of the third side b such that $b_2 = 2b_1$. Then sin A =
(a) $\sqrt{\frac{9a^2 - c^2}{8a^2}}$ (b) $\sqrt{\frac{9a^2 - c^2}{8c^2}}$
(c) $\sqrt{\frac{9a^2 + c^2}{8a^2}}$ (d) $\sqrt{\frac{9a^2 + c^2}{8c^2}}$

25. In a ΔABC, $a^2 \sin 2C + c^2 \sin 2A =$
(a) Δ (b) 2Δ (c) 3Δ (d) 4Δ

26. The product of the arithmetic mean of the lengths of the sides of a triangle and harmonic mean of the lengths of the altitudes of the triangle is equal to
(a) Δ (b) 2Δ (c) 3Δ (d) 4Δ

27. In a ΔABC, if AB = 5 cm, BC = 13 cm and CA = 12 cm, then the distance of vertex A from the side BC is (in cm)
(a) $\frac{25}{13}$ (b) $\frac{60}{13}$ (c) $\frac{65}{12}$ (d) $\frac{144}{13}$

28. In a ΔABC, if $b = 2$, $c = \sqrt{3}$ and $\angle A = \frac{\pi}{6}$, then value of R is equal to
(a) $\frac{1}{2}$ (b) 1 (d) 2 (d) $\frac{1}{4}$

29. In a triangle ABC, $a2\cos 2B + b2\cos 2A + 2ab\cos(A - B) =$
(a) a^2 (b) c^2 (c) b^2 (d) $a^2 + b^2$

30. If in a triangle ABC, $b = \sqrt{3}$, $c = 1$ and $B - C = 90°$ then $\angle A$ is
(a) 30° (b) 45° (c) 75° (d) 15°

31. If i a triangle ABC, $(s - a)(s - b) = s(s - c)$, then angle C is equal to
(a) 90° (b) 45° (c) 30° (d) 60°

32. In ΔABC, if $\sin^2\frac{A}{2}, \sin^2\frac{B}{2}, \sin^2\frac{C}{2}$ be in H. P. then a, b, c will be in
(a) A. P. (b) G. P. (c) H. P. (d) A.G.P.

33. In any triangle ABC, $\frac{\tan\frac{A}{2} - \tan\frac{B}{2}}{\tan\frac{A}{2} + \tan\frac{B}{2}} =$
(a) $\frac{a-b}{a+b}$ (b) $\frac{a-b}{c}$ (c) $\frac{a-b}{a+b+c}$ (d) $\frac{c}{a+b}$

34. In any triangle ABC, the value of $a(b^2 + c^2)\cos A + b(c^2 + a^2)\cos B + c(a^2 + b^2)\cos C$ is
(a) $3abc^2$ (b) $3a^2bc$ (c) $3abc$ (d) $3ab^2c$

35. In a triangle ABC, AD is altitude from A. Given b > c, $\angle C = 23°$ and $AD = \frac{abc}{b^2 - c^2}$, then $\angle B =$
(a) 67° (b) 44° (c) 113° (d) 157°

36. In a ΔABC if the sides are $a = 3$, $b = 5$ and $c = 4$, then $\sin\frac{B}{2} + \cos\frac{B}{2}$ is equal to
(a) $\sqrt{2}$ (b) $\frac{\sqrt{3}+1}{2}$ (c) $\frac{\sqrt{3}-1}{2}$ (d) 1

37. Which of the following is true in a triangle ABC
(a) $(b + c)\sin\frac{B - C}{2} = 2a\cos\frac{A}{2}$
(b) $(b + c)\cos\frac{A}{2} = 2a\sin\frac{B - C}{2}$
(c) $(b - c)\cos\frac{A}{2} = a\sin\frac{B - C}{2}$
(d) $(b - c)\sin\frac{B - C}{2} = 2a\cos\frac{A}{2}$

38. In a ΔABC, if $3a = b + c$, then the value of $\cot\frac{B}{2}\cot\frac{C}{2}$ is
(a) 1 (b) 2 (c) $\sqrt{3}$ (d) $\sqrt{2}$

39. In ΔABC, $2R^2 \sin A \sin B \sin C =$
(a) s^2 (b) $ab + bc + ca$
(c) Δ (d) Zero

40. In an equilateral triangle the inradius and the circum-radius are connected by
(a) $r = 4R$ (b) $r = R/2$
(c) $r = R/3$ (d) $r = R/4$

41. If the sides of the triangle are 5K, 6K, 5K and radius of incircle is 6 then value of K is equal to
(a) 4 (b) 5 (c) 6 (d) 7

42. The circum-radius of the triangle whose sides are 13, 12 and 5 is
(a) 15 (b) 13/2 (c) 15/2 (d) 6

43. In a ΔABC, if $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$, and the side $a = 2$, then area of the triangle is
(a) 1 (b) 2 (c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$

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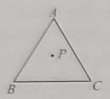
44. We are given and such that is acute and . Then -
(a) No triangle is possible
(b) One triangle is possible
(c) Two triangles are possible
(d) A right angled triangle is possible

45. In the adjacent figure 'P' is any interior point of the equilateral triangle ABC of side length 2 unit



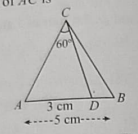
If x_a, x_b and x_c represent the distance of P from the sides BC, CA and AB respectively then $x_a + x_b + x_c$ is equal to -

(a) 6 (b) $\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $2\sqrt{3}$

46. If two sides a, b and the angle A be such that two triangles are formed, then the sum of the two values of the third side is

(a) $b^2 - a^2$ (b) $2b\cos A$ (c) $2b\sin A$ (d) $\frac{b-c}{b+c}$

47. In the figure, ABC is triangle in which $C = 90°$ and $AB = 5$ cm, D is a point on AB show that $AD = 3$ cm and $\angle ACD = 60°$. Then th length of AC is



(a) $5\sqrt{\frac{3}{7}}$ cm (b) $\sqrt{\frac{3}{7}}$ cm

(c) $\frac{3}{\sqrt{7}}$ cm (d) None of these

PRABAL (JEE MAIN LEVEL)

1. In a ΔABC, A: B: C = 3: 5: 4. Then $a + b + c\sqrt{2}$ is equal to
(a) $2b$ (b) $2c$ (c) $3b$ (d) $3a$

2. If in a ΔABC, $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$, then the triangle is:
(a) Right angled (b) Isosceles
(c) Equilateral (d) Obtuse angled

3. In a ΔABC $\frac{bc\sin^2 A}{\cos A + \cos B\cos C}$ is equal to
(a) $b^2 + c^2$ (b) bc
(c) a^2 (d) $a^2 + bc$

4. In a triangle ABC, $(a + b + c)(b + c - a) = kbc$, if
(a) $k < 0$ (b) $k > 6$
(c) $0 < k < 4$ (d) $k > 4$

5. In a ΔABC, $A = \frac{2\pi}{3}$, $b - c = 3\sqrt{3}$ cm and area $(\Delta ABC) = \frac{9\sqrt{3}}{2}$ cm^2. Then 'a' is
(a) $6\sqrt{3}$ cm (b) 9 cm
(c) 18 cm (d) None of these

6. If AD, BE and CF are the medians of ΔABC, then $(AD^2 + BE^2 + CF^2) : (BC^2 + CA^2 + AB^2)$ is equal to
(a) 4 : 3 (b) 3 : 2
(c) 3 : 4 (d) 2 : 3

7. The distance between the middle point of BC and the foot of the perpendicular form A is
(a) $\frac{-a^2 + b^2 + c^2}{2a}$ (b) $\frac{b^2 - c^2}{2a}$
(c) $\frac{b^2 + c^2}{\sqrt{bc}}$ (d) None of these

8. In a acute angled triangle ABC, AP is the altitude. Circle drawn with AP as its diameter cuts the sides AB and AC at D and E respectively, then length DE is equal to
(a) $\frac{\Delta}{2R}$ (b) $\frac{\Delta}{3R}$
(c) $\frac{\Delta}{4R}$ (d) $\frac{\Delta}{R}$

9. If in a ΔABC, $\Delta = a^2 - (b - c)^2$, then $\tan A =$
(a) 15/16 (b) 8/15
(c) 8/17 (d) 1/2

10. Let f, g, h be the lengths of the perpendiculars from the circumcentre of the ΔABC on the sides a, b and c respectively. If $\frac{a}{f} + \frac{b}{g} + \frac{c}{h} = \lambda\frac{abc}{fgh}$ then the value of λ is
(a) 1/4 (b) 1/2
(c) 1 (d) 2

11. If in a triangle ABC, right angle at B, $s - a = 3$ and $s - c = 2$, then
(a) $a = 2, c = 3$ (b) $a = 3, c = 4$
(c) $a = 4, c = 3$ (d) $a = 6, c = 8$

12. If in a triangle ABC, $b\cos^2\frac{A}{2} + a\cos^2\frac{B}{2} = \frac{3}{2}c$, then a, c, b are:
(a) in A.P. (b) in G.P.
(c) in H.P. (d) None of these

13. In a ΔABC if $b + c = 3a$, then $\cot\frac{B}{2}\cdot\cot\frac{C}{2}$ has the value equal to:
(a) 4 (b) 3
(c) 2 (d) 1

14. If in a ΔABC, $\angle A = \frac{\pi}{2}$, then $\tan\frac{C}{2}$ is equal to
(a) $\frac{a-c}{2b}$ (b) $\frac{a-b}{2c}$
(c) $\frac{a-c}{b}$ (d) $\frac{a-b}{c}$

15. If R denotes circumradius, then in ΔABC, $\frac{b^2 - c^2}{2a\,R}$ is equal to
(a) cos (B – C) (b) sin (B – C)
(c) cos B – cos C (d) None of these

16. In a ΔABC, the value of $\frac{a\cos A + b\cos B + c\cos C}{a + b + c}$ is equal to:
(a) $\frac{r}{R}$ (b) $\frac{R}{2r}$
(c) $\frac{R}{r}$ (d) $\frac{2r}{R}$

17. If the sides of a triangle are 3 : 7 : 8, then R : r is equal to
(a) 2 : 7 (b) 7 : 2
(c) 3 : 7 (d) 7 : 3

18. In a right angled triangle R is equal to
(a) $\frac{s + r}{2}$ (b) $\frac{s - r}{2}$
(c) $s - r$ (d) $\frac{s + r}{a}$

19. If the area of triangle is 100 sq. cm, r_1 = 10 cm and r_2 = 50 cm, then the value of $(b - a)$ is equal to
(a) 20 (b) 16
(c) 8 (d) 4

20. In a ΔABC, the inradius and three exradii are r, r_1, r_2 and r_3 respectively. In usual notations the value of $r \cdot r_1 \cdot r_2 \cdot r_3$ is equal to
(a) 2Δ (b) Δ^2
(c) $\frac{abc}{4R}$ (d) None of these

21. In a ΔABC if $r_1 > r_2 > r_3$, then
(a) $a > b > c$ (b) $a < b < c$
(c) $a > b$ and $b < c$ (d) $a < b$ and $b > c$

22. In a triangle ABC, right angled at B, the inradius is
(a) $\frac{AB + BC - AC}{2}$ (b) $\frac{AB + AC - BC}{2}$
(c) $\frac{AB + BC + AC}{2}$ (d) None of these

23. If H is the orthocentre of a triangle ABC, then the radii of the circle circumscribing the triangles BHC, CHA and AHB are respectively equal to
(a) R, R, R (b) $\sqrt{2}$R, $\sqrt{2}$R, $\sqrt{2}$R
(c) 2R, 2R, 2R (d) $\frac{2}{R}, \frac{2}{R}, \frac{2}{R}$

24. In a triangle ABC, if $\frac{a-b}{b-c} = \frac{s-a}{s-c}$, then r_1, r_2, r_3 are in
(a) A.P. (b) G.P.
(c) H.P. (d) None of these

25. A triangle is inscribed in a circle. The vertices of the triangle divide the circle into three arcs of length 3, 4 and 5 units. Then area of the triangle is equal to
(a) $\frac{9\sqrt{3}(1+\sqrt{3})}{\pi^2}$ (b) $\frac{9\sqrt{3}(\sqrt{3}-1)}{\pi^2}$
(c) $\frac{9\sqrt{3}(1+\sqrt{3})}{2\pi^2}$ (d) $\frac{9\sqrt{3}(\sqrt{3}-1)}{2\pi^2}$

26. If in a ΔABC, $\frac{r}{r_1} = \frac{1}{2}$, then the value of $\tan\frac{A}{2}\left(\tan\frac{B}{2} + \tan\frac{C}{2}\right)$ is equal to
(a) 2 (b) $\frac{1}{2}$
(c) 1 (d) None of these

27. If in a triangle ABC, the line joining the circumcentre and incentre is parallel to BC, then cos B + cos C is equal to
(a) 0 (b) 1
(c) 2 (d) None of these

28. If the incircle of the Δ ABC touches its sides respectively at L, M and N and if x, y, z be the circumradii of the triangles MIN, NIL and LIM where I is the incentre then the product xyz is equal to

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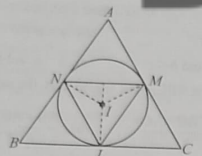

(a) $R r^2$ (b) $r R^2$ (c) $\frac{1}{2} R r^2$ (d) $\frac{1}{2} r R^2$

29. The perimeter of a triangle ABC is 6 times the arithmetic mean of the sines of its angles. If the side a is 1, then the angle A is

(a) $\pi/6$ (b) $\pi/3$ (c) $\pi/2$ (d) π

30. If a^2, b^2, c^2 are in A.P, then $\cot A, \cot B, \cot C$ are in

(a) A.P (b) G.P
(c) H.P (d) None of these

31. The area of the circle and the regular polygon of n sides and of equal perimeter are in the ratio of

(a) $\tan(\pi/n) : \pi/n$ (b) $\cos(\pi/n) : \pi/n$
(c) $\sin(\pi/n) : \pi/n$ (d) $\cot(\pi/n) : \pi/n$

32. In a triangle ABC, $(a+b+c)(b+c-a) = \lambda bc$ if

(a) $\lambda < 0$ (b) $\lambda > 0$
(c) $0 < \lambda < 4$ (d) $\lambda > 4$

33. In a triangle ABC, AD is the altitude from A. Given $b > c$, $C = 23°$ and $AD = \frac{abc}{b^2 - c^2}$ then $\angle B$ is equal to

(a) 23° (b) 113° (c) 67° (d) 90°

34. In any triangle ABC, $a^3\cos(B-C) + b^3\cos(C-A) + c^3\cos(A-B)$ is equal to

(a) $6abc$ (b) $9abc$
(c) $3abc$ (d) None of these

35. In a triangle ABC, $\sqrt{a} + \sqrt{b} - \sqrt{c}$ is

(a) Always positive
(b) Always negative
(c) Positive only when c is smallest
(d) None of these

36. In a triangle with sides a, b, and c, a semicircle touching the sides AC and CB is inscribed whose diameter lies on AB. Then, the radius of the semicircle is

(a) $a/2$
(b) Δ/s
(c) $\frac{2\Delta}{a+b}$
(d) $\frac{2abc}{(s)(a+b)}\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

37. If in ΔABC, $\sec A$, $\sec B$, $\sec C$ are in Harmonic progression then

(a) a, b, c, are in harmonic progression.
(b) $\cot\frac{A}{2}, \cot\frac{B}{2}, \cot\frac{C}{2}$ are in geometric progression
(c) r_1, r_2, r_3 are in arithmetic progression
(d) $\cot\frac{A}{2}, \cot\frac{B}{2}, \cot\frac{C}{2}$ are in arithmetic progression

38. If $\sin A$ and $\sin B$ of a triangle ABC satisfy $c^2x^2 - c(a+b)x + ab = 0$, then the triangle is

(a) Equilateral (b) Isosceles
(c) Right angled (d) Acute angled

39. The sides of a triangle are a, b and $\sqrt{a^2+ab+b^2}$, then the greatest angle is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$
(c) $\frac{2\pi}{3}$ (d) None of these

40. In a triangle ABC, R = circumradius and r = inradius. The value of $\frac{a\cos A + b\cos B + c\cos C}{a+b+c}$ is equal to

(a) $\frac{R}{r}$ (b) $\frac{R}{2r}$ (c) $\frac{r}{R}$ (d) $\frac{2r}{R}$

41. The distance of the circumcentre of the acute angled ΔABC from the sides BC, CA and AB are in the ratio

(a) $a \sin A : b \sin B : c \sin C$
(b) $\cos A : \cos B : \cos C$
(c) $a \cot A : b \cot B : c \cot C$
(d) None of these

42. If p_1, p_2 and p_3 are respectively the lengths of perpendiculars from the vertices of a triangle ABC to the opposite sides then the value of $p_1 p_2 p_3$ is

(a) $\frac{a^2b^2c^2}{8R^2}$ (b) $\frac{a^2b^2c^2}{8R^3}$
(c) $\frac{a^2b^2c^2}{8R^4}$ (d) $\frac{a^2b^2c^2}{4R^2}$

43. Suppose the angles of a triangle ABC are in A.P. and sides b and c satisfy $b : c = \sqrt{3} : \sqrt{2}$ then the angle A equals

(a) 45° (b) 60° (c) 75° (d) 90°

44. If a^2, b^2, c^2 are the roots of the equation $x^3 - Px^2 + Qx - R = 0$ where a, b, c be the sides of a triangle ABC then the value of $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c}$ equals to

(a) $\frac{P}{\sqrt{R}}$ (b) $\frac{P}{2\sqrt{R}}$
(c) $\frac{P}{4\sqrt{R}}$ (d) None of these

PARIKSHIT (JEE ADVANCED LEVEL)

SINGLE CORRECT TYPE QUESTIONS

1. If the angles of a triangle are in the ratio 4 : 1 : 1, then the ratio of the longest side to the perimeter is

(a) $\sqrt{3} : (2+\sqrt{3})$ (b) 1 : 6
(c) $1 : 2+\sqrt{3}$ (d) 2 : 3

2. In any ΔABC having sides a, b, c opposite to angles A, B, C respectively.

(a) $a\sin\left(\frac{B-C}{2}\right) = (b-c)\cos\frac{A}{2}$
(b) $a\cos\frac{A}{2} = (b-c)\sin\frac{B-C}{2}$
(c) $a\cos\frac{A}{2} = (b+c)\sin\frac{B+C}{2}$
(d) $a\sin\frac{B+C}{2} = (b+c)\cos\frac{A}{2}$

3. Let PQR be a triangle of area Δ with $a = 2$, $b = \frac{7}{2}$ and $c = \frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals

(a) $\frac{3}{4\Delta}$ (b) $\frac{45}{4\Delta}$
(c) $\left(\frac{3}{4\Delta}\right)^2$ (d) $\left(\frac{45}{4\Delta}\right)^2$

4. If a, b, c, are the sides of a triangle, then the minimum value of $\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c}$ is equal to

(a) 3 (b) 6 (c) 9 (d) 12

5. Triangle ABC is right angled at A. The points P and Q are on the hypotenuse BC such that $BP = PQ = QC$. If $AP = 3$ and $AQ = 4$ then the length BC is equal to

(a) $\sqrt{27}$ (b) $\sqrt{36}$
(c) $\sqrt{45}$ (d) $\sqrt{54}$

6. A sector $OABO$ of central angle θ is constructed in a circle with centre O and of radius 6. The radius of the circle that is circumscribed about the triangle OAB, is

(a) $6\cos\frac{\theta}{2}$ (b) $6\sec\frac{\theta}{2}$
(c) $3\left(\cos\frac{\theta}{2} + 2\right)$ (d) $3\sec\frac{\theta}{2}$

7. In a ΔABC, $a = a_1 = 2$, $b = a_2$, $c = a_3$ such that $a_{p+1} = \frac{5^p}{3^{2-p}} a_p \left(2^{2-p} - \frac{4p-2}{5^p} a_p\right)$ where $p = 1, 2$ then

(a) $r_1 = r_2$ (b) $r_3 = 2r_1$
(c) $r_2 = 2r_1$ (d) $r_2 = 3r_1$

8. If 'O' is the circumcentre of the ΔABC and R_1, R_2 and R_3 are the radii of the circumcircles of triangles OBC, OCA and OAB respectively then $\frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_3}$ has the value equal to:

(a) $\frac{abc}{2R^3}$ (b) $\frac{R^3}{abc}$
(c) $\frac{4\Delta}{R^2}$ (d) $\frac{\Delta}{4R^2}$

9. Let L and M be the respective intersections of the internal and external angle bisectors of the triangle ABC at C and the side AB produced. If $CL = CM$, then the value of $(a^2 + b^2)$ is (where a and b have their usual meanings)

(a) $2R^2$ (b) $2\sqrt{2}R^2$
(c) $4R^2$ (d) $4\sqrt{2}R^2$

MULTIPLE CORRECT TYPE QUESTIONS

10. Which of the following holds good for any triangle ABC?

(a) $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2+b^2+c^2}{2abc}$
(b) $\frac{\sin A}{a} + \frac{\sin B}{b} + \frac{\sin C}{c} = \frac{3}{2R}$
(c) $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$
(d) $\frac{\sin 2A}{a^2} = \frac{\sin 2B}{b^2} = \frac{\sin 2C}{c^2}$

11. If in a ΔABC, $a = 5$, $b = 4$ and $\cos(A - B) = \frac{31}{32}$, then

(a) $c = 6$ (b) $\sin A = \left(\frac{5\sqrt{7}}{16}\right)$
(c) area of $\Delta ABC = \frac{15\sqrt{7}}{4}$ (d) None of these

12. In a triangle ABC, with usual notations the length of the bisector of internal angle A is:

(a) $\frac{2bc\cos\frac{A}{2}}{b+c}$ (b) $\frac{2bc\sin\frac{A}{2}}{(b+c)}$
(c) $\frac{abc\,\mathrm{cosec}\frac{A}{2}}{2R(b+c)}$ (d) $\frac{2\Delta}{b+c}\mathrm{cosec}\frac{A}{2}$

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13. If $r_1 = 2r_2 = 3r_3$, then

(a) $\frac{a}{b} = \frac{4}{5}$ (b) $\frac{a}{b} = \frac{5}{4}$

(c) $\frac{a}{c} = \frac{3}{5}$ (d) $\frac{a}{c} = \frac{5}{3}$

14. In a ΔABC, following relations holds good. In which case(s) the triangle is a right angled triangle?

(a) $r_2 + r_3 = r_1 - r$ (b) $a^2 + b^2 + c^2 = 8R^2$

(c) $r_1 = s$ (d) $2R = r_1 - r$

15. If in a triangle ABC, cos A cos B + sin A sin B sin C = 1, then the triangle is

(a) Isosceles (b) Right angled

(c) Equilateral (d) None of these

16. AD, BE and CF are the perpendiculars from the angular points of an acute angled Δ ABC upon the opposite sides, then:

(a) $\frac{\text{Perimeter of } \Delta DEF}{\text{Perimeter of } \Delta ABC} = \frac{r}{R}$

(b) Area of $\Delta DEF = 2\Delta \cos A \cos B \cos C$

(c) Area of $\Delta AEF = \Delta \cos^2 A$

(d) Circum-radius of $\Delta DEF = \frac{R}{2}$

17. The product of the distances of the incentre from the angular points of a Δ ABC is:

(a) $4R^2 r$ (b) $4Rr^2$

(c) $\frac{(abc)R}{s}$ (d) $\frac{(abc)r}{s}$

18. In a triangle ABC, points D and E are taken on side BC such that BD = DE = EC. If angle ADE = angle AED = θ, then:

(a) $\tan\theta = 3\tan B$ (b) $3\tan\theta = \tan C$

(c) $\frac{6\tan\theta}{\tan^2\theta - 9} = \tan A$ (d) angle B = angle C

19. With usual notations, in a Δ ABC the value of $\Pi(r_1 - r)$ can be simplified as:

(a) $abc\,\Pi \tan\frac{A}{2}$ (b) $4rR^2$

(c) $\frac{(abc)^2}{R(a+b+c)^2}$ (d) $4Rr^2$

20. In a ΔABC, $A = \frac{\pi}{3}$ and $b : c = 2 : 3$.

If $\tan\alpha = \frac{\sqrt{3}}{5}, 0 < \alpha < \frac{\pi}{2}$ then-

(a) $B = 60° + \alpha$ (b) $C = 60° + \alpha$

(c) $B = 60° - \alpha$ (d) $C = 60° - \alpha$

21. If in a ΔABC, $a = 6$, $b = 3$ and $\cos(A - B) = \frac{4}{5}$ then-

(a) $C = \frac{\pi}{4}$ (b) $A = \sin^{-1}\frac{2}{\sqrt{5}}$

(c) ar $(\Delta ABC) = 9$ (d) None of these

COMPREHENSION BASED QUESTIONS

Comprehension–1 (Q. 22 to 24): The internal bisectors of the angle of ΔABC meet the sides BC, CA and AB at D, E and F respectively. Then

On the basis of above passage, answer the following questions:

22. The length BE is given by

(a) $\frac{2bc}{b+c}\cos\left(\frac{A}{2}\right)$ (b) $\frac{bc}{b+c}\cos\left(\frac{A}{2}\right)$

(c) $\frac{2ac}{a+c}\cos\left(\frac{B}{2}\right)$ (d) $\frac{ca}{c+a}\cos\left(\frac{B}{2}\right)$

23. $\frac{1}{AD}\cos\frac{A}{2} + \frac{1}{BE}\cos\frac{B}{2} + \frac{1}{CF}\cos\frac{C}{2} =$

(a) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ (b) abc

(c) $a + b + c$ (d) $ab + bc + ca$

24. Area of ΔDBF is given by

(a) $\frac{ac\sin B}{2(a+b)}$ (b) $\frac{\Delta ac}{(a+b)(b+c)}$

(c) $\frac{ac\sin B}{(a+b)(b+c)}$ (d) $\frac{ab\sin C}{(a+b)}$

Comprehension–2 (Q. 25 to 27): Consider a triangle ABC with $b = 3$. Altitude from the vertex B meets the opposite side in D which divides AC internally in the ratio 1 : 2. A circle of radius 1 passes through the point A and D and touches the circumcircle of the triangle BCD at D.

25. If E is the centre of the circle with radius 2 then angle EDB equals

(a) $\sin^{-1}\left(\frac{\sqrt{15}}{4}\right)$ (b) $\sin^{-1}\left(\frac{3}{4}\right)$

(c) $\sin^{-1}\left(\frac{1}{4}\right)$ (d) $\sin^{-1}\left(\frac{15}{16}\right)$

26. If F is the circumcentre of the triangle BDC then which one of the following does not hold good?

(a) $\angle FCD = \sin^{-1}\left(\frac{\sqrt{15}}{4}\right)$

(b) $\angle FDC = \cos^{-1}\left(\frac{1}{4}\right)$

(c) Triangle DFC is an isosceles triangle

(d) Area of ΔADE = (1/4)th of the area of ΔDBC

27. If R is the circumradius of the ΔABC, then R equal

(a) 4 (b) 6

(c) $2\left(\sqrt{\frac{61}{15}}\right)$ (d) $4\left(\sqrt{\frac{61}{15}}\right)$

MATCH THE COLUMN TYPE QUESTIONS

28. Match the following:

	Column-I		Column-II
A.	In a ΔABC, $a = 4$, $b = 3$ and the medians AA_1 and BB_1 are mutually perpendicular, then square of area of the ΔABC is equal to	p.	3
B.	If in an acute angled ΔABC, line joining the circumcentre and orthocentre is parallel to side AC, then value of tan A.tan C is equal to	q.	7
C.	In a ΔABC, $a = 5$, $b = 4$ and $\tan\frac{C}{2} = \sqrt{\frac{7}{9}}$, then side 'c' is equal to	r.	6
D.	In a ΔABC, $2a^2 + 4b^2 + c^2 = 4ab + 2ac$, then value of (8 cos B) is equal to	s.	11

(a) (A) → (s), (B) → (p), (C) → (r), (D) → (q)

(b) (A) → (q), (B) → (p), (C) → (s), (D) → (r)

(c) (A) → (r), (B) → (p), (C) → (s), (D) → (q)

(d) (A) → (q), (B) → (s), (C) → (q), (D) → (p)

29. In a triangle ABC, AD is perpendicular to BC and DE is perpendicular to AB.

	Column-I		Column-II
A.	Area of ΔADB	p.	$(b^2/4)\sin 2C$
B.	Area of ΔADC	q.	$(c^2/4)\cos^2 B \sin 2B$
C.	Area of ΔADE	r.	$(c^2/4)\sin 2B$
D.	Area of ΔBDE	s.	$(c^2/4)\sin^2 B \sin 2B$

(a) (A) → (s), (B) → (p), (C) → (r), (D) → (q)

(b) (A) → (q), (B) → (p), (C) → (s), (D) → (r)

(c) (A) → (r), (B) → (p), (C) → (s), (D) → (q)

(d) (A) → (q), (B) → (p), (C) → (s), (D) → (p)

INTEGER TYPE QUESTIONS

30. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. Suppose $a = 2$, $b = 3$, $c = 4$ and H be the orthocentre. Find $\frac{\sqrt{15}}{2}(HA)$.

31. In a triangle ABC, if $\frac{\sin A}{4} = \frac{\sin B}{5} = \frac{\sin C}{6}$, then the value of $\cos A + \cos B + \cos C = \frac{K}{M}$ then $\left(\frac{K}{M}\right) =$ _____

(where (·) denotes least integer function)

32. If two adjacent sides of a cyclic quadrilateral are 2 and 5 and the angle between them 60°. If the third side is 3, the remaining fourth side is

33. In triangle ABC, E is the midpoint of side BC and D is on side AC. Let the length of side AC is unity and $\angle BAC = 60°$, $\angle ABC = 100°$, $\angle ACB = 20°$, $\angle DEC = 80°$. If the area of the triangle ABC plus twice the area of triangle CDE equals $\sqrt{m/n}$ where m and n are coprime then find the value of $(m + n)$.

34. If the quadratic equation $ax^2 + bx + c = 0$ has equal roots where a, b and c denote the lengths of the sides opposite to vertices A, B and C of a triangle ABC respectively, then find the sum of integers in the range of $\left(\frac{\sin A}{\sin C} + \frac{\sin C}{\sin A}\right)$.

PYQ's (PAST YEAR QUESTIONS)

SINE RULE, COSINE RULE, PROJECTION FORMULA, NAPIER FORMULA

1. For a triangle ABC, the value of $\cos 2A + \cos 2B + \cos 2C$ is least. If its inradius is 3 and incentre is M, then which of the following is NOT correct? **[1 Feb, 2023 (Shift-I)]**

(a) Perimeter of ΔABC is $18\sqrt{3}$

(b) $\sin 2A + \sin 2B + \sin 2C = \sin A + \sin B + \sin C$

(c) $\overrightarrow{MA} \cdot \overrightarrow{MB} = -18$

(d) Area of ΔABC is $\frac{27\sqrt{3}}{2}$

Paragraph

Consider an obtuse angled triangle ABC in which the difference between the largest and the smallest angle is $\frac{\pi}{2}$ and whose sides are in arithmetic progression. Suppose that the vertices of this triangle lie on a circle of radius 1.

2. Let a be the area of the triangle ABC. Then the value of $(64a)^2$ is **[JEE Adv, 2023]**

3. Then the inradius of the triangle ABC is **[JEE Adv, 2023]**

BOOKWORMCBSE

4. Consider a triangle PQR having sides of lengths p, q and r opposite to the angles P, Q and R, respectively. Then which of the following statements is (are) TRUE?

(a) $\cos P \geq 1 - \frac{p^2}{2qr}$ **[JEE Adv, 2021]**

(b) $\cos R \geq \left(\frac{q-r}{p+q}\right)\cos P + \left(\frac{p-r}{p+q}\right)\cos Q$

(c) $\frac{q+r}{p} < 2\frac{\sqrt{\sin Q \sin R}}{\sin P}$

(d) If $p < q$ and $p < r$, then $\cos Q > \frac{p}{r}$ and $\cos R > \frac{p}{q}$

5. Given $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$ for ΔABC with usual notation.

If $\frac{\cos A}{\alpha} = \frac{\cos B}{\beta} = \frac{\cos C}{\gamma}$, then the ordered tried (α, β, γ) has a value: **[11 Jan, 2019 (Shift-II)]**

(a) (7, 19, 25) (b) (3, 4, 5)

(c) (5, 12 ,13) (d) (19, 7, 25)

6. If 5, $5r$, $5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to: **[10 Jan, 2019 (Shift-I)]**

(a) $\frac{3}{4}$ (b) $\frac{5}{4}$ (c) $\frac{7}{4}$ (d) $\frac{3}{2}$

7. If the lengths of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then a ratio of lengths of the sides of this triangle is: **[8 April, 2019 (Shift-II)]**

(a) 5 : 9 : 13 (b) 5 : 6 : 7

(c) 4 : 5 : 6 (d) 3 : 4 : 5

8. Let x, y and z be positive real numbers. Suppose x, y and z are lengths of the sides of a triangle opposite to its angles X, Y and Z, respectively. If $\tan\frac{X}{2} + \tan\frac{Z}{2} = \frac{2y}{x+y+z}$, then which of the following statements is/are TRUE? **[JEE Adv, 2019]**

(a) $2Y = X + Z$ (b) $Y = X + Z$

(c) $\tan\frac{X}{2} = \frac{x}{y+z}$ (d) $x^2 + z^2 - y^2 = xz$

TRIGONOMETRIC FUNCTION OF HALF ANGLES, M-N RULES

9. If the line $x = y = z$ intersects the line $x\sin A + y\sin B + z\sin C - 18 = 0 = x\sin 2A + y\sin 2B + z\sin 2C - 9$, where A, B, C are the angles of a triangle ABC, then $80\left(\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$ is equal to **[15 April, 2023 (Shift-I)]**

10. The value of $2\sin\left(\frac{\pi}{8}\right)\sin\left(\frac{2\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)\sin\left(\frac{5\pi}{8}\right)\sin\left(\frac{6\pi}{8}\right)\sin\left(\frac{7\pi}{8}\right)$ is: **[26 Aug, 2021 (Shift-II)]**

(a) $\frac{1}{8\sqrt{2}}$ (b) $\frac{1}{4\sqrt{2}}$ (c) $\frac{1}{4}$ (d) $\frac{1}{8}$

RADIUS OF INCIRCLE, ESCRIBED CIRCLE, ORTHOCENTRE AND PEDAL TRIANGLE, EXCENTRAL TRIANGLE

11. Let a, b and c be the length of sides of a triangle ABC such that $\frac{a+b}{7} = \frac{b+c}{8} = \frac{c+a}{9}$. If r and R are the radius of incircle and radius of circumcircle of the triangle ABC, respectively, then the value of $\frac{R}{r}$ is equal to **[25 June, 2022 (Shift-I)]**

(a) $\frac{5}{2}$ (b) 2 (c) $\frac{3}{2}$ (d) 1

12. If in a triangle ABC, $AB = 5$ units, $\angle B = \cos^{-1}\left(\frac{3}{5}\right)$ and radius of circumcircle of ΔABC is 5 units, then the area (in sq. units) of ΔABC is: **[20 July, 2021 (Shift-I)]**

(a) $6 + 8\sqrt{3}$ (b) $8 + 2\sqrt{2}$

(c) $10 + 6\sqrt{2}$ (d) $4 + 2\sqrt{3}$

13. Let in a right angled triangle, the smallest angle be θ. If a triangle formed by taking the reciprocal of its sides is also a right angled triangle, then $\sin\theta$ is equal to: **[20 July, 2021 (Shift-II)]**

(a) $\frac{\sqrt{2}-1}{2}$ (b) $\frac{\sqrt{5}+1}{4}$

(c) $\frac{\sqrt{5}-1}{2}$ (d) $\frac{\sqrt{5}-1}{4}$

14. In ΔABC, the lengths of sides AC and AB are 12 cm and 5 cm, respectively. If the area of ΔABC is 30cm^2 and R and r are respectively the radii of circumcircle and incircle of ΔABC, then the value of $2R + r$ (in cm) is equal to ____ **[16 March, 2021 (Shift-II)]**

15. In a triangle, the sum of lengths of two sides is x and the product of the lengths of the same two sides is y. If $x^2 - c^2 = y$, where c is the length of the third side of the triangle, then the circumradius of the triangle is: **[11 Jan, 2019 (Shift-I)]**

(a) $\frac{3}{2}y$ (b) $\frac{c}{\sqrt{3}}$ (c) $\frac{c}{3}$ (d) $\frac{y}{\sqrt{3}}$

16. Let PQR be a triangle of area Δ with $a = 2$, $b = \frac{7}{2}$ and $c = \frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals **[JEE Adv, 2013]**

(a) $\frac{3}{4\Delta}$ (b) $\frac{45}{4\Delta}$ (c) $\left(\frac{3}{4\Delta}\right)^2$ (d) $\left(\frac{45}{4\Delta}\right)^2$

17. In a triangle PQR, P is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides PQ, QR and RP at N, L and M respectively, such that the lengths of PN, QL and RM are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are) **[JEE Adv, 2013]**

(a) 16 (b) 18 (c) 24 (d) 22

18. In a triangle the sum of two sides is x and the product of the same two sides is y. If $x^2 - c^2 = y$, where c is the third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is **[JEE Adv, 2014]**

(a) $\frac{3y}{2x(x+c)}$ (b) $\frac{3y}{2c(x+c)}$

(c) $\frac{3y}{4x(x+c)}$ (d) $\frac{3y}{4c(x+c)}$

19. In a triangle PQR, let $\angle PQR = 30°$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10, respectively. Then, which of the following statement(s) is(are) TRUE? **[JEE Adv, 2018]**

(a) $\angle QPR = 45°$

(b) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120°$

(c) The radius of the incircle of the triangle PQR is $10\sqrt{3} - 15$

(d) The area of the circumcircle of the triangle PQR is 100π.

20. In a non-right-angled triangle ΔPQR, let p, q, r denote the lengths of the sides opposite to the angles at P, Q, R respectively. The median from R meets the side PQ at S, the perpendicular from P meets the side QR at E, and RS and PE intersect at O. If $p = \sqrt{3}$, $q = 1$, and the radius of the circumcircle of the ΔPQR equals 1, then which of the following options is/are correct? **[JEE Adv, 2019]**

(a) Area of $\Delta SOE = \frac{\sqrt{3}}{12}$

(b) Radius of incircle of $\Delta PQR = \frac{\sqrt{3}}{2}(2-\sqrt{3})$

(c) Length of $RS = \frac{\sqrt{7}}{2}$

(d) Length of $OE = \frac{1}{6}$

BOOKWORMCBSE

Answer Key

CONCEPT APPLICATION

1. (d) 2. (d) 3. (a) 4. (d) 5. (c) 6. (c) 8. A.P. 9. [30°] 10. (c)
11. $AD = 2, BC = CD = 1$ 13. (a) 14. (c) 15. (d) 17. [30°] 18. [5] 19. [1]
20. $\left[6+8\sqrt{3}\right]$ 21. 4m 22. $\left[12+7\sqrt{3}\right]$ 23. (a) 24. (c) 30. $HD = 4R \cos B \cos C$ 31. [60°]

SCHOOL LEVEL PROBLEMS

1. (c) 2. (c) 3. (a) 4. (b) 5. (d) 6. (a) 7. (b) 8. (a) 9. (c) 22. (b)
23. (b) 24. (b) 25. (c)

PRARAMBH (TOPICWISE)

1. (b) 2. (a) 3. (a) 4. (b) 5. (b) 6. (b) 7. (d) 8. (b) 9. (b) 10. (a)
11. (c) 12. (c) 13. (c) 14. (a) 15. (d) 16. (b) 17. (a) 18. (c) 19. (a) 20. (b)
21. (d) 22. (a) 23. (b) 24. (b) 25. (d) 26. (b) 27. (b) 28. (b) 29. (b) 30. (a)
31. (a) 32. (c) 33. (b) 34. (c) 35. (c) 36. (a) 37. (c) 38. (b) 39. (c) 40. (b)
41. (a) 42. (b) 43. (d) 44. (a) 45. (b) 46. (b) 47. (a)

PRABAL (JEE MAIN LEVEL)

1. (c) 2. (c) 3. (c) 4. (c) 5. (b) 6. (c) 7. (b) 8. (d) 9. (b) 10. (a)
11. (b) 12. (a) 13. (c) 14. (d) 15. (b) 16. (a) 17. (b) 18. (b) 19. (c) 20. (b)
21. (a) 22. (a) 23. (a) 24. (a) 25. (a) 26. (b) 27. (b) 28. (c) 29. (a) 30. (a)
31. (a) 32. (c) 33. (b) 34. (c) 35. (a) 36. (c) 37. (c) 38. (c) 39. (c) 40. (c)
41. (b) 42. (b) 43. (c) 44. (b)

PARIKSHIT (JEE ADVANCED LEVEL)

1. (a) 2. (a) 3. (c) 4. (a) 5. (c) 6. (d) 7. (d) 8. (c) 9. (c) 10. (a,b)
11. (a,b,c) 12. (a,c,d) 13. (b,d) 14. (a,b,c,d) 15. (a,b) 16. (a,b,c,d) 17. (b,d) 18. (a,c,d) 19. (a,c,d) 20. (b,c)
21. (b,c) 22. (c) 23. (a) 24. (b) 25. (a) 26. (d) 27. (c) 28. (a) 29. (c) 30. [7]
31. [2] 32. [2] 33. [67] 34. [12]

PYQ's (PAST YEAR QUESTIONS)

1. (a) 2. [1008.00] 3. [0.25] 4. (a, b) 5. (a) 6. (c) 7. (c) 8. (b,c) 9. [5]
10. (d) 11. (a) 12. (a) 13. (c) 14. [15] 15. (b) 16. (c) 17. (b,d) 18. (b)
19. (b, c, d) 20. (b, c, d)

CHAPTER 21

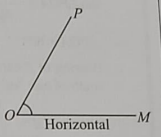
Height & Distances

INTRODUCTION

This chapter deals with the applications of trigonometry in practical situations concerning measurement of heights and distances, which are otherwise not directly measurable. We need to first define certain terms and state some properties before applying the principles of trigonometry.

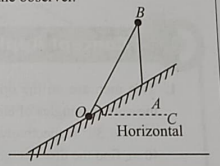
ANGLE OF ELEVATION

Consider a point P (at higher level) being observed from a point O (usually called observer) at a lower horizontal level. Draw a horizontal line OM through O, OP is called the line of sight and $\angle POM$ is called the angle of elevation of point P as seen from O.



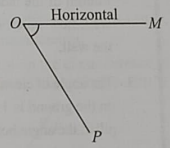
Unless otherwise stated, the height of the observer is neglected because mostly the heights and distances to be measured are very large compared to the height of the observer.

Suppose, there is a flagstaff AB mounted on a sloping level and its top B is observed from O. Then the angle $\angle AOB$ is not the angle of elevation of B because it is not measured w.r.t the horizontal at O. The correct angle of elevation is $\angle COB$.



ANGLE OF DEPRESSION

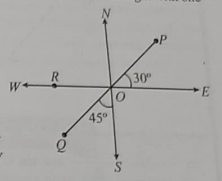
Consider a point P being observed from a point O at a higher horizontal level. $\angle POM$ here is called the angle of depression of the point P as seen from O. This is also measured w.r.t the horizontal at O.



DIRECTIONS OF A POINT

When the observer (O) and the object (P) are at the same horizontal level, to specify the direction or location of P w.r.t O, we take the help of cardinal directions North, East, West and South. Bearing of a point is defined as the angle made by the line of sight with one of the principal directions North, East, West or South.

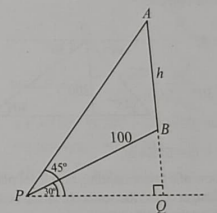
e.g., Bearing of point P can be stated as 30° north of east or 60° east of north, which can be symbolically written as $E30°N$ or $N60°E$. Bearing of Q is $W45°S$ or $S45°W$ or simply WS or SW and that of R is W.



Train Your Brain

Example 1: The angle of elevation of the top of a temple from a point on a hill of slope $\frac{1}{\sqrt{3}}$ is 45°. If the point is at a distance of 100 m from the temple along the hill, find the height of the temple.

Sol.



Given $\tan\angle BPQ = \frac{1}{\sqrt{3}} \Rightarrow \angle BPQ = 30°$

$\angle APQ = 45°$ and $PB = 100$

Now, $BQ = 100 \sin 30° = 50$ m

and $PQ = 100 \cos 30° = 50\sqrt{3}$ m

So, $\Delta APQ \Rightarrow \tan 45° = \frac{AQ}{PQ}$

$\Rightarrow \quad h + 50 = 50\sqrt{3} \Rightarrow h = 50(\sqrt{3} - 1) = 36.6$ m.

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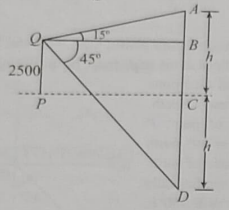
Example 2: The angle of elevation of a stationary cloud from a point 2500 metres above a lake is 15° and the angle of depression of its reflection in the lake is 45°. What is the height of the cloud above the lake level?

Sol. $\tan 15° = \frac{AB}{BQ}$

$\Rightarrow \tan(45°-30°) = \frac{h-2500}{BQ}$ ... (*i*)



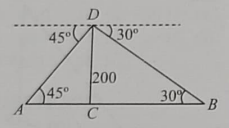
$\Rightarrow \frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}} = \frac{h-2500}{BQ}$ and $\tan 45° = \frac{BD}{BQ}$

$\Rightarrow 1 = \frac{h+2500}{BQ}$ ... (*ii*)

(i) ÷ (ii) $\Rightarrow \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{h-2500}{h+2500} \Rightarrow h = 2500\sqrt{3}$ m

Example 3: An observer on the top of a cliff 200 m above the sea level observes the angle of depression of two objects on the opposite sides of the cliff to be 45° and 30°. Find the distance between the objects.

Sol.



$AC = 200 \cot 45°$

$= 200$ m

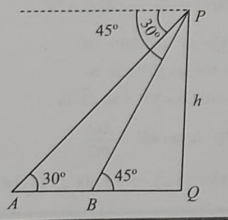
$BC = 200 \cot 30°$

$= 200\sqrt{3}$ m

$\Rightarrow AB = 200(\sqrt{3}+1) = 546.4$ m.

Example 4: The shadow of a tower standing on a level plane is found to be 60 m longer when the sun's elevation is 30° then when it is 45°. Find the height of tower.

Sol.



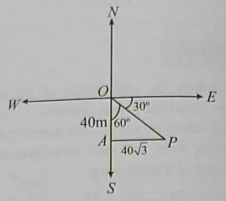
$AB = 60\ m = AQ - BQ$

$= h \cot 30° - h \cot 45°$

$\Rightarrow h = \frac{60}{\sqrt{3}-1}\text{m} = \frac{60(\sqrt{3}+1)}{3-1} = 81.96$ m

Example 5: Find the distance and bearing of a point situated 40 m south and $40\sqrt{3}$ m east of the reference point.

Sol.



$\tan\angle POA = \frac{40\sqrt{3}}{40} = \sqrt{3}$

$\Rightarrow \angle POA = 60°$

$\therefore$ Bearing of P can be written as $E30°S$ or $S60°E$ or 30° south of east 'or' 60° east of south.

Distance $OP = \sqrt{(40)+\left(40\sqrt{3}\right)^2} = 80$ m

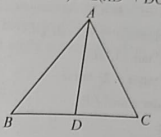
Concept Application

1. Two men are on the opposite side of a tower. They measure the angles of elevation of the top of the tower 45° and 30° respectively. If the height of the tower is 40 m, find the distance between the men.
2. A ladder 5 metre long leans against a vertical wall. The bottom of the ladder is 3 metre far from the wall. If the bottom of the ladder is pulled 1 metre farther from the wall, how much does the top of the ladder slide down the wall.
3. The angle of elevation of the top of a pillar at any point A on the ground is 15°. On walking 40 metres towards the pillar, the angle become 30°. The height of the pillar is.
4. The shadow of a tower standing on a level ground is found to be 40 m longer when the sun's altitude is 30° than when it is 60°. The height of the tower is.

APOLLONIUS THEOREM

If AD is the median of a ΔABC, then

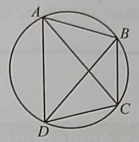
$AB^2 + AC^2 = 2(AD^2 + BD^2)$ or $2(AD^2 + DC^2)$



PTOLEMY'S THEOREM

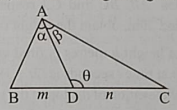
In a cyclic quadrilateral $ABCD$

$AC.BD = AB.CD + AD.BC$



m-n THEOREM

If D be the point on the side BC of a ΔABC which divides the side BC in the ratio $m : n$, then according the figure



we have,

(*i*) $(m+n)\cot\theta = m\cot\alpha - n\cot\beta$

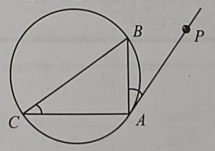
(*ii*) $(m+n)\cot\theta = n\cot B - m\cot C$

PROPERTIES OF CIRCLES

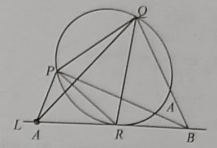
(*i*) If AB subtends equal angles at two points P and Q, the points A, B, P and Q are concyclic. ($\because$ angles on the same segment of a circle are equal)

(*ii*) Angle subtended by a chord at the center is twice the angle subtended at any point on the circumference.

(*iii*) Let AP be the tangent at a point A on the circumference of a circle passing through A, B and C. Then $\angle BAP = \angle ACB$.



(*iv*) A line segment PQ subtends various angles at different points on another line L. Obviously, at some point (say R) PQ will subtend the greatest angle also. Then, the circle passing through P, Q, R touches the line L at R.

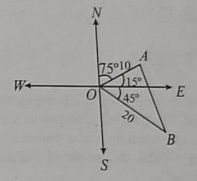


Three dimensional problems in heights and distances are generally solved by decomposing the picture into two 2-dimensional views [top (or plane) view and front view] and then principles of plane trigonometry are used.

Train Your Brain

Example 6: Two cars leave a place at the same time. One travels 10 km in the direction $N75°E$ and the other 20 km in the south-east direction. What is the final distance between the cars?

Sol.



$\cos\angle AOB = \frac{OA^2 + OB^2 - AB^2}{2(OA)(OB)}$

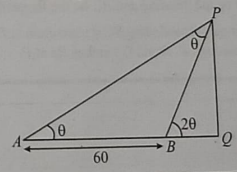
$\Rightarrow AB^2 = (10)^2 + (20)^2 - 2(10)(20)\cos 60°$

$= 100 + 400 - 200 = 300$

$\Rightarrow AB = 10\sqrt{3}$ km

Example 7: A man observes the angle of elevation of the top of a tower to be θ°. He moves a distance 60 m towards the tower and finds that the elevation has doubled. Find the value of sin2θ. If height of the tower is 50 m.

Sol.



BOOKWORMCBSE

In ΔPAB, $\angle ABP = \pi - 2\theta$ and $\angle APB = \theta$ (exterior angle is the sum of two opposite interior angles)

Applying sine rule,

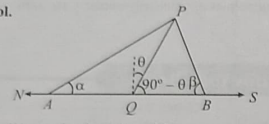
$$\frac{\sin\theta}{AB} = \frac{\sin(\pi - 2\theta)}{AP}$$

$$\Rightarrow \frac{\sin\theta}{60} = \frac{\sin 2\theta}{\frac{PQ}{\sin\theta}} = \frac{\sin 2\theta \cdot \sin\theta}{50}$$

$$\Rightarrow \sin 2\theta = \frac{5}{6}$$

Example 8: A chimney leans towards the south. At equal distances due north and south of it the angles of elevation of the top of chimney are α and β respectively. Find the inclination of the chimney to the vertical.

Sol.



We require the value of θ.

Applying $m - n$ theorem to ΔPAB, We have

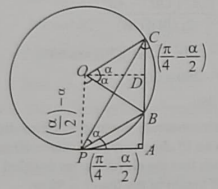
$$(1+1)\cot(90^\circ - \theta) = 1.\cot\alpha - 1.\cot\beta$$

$$\Rightarrow \tan\theta = \frac{\cot\alpha - \cot\beta}{2}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{\cot\alpha - \cot\beta}{2}\right)$$

Example 9: A man is moving towards a building on which a flagstaff is mounted. The flagstaff subtends its maximum angle α at the man's eye when he is at a distance 'c' from the base of building. Obtain the height of building and flagstaff in terms of c and α.

Sol.



Let AB be the building and BC be the flagstaff.

Since angle subtended by BC is maximum at P, a circle drawn through P, B and C touches PA at P.

The perpendicular bisector of chord BC and a line perpendicular to PA at P intersect at the center O of the circle. Consider chord BC.

$\because \angle BPC = \alpha \Rightarrow \angle BOC = 2\alpha \Rightarrow \angle COD = \alpha$

$\therefore$ Height of flagstaff $= BC = 2CD$

$= 2OD \tan\angle COD$

$= 2PA \tan\alpha = 2c\tan\alpha$

Now consider chord PB.

$\because \angle POB = \frac{\pi}{2} - \alpha$

$\Rightarrow \angle PCB = \frac{\pi}{4} - \frac{\alpha}{2}$

$\Rightarrow \angle BPA = \frac{\pi}{4} - \frac{\alpha}{2}$ (alternate segment theorem)

$\therefore$ Height of building $= AB = AP\tan\angle BPA = C\tan\left(\frac{\pi}{4} - \frac{\alpha}{2}\right)$

Concept Application

5. A vertical lamp post stands at the mid-point of the edge BC of a park in the shape of a triangle. The angle of elevation of the top of the lamp post from A is 30°. If the sides AB, BC and CA respectively are 40m, $20\sqrt{6}$ m and 20m, obtain the height of lamp post.

6. A rocket of height h metres is fired vertically upwards. Its velocity at time t seconds is $(2t+3)$ metres/second. If the angle of elevation of the top of the rocket from a point on the ground after 1 second of firing is $\pi/6$ and after 3 seconds it is $\pi/3$ then the distance of the point from the rocket is.

7. A piece of paper in the shape of a sector of a circle of radius 10 cm and of angle 216° just covers the lateral surface of a right circular cone of vertical angle 2θ. Then $\sin\theta$ is.

8. A man standing between two vertical posts finds that the angle subtended at his eyes by the tops of the posts is a right angle. If the heights of the two posts are two times and four times the height of the man, and the distance between them is equal to the length of the longer post, then the ratio of the distances of the man from the shorter and the longer post is.

AARAMBH (SOLVED EXAMPLES)

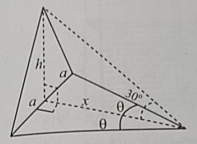
1. An isosceles triangle of wood is placed in a vertical plane, vertex upwards and faces the sun. If $2a$ be the base of the triangle, h its height and 30° the altitude of the sun, then the tangent of the angle at the apex of the shadow is

(a) $\frac{2ah\sqrt{3}}{3h^2 - a^2}$ (b) $\frac{2ah\sqrt{3}}{3h^2 + a^2}$

(c) $\frac{ah\sqrt{3}}{h^2 - a^2}$ (d) None of these

Sol. $x = h\cot 30^\circ = h\sqrt{3}$

$\tan\theta = \frac{a}{x} = \frac{a}{h\sqrt{3}}$



$$\Rightarrow \tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta} = \frac{2a}{h\sqrt{3}} \cdot \frac{3h^2}{3h^2 - a^2} = \frac{2ah\sqrt{3}}{3h^2 - a^2}$$

Therefore, option (*a*) is the correct answer.

2. A person standing at the foot of a tower walks a distance $3a$ away from the tower and observes that the angle of elevation of the top of the tower is α. He then walks a distance $4a$ perpendicular to the previous direction and observes the angle of elevation to be β. The height of the tower is

(a) $3a\tan\alpha$ (b) $5a\tan\beta$

(c) $4a\tan\beta$ (d) $7a\tan\beta$

Sol. $QB^2 = QA^2 + AB^2$

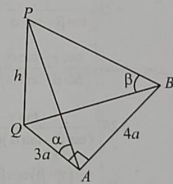
$\Rightarrow (h\cot\beta)^2 = (3a)^2 + (4a)^2$

$\Rightarrow h^2 = 25a^2\tan^2\beta$

$\Rightarrow h = 5a\tan\beta$

Also, $3a = h\cot\alpha$

$\Rightarrow h = 3a\tan\alpha$

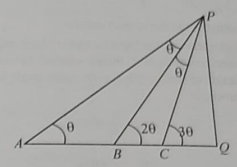


Therefore, option (*a*, *b*) are the correct answers.

3. A tower subtends angle θ, 2θ and 3θ at 3 points A, B, C respectively, lying on a horizontal line through the foot of the tower then the ratio $\frac{AB}{BC}$ equals to

(a) $\frac{\sin 3\theta}{\sin\theta}$ (b) $\frac{\sin\theta}{\sin 3\theta}$ (c) $\frac{\cos 3\theta}{\cos\theta}$ (d) $\frac{\tan\theta}{\tan 3\theta}$

Sol.



$\frac{AB}{BC} = \frac{BP}{BC} = \frac{\sin(180^\circ - 3\theta)}{\sin\theta}$

(From isosceles triangle ABP and sine rule in ΔBCP)

$\Rightarrow \frac{AB}{BC} = \frac{\sin 3\theta}{\sin\theta}$

Therefore, option (*a*) is the correct answer.

4. A 6-ft-tall man finds that the angle of elevation of the top of a 24-ft-high pillar and the angle of depression of its base are complementary angles. The distance of the man from the pillar is

(a) $2\sqrt{3}$ ft

(b) $8\sqrt{3}$ ft

(c) $6\sqrt{3}$ ft

(d) None of these

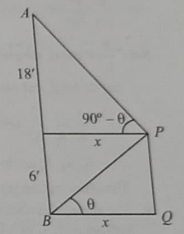
Sol. Here, $\tan\theta = \frac{6}{x}$,

$\tan(90^\circ - \theta) = \frac{18}{x}$

$\therefore \frac{6}{x} \cdot \frac{18}{x} = 1$

or $x^2 = 6 \times 18$

or $x = 6\sqrt{3}$



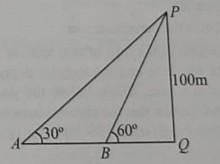
Therefore, option (*c*) is the correct answer.

5. A man from the top of a 100 metres high tower sees a car moving towards the tower at an angle of depression of 30°. After some time, the angle of depression becomes 60°. The distance (in metres) travelled by the car during this time is

(a) $100\sqrt{3}$ (b) $\frac{200\sqrt{3}}{3}$

(c) $\frac{100\sqrt{3}}{3}$ (d) $200\sqrt{3}$

Sol.



BOOKWORMCBSE

$AB = 100\cot 30° - 100\cot 60°$

$100\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{200}{\sqrt{3}} = \frac{200\sqrt{3}}{3}$

Therefore, option (*b*) is the correct answer.

6. The angle of elevation of the top of a tower observed from each of the three points *A*, *B*, *C* on the ground, forming an equilateral triangle of side length *a*, is the same angle α. The height of the tower is

(*a*) $a \sin\alpha$ (*b*) $\frac{a}{\sqrt{3}}\sin\alpha$

(*c*) $a\tan\alpha$ (*d*) $\frac{a}{\sqrt{3}}\tan\alpha$

Sol. Obviously foot of the tower is the circumcentre of the triangle *ABC*. The circum radius of the circumcircle of

$\Delta ABC = \frac{a}{2\sin 60°} = \frac{a}{\sqrt{3}}$

Hence height of the tower $= \frac{a}{\sqrt{3}}\tan\alpha$

Therefore, option (*d*) is the correct answer.

7. Three vertical poles of heights h_1, h_2 and h_3 at the vertices *A*, *B* and *C* of a ΔABC subtend angles α, β and γ respectively at the circumcentre of the triangle. If cotα, cotβ and cotγ are in *AP* then h_1, h_2, h_3 are in

(*a*) *AP* (*b*) *GP*

(*c*) *HP* (*d*) None of these

Sol. $\frac{R}{h_1} = \cot\alpha$, etc., where *R* is the circumradius

cotα, cotβ, cotγ are in *A.P.*

$\Rightarrow \frac{1}{h_1}, \frac{1}{h_2}, \frac{1}{h_3}$ are in *A.P.*

$\Rightarrow h_1, h_2, h_3$ are in *H.P.*

Therefore, option (*c*) is the correct answer.

8. The angle of elevation of the top of a hill from each of the vertices *A*, *B*, *C* of a horizontal triangle is α. The height of the hill is

(*a*) *b* tanα. cosec β (*b*) $\frac{1}{2}$ a tanα cosec*A*

(*c*) $\frac{1}{2}$ *c* tanα.cosec*C* (*d*) None of these

Sol. The distance of the foot from each vertex = $h\cot\alpha$

$\therefore$ the foot is at the circumcentre of the triangle.

So, $R = h\cot\alpha$

$\therefore h = R\tan\alpha = \frac{a}{2\sin A}\tan\alpha$

Therefore, option (*b*) is the correct answer.

9. If the angular elevations of the tops of two spheres which appear in a straight line is α and the angular depression of their reflections in a lake, *h* feet below the point of observation are β and γ, show that the distance between the spheres is $2h\cos^2\alpha \sin(\gamma-\beta)\operatorname{cosec}(\beta-\alpha)\operatorname{cosec}(\gamma-\alpha)$ ft where $\gamma > \beta$.

Sol. Let *P* and *Q* be the tops of two spheres, *P'* and *Q'* be their reflections. From question, $OA = h$

Let $BP = BP' = h_1$

$CQ = CQ' = h_2$

Let the distance between the spheres be *x*.

We have to find *x*.

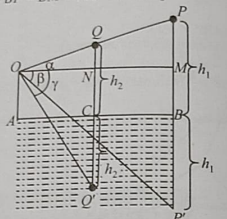
Clearly $x = MN = OM - ON$

From right angled triangle *OMP'*

$\tan\beta = \frac{P'M}{OM} = \frac{h_1 + h}{OM}$ or,

$OM\tan\beta = h_1 + h = (h + PM) + h$

$[\because h_1 = BP = BM + PM = h + PM]$



or, $OM\tan\beta = 2h + PM = 2h + OM\tan\alpha$

$\left[\because \text{From triangle PMO, } \tan\alpha = \frac{PM}{OM}\right]$

or, $OM(\tan\beta - \tan\alpha) = 2h \Rightarrow OM = \frac{2h}{\tan\beta - \tan\alpha}$ ...(i)

Similarly $ON = \frac{2h}{\tan\gamma - \tan\alpha}$

Hence $x = OM - ON$

$= \frac{2h}{\tan\beta - \tan\alpha} - \frac{2h}{\tan\gamma - \tan\alpha}$

$= 2h\left[\frac{1}{\tan\beta - \tan\alpha} - \frac{1}{\tan\gamma - \tan\alpha}\right]$

$= 2h\left[\frac{\tan\gamma - \tan\alpha - \tan\beta + \tan\alpha}{(\tan\beta - \tan\alpha)(\tan\gamma - \tan\alpha)}\right]$

$= 2h\left[\frac{\tan\gamma - \tan\beta}{(\tan\beta - \tan\alpha)(\tan\gamma - \tan\alpha)}\right]$

$= 2h\left[\frac{\sin(\gamma-\beta)\cos\beta\cos\alpha\cos\gamma\cos\alpha}{\cos\gamma.\cos\beta\sin(\beta-\alpha)\sin(\gamma-\alpha)}\right]$

$= 2h\cos^2\alpha \sin(\gamma-\beta)\operatorname{cosec}(\beta-\alpha)\operatorname{cosec}(\gamma-\alpha)$ ft.

10. A pole stands vertically on the center of a square. When α is the elevation of the sun its shadow just reaches the side of the square and is at a distance *x* and *y* from the ends of that side. Show that the height of the pole is $\sqrt{\frac{x^2+y^2}{2}}\tan\alpha$.

Sol. Let *O* be the center of the square, *OP* the pole. Shadow of the pole *OP* is *OQ*. From question $BQ = y$ and $CQ = x$.

Then, $BC = x + y$

Let $OR \perp BC$

$\therefore OR = \frac{x+y}{2}$ and $BR = \frac{x+y}{2}$

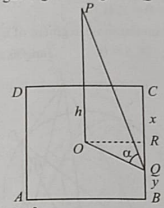
$RQ = \frac{x+y}{2} - y = \frac{x-y}{2}$

Let *h* be the height of the pole.

From right angled triangle *POQ*,

$\tan\alpha = \frac{h}{OQ}$ $\therefore OQ = h\cot\alpha$

Now, from right angled triangle *ORQ*.



$OQ^2 = OR^2 + RQ^2$

or, $h^2\cot^2\alpha = \left(\frac{x+y}{2}\right)^2 + \left(\frac{x-y}{2}\right)^2$

or, $h^2\cot^2\alpha = \frac{2(x^2+y^2)}{4}$ $\therefore h = \sqrt{\frac{x^2+y^2}{2}}.\tan\alpha$

11. A man observes a tower *AB* of height *h* from a point *P* on the ground. He moves a distance *d* towards the foot of the tower and finds that the angle of elevation has doubled. He further moves a distance $\frac{3}{4}d$ in the same direction and finds that the angle of elevation is three times that at *P*. Prove that $36h^2 = 35d^2$.

Sol. Let $AB = h$, $PQ = d$ and $QR = \frac{3}{4}d$.

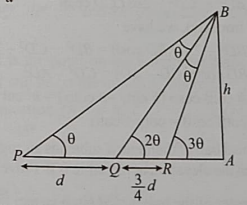
$\because \angle BPQ = \angle PBQ = \theta$

$\therefore PQ = QB = d$

[Here angles are given, hence we select right angled triangles].

From right angled triangle *BAQ*.

$\sin 2\theta = \frac{h}{d}$ $\therefore h = d\sin 2\theta$



or, $h = 2d\sin\theta\cos\theta$ ...(*i*)

Now, applying sine rule in triangle *BQR*,

$\frac{\frac{3}{4}d}{\sin\theta} = \frac{d}{\sin(180° - 3\theta)}$

or, $\frac{3}{4\sin\theta} = \frac{1}{3\sin\theta - 4\sin^3\theta}$

or, $9 - 12\sin^2\theta = 4$ $[\because \sin\theta \neq 0]$

or, $\sin^2\theta = \frac{5}{12}$

$\therefore \cos^2\theta = \frac{7}{12}$

From (i), we have $h^2 = 4d^2\sin^2\theta\cos^2\theta$

or, $h^2 = 4d^2.\frac{5}{12}.\frac{7}{12}$ or, $36h^2 = 35d^2$

12. A man observes that when he moves up a distance *c* meters on a slope, the angle of depression of a point on the horizontal plane from the base of the slope is 30°; and when moves up further a distance *c* metres, then angle of depression of the point is 45°. Obtain the angle of inclination of the slope with the horizontal.

Sol. Let the slope be *OB*, making an angle θ with the horizontal *OQ*, i.e. $\angle BOQ = \theta$.

Let *P* be the point on the horizontal plane through the base of the slope.

From question $OA = c$ and $AB = c$.

Also $\angle XAP = 30° = \angle APO$

and $\angle YBP = 45° = \angle BPO$.

Clearly, $\angle XAO = \angle AOQ = \theta$

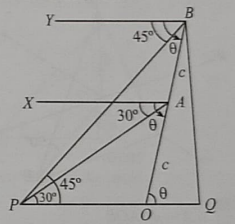
$\therefore \angle PAO = \theta - 30°$

and $\angle YBO = \angle BOQ = \theta$

$\therefore \angle PBO = \theta - 45°$

Suppose $OP = x$. We have to find θ.

Note: since given angles are not contained in right angled triangles we will use sine rule.



Now, applying sine rule in triangle *POA*.

$\frac{x}{\sin(\theta - 30°)} = \frac{c}{\sin 30°}$

$\therefore x = \frac{c\sin(\theta - 30°)}{\sin 30°}$ ...(*i*)

BOOKWORMCBSE

Applying sine rule in triangle POB,

$$\frac{x}{\sin(\theta-45^\circ)}=\frac{2c}{\sin 45^\circ}$$

$$\therefore x=\frac{2c\sin(\theta-45^\circ)}{\sin 45^\circ} \quad \ldots (ii)$$

From (i) and (ii), we get

$$\frac{c\sin(\theta-30^\circ)}{\sin 30^\circ}=\frac{2c\sin(\theta-45^\circ)}{\sin 45^\circ}$$

or, $\frac{1}{\sqrt{2}}\left(\sin\theta\cdot\frac{\sqrt{3}}{2}-\cos\theta\cdot\frac{1}{2}\right)$

$$=2\cdot\frac{1}{2}\left(\sin\theta\cdot\frac{1}{\sqrt{2}}-\cos\theta\frac{1}{\sqrt{2}}\right)$$

$$\tan\theta=\frac{1}{2-\sqrt{3}} \Rightarrow \theta=\tan^{-1}\left(\frac{1}{2-\sqrt{3}}\right)$$

13. A tower is observed from two stations A and B, where B is east of A at a distance 100 metres. The tower is due north of and due north west of B. The angle of elevation of the tower from A and B are complementary. Find the height of the tower.

Sol. Let OP be the tower of height h. A is a point due south of the tower and the angle of elevation of the tower at B will be $(90^\circ-\theta)$ (complementary of θ).

i.e. $\angle PAO=\theta$; $\angle PBO=90^\circ-\theta$

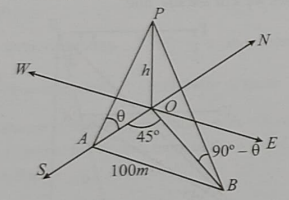
From right angled triangle POA.

$$\tan\theta=\frac{h}{OA} \quad \ldots (i)$$

From right angle triangle POB,

$$\tan(90^\circ-\theta)=\frac{h}{OB} \text{ or, } \tan\theta=\frac{OB}{h} \quad \ldots (ii)$$

From (i) and (ii), we have



$$\frac{h}{OA}=\frac{OB}{h} \therefore h=\sqrt{OA.OB} \quad \ldots (iii)$$

Since BO is north-west, OB will be south-east

$\therefore \angle AOB=45^\circ=\angle OBA$ $[\because \angle OAB=90^\circ]$

$\Rightarrow OA=AB=100$ m

Now, From right angled triangle OAB,

$OB^2=OA^2+AB^2=100^2+100^2$

$\therefore OB=100\sqrt{2}$ m From (iii)

$h=\sqrt{100.100\sqrt{2}}=100(2)^{\frac{1}{4}}$ metres

14. A tower stands on the edge of the circular lake $ABCD$. The foot of the tower is at D and the angle of elevation of the top at A, B, C are respectively α, β, γ. If $\angle ACB=\theta=\angle BAC$. Show that $2\cos\theta\cot\beta=\cot\alpha+\cot\gamma$

Sol. Let A, B, C be there points on the circular lake and DP be the tower of height h standing at D.

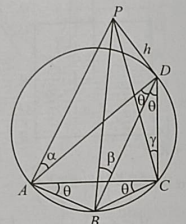
$\angle PAD=\alpha$, $\angle PBD=\beta$

and $\angle PCD=\gamma$

Also $\angle BAC=\theta=\angle ACB$

Since, angles on the same segment of a circle are equal.

$\therefore \angle ADB=\angle ACB=\theta$ [angles on the segment AB]



From right angled ΔPDA,

$$\tan\alpha=\frac{h}{AD} \quad \therefore AD=h\cot\alpha \quad \ldots (i)$$

Similarly, $BD=h\cot\beta$ $\ldots (ii)$

and $CD=h\cot\gamma$ $\ldots (iii)$

In triangle ABC,

$\angle BAC=\angle ACB=\theta$

$\therefore AB=BC$

or, $AB^2=BC^2$ $\ldots (iv)$

Applying cosine rule in triangle ADB.

$$\cos\theta=\frac{AD^2+BD^2-AB^2}{2AD.BD}$$

or, $AB^2=AD^2+BD^2-2AD.BD\cos\theta$

Similarly from triangle BDC,

$BC^2=BD^2+CD^2-2BD.CD\cos\theta$

Hence from (iv), we have

$AD^2+BD^2-2AD.BD\cos\theta=BD^2+CD^2-2BD.CD\cos\theta$

or, $2BD\cos\theta\,[CD-AD]=CD^2-AD^2$

or, $2.h\cot\beta\cos\theta\,(h\cot\gamma-h\cot\alpha]=h^2\cot^2\gamma-h^2\cot^2\alpha$

or, $2\cot\beta\cos\theta=\cot\gamma+\cot\alpha$

15. A spherical ball of diameter d subtends an angle α at a man's eye when the elevation of its centre is β. Prove that the height of the center of the ball is $\frac{1}{2}d\sin\beta\,\text{cosec}\left(\frac{\alpha}{2}\right)$.

Sol. Let O be the center of the ball and E, the man's eye. EA and EB are the tangents covering the ball and hence

$\angle AEB=\alpha$

Draw $OO'\perp EO'$

$\angle OEO'=\beta$

(From question)

Let $OO'=h$

From right angled triangle $OO'E$,

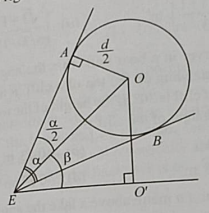
$$\sin\beta=\frac{h}{EO} \quad \ldots (i)$$

$\therefore h=EO\sin\beta$

From geometry, it is clear that

$\angle AEO=\angle BEO=\frac{\alpha}{2}$ $[\angle AEB=\alpha]$

Now, From right angled triangle OAE,



$$\sin\frac{\alpha}{2}=\frac{\frac{d}{2}}{OE} \text{ or, } OE=\frac{1}{2}d\,\text{cosec}\frac{\alpha}{2} \quad \ldots (ii)$$

From (i) and (ii), $h=\frac{1}{2}d\ \sin\beta\ \text{cosec}\ \frac{\alpha}{2}$

16. A right circular cylindrical tower of height h and radius r stands on a horizontal plane. Let A be a point in the horizontal plane and PQR be the semi-circular edge of the top of the tower such that Q is the point in it nearest to A. The angles of elevation of the points P and Q are 45° and 60° respectively.

Show that $\frac{h}{r}=\frac{\sqrt{3}\left(1+\sqrt{5}\right)}{2}$.

Sol. Let C be the center of the top of the circular tower.

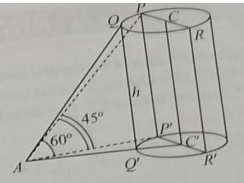
Draw QQ', PP' CC' and RR' perpendicular to the horizontal plane.

Since A is the point on the horizontal plane nearest to Q, hence A will be on the line $Q'A$, where $Q'A\perp QQ'$.

From question $QQ'=h$, $C'Q'=r$, $\angle QAQ'=60^\circ$ and $\angle PAP'=45^\circ$

From right angled triangle $QQ'A$

$$\tan60^\circ=\frac{h}{AQ'}, \quad \therefore AQ'=\frac{h}{\sqrt{3}} \quad \ldots (i)$$



From right angle triangle $PP'A$,

$$\tan45^\circ=\frac{h}{AP'}, \quad \therefore AP'=h \quad \ldots (ii)$$

Now, $AC'=AQ'+Q'C'=\frac{h}{\sqrt{3}}+r$ $\ldots (iii)$

$C'P'=r$ $\ldots (iv)$

and $\angle AC'P'=90^\circ$

From right angled triangle $AC'P'$, $AP'^2=AC'^2+C'P'^2$

or, $h^2=\left(\frac{h}{\sqrt{3}}+r\right)^2+r^2$

or, $h^2=\frac{h^2}{3}+2\frac{h}{\sqrt{3}}.r+r^2+r^2$

or, $2h^2-2\sqrt{3}\ hr-6r^2=0$ or, $h^2-\sqrt{3}\ rh-3r^2=0$

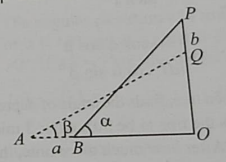
$$\therefore h=\frac{\sqrt{3}r\pm\sqrt{3r^2-4.1\left(-3r^2\right)}}{2}$$

or, $h=\frac{\sqrt{3}r\left(1\pm\sqrt{5}\right)}{2}$

As $h>0 \Rightarrow \frac{h}{r}=\frac{\sqrt{3}\left(1+\sqrt{5}\right)}{2}$

17. A ladder rests against a wall at an angle α to the horizontal. Its foot is pulled away from the wall through a distance a, so that it slides a distance b down the wall making an angle β with the horizontal. Show that $a=b\tan\frac{1}{2}(\alpha+\beta)$

Sol.



Let l be the length of the ladder.

$a=OA-OB=l\cos\beta-l\cos\alpha$

$b=OP-OQ=l\sin\alpha-l\sin\beta$

$$\Rightarrow \frac{a}{b}=\frac{\cos\beta-\cos\alpha}{\sin\alpha-\sin\beta}=\frac{2\sin\frac{\alpha+\beta}{2}\cdot\sin\frac{\alpha-\beta}{2}}{2\sin\frac{\alpha-\beta}{2}\cos\frac{\alpha+\beta}{2}}$$

$$\Rightarrow a=b\tan\frac{\alpha+\beta}{2}$$

BOOKWORMCBSE

1. The angle of elevation of the top of a tower at point on the ground is 30°. If on walking 20 metres toward the tower, the angle of elevation become 60°, then the height of the tower is

(a) 10 meter (b) $\frac{10}{\sqrt{3}}$ meter

(c) $10\sqrt{3}$ meter (d) $100\sqrt{3}$ meter

2. The angle of elevation of the top of the tower observed from each of the three points A, B, C on the ground, forming a triangle is the same angle α. If R is the circum-radius of the triangle ABC, then the height of the tower is

(a) R sin α (b) R cos α

(c) R cot α (d) R tan α

3. A person standing on the bank of a river observes that the angle subtended by a tree on the opposite bank is 60°. When he retires 40 meters from the bank, he finds the angle to be 30°. The breadth of the river is

(a) 20 m (b) 40 m

(c) 30 m (d) 60 m

4. A vertical pole consists of two parts, the lower part being one third of the whole. At a point in the horizontal plane through the base of the pole and distance 20 meters from it, the upper part of the pole subtends an angle whose tangent is $\frac{1}{2}$. The possible heights of the pole are

(a) 20 m and $20\sqrt{3}$ m (b) 20 m and 60 m

(c) 16 m and 48 m (d) 20 m and 16 m

5. From a 60 meter high tower angles of depression of the top and bottom of a house are α and β respectively. If the height of the house is $\frac{60\sin(\beta-\alpha)}{x}$, then x =

(a) sin α sin β (b) cos α cos β

(c) sin α cos β (d) cos α sin β

6. An observer on the top of a tree, finds the angle of depression of a car moving towards the tree to be 30°. After 3 minutes this angle becomes 60°. After how much more time, the car will reach the tree

(a) 4 min. (b) 4.5 min.

(c) 1.5 min. (d) 2 min.

7. A house of height 100 metres subtends a right angle at the window of an opposite house. If the height of the window be 64 metres, then the distance between the two houses is

(a) 48 m (b) 36 m

(c) 54 m (d) 72 m

8. The length of the shadow of a pole inclined at 10° to the vertical towards the sun is 2.05 metres, when the elevation of the sun is 38°. The length of the pole is

(a) $\frac{2.05\sin 38^\circ}{\sin 42^\circ}$ (b) $\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)30$ m

(c) $\frac{2.05\cos 38^\circ}{\cos 42^\circ}$ (d) $\frac{2.05\cos 42^\circ}{\sin 38^\circ}$

9. From the top of a light house 60 meters high with its base at the sea level, the angle of depression of a boat is 15°. The distance of the boat from the foot of light house is

(a) $\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)60$ m (b) $\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)60$ m

(c) $\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$ m (d) $\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)30$m

10. An observer in a boat finds that the angle of elevation of a tower standing on the top of a cliff is 60° and that of the bottom of cliff is 30°. If the height of the tower be 60 meters, then the height of the cliff is

(a) 30 m (b) $40\sqrt{3}$ m

(c) $20\sqrt{3}$ m (d) $40\sqrt{3}$ m

11. From a point a metre above a lake the angle of elevation of a cloud is α and the angle of depression of its reflection is β. The height of the cloud is

(a) $\frac{a\sin(\alpha+\beta)}{\sin(\alpha-\beta)}$ metre (b) $\frac{a\sin(\alpha+\beta)}{\sin(\beta-\alpha)}$ metre

(c) $\frac{a\sin(\beta-\alpha)}{\sin(\alpha+\beta)}$ metre (d) $\frac{2a\sin(\beta-\alpha)}{\sin(\alpha+\beta)}$

12. The shadow of a tower is found to be 60 metre shorter when the sun's altitude changes from 30° to 60°. The height of the tower from the ground is approximately equal to

(a) 62 m (b) 301 m

(c) 101 m (d) 52 m

13. If the angles of elevation of two towers from the middle point of the line joining their feet be 60° and 30° respectively, then the ratio of their heights is

(a) 2 : 1 (b) 1 : $\sqrt{2}$

(c) 3 : 1 (d) 1 : $\sqrt{3}$

14. Some portion of a 20 meters long tree is broken by the wind and the top struck the ground at an angle of 30°. The height of the point where the tree is broken is

(a) 10 m (b) $(2\sqrt{3}-3)20$ m

(c) $\frac{20}{3}$m (d) $10\sqrt{3}$ m

15. The base of a cliff is circular. From the extremities of a diameter of the base the angles of elevation of the top of the cliff are 30° and 60°. If the height of the cliff be 500 metres, then the diameter of the base of the cliff is

(a) $1000\sqrt{3}$ m (b) $\frac{2000}{\sqrt{3}}$m

(c) $\frac{1000}{\sqrt{3}}$m (d) $2000\sqrt{2}$ m

16. The angle of elevation of the top of a tower from the top of a house is 60° and the angle of depression of its base is 30°. If the horizontal distance between the house and the tower be 12 m, then the height of the tower is

(a) $48\sqrt{3}$ m (b) $16\sqrt{3}$ m (c) $24\sqrt{3}$ m (d) $\frac{16}{\sqrt{3}}$m

17. A man whose eye level is 1.5 metres above the ground observes the angle of elevation of a tower to be 60°. If the distance of the man from the tower be 10 meters, the height of the tower is

(a) $(1.5+10\sqrt{3})$m (b) $10\sqrt{3}$m

(c) $\left(1.5+\frac{10}{\sqrt{3}}\right)$m (d) $100\sqrt{3}$m

18. A tower subtends an angle of 30° at a point distant d from the foot of the tower and on the same level as the foot of the tower. At a second point h vertically above the first, the depression of the foot of the tower is 60°. The height of the tower is

(a) $\frac{h}{3}$ (b) $\frac{h}{3d}$ (c) $3h$ (d) $\frac{3h}{d}$

19. A tree is broken by wind, its upper part touches the ground at a point 10 metres from the foot of the tree and makes an angle of 45° with the ground. The total length of tree is

(a) 15 metres (b) 20 metres

(c) $10(1+\sqrt{2})$ metres (d) $10\left(1+\frac{\sqrt{3}}{2}\right)$ metres

20. The angle of elevation of a stationary cloud from a point 2500 m above a lake is 15° and the angle of depression of its reflection In the lake is 45°. The height of cloud above the lake level is

(a) $2500\sqrt{3}$ metres (b) 2500 metres

(c) $500\sqrt{3}$ metres (d) 5000 metres

21. From an aeroplane vertically over a straight horizontally road, the angles of depression of two consecutive mile stones on opposite sides of the aeroplane are observed to be α and β, then the height in miles of aeroplane above the road is

(a) $\frac{\tan\alpha.\tan\beta}{\cot\alpha+\cot\beta}$ (b) $\frac{\tan\alpha+\tan\beta}{\tan\alpha.\tan\beta}$

(c) $\frac{\cot\alpha+\cot\beta}{\tan\alpha.\tan\beta}$ (d) $\frac{\tan\alpha.\tan\beta}{\tan\alpha+\tan\beta}$

22. A flag-post 20 m high standing on the top of a house subtends an angle whose tangent is $\frac{1}{6}$ at a distance 70 m from the foot of the house. The height of the house is

(a) 30 m (b) 60 m

(c) 50 m (d) 20 m

23. A balloon is coming down at the rate of 4 m/min. and its angle of elevation is 45° from a point on the ground which has been reduced to 30° after 10 minutes. Balloon will be on the ground at a distance of how many meters from the observer

(a) $20\sqrt{3}$ m (b) $20(3+\sqrt{3})$m

(c) $10(3+\sqrt{3})$m (d) $10(3-\sqrt{3})$m

24. AB is a vertical pole resting at the end A on the level ground. P is a point on the level ground such that $AP = 3\,AB$. If C is the mid-point of AB and CB subtends an angle β at P, the value of tan β is

(a) $\frac{18}{19}$ (b) $\frac{3}{19}$

(c) $\frac{1}{6}$ (d) $\frac{1}{64}$

25. Two straight roads intersect at an angle of 60°. A bus on one road is 2 km away from the intersection and a car on the other road is 3 km away from the intersection. Then the direct distance between the two vehicles is

(a) 1 km (b) $\sqrt{2}$ km

(c) 4 km (d) $\sqrt{7}$ km

26. The angle of elevation of a cliff at a point A on the ground and a point B, 100 m vertically at A are α and β respectively. The height of the cliff is

(a) $\frac{100\cot\alpha}{\cot\alpha-\cot\beta}$ (b) $\frac{100\cot\beta}{\cot\alpha-\cot\beta}$

(c) $\frac{100\cot\beta}{\cot\beta-\cot\alpha}$ (d) $\frac{100\cot\beta}{\cot\beta+\cot\alpha}$

27. The angular elevation of a tower CD at a point A due south of it is 60° and at a point B due west of A, the elevation is 30°. If AB = 3 km, the height of the tower is

(a) $2\sqrt{3}$ km (b) $2\sqrt{6}$ km

(c) $\frac{3\sqrt{3}}{2}$km (d) $\frac{3\sqrt{6}}{4}$km

28. The angles of elevation of the top of a tower (a) from the top (b) and bottom (d) at a building of height a are 30° and 45° respectively. If the tower and the building stand at the same level, then the height of the tower is

(a) $a\sqrt{3}$ (b) $\frac{a\sqrt{3}}{\sqrt{3}-1}$

(c) $\frac{a(3+\sqrt{3})}{2}$ (d) $a(\sqrt{3}-1)$

BOOKWORMCBSE

29. The top of a hill observed from the top and bottom of a building of height h is at the angle of elevation p and q respectively. The height of the hills is

(a) $\frac{h\cot q}{\cot q-\cot p}$ (b) $\frac{h\cot p}{\cot p-\cot q}$

(c) $\frac{h\tan p}{\tan p-\tan q}$ (d) $\frac{h\cot p}{\cot q+\cot p}$

30. For a man, the angle of elevation of the highest point of the temple situated east of him is 60°. On walking 240 metres to north, the angle of elevation is reduced to 30°, then the height of the temple is

(a) $60\sqrt{6}$ m (b) 60 m

(c) $50\sqrt{3}$ m (d) $30\sqrt{6}$ m

31. A tower subtends angles α, 2α, 3α respectively at points A, B and C, all lying on a horizontal line through the foot of the tower. Then AB/BC =

(a) $\frac{\sin 3\alpha}{\sin 2\alpha}$ (b) $1+2\cos 2\alpha$

(c) $2+\cos 3\alpha$ (d) $\frac{\sin 2\alpha}{\sin \alpha}$

32. Two pillars of equal height stand on either side of a road, which is 60 metres wide. At a point in the roadway between the pillars, the elevation of the top of pillars are 60° and 30°. The height of the pillars is

(a) $15\sqrt{3}$ m (b) $\frac{15}{\sqrt{3}}$ m

(c) 15 m (d) 20 m

33. A ladder rests against a wall making an angle α with the horizontal. The foot of the ladder is pulled away from the wall through a distance x, so that it slides a distance y down the wall making an angle β with the horizontal. The correct relation is

(a) $x=y\tan\frac{\alpha+\beta}{2}$ (b) $y=x\tan\frac{\alpha+\beta}{2}$

(c) $x=y\tan(\alpha+\beta)$ (d) $y=x\tan(\alpha+\beta)$

PRABAL (JEE MAIN LEVEL)

1. A pole 25m long stands on the top of a tower 225m high. If θ is the angle subtended by the pole at a point on the ground which is at a distance of 2.25 km from the foot of the tower, then tanθ is equal to

(a) $\frac{1}{90}$ (b) $\frac{1}{91}$ (c) $\frac{1}{10}$ (d) $\frac{1}{9}$

2. The angle of elevation of the top of a vertical pole when observed from each vertex of a regular hexagon is $\frac{\pi}{3}$. If the area of the circle circumscribing the hexagon be A metre^2 then the area of the hexagon is

(a) $\frac{3\sqrt{3}}{8}A$ metre^2 (b) $\frac{\sqrt{3}}{\pi}A$ metre^2

(c) $\frac{3\sqrt{3}}{4\pi}A$ metre^2 (d) $\frac{3\sqrt{3}}{2\pi}A$ metre^2

3. A vertical pole PO is standing at the center O of a square $ABCD$. If AC subtends an angle 90° at the top P of the pole then the angle subtended by a side of the square at P is

(a) 45° (b) 30° (c) 60° (d) 90°

4. The angles of elevation of the top of a tower standing on a horizontal plane, from two points on a line passing through its foot at distances a and b respectively, are complementary angles. If the line joining the two points subtends an angle θ at the top of the tower, then

(a) $\sin\theta=\frac{a-b}{a+b}$ (b) $\tan\theta=\frac{2\sqrt{ab}}{a-b}$

(c) $\sin\theta=\frac{a+b}{a-b}$ (d) $\cos\theta=\frac{2\sqrt{ab}}{a-b}$

5. A vertical lamp-post, 6 m high, stands at a distance 2 m from a wall, 4 m high. A 1.5-m-tall man starts to walk away from the wall on the other side of the wall, in line with the lamp-post. The maximum distance to which the man can walk remaining in the shadow is

(a) $\frac{5}{2}$m (b) $\frac{3}{2}$m (c) 4 m (d) $\frac{4}{3}$m

6. A circular ring of radius 3 cm is suspended horizontally from a point 4 cm vertically above the center by 4 strings attached at equal intervals to its circumference. If the angle between two consecutive strings be θ then cos θ is

(a) $\frac{4}{5}$ (b) $\frac{4}{25}$ (c) $\frac{16}{25}$ (d) $\frac{5}{26}$

7. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar is

(a) $\sqrt{3}:1$ (b) $1:3$

(c) $1:\sqrt{3}$ (d) $\sqrt{3}:2$

8. As seen from A, due west of a hill HL itself leaning to the angle of elevation of top H of the hill is 60°; and after walking a distance of one kilometer along an incline of 30° to a point B, it was seen that the hill LH was printed at right angles to AB, the height LH of the hill is

(a) $\frac{1}{\sqrt{3}}$km (b) $\sqrt{3}$

(c) $2\sqrt{3}$ km (d) $\frac{2}{\sqrt{3}}$km

9. ABC is a triangular park with $AB=AC=100$ metres. A clock tower is situated at the mid point of BC. The angles of elevation of the top of the tower at A and B are $\cot^{-1}3.2$ and $\operatorname{cosec}^{-1}2.6$ respectively. The height of the tower is

(a) 16 m (b) 25 m

(c) 50 m (d) 75 m

10. The angles of elevation of the top of a tower from the top and bottom of a building of height 'a' are 30° and 45°, respectively. If the tower and the building stand at the same level, the height of the tower is

(a) $a\sqrt{3}$ (b) $\frac{3\sqrt{3}a}{2}$

(c) $\frac{\sqrt{3}a}{\sqrt{3}-1}$ (d) $a(\sqrt{3}+1)$

11. A flag staff of 5 m high stands on a building of 25 m high. At an observer at a height of 30 m. the flag staff and the building subtend equal angles. The distance of the observer from the top of the flag staff is

(a) $\frac{5\sqrt{3}}{2}$ (b) $5\sqrt{\frac{3}{2}}$

(c) $5\sqrt{\frac{2}{3}}$ (d) $5\sqrt{\frac{2}{5}}$

12. If a flag-staff of 6 metres high placed on the top of a tower throws a shadow of $2\sqrt{3}$ metres along the ground then the angle (in degrees) that the sun makes with the ground is

(a) 60° (b) 30°

(c) 45° (d) 90°

13. The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance of 100 m. from its base is 45°. If the angle of elevation of the top of the complete pillar at the same point is to be 60°, then the height of the incomplete pillar is to be increased by

(a) $50\sqrt{2}$ m (b) 100 m

(c) $100(\sqrt{3}-1)$ m (d) $100(\sqrt{3}+1)$ m

14. The angles of elevation of a cliff at a point A on the ground and a point B, 100 meters vertically at A are α and β respectively. The height of the cliff is

(a) $\frac{100\cot\alpha}{\cot\alpha-\cot\beta}$ (b) $\frac{100\cot\beta}{\cot\alpha-\cot\beta}$

(c) $\frac{100\cot\beta}{\cot\beta-\cot\alpha}$ (d) $\frac{100\cot\beta}{\cot\beta+\cot\alpha}$

15. The angle of elevation of a cloud from a point h mt. above is θ and the angle of depression of its reflection in the lake is ϕ. Then the height is

(a) $\frac{h\sin(\phi-\theta)}{\sin(\phi+\theta)}$ (b) $\frac{h\sin(\phi+\theta)}{\sin(\phi-\theta)}$

(c) $\frac{h\sin(\theta+\phi)}{\sin(\theta-\phi)}$ (d) $\frac{h\sin(\phi-\theta)}{\sin(\theta-\phi)}$

16. On the level ground the angle of elevation of the top of a tower is 30°. On moving 20 m. nearer the tower, the angle of elevation is found to be 60°. The height of the tower is

(a) 10 m (b) 20 m

(c) $10\sqrt{3}$ m (d) $100\sqrt{3}$ m

17. Each side of a square subtends an angle of 60° at the top of a tower h metres high standing in the centre of the square. If a is the length of each side of the square, then

(a) $2a^2=h^2$ (b) $2h^2=a^2$

(c) $3a^2=2h^2$ (d) $3h^2=2a^2$.

18. From the top of light house 60 metres high with its base at the sea level, the angle of depression of a boat is 15°. The distance of the boat from the foot of the light house is

(a) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$.60 metres (b) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$.60 metres

(c) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$metres (d) $\sqrt{3}+1$ metres

19. A person standing on the bank of a river observes that the angle subtended by a tree on the opposite bank is 60°, when he retires 40 metres from the bank he finds the angle to be 30°. Then the breadth of the river is

(a) 40 m (b) 60 m (c) 20 m (d) 30 m

20. AB is a vertical pole. The end A is on the level ground. C is the middle point of AB. P is a point on the level ground. The portion BC subtends an angle β at P. If $AP=n\,AB$, then tan β =

(a) $\frac{n}{2n^2+1}$ (b) $\frac{n}{n^2-1}$

(c) $\frac{n}{n^2+1}$ (d) $\frac{n^2}{n+1}$

21. A tree is broken by wind, its upper part touches the ground at a point 10 metres from the foot of the tree and makes an angle of 45° with the ground. The entire length of the tree is

(a) 15 metres (b) 20 metres

(c) $10(1+\sqrt{2})$ metres (d) $\left(1+\frac{\sqrt{3}}{2}\right)$metres

22. An aeroplane flying at a height of 300 metres above the ground passes vertically above another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. The height of the lower plane from the ground (in metres) is

(a) $100\sqrt{3}$ (b) $\frac{100}{\sqrt{3}}$

(c) 50 (d) $150(\sqrt{3}+1)$

23. A tower subtends an angle α at a point A in the plane of its base and the angle of depression of the foot of the tower at a point b ft. just above A is β. Then height of the tower is

(a) $b\tan\alpha\cot\beta$ (b) $b\cot\alpha\tan\beta$

(c) $b\tan\alpha\tan\beta$ (d) $b\cot\alpha\cot\beta$

BOOKWORMCBSE

PYQ's (PAST YEAR QUESTIONS)

HEIGHTS AND DISTANCES

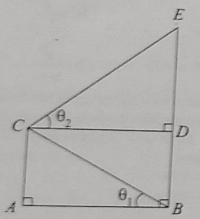
1. From the top A of a vertical wall AB of height 30 m, the angles of depression of the top P and bottom Q of a vertical tower PQ are 15° and 60° respectively. B and Q are on the same horizontal level. If C is a point on AB such that $CB = PQ$, then the area (in m^2) of the quadrilateral $BCPQ$ is equal to
[6 April, 2023 (Shift-I)]

(a) $600(\sqrt{3}-1)$ (b) $300(\sqrt{3}+1)$
(c) $200(3-\sqrt{3})$ (d) $300(\sqrt{3}-1)$

2. The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due west of the tower is 30°. If the height of the tower is 5 meters, then the distance (in meters) between the two persons is equal to
[11 April, 2023 (Shift-II)]

(a) 10 (b) 5 (c) $5\sqrt{5}$ (d) $\frac{5}{2}\sqrt{5}$

3. In the figure, $\theta_1+\theta_2=\frac{\pi}{2}$ and $\sqrt{3}(BE)=4(AB)$. If the area of ΔCAB is $2\sqrt{3}-3$ unit^2, when $\frac{\theta_2}{\theta_1}$ is the largest, then the perimeter (in unit) of ΔCED is equal to _____
[10 April, 2023 (Shift-II)]



4. The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is $\cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$. If the distance of the point B from the tower is 15 units, then cot α is equal to:
[29 July, 2022 (Shift-I)]

(a) $\frac{6}{5}$ (b) $\frac{9}{5}$ (c) $\frac{4}{3}$ (d) $\frac{7}{3}$

5. Let a vertical tower AB of height $2h$ stands on a horizontal ground. Let from a point P on the ground a man can see upto height h of the tower with an angle of elevation 2α. When from P, he moves a distance d in the direction of AP, he can see the top B of the tower with an angle of elevation α. If $d=\sqrt{7}h$, then tan α is equal to **[27 July, 2022 (Shift-I)]**

(a) $\sqrt{5}-2$ (b) $\sqrt{3}-1$
(c) $\sqrt{7}-2$ (d) $\sqrt{7}-\sqrt{3}$

6. From the base of a pole of height 20 meter, the angle of elevation of the top of a tower is 60°. The pole subtends an angle 30° at the top of the tower. Then the height of the tower is: **[29 June, 2022 (Shift-...)]**

(a) $15\sqrt{3}$ (b) $20\sqrt{3}$
(c) $20+10\sqrt{3}$ (d) 30

7. The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45°. If R be a point on AQ and from a point B, vertically above A, the angle of elevation of P is 60°. If $\angle BAQ = 30°$, AB and the area of the trapezium $PQRB$ is α, then the ordered pair (d, α) is: **[27 July, 2022 (Shift-...)]**

(a) $\left(10(\sqrt{3}-1),25\right)$ (b) $\left(10(\sqrt{3}-1),\frac{25}{2}\right)$
(c) $\left(10(\sqrt{3}+1),25\right)$ (d) $\left(10(\sqrt{3}+1),\frac{25}{2}\right)$

8. A horizontal park is in the shape of a triangle OAB in which $AB = 16$. A vertical lamp post OP is erected at the point O such that $\angle PAO = \angle PBO = 15°$ and $\angle PCO = 45°$, where C is the midpoint of AB. Then $(OP)^2$ is equal to **[28 July, 2022 (Shift-...)]**

(a) $\frac{32}{\sqrt{3}}(\sqrt{3}-1)$ (b) $\frac{32}{\sqrt{3}}(2-\sqrt{3})$
(c) $\frac{16}{\sqrt{3}}(\sqrt{3}-1)$ (d) $\frac{16}{\sqrt{3}}(2-\sqrt{3})$

9. Let AB and PQ be two vertical poles, 160m apart from each other. Let C be the middle point of B and Q, which are the feet of these two poles. Let $\frac{\pi}{8}$ and θ be the angles of elevation from C to P and A, respectively. If the height of pole PQ is twice the height of pole AB, then $\tan^2\theta$ is equal to **[28 June, 2022 (Shift-...)]**

(a) $\frac{3-2\sqrt{2}}{2}$ (b) $\frac{3+\sqrt{2}}{2}$
(c) $\frac{3-2\sqrt{2}}{4}$ (d) $\frac{3-\sqrt{2}}{4}$

10. A tower PQ stands on a horizontal ground with the foot Q on the ground. The point R divides the tower in two parts such that $QR = 15$ m. If from a point A on the ground the angle of elevation of R is 60° and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is: **[25 July, 2022 (Shift-...)]**

(a) $5(2\sqrt{3}+3)$m (b) $5(\sqrt{3}+3)$m
(c) $10(\sqrt{3}+1)$m (d) $10(2\sqrt{3}+1)$m

11. A pole stands vertically inside a triangular park ABC. Let the angel of elevation of the top of the pole from each corner of the park be $\frac{\pi}{3}$. If the radius of the circumcircle of ΔABC is 2, then the height of the pole is equal to:
[18 March, 2021 (Shift-II)]

(a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$ (c) $2\sqrt{3}$ (d) $\frac{2\sqrt{3}}{3}$

12. A spherical gas balloon of radius 16 meter subtends an angle 60° at the eye of the observer A while the angle of elevation of its center from the eye of A is 75°. Then the height (in meter) of the top most point of the balloon from the level of the observer's eye is: **[25 July, 2021 (Shift-I)]**

(a) $8\left(2+2\sqrt{3}+\sqrt{2}\right)$ (b) $8\left(\sqrt{6}+\sqrt{2}+2\right)$
(c) $8\left(\sqrt{6}-\sqrt{2}+2\right)$ (d) $8\left(\sqrt{2}+2+\sqrt{3}\right)$

13. Two vertical poles are 150m apart and height of one is three times that of other. If from the middle point of the line joining their feet, an observer finds the angle of elevation of their tops to be complementary, then height of shorter pole (in m) is: **[24 Feb, 2021 (Shift-I)]**

(a) 30 (b) 25 (c) $20\sqrt{3}$ (d) $25\sqrt{3}$

14. A man is observing from the top of tower a boat speeding towards the tower from a certain point A, with uniform speed. At that point, angle of depression of boat with man's eye is 30° (ignore man's height). After Sailing for 20 seconds, towards the base of tower (same level as of water), the boat has reached a point B, where angle of depression is 45°. Then time (in seconds) taken by boat from B to reach base of tower is:
[25 Feb, 2021 (Shift-I)]

(a) $10(\sqrt{3}+1)$ (b) $10(\sqrt{3}-1)$
(c) 10 (d) $10(\sqrt{3})$

15. Two poles, AB of length a metres and CD of length $a+b$ ($a \neq b$) metres are erected at same horizontal level with bases at B and D. If $BD = x$ and $\tan\angle ACB = \frac{1}{2}$, then,
[27 Aug, 2021 (Shift-II)]

(a) $x^2+2(a+2b)x-b(a+b)=0$
(b) $x^2+2(a+2b)x+a(a+b)=0$
(c) $x^2-2ax+b(a+b)=0$
(d) $x^2-2ax+a(a+b)=0$

16. The angle of elevation of jet plane from a point A on the ground is 60° After a flight of 20 seconds at the speed of 432 km/hr, the angle of elevation changes to 30°. If the jet plane is flying at a constant height, then it's height is:
[24 Feb, 2021 (Shift-II)]

(a) $3600\sqrt{3}$ m (b) $1200\sqrt{3}$ m
(c) $1800\sqrt{3}$ m (d) $2400\sqrt{3}$ m

17. A vertical pole fixed to the horizontal ground is divided in the ratio 3 : 7 by a mark on it with lower part shorter than the upper part. If the two parts subtend equal angles at a point on the ground 18 m away from the base of the pole, then the height of the pole (in meters) is **[31 Aug, 2021 (Shift-I)]**

(a) $8\sqrt{10}$ (b) $12\sqrt{15}$ (c) $12\sqrt{10}$ (d) $6\sqrt{10}$

18. The angle of elevation of the summit of a mountain from a point on the ground is 45°. After climbing up one km towards the summit at an inclination of 30° from the ground, the angle of elevation of the summit is found to be 60°. Then the height (in km) of the summit from the ground is:
[6 Sep, 2020 (Shift-II)]

(a) $\frac{1}{\sqrt{3}+1}$ (b) $\frac{1}{\sqrt{3}-1}$ (c) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (d) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

19. Two vertical poles $AB = 15$ m and $CD = 10$ m are standing apart on a horizontal ground with points A and C on the ground. If P is the point of intersection of BC and AD, then the height of P (in m) above the line AC is:
[4 Sep, 2020 (Shift-I)]

(a) 6 (b) 20/3 (c) 10/3 (d) 5

20. The angle of elevation of a cloud C from a point P, 200 m above a still lake is 30°. If the angle of depression of the image of C in the lake from the point P is 60°, then PC (in m) is equal to: **[4 Sep, 2020 (Shift-II)]**

(a) 400 (b) $400\sqrt{3}$
(c) 100 (d) $200\sqrt{3}$

21. The angle of elevation of the top of a hill from a point on the horizontal plane passing through the foot of the hill is found to be 45°. After walking a distance of 80 meters towards the top, up a slope inclined at an angle of 30° to the horizontal plane, the angle of elevation of the top of the hill becomes 75°. Then the height of the hill (in meters) is:
[6 Sep, 2020 (Shift-I)]

(a) 20 (b) 80 (c) 40 (d) 70

22. ABC is a triangular park with AB = AC = 100 metres. A vertical tower is situated at the mid point of BC. If the angles of elevation of the top of the tower at A and B are $\cot^{-1}(3\sqrt{2})$ and $\text{cosec}^{-1}(2\sqrt{2})$ respectively, then the height of the tower (in metres) is: **[10 April, 2019 (Shift-I)]**

(a) $10\sqrt{5}$ (b) $\frac{100}{3\sqrt{3}}$ (c) 20 (d) 25

23. Two poles standing on a horizontal ground are of heights 5m and 10 m respectively. The line joining their tops makes an angle of 15° with ground. Then the distance (in m) between the poles, is: **[9 April, 2019 (Shift-II)]**

(a) $\frac{5}{2}(2+\sqrt{3})$ (b) $5(\sqrt{3}+1)$
(c) $5(2+\sqrt{3})$ (d) $10(\sqrt{3}-1)$

24. The angle of elevation of the top of vertical tower standing on a horizontal plane is observed to be 45° from a point A on the plane. Let B be the point 30 m vertically above the point. If the angle of elevation of the top of the tower from B be 30°, then the distance (in m) of the foot of the tower from the point A is: **[12 April, 2019 (Shift-II)]**

(a) $15(3-\sqrt{3})$ (b) $15(3+\sqrt{3})$

(c) $15(1+\sqrt{3})$ (d) $15(5-\sqrt{3})$

25. Consider a triangular plot ABC with sides $AB = 7$ m, $BC = 5$m and $CA = 6$m. A vertical lamp-post at the mid point D of AC subtends an angle 30° at B. The height (in m) of the lamp-post is: **[10 Jan, 2019 (Shift-I)]**

(a) $\frac{3}{2}\sqrt{21}$ (b) $\frac{2}{3}\sqrt{21}$ (c) $2\sqrt{21}$ (d) $7\sqrt{21}$

26. Two vertical poles of heights, 20 m and 80 m stand on a horizontal plane. The height (in meters) of the point of intersection of the lines joining the top of each pole to the foot of the other, from this horizontal plane is: **[8 April, 2019 (Shift-I)]**

(a) 12 (b) 15

(c) 16 (d) 18

27. If the angle of elevation of a cloud from a point P is 25 m above a lake be 30° and the angle of depression reflection of the cloud in the lake from P be 60°, then height of the cloud (in meters) from the surface of the is **[12 Jan, 2019 (Shift-I)]**

(a) 60 (b) 50

(c) 45 (d) 42

Answer Key

CONCEPT APPLICATION

1. 109.28 m 2. 1 m 3. 20 m 4. $20\sqrt{3}$ m 5. $h = \frac{20}{\sqrt{3}}$ m.

6. $AP = 7\sqrt{3}$ m 7. $\sin\theta = \frac{3}{5}$ 8. 3 : 1 and 1 : 3

PRARAMBH (TOPICWISE)

1. (c)	2. (d)	3. (a)	4. (b)	5. (d)	6. (c)	7. (a)	8. (a)	9. (b)	10. (a)
11. (b)	12. (d)	13. (c)	14. (c)	15. (b)	16. (b)	17. (a)	18. (a)	19. (c)	20. (a)
21. (d)	22. (c)	23. (b)	24. (b)	25. (d)	26. (c)	27. (d)	28. (c)	29. (b)	30. (a)
31. (b)	32. (a)	33. (a)							

PRABAL (JEE MAIN LEVEL)

1. (b)	2. (d)	3. (c)	4. (a)	5. (a)	6. (c)	7. (c)	8. (a)	9. (b)	10. (c)
11. (b)	12. (a)	13. (c)	14. (c)	15. (b)	16. (c)	17. (b)	18. (b)	19. (c)	20. (a)
21. (c)	22. (a)	23. (a)							

PYQ's (PAST YEAR QUESTIONS)

1. (a)	2. (a)	3. [6]	4. (a)	5. (c)	6. (d)	7. (a)	8. (b)	9. (c)	10. (a)
11. (c)	12. (b)	13. (d)	14. (a)	15. (c)	16. (b)	17. (c)	18. (b)	19. (a)	20. (a)
21. (b)	22. (c)	23. (c)	24. (b)	25. (b)	26. (c)	27. (b)			

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